

PrimeSim™ Continuum Reference Manual: Device Models

Version W-2024.09, September 2024



Copyright and Proprietary Information Notice

© 2024 Synopsys, Inc. This Synopsys software and all associated documentation are proprietary to Synopsys, Inc. and may only be used pursuant to the terms and conditions of a written license agreement with Synopsys, Inc. All other use, reproduction, modification, or distribution of the Synopsys software or the associated documentation is strictly prohibited.

Destination Control Statement

All technical data contained in this publication is subject to the export control laws of the United States of America. Disclosure to nationals of other countries contrary to United States law is prohibited. It is the reader's responsibility to determine the applicable regulations and to comply with them.

Disclaimer

SYNOPSYS, INC., AND ITS LICENSORS MAKE NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE.

Trademarks

Synopsys and certain Synopsys product names are trademarks of Synopsys, as set forth at <https://www.synopsys.com/company/legal/trademarks-brands.html>. All other product or company names may be trademarks of their respective owners.

Free and Open-Source Licensing Notices

If applicable, Free and Open-Source Software (FOSS) licensing notices are available in the product installation.

Third-Party Links

Any links to third-party websites included in this document are for your convenience only. Synopsys does not endorse and is not responsible for such websites and their practices, including privacy practices, availability, and content.

www.synopsys.com

Contents

New in This Release	12
Related Products, Publications, and Trademarks	12
Conventions	12
Customer Support	13
Statement on Inclusivity and Diversity	14

1. Overview of Models	15
Using Models to Define Netlist Elements	16
Supported Models for Specific Simulators	17
Selecting Models	17
Subcircuits	17
Safe Operating Area Warnings	19
Model LEVEL Parameter Must be a Constant Value	19
Use of Example Syntax	19

2. Passive Device Models	20
Resistor Device Model and Equations	21
Wire RC Model	21
Resistor and Wire RC Model Parameter Example	22
Resistor Syntax	24
Resistor Model Selector	24
Resistor Model Equations	25
Wire Resistance Calculation	25
Wire Capacitance Calculation	26
Resistor Noise Equation	27
Noise Parameter for Resistors	28
Evaluating Flicker Noise of Resistors	28
LEVEL2 CMC R2 Resistor Model	31
CMC R2 Resistor Model — Version Updates	32
CMC R2 Resistor Model 1.0.1 Updates	32
LEVEL 2 Model Parameters	32

Contents

LEVEL 5 CMC R3 Resistor Model	35
LEVEL=5 Model Updates	35
LEVEL 5: V1.1.2 Model Updates	36
LEVEL 5: V1.1.1 Model Updates	36
LEVEL 5: V1.1.0 Model Updates	36
LEVEL 5: V1.0.0 Model Updates	36
Level 5 Instance and Model Parameters	37
CMC R3 Usage	43
CMC R3 Resistor Model Version Update	44
Capacitor Device Model and Equations	44
Capacitor Device Equations	45
Effective Capacitance Calculation	45
Capacitance Voltage Equation	46
Inductor Device Model and Equations	47
Inductor Core Models	47
Jiles-Atherton Ferromagnetic Core Model	50
Discontinuities in Inductance Due to Hysteresis	53
Optimizing the Extraction of Parameters	54
3. Diode Models	56
Diode Types	57
Using Diode Model Statements	57
Setting Control Options	58
Bypassing Latent Devices (PrimeSim Continuum only)	58
Setting Scaling Options	58
Using the Capacitor Equation Selector Option — DCAP	59
Using Control Options for Convergence	59
Specifying Junction Diode Models	59
Using the Junction Model Statement	61
Using Junction Model Parameters	61
Geometric Scaling for Diode Models	66
LEVEL=1 Scaling	66
LEVEL=3 Scaling	66
Defining Diode Models	67
Diode Current	68
Using Diode Equivalent Circuits	68
Determining Temperature Effects on Junction Diodes	70
Using Junction Diode Equations	72
Using Junction DC Equations	73

Contents

Using Diode Capacitance Equations	75
Using Noise Equations	77
Temperature Compensation Equations	77
Using the JUNCAP Models	80
JUNCAP1 Model	81
Model Parameters	82
Theory	84
JUNCAP Model Equations	85
Nomenclature	85
ON/OFF Condition	86
DC Operating Point Output	86
Temperature, Geometry, and Voltage Dependence	87
JUNCAP Capacitor and Leakage Current Model	88
JUNCAP2 Model - Level 6	91
Physical Effects	91
JUNCAP2 Model Updates	91
Usage in the PrimeSim Continuum Simulator Solution	92
Model Parameters	93
Using the Fowler-Nordheim Diode, Level 2	96
Fowler-Nordheim Diode Model Parameters LEVEL=2	96
Using Fowler-Nordheim Diode Equations	97
Fowler-Nordheim Diode Capacitances	97
Philips D500 Model (Advanced Diode Model), Level 5	97
Using the Philips D-500 Model	98
Equivalent Circuits and Equations	99
DIODE_CMC Model, Level 7	105
Diode_CMC Version Updates	106
Diode_CMC 2.0.0 Updates	106
Instance Parameters	106
Model Parameters	107
DIODE_CMC - Reduced Parameters	108
JUNCAP2 - Express Parameters	112
Diode_CMC Recovery and High-Injection Equation Parameters	113
Converting National Semiconductor Models	114
Using the Scaled Diode Subcircuit Definition	114
DC Operating Point Output of Diodes	115
4. JFET and MESFET Models	116
Overview of JFETs	117

Contents

Specifying a Model	117
Bypassing Latent Devices (PrimeSim HSPICE Only)	118
Overview of Capacitor Model	119
Model Applications	120
Convergence	121
Capacitor Equations	121
JFET and MESFET Equivalent Circuits	121
Scaling	121
JFET Current Conventions	121
JFET Equivalent Circuits	122
Transconductance	122
Output Conductance	122
JFET and MESFET Model Statements	126
Syntax	126
JFET and MESFET Model Parameters	127
ACM (Area Calculation Method) Parameter Equations	132
JFET and MESFET Capacitances	134
Gate Capacitance CAPOP=0	134
Gate Capacitance CAPOP=1	136
Gate Capacitance CAPOP=2	136
Capacitance Comparison (CAPOP=1 and CAPOP=2)	137
JFET and MESFET DC Equations	139
DC Model LEVEL=1	139
DC Model LEVEL=2	140
DC Model LEVEL=3	140
JFET and MESFET Noise Models	142
Noise Equations	142
JFET and MESFET Temperature Equations	143
Temperature Compensation Equations	145
Energy Gap Temperature Equations	145
Saturation Current Temperature Equations	146
Gate Capacitance Temperature Equations	146
Threshold Voltage Temperature Equation	147
Mobility Temperature Equation	148
Parasitic Resistor Temperature Equations	148
TriQuint (TOM) Extensions to LEVEL=3	148
LEVEL=7 TOM3 (TriQuint's Own Model III)	149
Using the TOM3 Model	150

Contents

Model Description	150
DC Equations	150
Capacitance Equations	152
LEVEL=8 Materka Model	154
Using the Materka Model	155
DC Model	155
Gate Capacitance Model	156
Noise Model	157
Level=100 TSMC JFET Model	158
JFET Level=100 Model — General Syntax	158
Instance Parameters	159
Implementation History of JFET Level=100	159
References	159
5. BJT Models	161
Overview of BJT Models	162
Selecting Models	162
BJT Control Options	163
Convergence	163
BJT Model Statement	163
BJT Basic Model Parameters	164
Bypassing Latent Devices (PrimeSim HSPICE Only)	165
Parameters	165
BJT Model Temperature Effects	170
BJT Device Equivalent Circuits	175
Scaling	175
BJT Current Conventions	176
BJT Equivalent Circuits	176
BJT Model Equations (NPN and PNP)	186
Transistor Geometry in Substrate Diodes	186
DC Model Equations	187
Substrate Current Equations	189
Base Charge Equations	189
Variable Base Resistance Equations	190
BJT Capacitance Equations	191
Base-Emitter Capacitance Equations	191
Determining Base-Emitter Diffusion Capacitance	192

Contents

Determining Base-Emitter Depletion Capacitance	192
Determining Base Collector Capacitance	193
Determining Base Collector Diffusion Capacitance	193
Determining Base Collector Depletion Capacitance	193
External Base — Internal Collector Junction Capacitance	194
Substrate Capacitance	195
Substrate Capacitance Equation: Lateral	195
Substrate Capacitance Equation: Vertical	195
Excess Phase Equation	195
Defining BJT Noise Equations	196
Defining Noise Equations	196
BJT Temperature Compensation Equations	197
Energy Gap Temperature Equations	197
Saturation/Beta Temperature Equations, TLEV=0 or 2	197
Saturation and Temperature Equations, TLEV=1	198
Saturation Temperature Equations, TLEV=3	199
Capacitance Temperature Equations	200
Parasitic Resistor Temperature Equations	202
BJT LEVEL=2 Temperature Equations	203
BJT Quasi-Saturation Model	203
Defining Noise Equations	205
Epitaxial Current Source I_{epi}	206
Epitaxial Charge Storage Elements C_i and C_x	207
Converting National Semiconductor Models	208
Defining Scaled BJT Subcircuits	209
VBIC Bipolar Transistor Model	210
Defining Noise Equations	210
History of VBIC	212
VBIC Parameters	212
Noise Analysis	217
Defining Noise Equations	217
Self-heating and Excess Phase	219
Notes on Using VBIC	219
LEVEL=6 Philips Bipolar Model (MEXTRAM)	220
LEVEL=6 Element Syntax	220
Version Control of MEXTRAM Model	221
Mextram Level 503	221
Defining Noise Equations	222

Contents

Model Parameters (503)	223
MEXTRAM Level 504	226
Mextram 504 Update Releases	228
Notes on Using MEXTRAM 503 or 504 Devices	229
LEVEL=6 Model Parameters (504)	230
MEXTRAM Level 505	239
Mextram 505 Update Releases	240
LEVEL=6 Model Parameters (505)	243
LEVEL=8 HICUM Model	248
HICUM Model Advantages	249
PrimeSim Continuum HICUM Model vs. Public HICUM Model	251
HICUM/L2 Version 3.1 Updates	251
HICUM/L2 Version 3.0 Updates	252
HICUM/L2 Version 2.4 Updates	252
HICUM/L2 Version 2.34 Updates	252
HICUM/L2 Version 2.33 Updates	252
LEVEL=8 Element Syntax	252
Syntax	253
HICUM LEVEL=2 Circuit Diagram	254
Input Netlist	254
LEVEL=8 Model Parameters	256
Internal Transistors	259
Peripheral Elements	262
External Elements	263
LEVEL=9 VBIC99 Model	268
Level 9 Usage Notes	268
LEVEL=9 Element Syntax	268
Effects of VBIC99	269
Model Implementation	270
LEVEL=9 Model Parameters	270
LEVEL=10 Philips MODELLA Bipolar Model	275
Equivalent Circuits	280
DC Operating Point Output	281
Model Equations	283
Early Factors	283
Currents	283
Base Current	285
Substrate current	286
Charges	286
Series Resistances	288

Contents

Noise Equations	289
Temperature Dependence of Parameters	290
Series Resistance	290
Depletion Capacitances	290
Temperature Dependence of Other Parameters	291
LEVEL=11 UCSD HBT Model	292
Usage Notes	292
LEVEL=11 Element Syntax	293
Model Equations	297
Current Flow	297
Charge Storage	299
Noise	305
Equivalent Circuit	306
Example Model Statement for BJT LEVEL=11	306
LEVEL=13 HICUM/L0 Model	307
HICUM/L0 Model Advantages	307
HICUM/L0 Model vs. HICUM LEVEL=2 Model	308
HICUM/L0 Version 2.10 Updates	308
HICUM/L0 Version 2.00 Updates	308
HICUM/L0 Version 1.32 Updates	308
HICUM/L0 Version 1.31 Updates	309
HICUM/L0 Version 1.3 Updates	309
LEVEL=13 Element Syntax	310
LEVEL=13 Model Parameters	310
References	315
6. Magnetic Tunnel Junction Model	316
Overview of MTJ Model	316
Level1 BSIM BDMC STT-MRAM Model	316
Using the STT-MRAM Model	316
Model Revisions	317
STT-MRAM Model — General Syntax	317
STT-MRAM Model Parameters and Equations	317
A. Finding Device Libraries	318
Overview of Library Listings	318
Analog Device Models	319

Contents

Behavioral Device Models	322
Bipolar Transistor Models	323
Diode Models	324
JFET and MESFET Models	327
Transmission Line Models	329

About the PrimeSim Continuum Reference Manual: Device Models

The *PrimeSim Continuum Reference Manual: Device Models* describes standard models that you can use when simulating your circuit designs in the PrimeSim Continuum solution.

This preface includes the following sections:

- [New in This Release](#)
- [Related Products, Publications, and Trademarks](#)
- [Conventions](#)
- [Customer Support](#)
- [Statement on Inclusivity and Diversity](#)

New in This Release

Information about new features, enhancements, and changes, known limitations, and resolved Synopsys Technical Action Requests (STARs) is available in the PrimeSim HSPICE Release Notes on the SolvNetPlus site.

Related Products, Publications, and Trademarks

For additional information about the PrimeSim Continuum solution, see the documentation on the Synopsys SolvNetPlus support site at the following address:

<https://solvnetplus.synopsys.com>

Conventions

The following conventions are used in Synopsys documentation.

Convention	Description
Courier	Indicates syntax, such as <code>write_file</code> .
<i>Courier italic</i>	Indicates a user-defined value in syntax, such as <code>write_file design_list</code>

Convention	Description
Courier bold	Indicates user input—text you type verbatim—in examples, such as <code>prompt> write_file top</code>
Purple	• Within an example, indicates information of special interest. • Within a command-syntax section, indicates a default, such as <code>include_enclosing = true false</code>
[]	Denotes optional arguments in syntax, such as <code>write_file [-format fmt]</code>
...	Indicates that arguments can be repeated as many times as needed, such as <code>pin1 pin2 ... pinN.</code>
 	Indicates a choice among alternatives, such as <code>low medium high</code>
\	Indicates a continuation of a command line.
/	Indicates levels of directory structure.
Bold	Indicates a graphical user interface (GUI) element that has an action associated with it.
Edit > Copy	Indicates a path to a menu command, such as opening the Edit menu and choosing Copy .
Ctrl+C	Indicates a keyboard combination, such as holding down the Ctrl key and pressing C.

Customer Support

Customer support is available through SolvNetPlus.

Accessing SolvNetPlus

The SolvNetPlus site includes a knowledge base of technical articles and answers to frequently asked questions about Synopsys tools. The SolvNetPlus site also gives you access to a wide range of Synopsys online services including software downloads, documentation, and technical support.

To access the SolvNetPlus site, go to the following address:

<https://solvnetplus.synopsys.com>

If prompted, enter your user name and password. If you do not have a Synopsys user name and password, follow the instructions to sign up for an account.

If you need help using the SolvNetPlus site, click REGISTRATION HELP in the top-right menu bar.

Contacting Customer Support

To contact Customer Support, go to <https://solvnetplus.synopsys.com>.

Statement on Inclusivity and Diversity

Synopsys is committed to creating an inclusive environment where every employee, customer, and partner feels welcomed. We are reviewing and removing exclusionary language from our products and supporting customer-facing collateral. Our effort also includes internal initiatives to remove biased language from our engineering and working environment, including terms that are embedded in our software and IPs. At the same time, we are working to ensure that our web content and software applications are usable to people of varying abilities. You may still find examples of non-inclusive language in our software or documentation as our IPs implement industry-standard specifications that are currently under review to remove exclusionary language.

1

Overview of Models

Describes the elements and models you can use to create a netlist in the PrimeSim Continuum solution.

The PrimeSim Continuum solution ships many examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

A circuit *netlist* describes the basic functionality of an electronic circuit that you are designing. In the PrimeSim Continuum solution format, a netlist consists of a series of *elements* that define the individual components of the overall circuit. You can use your PrimeSim HSPICE -format netlist to simulate your circuit to help you verify, analyze, and debug your design, before you turn that design into an actual electronic circuit.

Your netlist can include several types of elements:

- Passive elements, see [Passive Device Models](#)
 - Resistors
 - Capacitors
 - Inductors
 - Mutual Inductors
- Active elements:
 - Diodes, see [Diode Models](#)
 - Junction Field Effect Transistors (JFETs), see [JFET and MESFET Models](#)
 - Metal Semiconductor Field Effect Transistors (MESFETs), see [JFET and MESFET Models](#)
 - Bipolar Junction Transistors (BJTs), see [BJT Models](#)

- Transmission lines (see the *PrimeSim™ HSPICE® User Guide: Signal Integrity Modeling and Analysis*):
 - S-Element
 - T-Element
 - U-Element
 - W-Element
- Metal Oxide Semiconductor Field Effect Transistors (MOSFETs), see the *PrimeSim™ Continuum Reference Manual: MOSFET Models*.

This chapter discusses these topics:

- [Using Models to Define Netlist Elements](#)
- [Supported Models for Specific Simulators](#)
- [Subcircuits](#)
- [Safe Operating Area Warnings](#)
- [Model LEVEL Parameter Must be a Constant Value](#)
- [Use of Example Syntax](#)

Using Models to Define Netlist Elements

A series of standard models have been provided with the PrimeSim Continuum solution. Each model is like a template that defines various versions of each supported element type used in the PrimeSim HSPICE solution -format netlist. Individual elements in your netlist can refer to these standard models for their basic definitions. When you use these models, you can quickly and efficiently create a netlist and simulate your circuit design.

Eight different versions (or *levels*) of JFET and MESFET models for use with the PrimeSim Continuum solution are supplied. An individual JFET or MESFET element in your netlist can refer to one of these models for its definition. That is, you do not need to define all of the characteristics (called *parameters*) of each JFET or MESFET element within your netlist.

Referring to standard models in this way reduces the amount of time required to:

- Create the netlist.
- Simulate and debug your circuit design.
- Turn your circuit design into actual circuit hardware.

Within your netlist, each element that refers to a model is known as an *instance* of that model. When your netlist refers to predefined device models, you reduce both the time required to create and simulate a netlist, and the risk of errors, compared to fully defining each element within your netlist.

Supported Models for Specific Simulators

This manual describes individual models that have been provided. The PrimeSim Continuum solution supports a specific subset of the available models. This manual describes the Synopsys device models for passive and active elements. You can include these models in your PrimeSim Continuum-format netlists.

Selecting Models

To specify a device in your netlist, use both an element and a model statement. The element statement uses the simulation device model name to reference the model statement.

The following example uses the `MOD1` name to refer to a BJT model. The example uses a NPN model type to describe a NPN transistor.

```
Q3 3 2 5 MOD1 <parameters>
.MODEL MOD1 NPN <parameters>
```

You can specify parameters in both element and model statements. If you specify the same parameter in both an element and a model, then the element parameter (local to the specific instance of the model) always overrides the model parameter (global default for all instances of the model, if you do not define the parameter locally).

The model statement specifies the type of device—for example, a BJT, the device type might be NPN or PNP.

Subcircuits

`Xsubcircuit_name` adds an instance of a subcircuit to your netlist. You must already have defined that subcircuit in your netlist by using a `.MACRO` or a `.SUBCKT` command.

If you initialize a non-existent subcircuit node, the PrimeSim Continuum solution generates a warning message. This can occur if you use an existing `.ic` file (initial conditions) to initialize a circuit that you modified since you created the `.ic` file.

Syntax

```
X[subcircuit_name] n1 [n2 n3 ...] subnam
[parnam = val &] [M = val] [S=val] [DTEMP=val]
```

Argument	Definition
<i>Xsubcircuit_name</i>	Subcircuit element name. Must begin with a X, followed up to 15 alphanumeric characters.
<i>n1 ...</i>	Node names for external reference.
<i>subnam</i>	Subcircuit model reference name.
<i>parnam</i>	A parameter name set to a value (<i>val</i>) for use only in the subcircuit. It overrides the parameter value in the subcircuit definition, but is overridden by a value set in a <code>.PARAM</code> statement.
<i>M</i>	Multiplier. Makes the subcircuit appear as M subcircuits in parallel. You can use this multiplier to characterize circuit loading. The PrimeSim Continuum solution does not need additional calculation time to evaluate multiple subcircuits. Do not assign a negative value or zero as the M value.
<i>S</i>	Scales a subcircuit. For more information about the S parameter, see S (Scale) Parameter in the <i>PrimeSim HSPICE User Guide: Basic Simulation and Analysis</i> . This keyword functions only if you set the <code>.OPTION HIER_SCALE</code> command.
<i>DTEMP</i>	Element temperature difference with respect to the circuit temperature in Celsius. Default value=0.0. This argument sets a different temperature in subcircuits than the global temperature. This keyword functions only when the you set the <code>.OPTION XDTEMP</code> command.

Example 1

The following example calls a subcircuit model named `MULTI`. It assigns the `WN = 100` and `LN = 5` parameters in the `.SUBCKT` statement (not shown). The subcircuit name is `x1`. All subcircuit names must begin with `x`.

```
X1 2 4 17 31 MULTI WN = 100 LN = 5
```

Example 2

This example defines a subcircuit named `YYY`. The subcircuit consists of two 1 ohm resistors in series. The `.IC` statement uses the `VCC` passed parameter to initialize the `NODEX` subcircuit node.

```
.SUBCKT YYY NODE1 NODE2 VCC = 5V
.IC NODEX = VCC
  R1 NODE1 NODEX 1
  R2 NODEX NODE2 1
.EOM
XXXX 5 6 YYY VCC = 3V
```

Safe Operating Area Warnings

The following warning message is issued when terminal voltages of a device (MOSFET, BJT, Diode, Resistor, Capacitor, and so on) exceed their safe operating area (SOA):

```
**warning** (filename:line number)  resulted during SOA check <node  
voltage name> (=val) of <device/element name> has exceeded <node voltage  
name>_max (=val)
```

To turn the SOA warning message off, use `.option WARN=0`.

See the following control options for details:

- `.OPTION WARN`
- `.OPTION MAXWARNS`

Model LEVEL Parameter Must be a Constant Value

The LEVEL parameter may not be entered as an expression or by using single quotation marks. Such action results in the following error message:

```
**error** (inv.sp:31)  level should be defined as a constant value and it  
cannot be an expression or in single quotes.
```

Use of Example Syntax

To copy and paste proven syntax, use the demonstration files shipped with your installation of the PrimeSim Continuum solution (see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists*). Attempting to copy and paste from the book or help documentation may present unexpected results, as text used in formatting may include hidden characters or white space, for visual clarity.

2

Passive Device Models

Describes passive devices you can include in the PrimeSim HSPICE -format netlist, including resistor, inductor, and capacitor models.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Input Demonstration Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

You can use the set of passive model definitions in conjunction with element definitions to construct a wide range of board and integrated circuit-level designs. Passive device models let you include the following in any analysis:

- Transformers
- PC board trace interconnects
- Coaxial cables
- Transmission lines

The wire element model is specifically designed to model the RC delay and RC transmission line effects of interconnects, at both the IC level and the PC board level.

To aid in designing power supplies, a mutual-inductor model includes switching regulators and several other magnetic circuits including a magnetic-core model and element. To specify precision modeling of passive elements, you can use the following types of model parameters:

- Geometric
- Temperature
- Parasitic

These topics are covered in the following sections:

- [Resistor Device Model and Equations](#)
- [Noise Parameter for Resistors](#)
- [Evaluating Flicker Noise of Resistors](#)

- [LEVEL2 CMC R2 Resistor Model](#)
- [LEVEL 5 CMC R3 Resistor Model](#)
- [Capacitor Device Model and Equations](#)
- [Inductor Device Model and Equations](#)

Resistor Device Model and Equations

The following section describes equations for Wire RC and Resistor models.

For a listing of output templates for resistor models see Element Template Listings (PrimeSim HSPICE Only), Table 33, in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

Wire RC Model

You can use the `.MODEL` statement to include a Wire RC model in your PrimeSim Continuum netlist and evaluate both thermal and flicker noise in the PrimeSim Continuum solution. For a general description of the `.MODEL` statement, see `.MODEL` in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

Syntax

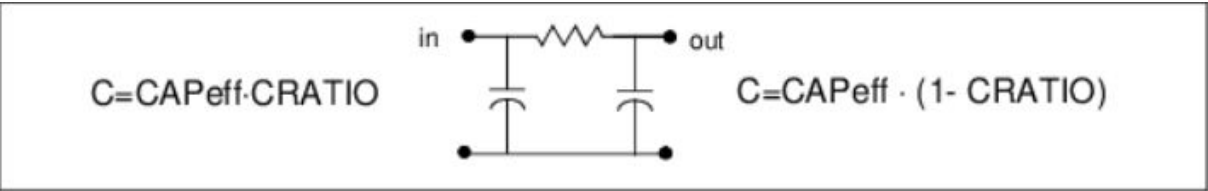
```
.MODEL MNAME R KEYWORD=value [CRATIO=val]
```

The wire element RC model is a CRC (pi) model. Use the `CRATIO` wire model parameter to allocate the parasitic capacitance of the wire element (between the input capacitor and the output capacitor of the model). This allows for symmetric node impedance for bidirectional circuits, such as buses.

Parameter	Description
mname	Model name. Elements use this name to reference the model.
R	Specifies a wire model.
keyword	Any model parameter name.

Parameter	Description
CRATIO	Ratio to allocate the total wire element parasitic capacitance. This is the capacitance between the capacitor connected to the input node and the capacitor connected to the output node of the wire element pi model. You can assign a value between 0 and 1 to CRATIO. Default=0.5. • 0 Assigns all of the parasitic capacitance (CAP_{eff}) to the output node. • CRATIO values smaller than 0.5 assign more of the capacitance to the output node than to the input node. • Values greater than 0.5 assign more of the capacitance to the input node than to the output node. • 0.5 Assigns half of the parasitic capacitance to the input node and half to the output node. • 1 Assigns all of the parasitic capacitance to the input node. If you set a CRATIO value outside the 0 to 1.0 range, the simulator displays a warning, sets CRATIO to 0.5, and continues the analysis.

Figure 1 Wire Model Example



A wire-model resistor behaves like an elementary transmission line (see T-element (Ideal Transmission Lines) in the *PrimeSim Continuum User Guide: Elements*), if the `.MODEL` statement specifies an optional capacitor (from the n2 node to a bulk or ground node). The bulk node functions as a ground plane for the wire capacitance.

A wire has a drawn length and a drawn width. The resistance of the wire is the effective length multiplied by `RSH` and divided by the effective width.

Resistor and Wire RC Model Parameter Example

To avoid syntactic conflicts, if a resistor model uses the same name as a parameter for `rval` in the element statement, then the simulation uses the model name.

In the following example, `R1` assumes that `REXX` refers to the model, and not to the parameter.

```
.PARAMETER REXX=1
R1 1 2 REXX
.MODEL REXX R RES=1
```

Wire RC Model Parameter Syntax

```
Rxxx n1 n2 [bulk] [mname Rval] [TC1 TC2 TC] [SCALE=val] [M=val]
+ [AC=val] [DTEMP=val] [L=val] [W=val] [C=val]
+ [NOISE=val]
```

```
Rxxx n1 n2 [mname] [R=]resistance [TC1=val]
+ [TC2=val] [TC=val] [SCALE=val] [M=val]
+ [AC=val] [DTEMP=val] [L=val] [W=val] [C=val]
+ [NOISE=val]
```

Table 1 Wire Model Parameters

Name (Alias)	Units	Default	Description
BULK	gnd		Default reference node for capacitance. <i>[bulk]</i> node is optional.
CAP	F	0	Default capacitance.
CAPSW	F/m	0	Sidewall fringing capacitance.
COX	F/m ²	0	Bottomwall capacitance.
DI	-	0	Relative dielectric constant.
DLR	m	0	Difference between the drawn length and the actual length (for resistance calculation only). The capacitance calculation uses DW. $DLR_{eff}=DLR \cdot SCALM$
DW	m	0	Difference between the drawn width and the actual width. $DW_{eff}=DW \cdot SCALM$
L	m	0	Default length of the wire. $L_{scaled}=L \cdot SHRINK \cdot SCALM$
LEVEL	-	-	Model selector (not used).
RAC	ohm	-	Default AC resistance (the RACeff default is Reff).
RES	ohm	0	Default resistance.
RSH	-	0	Sheet resistance/square.
SHRINK	-	1	Shrink factor.
TC1C	1/deg	0	First-order temperature coefficient for capacitance.
TC2C	1/deg ²	0	Second-order temperature coefficient for capacitance.
TC1R	1/deg	0	First-order temperature coefficient for resistance.
TC2R	1/deg ²	0	Second-order temperature coefficient for resistance.
VC1R	1/volt	0	First-order voltage-bias coefficient for resistance.
VC2R	1/volt ²	0	Second-order voltage-bias coefficient for resistance.

Table 1 Wire Model Parameters (Continued)

Name (Alias)	Units	Default	Description
THICK	m	0	Dielectric thickness.
TREF	deg C	TNOM	Temperature reference for model parameters.
W	m	0	Default width of the wire. $W_{scaled} = W \cdot SHRINK \cdot SCALM$

Wire RC model can also be enabled without the `.MODEL` statement. The topology is the same as the one with the `.MODEL` statement.

Note:

In this case, the CRATIO is fixed to 0.5 and the reference node is ground.

Wire RC without the `.MODEL` statement example:

```
Rxxx n1 n2 [bulk] R=val C=val
```

Resistor Syntax

```
Rxxx n1 n2 mname [R=]resistance [TC1=val]+ [TC2=val] [VC1=val] [VC2=val]
[SCALE=val] [M=val] [AC=val]+ [DTEMP=val] [L=val] [W=val] [C=val]
[NOISE=val]
```

Supported instance parameters above include: R, TC1, TC2, SCALE, M, AC, DTEMP, L, W, C, and NOISE.

TC1 and TC2 are aliases for TC1R and TC2R. Resistor syntax is described in Resistors in *PrimeSim Continuum® User Guide: Elements*.

Resistor Model Selector

For multiple resistor models, you can use the automatic model selector in the PrimeSim Continuum solution to find the proper model for each resistor.

The model selector syntax is based on a common model root name with a unique extension for each model.

Example

```
.model REXX.1 R LMIN=0.5 LMAX=0.7 WMIN=0.1 WMAX=0.5 RES=1.2
.model REXX.2 R LMIN=0.7 LMAX=0.9 WMIN=0.1 WMAX=0.5 RES=1.3
```


You can then use the standard resistor model call to map the models to an element declaration:

```
R1 1 2 REXX L=0.6 W=0.5
```

The resistor model selector uses the following criteria:

```
LMIN <= L < LMAX  
WMIN <= W < WMAX
```

Resistor Model Equations

This section contains equations for different characteristics of resistors.

Wire Resistance Calculation

You can specify the wire width and length in both the element and model statements. The element values override the model values.

- To scale the element width and length, use `.OPTION SCALE` and the `SHRINK` model parameter.
- To scale the model width and length, use `.OPTION SCALM` and the `SHRINK` model parameter.

The following equations calculate the effective width and length:

$$W_{eff} = W_{scaled} \cdot 2 \cdot DW_{eff}$$

$$L_{eff} = L_{scaled} \cdot 2 \cdot DL_{Reff}$$

If you specify element resistance:

$$R_{eff} = \frac{R \cdot SCALE(element)}{M}$$

Otherwise, if

$$(w_{eff} \cdot l_{eff} \cdot rsh)$$

is greater than zero, then:

$$R_{eff} = \frac{L_{eff} \cdot RSH \cdot SCALE(element)}{M \cdot W_{eff}}$$

If $(w_{eff} \cdot l_{eff} \cdot rsh)$ is zero, then:

$$R_{eff} = \frac{RES \cdot SCALE(element)}{M}$$

If you specify AC resistance in the element, then:

$$RAC_{eff} = \frac{AC \cdot SCALE(element)}{M}$$

Otherwise, if the model specifies RAC , the RAC_{eff} equation uses RAC :

$$RAC_{eff} = \frac{RAC \cdot SCALE(element)}{M}$$

If the model does not specify either AC resistance or RAC , then the equation defaults to:

$$RAC_{eff} = R_{eff}$$

If the resistance is less than the $RESMIN$ option, then the RAC_{eff} equation resets it to the $RESMIN$ value, and issues a warning message.

$$RESMIN = \frac{1}{GMAX \cdot 1000 \cdot M}$$

Wire Capacitance Calculation

The effective length is the scaled drawn length, $l_{ess} (2 \cdot DL_{eff})$.

- L_{eff} represents the effective length of the resistor, from physical edge to physical edge.
- DW_{eff} is the distance from the drawn edge of the resistor to the physical edge of the resistor.

$$L_{eff} = L_{scaled} - 2 \cdot DW_{eff}$$

The effective width is the same as the width used in the resistor calculation.

$$W_{eff} = W_{scaled} - 2 \cdot DW_{eff}$$

If you specify the element capacitance, C :

$$CAP_{eff} = C \cdot SCALE(element) \cdot M$$

Otherwise, the equation calculates the capacitance from the L_{eff} , W_{eff} , and COX values:

$$CAP_{eff} = M \cdot SCALE(element) \cdot [L_{eff} \cdot W_{eff} \cdot COX] + 2 \cdot (L_{eff} + W_{eff}) \cdot CAPSW]$$

Computing the bottom-wall capacitance, COX , is based on a hierarchy of defaults and specified values, involving:

- Dielectric thickness ($THICK$)
- Relative dielectric constant (DI)

Whether you specify a `COX` value affects how the PrimeSim Continuum solution uses the equation:

- If you specify `COX=value`, then the equation uses the *value*.
- If you do not specify `COX`, but you do specify a value other than zero for `THICK` (the dielectric thickness):
 - If you specify a non-zero value for `DI=value`, then:

$$COX = \frac{DI \cdot \epsilon_0}{THICK}$$

- If you do not specify a `DI` value or if the value is zero, then:

$$COX = \frac{\epsilon_{OX}}{THICK}$$

The following values apply to the preceding two equations:

$$\epsilon_0 = 8.8542149e-12 \text{ F/meter}$$

$$\epsilon_{OX} = 3.453148e-11 \text{ F/meter}$$

- If you do not specify `COX` and `THICK = 0`, this is an error.
 - If you specify only the model capacitance (`CAP`), then:

$$CAP_{eff} = CAP \cdot SCALE(element) \cdot M$$

- If you specify the capacitance, but you do not specify the bulk node, then the resistor model does not evaluate the capacitance and issues a warning message.

Resistor Noise Equation

The following equation models the thermal noise of a resistor:

$$inr = \left(NOISE \cdot \frac{4kT}{R_{val}} \right)^{\frac{1}{2}}$$

In the preceding equation, `NOISE` is a model parameter (default=1). To eliminate the contribution of resistor noise, specify the `NOISE` parameter in a resistor model.

Parameter	Description
RX	Transfers the function of thermal noise to the output. This is not noise, but is a transfer coefficient, which reflects the contribution of thermal noise to the output. For example: $V(\text{output}) = I(\text{local}) \cdot r_x$ (from local to output) Where $V(\text{output})$ is the noise voltage at the output port, $I(\text{local})$ is the local noise current in the specific noise element. It is clear that r_x should have an unit of impedance, therefore it is referred as transimpedance. By summarizing all the contributions (power) from each independent noisy element, we can get the total noise contribution(power) at the output port.
TOT, V^2/Hz	Total output noise: $TOT = RX^2 \cdot inr^2$

Noise Parameter for Resistors

Resistor models generate electrical thermal noise. However, some tasks such as macro modeling require noiseless resistor models.

- If you set *noise*=1 (default) or if you do not specify the noise parameter, the PrimeSim Continuum solution models a resistor that generates noise.
- If you do not want the resistor model to generate thermal noise, set *noise*=0 in the instance statement (noiseless resistor model).

Example

This example is located in the following directory:

```
$installdir/demo/hspice/apps/noise_apps.sp
```

In this example, *rd* is a 1-ohm noiseless resistor that connects between node 1 and node 6.

Evaluating Flicker Noise of Resistors

This section describes the flicker noise equation and parameters. It also describes how to test for flicker noise in the PrimeSim Continuum solution.

The following equation supports the flicker noise model for resistors:

$$\text{Noise}(sid) = \frac{KF \times (I^{AF})}{(Leff^{Lf}) \times (Weff^{Wf}) \times (f^{Ef})}$$

Where:

Parameter	Description	Unit
f	Frequency	Hz
I	Current	A
L_{eff}	Effective length ($L_{drawn} - dL$)	m
$Noise(sid)$	Noise spectrum density	$A^2 \text{ Hz}$
W_{eff}	Effective width ($W_{drawn} - dw$)	m

The following table lists the parameters and descriptions for the flicker noise model:

Parameter	Description	Unit	Default	Bin
AF	exponent of current (> 0)	none	2	no
EF	exponent of frequency (> 0)	none	1	no
KF	flicker noise coefficient (≥ 0)	$A^{(2-AF)} \times m^{(LF+WF)} \times Hz^{(EF-1)}$	0	no
LF	exponent of effective length (> 0)	none	1	no
WF	exponent of effective width (> 0)	none	1	no

Example of Using Flicker Noise Parameters

To use the flicker noise parameters in a resistor model, specify the flicker noise parameters (KF , AF , LF , WF , and EF) in the resistor model card. For example:

```
.model Res1 R noise=1 kf=6.0e-28 af=2 lf=1 wf=1 ef=1 l=1u dlr=0.01u
w=10u dw=0.01u
```

If these parameters are not specified, the PrimeSim Continuum solution uses their default values, and the flicker noise is 0. Observe that these parameters all have limitations: $KF \geq 0$, AF , LF , WF , and $EF > 0$. If their values exceed the limitations, the PrimeSim Continuum solution issues warning messages. The parameters l , w , dl , and dw must also be specified in the model card for the PrimeSim Continuum solution to evaluate the effective length and width. If these parameters are not specified, their default value is 0.

This means that the effective length and width are 0, and there is a divide-by-zero error in the evaluation of the flicker noise equation. In this case, the PrimeSim Continuum solution issues an error message and abort the simulation.

Controlling the Evaluation of Noise

The PrimeSim Continuum solution uses the NOISE parameter in a model as a switch to control the evaluation of thermal and flicker noise. Its value is 0 or 1. For example:

```
.model Res1 R noise=1 kf=6.0e-28 af=2 lf=1 wf=1 ef=1 l=1u dlr=0.01u w=10u
dw=0.01u
.model Res2 R noise=0 kf=6.0e-28 af=2 lf=1 wf=1 ef=1 l=1u dlr=0.01u
w=10u dw=0.01u
R1 1 2 Res1 50
R2 2 3 Res2 50
```

In the above example, resistor R1 uses model Res1 which specifies noise=1, and resistor R2 uses model Res2 which specifies noise=0. The noise analysis, R1 has noise and R2 is noise-free. The default value of model parameter NOISE is 1.

Instance of Parameter NOISE: Noiseless Resistor

Sometimes a noiseless resistor is required. The resistor can be noiseless even if there are noise parameters set in the model. This is because the parameter NOISE can be both an instance parameter and a model parameter. If NOISE is set in both the instance and the model, the instance parameter overrides the model parameter. For example:

```
.model Res1 R noise=1 kf=6.0e-28 af=2 lf=1 wf=1 ef=1 l=1u dlr=0.01u
w=10u dw=0.01u
.model Res2 R noise=0 kf=6.0e-28 af=2 lf=1 wf=1 ef=1 l=1u dlr=0.01u
w=10u dw=0.01u
R1 1 2 Res1 50
R2 2 3 Res2 50
R3 3 4 Res1 50 noise=0
R4 4 5 Res2 50 noise=1
```

In the example, resistors R1 and R3 use model Res1 and resistors R2 and R4 use model Res2. R1 has noise and R2 is noiseless, since these resistors use the noise parameter defined in the model. Resistor R3 is noiseless and R4 has noise because the noise instance parameter is specified for each.

Test Case for Both Thermal and Flicker Noise

The following example uses the `.noise` analysis syntax to test for both thermal and flicker noise in a resistor.

```
v1 1 0 10 ac=1
R1 1 2 Res1 50 $noise=1
R2 2 3 Res2 50 $noise=1
R3 3 4 Res1 50 noise=0
R4 4 0 Res2 50 noise=1
```

```
.model Res1 R noise=1 kf=6.0e-28 af=2 lf=1 wf=1 ef=1
+ l=1u dlr=0.01u w=10u dw=0.01u
.model Res2 R noise=0 kf=6.0e-28 af=2 lf=1 wf=1 ef=1
+ l=1u dlr=0.01u w=10u dw=0.01u
.model Res3 R $noise=1
.model Res4 R $noise=1
.options post list
.ac dec 10 10k 100k
.noise v(2,3) v1 10
.print noise inoise onoise
.op
.end
```

In the *.list* file below, the output shown as *rs* is the thermal noise and $1/f$ is the flicker noise:

```
**** resistor squared noise voltages (sq v/hz)
element      0:r1      0:r2      0:r3      0:r4
rs           5.145e-20    0.        0.        5.145e-20
1/f          5.924e-20    0.        0.        5.924e-20
total        1.107e-19    0.        0.        1.107e-19
rx           12.5000     37.5000    12.5000    12.5000
```

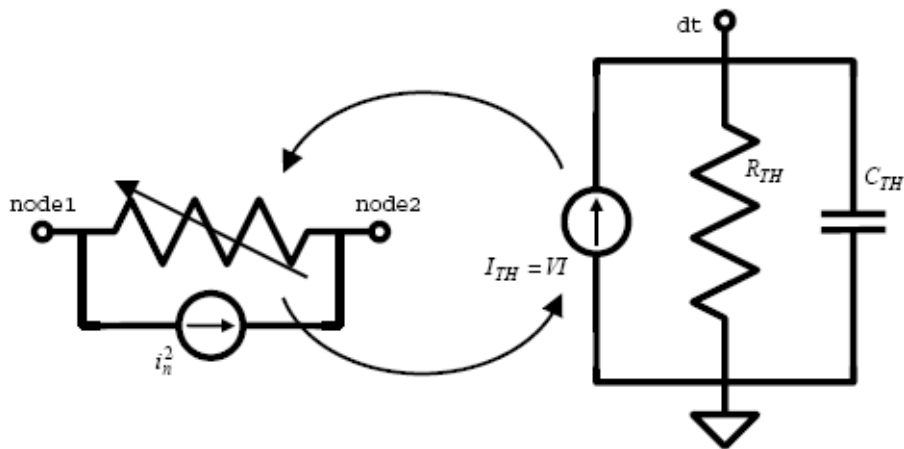
LEVEL2 CMC R2 Resistor Model

R2_CMC is a nonlinear 2-terminal resistor model. The model does not include parasitic capacitances. Optionally, the model can include self-heating. R2_CMC is well behaved and does not have the numerical problems that can arise in polynomial models. Although empirical, the form of the nonlinearity can be modelled reasonably well, especially for velocity saturation effects which are important in short resistors. The CMC_R2 model is implemented as LEVEL=2 in the RESISTOR models.

LEVEL=2 Element Syntax

```
Rxxx n1 n2 mname [M=val] instanceParameters
.MODEL mname r LEVEL = 2 [keyword = val] ...
```

Figure 2 R2_CMC Model Equivalent Network (thermal sub-network is optional)



CMC R2 Resistor Model — Version Updates

This section covers the following topics:

CMC R2 Resistor Model 1.0.1 Updates

CMC R2 Resistor Model 1.0.1 implements the following features:

- Added an optional temperature node for self-heating analysis.
- Revised flicker noise implementation for proper results in time-domain noise simulations.

LEVEL 2 Model Parameters

[Table 2](#) and [Table 3](#) describe the instance and model parameters, respectively.

Table 2 Instance Parameters

Name	Default	Min.	Max.	Units	Description
m	1	0	∞	-	Multiplicity factor (number in parallel)
w	1.0e-6	0.0	∞	m	Design width of resistor body
l	1.0e-6	0.0	∞	m	Design length of resistor body
r	100.0	0.0	∞	Ω	Resistance (per segment, total resistance is r/m)

Table 2 Instance Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
c1	1	0	1	-	Contact at terminal 1: 0=no 1=yes
c2	1	0	1	-	Contact at terminal 2: 0=no 1=yes
isnoisy	1	0	1	-	Switch for noise: 0=no 1=yes
sw_et	1	0	1	-	Switch for turning off self-heating: 0=exclude 1=include
trise	0.0	-	-	°C	Local temperature delta to ambient (before self-heating)

Table 3 Model Parameters

Name	Default	Min.	Max.	Units	Description
level	2	-	-	-	Model level
version	-	-	-	-	Model version
revision	-	-	-	-	Model revision (sub-version)
afn	2.0	0.0	∞	-	Flicker noise current exponent
bfm	1.0	0.0	∞	-	Flicker noise 1/f exponent
cth0	0.0	0.0	∞	sW / K	Thermal capacitance fixed component
ctha	0.0	0.0	∞	$(sW) / K\mu m^2$	Thermal capacitance area component
cthp	0.0	0.0	∞	$(sW) / K\mu m$	Thermal capacitance perimeter component
dxle	0.0	-	-	μm	Length delta for field calculation
gmin	1.0e-12	0.0	∞	-	
gth0	1.0e+6	0.0	∞	W / K	Thermal capacitance fixed component
gtha	0.0	0.0	∞	$W / K\mu m^2$	Thermal capacitance area component
gthp	0.0	0.0	∞	$W / K\mu m$	Thermal capacitance perimeter component
jmax	100.0	0.0		$A / \mu m$	Maximum current density
kfn	0.0	0.0	∞	-	Flicker noise coefficient (unit depends on afn)
lmax	9.9e99	0.0	∞	μm	Maximum allowed drawn length
lmin	0.0	0.0	∞	μm	Minimum allowed drawn length

Table 3 Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
p2	0.0	0.0	1.0-p3	-	Quadratic field coefficient factor: $E_{c2} = p2 * p3 / q3$
p3	0.0	0.0	1.0	-	Linear field coefficient factor: $E_{c2} = p2 * p3 / q3$
q2	0.0	0.0	∞	$\mu m / V$	1/field at which the quadratic field coefficient activates
q3	0.0	0.0	∞	$\mu m / V$	1/field at which the linear field coefficient activates
rsh	100.0	0.0	∞	Ω	Sheet resistance
rthresh	1.0e-3	0.0	∞	Ω	Threshold to switch to resistance form
scale	1.0	0.0	1.0	-	Scale factor for instance geometries
shrink	0.0	0.0	100.0	%	Shrink percentage for instance geometries
sw_efgeo	0	0	1	-	Switch for electric field geometry calculation: 0=design 1=effective
sw_fngeo	0	-	-	-	Switch for flicker noise geometry calculation: 0=design 1=effective
tc1	0.0	-	-	/K	Resistance linear TC
tc1kfn	0.0	-	-	-	Flicker noise coefficient linear TC
tc1l	0.0	-	-	$\mu m / K$	Resistance linear TC length coefficient
tc1w	0.0	-	-	$\mu m / K$	Resistance linear TC width coefficient
tc2	0.0	-	-	/K ²	Resistance quadratic TC
tc2l	0.0	-	-	$\mu m / K^2$	Resistance quadratic TC length coefficient
tc2w	0.0	-	-	$\mu m / K^2$	Resistance quadratic TC width coefficient
tmax	500.0	27.0	1000.0	°C	Maximum ambient temperature
tmaxclip	500.0	27.0	1000.0	°C	Clip maximum temperature
tmin	-100.0	-250.0	27.0	°C	Minimum ambient temperature
tminclip	-100.0	-250.0	27.0	°C	Clip minimum temperature
tnom	27.0	-250	1000.0	°C	Nominal (reference) temperature

Table 3 Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
wmax	9.9e99	0.0	∞	μm	Maximum allowed drawn width
wmin	0.0	0.0	∞	μm	Minimum allowed drawn width
xl	0.0			μm	Length offset (total)
xw	0.0			μm	Width offset (total)

Usage

With Model Card:

```
rinstanceName (node1 node2) modelName instanceParameters
.model modelName r2_cmc modelParameters
```

Without Model Card:

```
rname (node1 node2) r=resistanceValue [tc1=tc1Value] [tc2=tc2Value]
```

Example

```
r137 (n1 n2) rnpoly1 w=1u l=10u
.model rnpoly1 r2_cmc
+ rsh=100.0 xl=0.2u xw=-0.05u
+ p3=0.12 q3=1.63 p2=0.014 q2=3.79
r138 (n2 n3) r=110.0 tc1=1.0e-3
```

LEVEL 5 CMC R3 Resistor Model

R3_CMC model is a nonlinear 3-terminal resistor model that includes self-heating, velocity saturation, statistical variations, and parasitic capacitance and currents. The core depletion pinching model formulations is for p-n junctions of diffused resistors, but is also applicable for MOS behavior of polysilicon resistors. Since p-n junction pinching controls JFET device behavior, the R3_CMC model is also applicable to JFETs. The CMC_R3 model is implemented as LEVEL=5 in the RESISTOR models.

LEVEL=5 Model Updates

- [LEVEL 5: V1.1.2 Model Updates](#)
- [LEVEL 5:V1.1.1 Model Updates](#)

- [LEVEL 5:V1.1.0 Model Updates](#)
- [LEVEL 5:V1.0.0 Model Updates](#)

LEVEL 5: V1.1.2 Model Updates

Supports CMC R3 V1.1.2.

LEVEL 5: V1.1.1 Model Updates

Changes only to comments in code.

LEVEL 5: V1.1.0 Model Updates

Improved pinch-off and temperature modeling and fixed flicker noise.

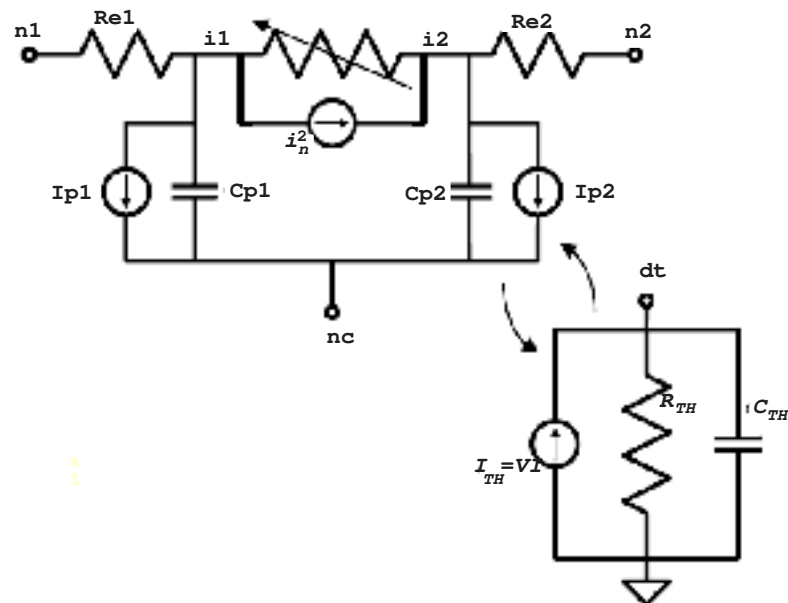
LEVEL 5: V1.0.0 Model Updates

V1.0.0 Initial release

LEVEL=5 Element Syntax

```
Rxxx n1 nc n2 mname [M=val] instanceParameters
.MODEL mname r LEVEL=5 [keyword = val] ...
```

Figure 3 R3_CMC Model Equivalent Network (thermal sub-network is optional)



Level 5 Instance and Model Parameters

Table 4, Table 5, and Table 6 describe instance, special model parameters, and model parameters, respectively.

Table 4 CMC R3 Instance Parameters

Name	Default	Min.	Max.	Units	Description
a1	0.0	0.0	∞	m ²	Area of node n1 partition
a2	0.0	0.0	∞	m ²	Area of node n2 partition
c1	0	0	∞	-	# contacts at n1 terminal
c2	0	0	∞	-	# contacts at n2 terminal
l	1.0e-6	0.0	∞	m	Design length of resistor body
m	1	0	∞	-	Multiplicity factor (number in parallel)
nsmm_l	0.0	-	-	-	Number of standard deviations of local variation for l
nsmm_rsh	0.0	-	-	-	Number of standard deviations of local variation for rsh
nsmm_w	0.0	-	-	-	Number of standard deviations of local variation for w
p1	0.0	0.0	∞	m	Perimeter of node n1 partition
p2	0.0	0.0	∞	m	Perimeter of node n2 partition
sw_et	1	0	1	-	Switch for turning off self-heating: 0=exclude 1=include
sw_mman	0	0	1	-	Switch for mismatch analysis: 0=no and 1=yes
sw_lin	0	-	-	-	Switch to force linearity: 0=no and 1=yes
swnoise	1	0	1	-	Switch for noise: 0=no 1=yes
trise	0.0	-	-	°C	Local temperature delta to ambient (before self-heating)
w	1.0e-6	0.0	∞	m	Design width of resistor body
wd	0.0	0.0	∞	m	Dogbone width (total; not per side)

Table 5 CMC R3 Special Model Parameters

Name	Default	Min.	Max.	Units	Description
version	-	-	-	-	Model version (major version, subversion, revision) To be compatible with previous versions, the subversion on decimal place of version=1.1 is equivalent to subversion=1.
subversion	-	-	-	-	Model subversion (major change)
revision	-	-	-	-	Model revision (implementation update)
atsl	0.0	-	-	$V * \mu m$	Saturation smoothing coefficient
dpw	0.0	-	-	$V * \mu m^{dpwe}$	Depletion potential width coefficient
dpwe	0.5	-	-	-	Depletion potential width exponent
dpl	0.0	-	-	$V * \mu m^{dple}$	Depletion potential length coefficient
dple	2.0	-	-	-	Depletion potential length exponent
dpwl	0.0	-	-	$V * \mu m^{(dpwe + dple)}$	Depletion potential width coefficient
gmin	1.0e-12	0.0	∞	S	Minimum conductance (all branches)
grpo	1e-12	0.0	0.1	-	Minimum body conductance at pinch-off (ratio w.r.t. V_{DS})
xvsat	0.0	-	-	-	Exponent for saturation current temperature dependence
tc1wl	0.0	-	-	$\mu m^2 / K$	Resistance linear TC coefficient
tc2wl	0.0	-	-	$\mu m^2 / K^2$	Resistance quadratic TC coefficient
tc1dp	0.0	-	-	$/ K$	Depletion potential linear TC coefficient
tc2dp	0.0	-	-	$/ K^2$	Depletion potential quadratic TC coefficient

Table 5 CMC R3 Special Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
tegh	0.0	$-\infty$	0.0	-	Thermal conductance exponent
sw_vsatt	0	-	-	-	Switch for vsat temperature 0=none and 1=link with resistance
sw_accpo	0	0	3	-	Switch for pinchoff model 0->3 increases accuracy of simulation time
level	5	-	-	-	Model level
imax	1.0	0.0	∞	-	Current at which to limit currents
jmax	100.0	0.0	-	$A / (\mu m)$	Maximum current density
lmax	9.9e99	lmin	∞	μm	Maximum allowed drain length
lmin	0.0	0.0	∞	μm	Minimum allowed drain length
rthresh	0.001	0.0	∞	Ω	Threshold to switch between $V=I \cdot R$ form
scale	1.0	0.0	1.0	-	Scale factor for instantaneous
shrink	0.0	0.0	100.0	%	Shrink percentage for various geometries
tmax	500.0	27.0	1000.0	$^{\circ}C$	Maximum ambient temperature
tmaxclip	500.0	27.0	1000.0	$^{\circ}C$	Clip maximum temperature
tmin	-100.0	-250.0	27.0	$^{\circ}C$	Minimum ambient temperature
tminclip	-100.0	-250.0	27.0	$^{\circ}C$	Clip minimum temperature
type	-1	-1	+1	-	Resistor type: -1=n-body, +1=p-body
vmax	9.9e99	0.0	∞	V	Maximum voltage with no current
wmax	9.9e99	wmin	∞	μm	Maximum allowed drain width
wmin	0.0	0.0	∞	μm	Minimum allowed drain width

Table 6 CMC R3 Special Model Parameters

Name	Default	Min.	Max.	Units	Description
afn	2.0	0.0	∞	-	Flicker noise current exponent
aja	-0.5	-	∞	V	Smoothing parameter for cja
ajp	-0.5	-		V	Smoothing parameter for cjp
ats	-		∞	-	Saturation smoothing parameter
bfm	1.0	0.0	∞	-	Flicker noise 1/f exponent
ca	0.0	0.0	∞	$F / (\mu m)^2$	Fixed capacitance per unit area
cja	0.0	0.0	∞	$F / (\mu m)^2$	Depletion capacitance per unit area
cjp	0.0	0.0	∞	$F / (\mu m)$	Depletion capacitance per unit perimeter
cp	0.0	0.0	∞	$F / (\mu m)$	Fixed capacitance per unit perimeter
cth0	0.0	0.0	∞	$(sW) / K$	Thermal capacitance fixed component
ctha	0.0	0.0	∞	$(sW) / (K(\mu m^2))$	Thermal capacitance area component
cthc	0.0	0.0	-	$(sW) / (K\mu m)$	Thermal capacitance contact component
cthp	0.0	0.0	∞	$(sW) / (K\mu m)$	Thermal capacitance perimeter component
dfinf	0.01	0.0001	10.0	$1 / (\sqrt{V})$	Depletion factor for wide/long device
dfl	0.0	-	-	$(\mu m) / (\sqrt{V})$	Depletion factor 1/l coefficient
dfw	0.0	-	-	$(\mu m) / (\sqrt{V})$	Depletion factor 1/w coefficient
dfwl	0.0	-	-	$\mu m^2 / (\sqrt{V})$	Depletion factor 1/(w*l) coefficient
dp	2.0	0.1	1000.0	V	Depletion potential
du	0.02	0.0	1000.0	-	Mobility reduction at ecom
dxlsat	0.0	-	-	μm	Additional length offset for velocity saturation calculation
ea	1.12	-	-	V	Activation voltage for diode temperature dependence

Table 6 CMC R3 Special Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
ecorn	0.4	0.01	ecrit	$V/\mu m$	Velocity saturation corner field
ecrit	4.0	0.02	1000.0	$V/\mu m$	Velocity saturation critical field
fc	0.9	0.0	0.99	-	Depletion capacitance linearization factor
fdrw	1.0	0.0	∞	μm	Finite doping width offset reference width
fdxwinf	0.0	-	-	μm	Finite doping width offset width value for wide devices
gth0	1.0e+6	0.0	∞	W / K	Thermal capacitance fixed component
gtha	0.0	0.0	∞	$W / (K\mu m^2)$	Thermal capacitance area component
gthp	0.0	0.0	∞	$W / (K\mu m)$	Thermal capacitance perimeter component
ibv	1.0e-6	0.0	∞	A	Current at breakdown
isa	0.0	0.0	∞	$A / (\mu m)^2$	Diode saturation current per unit area
isp	0.0	0.0	∞	$A / (\mu m)$	Diode saturation current per unit perimeter
kfn	0.0	0.0	∞	-	Flicker noise coefficient (unit depends on afn)
ma	0.33	0.0	1.0	-	Grading coefficient for cja
mp	0.33	0.0	1.0	-	Grading coefficient for cjp
na	1.0	0.0	∞	-	Ideality factor for isa
nbv	1.0	0.0	∞	-	Ideality factor for breakdown current
np	1.0	0.0	∞	-	Ideality factor for isp
nsig_l	0.0	-	-	-	Number of standard deviations of global variation for l
nsig_rsh	0.0	-	-	-	Number of standard deviations of global variation for rsh
nsig_w	0.0	-	-	-	Number of standard deviations of global variation for w

Table 6 CMC R3 Special Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
nst	1.0	0.1	5.0	-	Subthreshold slope parameter
nwxw	0.0	-	-	μm^2	Narrow width offset correction coefficient
pa	0.75	0.0	∞	V	Built-in potential for cja
pp	0.75	0.0	∞	V	Built-in potential for cjp
rc	0.0	0.0	∞	Ω	Resistance per contact
rcw	0.0	0.0	∞	$\Omega \mu m$	Width adjustment for contact resistance
rsh	100.0	0.0	∞	Ω	Sheet resistance
sig_l	0.0	0.0	∞	μm	Global variation standard deviation for l (absolute)
sig_rsh	0.0	0.0	∞	%	Local variation standard deviation for rsh (relative)
sig_w	0.0	0.0	∞	μm	Local variation standard deviation for w (absolute)
smm_l	0.0	0.0	∞	$\mu m^{1.5}$	Local variation standard deviation for l (absolute)
smm_rsh	0.0	0.0	∞	$\% \mu m$	local variation standard deviation for rsh (relative)
smm_w	0.0	0.0	∞	$\mu m^{1.5}$	Local variation standard deviation for rsh (relative)
sw_dfgeo	1	0	1	-	Switch for depletion factor geometry dependence: 0=drawn and 1=effective
sw_fngeo	0	0	1	-	Switch for flicker noise geometry calculation: 0=design 1=effective
sw_mmgeo	0	0	1	-	Switch for flicker noise geometry calculation: 0=drawnand 1=effective
tc1	0.0	-	-	1/K	Resistance linear TC
tc1kfn	0.0	-	-	-	Flicker noise coefficient linear TC
tc1l	0.0	-	-	$\mu m / K$	Resistance linear TC length coefficient
tc1nbv	0.0	-	-	-	Breakdown ideality factor linear TC

Table 6 CMC R3 Special Model Parameters (Continued)

Name	Default	Min.	Max.	Units	Description
tc1rc	0.0	-	-	1/K	Contact resistance linear TC
tc1vbv	0.0	-	-	V	Breakdown voltage linear
tc1w	0.0	-	-	$\mu m / K$	Resistance linear TC width coefficient
tc2	0.0	-	-	1/K ²	Resistance quadratic TC
tc2l	0.0	-	-	$(\mu m) / K^2$	Resistance quadratic TC length coefficient
tc2rc	0.0	-	-	1/K ²	Contact resistance quadratic TC
tc2vbv	0.0	-	-	V ²	Breakdown voltage quadratic TC
tc2w	0.0	-	-	$(\mu m) / K^2$	Resistance quadratic TC width coefficient
tnom	27.0	-250	1000.0	°C	Nominal (reference) temperature
vbv	0.0	0.0	∞	V	Breakdown voltage
wexw	0.0	-	-	μm	Webbing effect width offset correction coefficient (for dog-boned devices)
xis	3.0	-	-	-	Exponent for diode temperature dependence
xl	0.0	-	-	μm	Length offset (total)
xlw	0.0	-	-	-	Width dependence of length offset
xw	0.0	-	-	μm	Width offset (total)

CMC R3 Usage

With Model card:

```
rinstanceName ([node1 node2 node3] [mname] instanceParameters
.MODEL modelName r modelParameters
```

Without Model card

```
rname (node1 node2) r=resistanceValue [tc1=tc1Value] [tc2=tc2Value]
```

Example

```
r137 n1 n2 n3 rnpoly1 w=1u l=10u
.model rnpoly1 r
+ level=5 rsh=100.0
r138 n2 n3 r=110.0 tc1=1.0e-3
```

CMC R3 Resistor Model Version Update

- Supports CMC R3 version 1.1.2.

Capacitor Device Model and Equations

This section describes capacitor models and their equations.

For a listing of output templates for diode models see Element Template Listings (PrimeSim HSPICE Only), Table 34, in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

Capacitance Model

You can use the `.MODEL` statement to include a capacitance model in the PrimeSim Continuum solution netlist. For a general description of the `.MODEL` statement, see `.MODEL` in the *PrimeSim HSPICE Reference Manual: Commands and Control Options*.

Syntax

```
.MODEL mname C parameter=value
```

Parameter	Description
<i>mname</i>	Model name
C	Specifies a capacitance model
<i>parameter</i>	Any model parameter name

Table 7 Capacitance Parameters

Name (Alias)	Units	Default	Description
CAP	F	0	Default capacitance value.
CAPSW	F/m	0	Sidewall fringing capacitance.
COX	F/m ²	0	Bottomwall capacitance.

Table 7 Capacitance Parameters (Continued)

Name (Alias)	Units	Default	Description
DEL	m	0	Difference between the drawn width and the actual width or length. $DEL_{eff} = DEL \cdot SCALM$
DI	-	0	Relative dielectric constant.
L	m	0	Default length of the capacitor. $L_{scaled} = L \cdot SHRINK \cdot SCALM$
SHRINK	-	1	Shrink factor.
TC1	1/deg	0	First temperature coefficient for capacitance.
TC2	1/deg ²	0	Second temperature coefficient for capacitance.
VC1	1/volt	0	First-order voltage-bias coefficient for capacitance.
VC2	1/volt ²	0	Second-order voltage-bias coefficient for capacitance.
THICK	m	0	Insulator thickness.
TREF	deg C	TNOM	Reference temperature.
W	m	0	Default width of the capacitor. $W_{scaled} = W \cdot SHRINK \cdot SCALM$

Parameter Limit Checking

If a capacitive element value exceeds 0.1 F, then a warning message is written to the output listing file. This feature helps you to identify elements that have missing units or incorrect values, particularly if the elements are in automatically-produced netlists.

Capacitor Device Equations

Capacitor equations include effective capacitance and capacitance temperature.

Effective Capacitance Calculation

You can associate a model with a capacitor. You can specify some of the parameters in both the element and the model descriptions. The element values override the model values.

- To scale the element width and length, use `.OPTION SCALE` and the `SHRINK` model parameter.
- To scale the model width and length, use `.OPTION SCALM` and the `SHRINK` model parameter.

The following equations calculate the effective width and length:

$$W_{eff} = W_{scaled} \cdot 2 \cdot p_{DE} \cdot L_{eff}$$

$$L_{eff} = L_{scaled} \cdot 2 \cdot p_{DE} \cdot L_{eff}$$

If you specify the element capacitance:

$$CAP_{eff} = C \cdot SCALE(element) \cdot M$$

Otherwise, the equation calculates the capacitance from the L_{eff} , W_{eff} , and COX values:

$$CAP_{eff} = M \cdot SCALE(element) \cdot [L_{eff} \cdot W_{eff} \cdot COX + 2 \cdot (L_{eff} + W_{eff}) \cdot CAP_{SW}]$$

If you do not specify COX , but $THICK$ is not zero, then:

$$COX = \frac{DI \cdot \epsilon_o}{THICK} \text{ if } DI \text{ not zero}$$

$$COX = \frac{\epsilon_{OX}}{THICK} \text{ if } DI=0$$

The following values apply to the preceding equation:

$$\epsilon_o = 8.8542149e-12$$

$$\frac{F}{meter}$$

$$\epsilon_{OX} = 3.453148e-11$$

$$\frac{F}{meter}$$

If you specify only the model capacitance (CAP), then:

$$CAP_{eff} = CAP \cdot SCALE(element) \cdot M$$

Capacitance Voltage Equation

The following equation calculates the capacitance as a function of voltage across a given capacitor:

$$C(v) = C \cdot (1 + V \cdot VC1 + V^2 \cdot VC2)$$

Inductor Device Model and Equations

You can use several elements and models to analyze:

- Switching regulators
- Transformers
- Mutual inductive circuits

These elements include:

- Magnetic winding elements
- Mutual cores
- Magnetic core models

You can use the saturable core model for:

- Chokes
- Saturable transformers
- Linear transformers

To use the model, you must:

1. Provide a mutual core statement.
2. Use a `.MODEL` statement to specify the core parameters.
3. Use a magnetic winding element statement to specify the windings around each core element.

Inductor Core Models

Syntax

Magnetic Core

```
.MODEL mname L (<pname1 = val1>...)
```

Jiles-Atherton Ferromagnetic Core

```
.MODEL mname CORE (LEVEL=1 <pname1 = val1>...)
```

LEVEL=3 Resistor Model

```
.MODEL mname L LEVEL=3 <SCALE=val> <TNOM=val>
```

Example 1

```
.MODEL CHOKE L(BS=12K BR=10K HS=1 HCR=.2 HC=.3 AC=1. LC=3.)
```

To use this example, obtain the core model parameters from the manufacturer's data. [Figure 4 on page 49](#) illustrates the required b-h loop parameters for the model.

The model includes:

- Core area
- Length
- Gap size
- Core growth time constant

Example 2

This example is located in the following directory: `$installdir/demo/hspice/mag/bhloop.sp`

Table 8 *Magnetic Core Model Parameters*

Name (Alias)	Units	Default	Description
AC	cm · 2	1.0	Core area.
BS	Gauss	13000	Magnetic flux density, at saturation.
BR	Gauss	12000	Residual magnetization.
HC	Oersted	0.8	Coercive magnetizing force.
HCR	Oersted	0.6	Critical magnetizing force.
HS	Oersted	1.5	Magnetizing force, at saturation.
LC	cm	3.0	Core length.
LG	cm	0.0	Gap length.
TC	s	0.0	Core growth time constant.

Figure 4 Magnetic Saturable Core Model

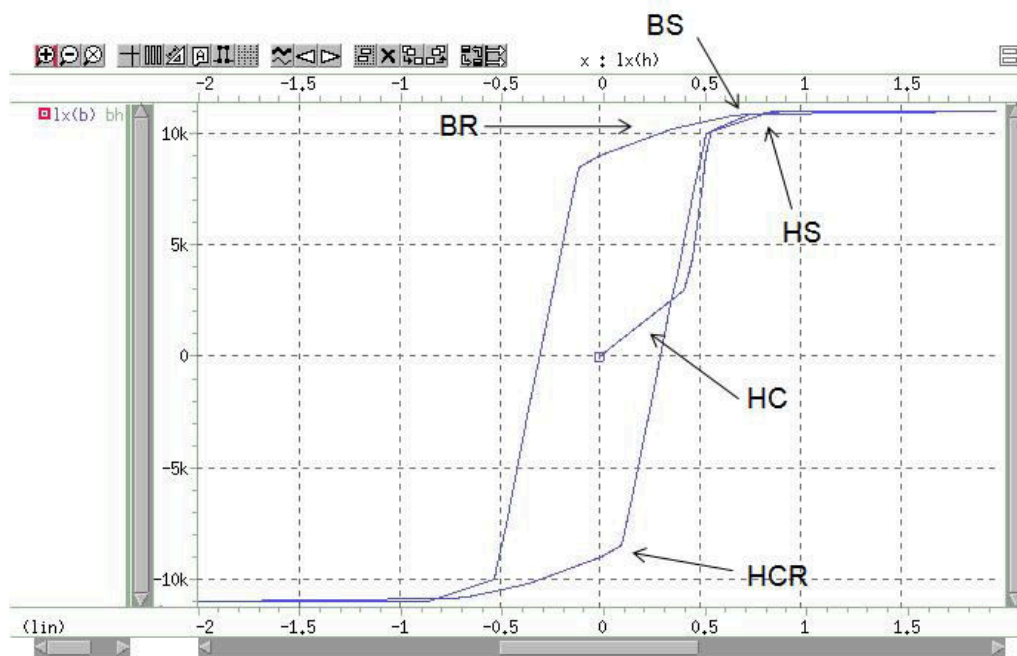


Table 9 Jiles-Atherton Core Model Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	2	Model selector. - LEVEL=1 selects the Jiles-Atherton model. - LEVEL=2 (default) selects the Pheno model, which is the original model.
A	amp/meter	1e3	Characterizes the shape of anhysteretic magnetization.
ALPHA	-	1e-3	Coupling between the magnetic domains.
AREA, (AC)	cm ²	1	Mean of the cross section for the magnetic core. AC is an alias of AREA.
C	-	0.2	Domain flexing parameter.
K	amp/meter	500	Domain of an isotropy parameter.
MS	amp/meter	1e6	Magnetization saturation.
PATH, (LC)	cm	3	Mean of the path length for the magnetic core. LC is an alias of PATH.

Table 10 Magnetic Core Element Outputs

Output Variable	Description
LX1	Magnetic field, h (oersted)
LX2	Magnetic flux density, b (gauss)
LX3	Slope of the magnetization curve, $\frac{d}{dh}m$
LX4	Bulk magnetization, m (amp/meter)
LX5	Slope of the anhysteretic magnetization curve, $\frac{d}{dh}m_{an}$
LX6	Anhysteretic magnetization, m_{an} (amp/meter)
LX7	Effective magnetic field, h_e (amp/meter)

Jiles-Atherton Ferromagnetic Core Model

The Jiles-Atherton ferromagnetic core model is based on domain wall motion, including both bending and translation. A modified Langevin expression describes the hysteresis-free (anhysteretic) magnetization curve. This leads to:

$$m_{an} = M\dot{S} \cdot \left(\coth \left(\frac{h_e}{A} \right) - \frac{A}{h_e} \right)$$

$$h_e = h + ALPHA \cdot m_{an}$$

Parameter	Description
m_{an}	Magnetization level, if the domain walls could move freely.
h_e	Effective magnetic field.
h	Magnetic field.
MS	This model parameter represents the saturation magnetization.
A	This model parameter characterizes the shape of anhysteretic magnetization.
$ALPHA$	This model parameter represents the coupling between the magnetic domains.

The preceding equation generates anhysteretic curves, if the `ALPHA` model parameter has a small value. Otherwise, the equation generates some elementary forms of hysteresis loops, which is not a desirable result.

The following equation calculates the slope of the curve, at zero (0):

$$\frac{d}{dh} m_{an} = \frac{1}{3 \cdot \frac{A}{MS} - ALPHA}$$

The slope must be positive; therefore, the denominator of the preceding equation must be positive. If the slope becomes negative, an error message appears.

Anhysteretic magnetization represents the global energy state of the material, if the domain walls could move freely, but the walls are displaced and bent in the material.

If you express the bulk magnetization (m) as the sum of an irreversible component (due to wall displacement), and a reversible component (due to domain wall bending), then:

$$\frac{d}{dh} m = \frac{(m_{an} - m)}{K} + C \cdot \left(\frac{d}{dh} m_{an} - \frac{d}{dh} m \right)$$

-or-

$$\frac{d}{dh} m = \frac{(m_{an} - m)}{(1 + C) \cdot K} + \frac{C}{1 + C} \cdot \frac{d}{dh} m_{an}$$

Solving the above differential equation provides the bulk magnetization value, m . The following equation uses m to compute the flux density (b):

$$b = \mu_0 \cdot (h + m)$$

The following values apply to the preceding equation:

- μ_0 The permeability of free space, is $4\pi \cdot 10^{-7}$
- The units of h and m are in amp/meter.
- The units of b are in Tesla (Wb/meter^2).

Example

This example demonstrates the effects of varying the `ALPHA`, `A`, and `K` model parameters, on the b - h curve.

- [Figure 5](#) shows b - h curves for three values of `ALPHA`.
- [Figure 6](#) shows b - h curves for three values of `A`.
- [Figure 7](#) shows b - h curves for three values of `K`.

Input File

This input file is located in the following directory: `$installdir/demo/hspice/mag/jfiles.sp`

Figure 5 Anhyseretic b-h Curve Variation: Slope and ALPHA Increase

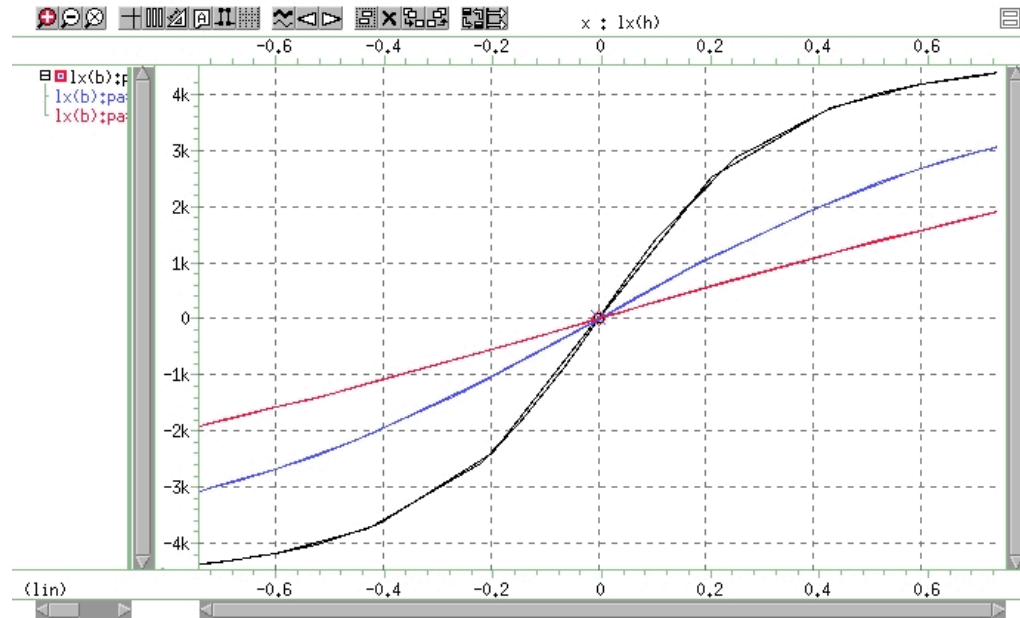
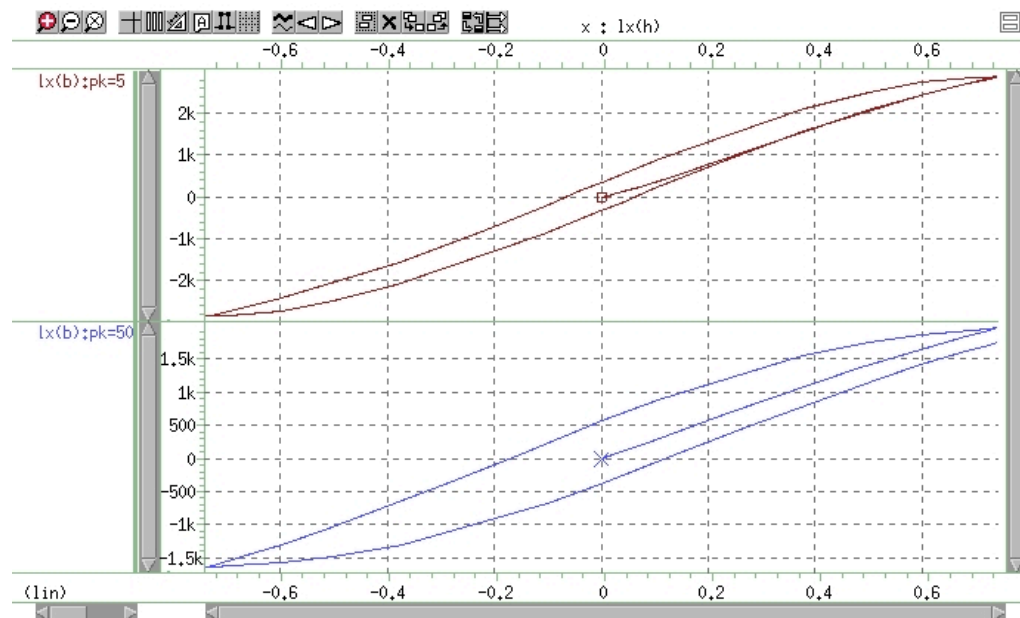


Figure 6 Anhyseretic b-h Curve Variation: Slope Decreases, A Increases



Discontinuities in Inductance Due to Hysteresis

This example creates multi-loop hysteresis b-h curves for a magnetic core. Discontinuities in the inductance, which are proportional to the slope of the b-h curve, can cause convergence problems. [Figure 7](#) and [Figure 8](#) demonstrate the effects of hysteresis on the inductance of the core.

This input file is located in the following directory: `$installdir/demo/hspice/mag/tj2b.sp`

Figure 7 Inductance Curve

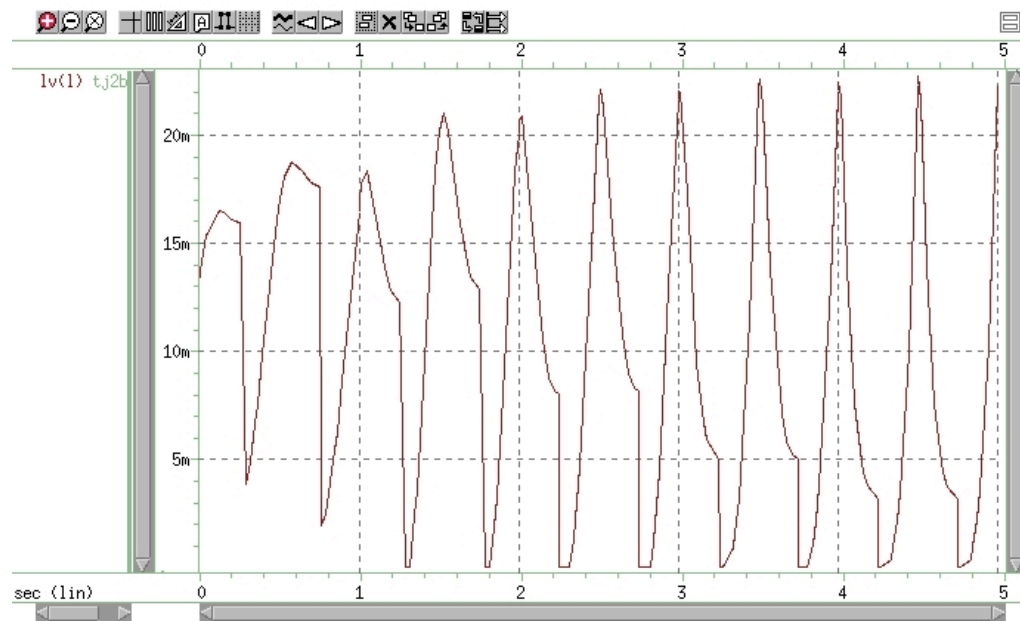
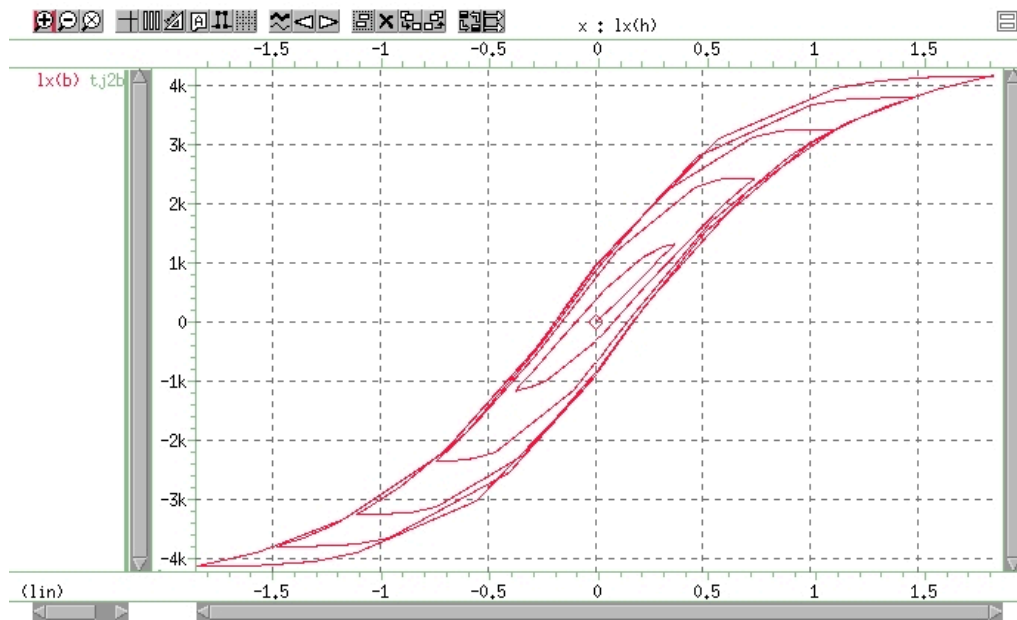


Figure 8 Hysteresis Curve



Optimizing the Extraction of Parameters

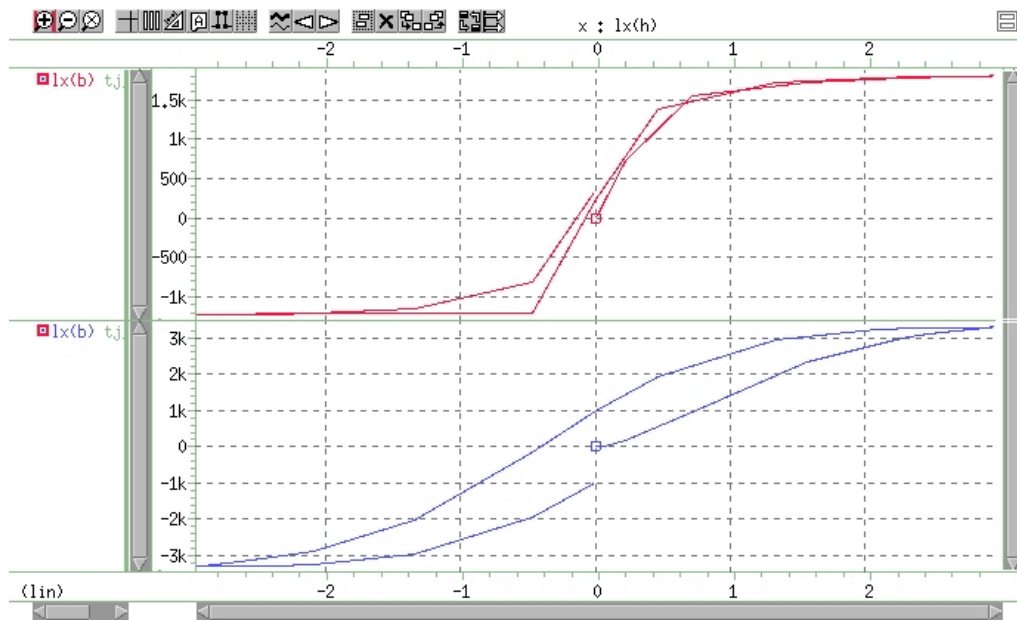
This example demonstrates how to optimize the process of extracting parameters from the Jiles-Atherton model. [Figure 9](#) shows the plots of the core output, before and after optimization.

Input File

This input file is located in the following directory: `$installdir/demo/hspice/mag/tj_opt.sp`

The `tj_opt.sp` file also contains the analysis results listing.

Figure 9 Output Curves, Before Optimization (top), and After Optimization (bottom)



3

Diode Models

Describes model parameters and scaling effects for geometric and non-geometric junction diodes.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

You use diode models to describe p-n junction diodes within MOS and bipolar integrated circuit environments and discrete devices. You can use four types of models (as well as a wide range of parameters) to model standard junction diodes:

- Zener diodes
- Silicon diffused junction diodes
- Schottky barrier diodes
- Nonvolatile memory diodes (tunneling current)

Note:

See MOSFET Diode Models in the *PrimeSim Continuum Reference Manual: MOSFET Models* for other MOSFET and standard discrete diodes.

Diode model types include the junction diode model and the Fowler-Nordheim model. Junction diode models have two variations: geometric and nongeometric.

For a listing of output templates for diode models, see Element Template Listings (PrimeSim SPICE Only), Table 43, in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

These topics are discussed in the following sections:

- [Diode Types](#)
- [Specifying Junction Diode Models](#)
- [Using the JUNCAP Models](#)
- [Using the Fowler-Nordheim Diode, Level 2](#)

- [Philips D500 Model \(Advanced Diode Model\), Level 5](#)
- [DIODE_CMC Model, Level 7](#)
- [Converting National Semiconductor Models](#)
- [DC Operating Point Output of Diodes](#)

Diode Types

Use the geometric junction diode to model:

- IC-based, standard silicon-diffused diodes
- Schottky barrier diodes
- Zener diodes

Use the geometric parameter to specify dimensions for PN junction poly and metal capacitance for a particular IC process technology.

Use the non-geometric junction diode to model discrete diode devices, such as standard and Zener diodes. You can use the non-geometric model to scale currents, resistances, and capacitances by using dimensionless area parameters.

The *Fowler-Nordheim* diode defines a tunneling current-flow through insulators. The model simulates diode effects in nonvolatile EEPROM memory.

These topics are presented in the following sections:

- [Using Diode Model Statements](#)
- [Setting Control Options](#)

Using Diode Model Statements

Use model and element statements to select the diode models. Use the `LEVEL` parameter (in model statements) to select the type of diode model:

- `LEVEL=1` selects the non-geometric, junction diode model
- `LEVEL=2` selects the Fowler-Nordheim diode model
- `LEVEL=3` selects the geometric, junction diode model.

To design Zener, Schottky barrier, and silicon diffused diodes, alter the model parameters for both `LEVEL=1` and `LEVEL=3`. `LEVEL=2` does not permit modeling of these effects. For Zener diodes, set the `BV` parameter for an appropriate Zener breakdown voltage.

If you do not specify the `LEVEL` parameter in the `.MODEL` statement, the model defaults to the non-geometric, junction diode model, `LEVEL=1`.

Use control options with the diode model, to:

- Scale model units
- Select diffusion capacitance equations
- Change model parameters

Setting Control Options

To set control options, use the `.OPTION` statement.

The following control options are related to the analysis of diode circuits and other models:

- `DCAP`
- `DCCAP`
- `GMIN`
- `GMINDC`
- `SCALE`
- `SCALM`

Bypassing Latent Devices (PrimeSim Continuum only)

Use `.OPTION BYPASS` (latency) to decrease simulation time in large designs. To speed up simulation time, this option does not recalculate currents, capacitances, and conductances, if the voltages at the terminal device nodes have not changed. `.OPTION BYPASS` applies to MOSFETs, MESFETs, JFETs, BJTs, and diodes.

Use `.OPTION BYPASS` to set `BYPASS`. `BYPASS` might reduce simulation accuracy for tightly-coupled circuits such as op-amps and high-gain ring oscillators.

Setting Scaling Options

- Use the `SCALE` element option to scale `LEVEL=2` and `LEVEL=3` diode element parameters.
- Use the `SCALM` (scale model) option to scale `LEVEL=2` and `LEVEL=3` diode model parameters.

`LEVEL=1` does not use `SCALE` or `SCALM`.

Include `SCALM=val` in the `.MODEL` statement (in a diode model) to override global scaling that uses the `.OPTION SCALM=val` statement.

Using the Capacitor Equation Selector Option — DCAP

- Use the `DCAP` option to select the equations used in calculating the depletion capacitance (`LEVEL=1` and `LEVEL=3`).
- Use the `DCCAP` option to calculate capacitances in DC analysis.

Include `DCAP=val` in the `.MODEL` statement for the diode to override the global depletion capacitance equation that the `.OPTION DCAP=val` statement selects.

Using Control Options for Convergence

Diode convergence problems often occur at the breakdown voltage region, when the diode is either over-driven or in the OFF condition.

To achieve convergence in such cases, do one of the following:

- Include a non-zero value in the model for the `RS` (series resistor) parameter.
- Increase `GMIN` (the parallel conductance that the PrimeSim Continuum solution automatically places in the circuit). You can specify `GMIN` and `GMINDC` in the `.OPTION` statement.

Table 11 shows the diode control options:

Table 11 Diode Control Options

Function	Control Options
Capacitance	DCAP, DCCAP
Conductance	GMIN, GMINDC
Geometry	SCALM, SCALE

Specifying Junction Diode Models

Use the diode element statement to specify the two types of junction diodes: geometric or non-geometric. Use a different element type format for the Fowler-Nordheim model.

Use the parameter fields in the diode element statement to define the following parameters of the diode model, specified in the `.MODEL` statement for the diode:

- Connecting nodes
- Initialization
- Temperature

- Geometric junction
- Capacitance parameters

Both `LEVEL=1` and `LEVEL=3` junction diode models share the same element parameter set. Poly and metal capacitor parameters of `LM`, `LP`, `WM` and `WP`, do not share the same element parameter.

Element parameters have precedence over model parameters, if you repeat them as model parameters in the `.MODEL` statement.

Parameters common to both element and model statements are:

`AREA`, `PJ`, `M`, `LM`, `LP`, `WM`, `WP`, `W`, `L`.

Table 12 *Junction Diode Element Parameters*

Function	Parameters
Netlist	Dxxx, n+, n-, mname
Initialization	IC, OFF
Temperature	DTEMP
Non-Geometric junction (LEVEL=1)	AREA, M, PJ
Geometric junction (LEVEL=3)	AREA, L, M, PJ, W
Geometric capacitance (LEVEL=3 only)	LM, LP, WM, WP

These topics are discussed in the following sections:

- [Using the Junction Model Statement](#)
- [Using Junction Model Parameters](#)
- [Geometric Scaling for Diode Models](#)
- [Defining Diode Models](#)
- [Determining Temperature Effects on Junction Diodes](#)
- [Using Junction Diode Equations](#)

Using the Junction Model Statement

You can use the `.MODEL` statement to include a junction model in your PrimeSim Continuum simulator netlist. For a general description of the `.MODEL` statement, see `.MODEL` in the *PrimeSim Continuum Reference Manual: Commands and Control Options*.

Syntax

```
.MODEL mname D LEVEL = val keyword = val ...
```

Parameter	Description
<i>mname</i>	Model name. The diode element uses this name to refer to the model.
<i>D</i>	Symbol that identifies a diode model.
<i>LEVEL</i>	Symbol that identifies a diode model. - LEVEL=1 =junction diode. - LEVEL=2 =Fowler-Nordheim diode. - LEVEL=3 =geometric processing for junction diode.
<i>keyword</i>	Model parameter keyword, such as CJO or IS.

Example

```
.MODEL D D (CO=2PF, RS=1, IS=1P)
.MODEL DFOWLER D (LEVEL=2, TOX=100, JF=1E-10, EF=1E8)
.MODEL DGEO D (LEVEL=3, JS=1E-4, JSW=1E-8)
.MODEL dln750a D
+ LEVEL=1 XP =0.0 EG =1.1
+ XOI =0.0 XOM =0.0 XM =0.0
+ WP =0.0 WM =0.0 LP =0.0
+ LM =0.0 AF =1.0 JSW =0.0
+ PB =0.65 PHP =0.8 M =0.2994
+ FC =0.95 FCS =0.4 MJSW=0.5
+ TT =2.446e-9 BV =4.65 RS =19
+ IS =1.485e-11 CJO =1.09e-9 CJP =0.0
+ PJ =0.0 N =1.615 IK =0.0
+ IKR =1.100e-2 IBV =2.00e-2
```

Using Junction Model Parameters

The diode element statement references the `.MODEL` statement. The `.MODEL` statement contains parameters that specify:

- Type of diode model (LEVEL=1, 2, or 3)
- DC

- Capacitance
- Temperature
- Resistance
- Geometry
- Noise

Table 13 Junction Diode Model Parameters (LEVEL=1 and 3)

Function	Parameters
model type	LEVEL
DC parameters	IBV, IK, IKR, IS, ISW, N, RS, VB, RS
geometric junction	AREA, M, PJ
geometric capacitance (LEVEL=3 only)	L, LM, LP, SHRINK, W, WM, WP, XM, XOJ, XOM, XP, XW
capacitance	CJ, CJP, FC, FCS, M, MJSW, PB, PHP, TT
noise	AK, KF

Table 14 Junction DC Parameters in LEVEL=1 and 3

Name (Alias)	Units	Default	Description
UPDATE	-	0.0	UPDATE = 1: Breakdown equation and temperature dependency enhancements for diode level 3. Related parameters: TBV1, TBV2, IBVL, NBVL.
AREA	-	1.0	Junction area. • For LEVEL=1 $AREA_{eff} = AREA \cdot M$, unitless • For LEVEL=3 $AREA_{eff} = AREA \cdot SCALM^2 \cdot SHRINK^2 \cdot M$, unit = meter² • If you specify W and L: $AREA_{eff} = W_{eff} \cdot L_{eff} \cdot M$, unit = meter²
EXPLI	amp/ AREA _{eff}	0	Current-explosion model parameter. The PN junction characteristics (above the explosion current) are linear with the slope at the explosion point. This speeds up the simulation and improves convergence. $EXPLI_{eff} = EXPLI \cdot AREA_{eff}$
EXPLIR	amp/ AREA _{eff}	EXPLI	Reverse mode current explosion model parameter. $EXPLIR_{eff} = EXPLIR \cdot AREA_{eff}$

Table 14 Junction DC Parameters in LEVEL=1 and 3 (Continued)

Name (Alias)	Units	Default	Description
IBV	amp/ AREAeff	1.0e-3	Current, at breakdown voltage: For LEVEL=3: $IBV_{eff} = IBV \cdot AREA_{eff} / SCALM2$
IK (IKF, JBF)	amp/AREA eff	0.0	Forward-knee current (intersection of the high- and low-current asymptotes). $IK_{eff} = IK \cdot AREA_{eff}$
IKR (JBR)	amp/AREA eff	0.0	Reverse-knee current (intersection of the high- and low-current asymptotes). $IKR_{eff} = IKR \cdot AREA_{eff}$
IS (JS)	amp/AREA eff	LEVEL 1= 1.0e-14 LEVEL 3= 0.0	Saturation current per unit area. If the IS value is less than EPSMIN, the program displays a warning message and resets the value of IS to EPSMIN. EPSMIN default=1.0e-28 If the value of IS is too large, the program displays a warning. - For LEVEL=1: $IS_{eff} = AREA_{eff} \cdot IS$ - For LEVEL=3: $IS_{eff} = AREA_{eff} \cdot IS / SCALM2$
JSW (ISP)	amp/ PJe _{ff}	0.0	Sidewall saturation current, per unit junction periphery. - For LEVEL=1: $JSW_{eff} = PJe_{ff} \cdot JSW$ - For LEVEL=3: $JSW_{eff} = PJe_{ff} \cdot JSW / SCALM$
JTUN	amp/AREA eff	0.0	Tunneling saturation current per area. - For LEVEL=1: $JTUN_{eff} = AREA_{eff} \cdot JTUN$ - For LEVEL=3: $JTUN_{eff} = AREA_{eff} \cdot JTUN / SCALM2$
JTUNSW	amp/PJe _{ff}	0.0	Sidewall tunneling saturation current per unit junction periphery. - For LEVEL=1: $JTUNSW_{eff} = PJe_{ff} \cdot JTUNSW$ - For LEVEL=3: $JTUNSW_{eff} = PJe_{ff} \cdot JTUNSW / SCALM$
L	-	-	Default length of the diode. $L_{eff} = L \cdot SHRINK \cdot SCALM + XW_{eff}$
LEVEL	-	1	Diode model selector. - LEVEL=1 or LEVEL=3 selects the junction diode model. - LEVEL=2 selects the Fowler-Nordheim model.
N	-	1.0	Emission coefficient.
NBV	-	-	N Breakdown emission coefficient. (If not specified, NBV is set to N.)
NTUN	-	30	Tunneling emission coefficient.

Table 14 Junction DC Parameters in LEVEL=1 and 3 (Continued)

Name (Alias)	Units	Default	Description
PJ (PD, PERI) Note: Only for level=3, PD and PERI alias PJ.	-	0.0	Junction periphery. - For LEVEL=1: $PJ_{eff} = PJ \cdot M$, unitless - For LEVEL=3: $PJ_{eff} = PJ \cdot SCALM \cdot M \cdot SHRINK$, meter - If you specify W and L: $PJ_{eff} = (2 \cdot W_{eff} + 2 \cdot L_{eff}) \cdot M$, meter
RS	ohms or $ohms/m^2$ (see the note following this table)	0.0	Ohmic series resistance. - For LEVEL=1: $RS_{eff} = RS/AREA_{eff}$ - For LEVEL=3: $RS_{eff} = RS \cdot SCALM^2/AREA_{eff}$
SHRINK	-	1.0	Shrink factor.
VB (BV, VAR, VRB)	V	0.0	Reverse breakdown voltage. 0.0 indicates an infinite breakdown voltage.
XW	-	-	Accounts for masking and etching effects. $XW_{eff} = XW \cdot SCALM$

Note:

If you use a diode model whose parameter `AREA` is not specified in the model, then `AREA` defaults to 1 and `RS` is in units of ohms. If you specify the `AREA` in square meters (m^2) in the netlist, the units of `RS` are $ohms/m^2$.

Table 15 Junction Capacitance Parameters

Name (Alias)	Units	Default	Description
CJ (CJA, CJO)	F / $AREA_{eff}$	0.0	Zero-bias junction capacitance, per unit-junction bottomwall area. - For LEVEL=1: $CJO_{eff} = CJO \cdot AREA_{eff}$ - For LEVEL=3: $CJ_{eff} = CJ \cdot AREA_{eff}/SCALM^2$
CJP (CJSW)	F / PJ_{eff}	0.0	Zero-bias junction capacitance, per unit-junction periphery (PJ). - For LEVEL=1: $CJP_{eff} = CJP \cdot PJ_{eff}$ - For LEVEL=3: $CJP_{eff} = CJP \cdot PJ_{eff}/SCALM$
FC	-	0.5	Coefficient for the formula that calculates the capacitance for the forward-bias depletion area.

Table 15 Junction Capacitance Parameters (Continued)

Name (Alias)	Units	Default	Description
FCS	-	0.5	Coefficient for the formula that calculates the capacitance for the forward-bias depletion periphery.
M (EXA, MJ)	-	0.5	Grading coefficient at area junction.
MJSW (EXP)	-	0.33	Grading coefficient at periphery junction.
PB (PHI, VJ, PHA)	V	0.8	Contact potential at area junction.
PHP	V	PB	Contact potential at periphery junction.
TT	s	0.0	Transit time.

Table 16 Metal and Poly Capacitor Parameters, LEVEL=3

Name (Alias)	Units	Default	Description
LM	m	0.0	Default length of metal. Use this parameter, if the element statement does not specify LM. $LM_{eff} = LM \cdot SCALM \cdot SHRINK$
LP	m	0.0	Default length of polysilicon. Use this parameter, if the element statement does not specify LP. $LP_{eff} = LP \cdot SCALM \cdot SHRINK$
WM	m	0.0	Default width of metal. Use this parameter, if the element statement does not specify WM. $WM_{eff} = WM \cdot SCALM \cdot SHRINK$
WP	m	0.0	Default width of polysilicon. Use this parameter, if the element statement does not specify WP. $WP_{eff} = WP \cdot SCALM \cdot SHRINK$
XM	m	0.0	Accounts for masking and etching effects in metal layer: $XM_{eff} = XM \cdot SCALM$
XOI	-	10k	Thickness of the poly to bulk oxide.
XOM	Å	10k	Thickness of the metal to bulk oxide.
XP	m	0.0	Accounts for masking and etching effects in poly layer: $XP_{eff} = XP \cdot SCALM$

Table 17 Noise Parameters for LEVEL=1 and 3

Name (Alias)	Units	Default	Description
AF	-	1.0	Flicker noise exponent.
KF	-	0.0	Flicker noise coefficient.

Geometric Scaling for Diode Models

LEVEL=1 Scaling

LEVEL=1 uses the AREA and M element parameters to scale the following element and model parameters: IK, IKR, JS, CJO, and RS. For AREA and M, default=1.

This element is not geometric, because it uses dimensionless values to measure both the area (AREA) and the periphery (PJ). The .OPTION SCALE and .OPTION SCALM statements do not affect these parameters.

Here is the process:

1. Diode models multiply the periphery junction parameter by M (the multiplier parameter) to scale a dimensionless periphery junction:

$$PJ_{\text{eff}} = PJ \cdot M$$

2. The diode models then use PJeff to scale CJP (the zero-bias junction capacitance), and the sidewall saturation current (JSW).

$$CJP_{\text{eff}} = PJ_{\text{eff}} \cdot CJP \quad JSW_{\text{eff}} = PJ_{\text{eff}} \cdot JSW$$

$$JTUNSW_{\text{eff}} = PJ_{\text{eff}} \cdot JTUNSW$$

3. The models use the AREA and M values to obtain AREAeff.

$$AREA_{\text{eff}} = AREA \cdot M$$

4. Models multiply CJO, IK, IKR, IBV, and IS by AREAeff to obtain their effective scaled values. However, diode models divide RS by AREAeff.

$$IK_{\text{eff}} = AREA_{\text{eff}} \cdot IK \quad IKR_{\text{eff}} = AREA_{\text{eff}} \cdot IKR \quad IS_{\text{eff}} = AREA_{\text{eff}} \cdot IS \quad RS_{\text{eff}} = RS / AREA_{\text{eff}}$$

$$CJO_{\text{eff}} = CJO \cdot AREA_{\text{eff}}$$

$$JTUN_{\text{eff}} = JTUN \cdot AREA_{\text{eff}}$$

LEVEL=3 Scaling

The SCALM, SCALE, SHRINK, and M parameters affect LEVEL=3 scaling.

- **SCALE** affects the following **LEVEL=3** element parameters:

AREA, LM, LP, PJ, WM, WP, W, L

- **SCALM** affects the following model parameters:

AREA, IBV, IK, IKR, IS, PJ, JSW, RS, CJO, CJP, LM, LP, WP, XM, XP, W, L, XW, JTUN, JTUNSW

If you include **AREA** as either an element parameter or a model parameter, then the program uses **SCALE** or **SCALM**. The following equations use the **AREA** element parameter, instead of the **AREA** model parameter.

If you specified the **AREA** and **PJ** model parameters, but not the element, then use **SCALM** as the scaling factor, instead of **SCALE**.

The following equations determine the parameters of the scaled effective area and of the periphery junction element:

$$AREA_{eff} = AREA \cdot M \cdot SCALE^2 \cdot SHRINK^2 \quad PJ_{eff} = PJ \cdot SCALE \cdot M \cdot SHRINK$$

If you specified **W** and **L**:

$$AREA_{eff} = W_{eff} \cdot L_{eff} \cdot M \quad PJ_{eff} = (2 \cdot W_{eff} + 2 \cdot L_{eff}) \cdot M$$

The following values apply to the preceding equations:

$$W_{eff} = W \cdot SCALE \cdot SHRINK + XW_{eff} \quad L_{eff} = L \cdot SCALE \cdot SHRINK + XL_{eff}$$

To find the value of **JSWeff** and **CJPeff**, use the following formula:

$$JSW_{eff} = PJ_{eff} \cdot (JSW/SCALM) \quad CJP_{eff} = PJ_{eff} \cdot (CJP/SCALM)$$

$$JTUNSW_{eff} = PJ_{eff} \cdot JTUNSW/SCALM$$

To determine the polysilicon and metal capacitor dimensions, multiply each by **SCALE** or by **SCALM**, if you used model parameters to specify these dimensions.

$$LM_{eff} = LM \cdot SCALE \cdot SHRINK \quad WM_{eff} = WM \cdot SCALE \cdot SHRINK \quad LP_{eff} = LP \cdot SCALE \cdot SHRINK \\ WP_{eff} = WP \cdot SCALE \cdot SHRINK \quad XP_{eff} = XP \cdot SCALM \quad XM_{eff} = XM \cdot SCALM$$

Use the following formulas to determine the effective scaled model parameters (**IBeff**, **IKeff**, **IKReff**, **IBVeff**, **RSeff**, and **CJO**):

$$IK_{eff} = AREA_{eff} \cdot IK \quad IKR_{eff} = AREA_{eff} \cdot IKR \quad IBV_{eff} = IBV \cdot AREA_{eff}/SCALM^2 \quad IS_{eff} = IS \cdot (AREA_{eff}/SCALM^2) \\ RS_{eff} = RS/(AREA_{eff} \cdot SCALM^2) \quad CJO_{eff} = AREA_{eff} \cdot (CJO/SCALM^2)$$

$$JTUN_{eff} = AREA_{eff} \cdot (JTUN/SCALM^2)$$

Defining Diode Models

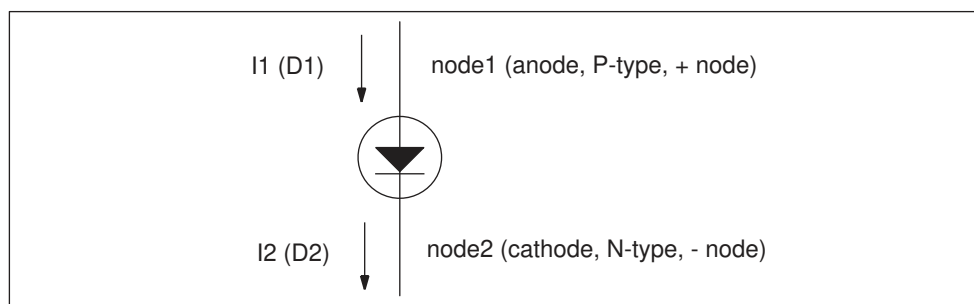
This section describes diode current and diode-equivalent circuits.

Diode Current

Figure 10 shows the direction of current flow through the diode. Use either `I (D1)` or `I1 (D1)` syntax to print the diode current.

If the voltage on node1 is 0.6V greater than the voltage on node2, then the diode is forward biased or turned on. The anode is the p-doped side of a diode and the cathode is the n-doped side.

Figure 10 Diode Current Convention



Using Diode Equivalent Circuits

Synopsys diode device models provide three equivalent circuits for diode analysis: transient, AC, and noise circuits. Components of these circuits form the basis for all element and model equations.

The fundamental component in the DC-equivalent circuit is the DC diode current (i_d). Noise and AC analyses do not use the actual i_d current; instead, these analyses use the partial derivative of i_d with respect to the v_d terminal voltage.

The conductance equation for this partial derivative is:

$$gd = \frac{\partial i_d}{\partial v_d}$$

The drain current (i_d) equation accounts for all basic DC effects of the diodes. The diode device models assume that capacitance effects are separate from the i_d equations.

Figure 11 *Equivalent Circuit for Diode in Transient Analysis*

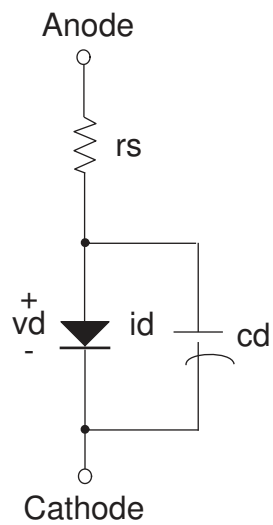


Figure 12 *Equivalent Circuit for Diode in AC Analysis*

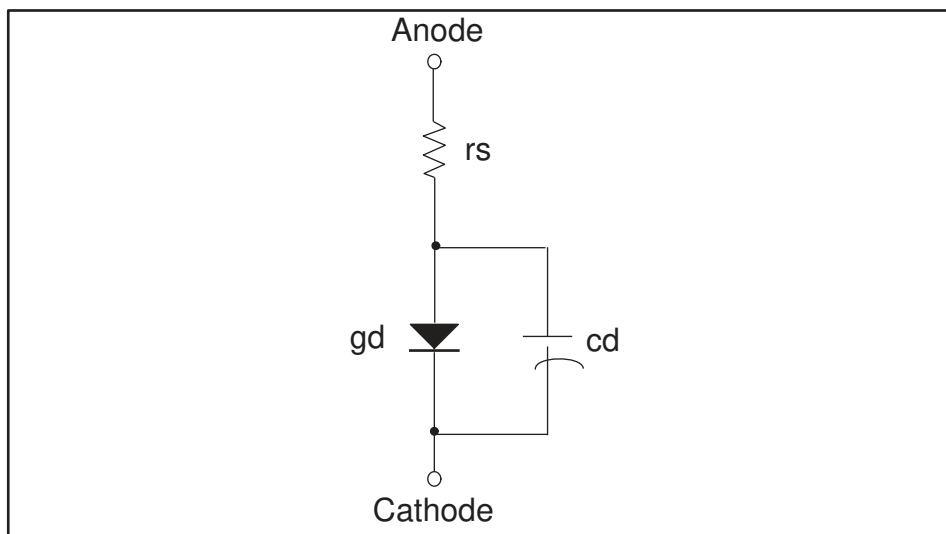
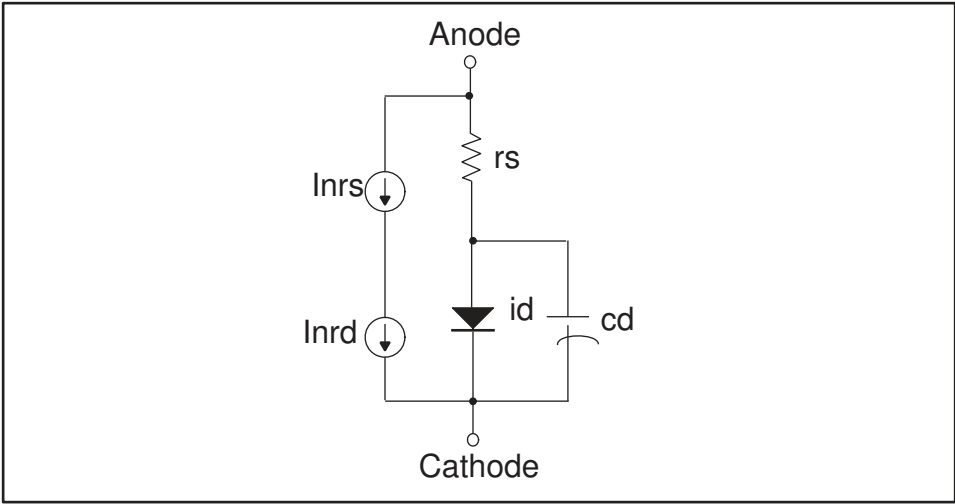


Figure 13 *Equivalent Circuit for Diode in AC Noise Analysis*



Determining Temperature Effects on Junction Diodes

`LEVEL=1` and `LEVEL=3` model statements contain parameters that calculate temperature effects. `TLEV` and `TLEVc` select different temperature equation to calculate temperature effects on:

- Energy gap
- Leakage current
- Breakdown voltage
- Contact potential
- Junction capacitance
- Grading

Table 18 *Junction Diode Temperature Parameters (LEVEL=1 and 3)*

Variable	Parameter
Resistance coefficient	TRS
Capacitance coefficient (area)	CTA
Capacitance coefficient (periphery)	CTP
Energy gap (pn junction)	EG

Table 18 Junction Diode Temperature Parameters (LEVEL=1 and 3)
(Continued)

Variable	Parameter
Energy gap (bandgap corrections)	GAP1, GAP2
Transit time coefficient	TTT1, TTT2
Reference temperature	TREF
Temperature selectors	TLEV, TLEVC
Miscellaneous	TM1, TM2, TPB, TPHP
Saturation current temperature	XTI

Table 19 Junction Diode Temperature Effects, LEVEL=1 and 3

Name (Alias)	Units	Default	Description
CTA (CTC)	1/°C	0.0	Temperature coefficient for area junction capacitance (CJ). If you set the TLEVC parameter to 1, CTA overrides the default temperature coefficient.
CTP	1/°C	0.0	Temperature coefficient for periphery junction capacitance (CJP). If you set TLEVC to 1, CTP overrides the default temperature coefficient.
EG	eV		Energy gap for pn junction diode. - For TLEV=0, 1, default=1.11. - For TLEV=2, default=1.16. 1.17 silicon 0.69 Schottky barrier diode 0.67 germanium 1.52 gallium arsenide
KEG	-	1	EG correction factor for tunneling
GAP1	eV/°C	7.02e-4	First bandgap correction factor. From Sze, alpha term. 7.02e-4 - silicon (old value) 4.73e-4 - silicon 4.56e-4 - germanium 5.41e-4 - gallium arsenide
GAP2	°C	1108	Second bandgap correction factor. From Sze, beta term. 1108 - silicon (old value) 636 - silicon 210 - germanium 204 - gallium arsenide

Table 19 Junction Diode Temperature Effects, LEVEL=1 and 3 (Continued)

Name (Alias)	Units	Default	Description
PT	-	3.0	PT is an alias for XTI.
TCV	1/°	0.0	Temperature coefficient of breakdown voltage.
TLEV	-	0.0	Diode temperature equation selector. Interacts with TLEVC.
TLEVC	-	0.0	Level selector for diode temperature, junction capacitance, and contact potentials. Interacts with TLEV.
TM1	1/°	0.0	First-order temperature coefficient for MJ.
TM2	1/° ²	0.0	Second-order temperature coefficient for MJ.
TPB (TVJ)	V/°	0.0	PB temperature coefficient. If you set the TLEVC parameter to 1 or 2, TPB overrides default temperature compensation.
TPHP	V/°	0.0	PHP temperature coefficient. If you set the TLEVC parameter to 1 or 2, TPHP overrides default temperature compensation.
TREF	°C	25.0	Model reference temperature (LEVEL=1 or 3 only).
TRS	1/°	0.0	Resistance temperature coefficient.
TTT1	1/°	0.0	First-order temperature coefficient for TT.
TTT2	1/° ²	0.0	Second-order temperature coefficient for TT.
XTI	-	3.0	Exponent for the saturation-current temperature. - Set XTI=3.0 for a silicon-diffused junction. - Set XTI=2.0 for a Schottky barrier diode. PT is an alias for XTI.
XTITUN	-	3.0	Exponent for the tunneling current temperature.

Using Junction Diode Equations

Table 20 shows the definitions of variables in diode equations.

Table 20 Equation Variable Definitions

Variable	Definition
cd	Total diode capacitance

Table 20 *Equation Variable Definitions (Continued)*

Variable	Definition
f	Frequency
gd	Diode conductance
id	Diode DC current
id1	Current, without high-level injection
ind	Equivalent noise current for a diode
inrs	Equivalent noise current for a series resistor
vd	Voltage, across the diode

[Table 21](#) shows the definitions of equation quantities.

Table 21 *Equation Quantity Definition*

Quantity	Definition
tox	3.453143e-11 F/m
k	1.3806226e-23 (Boltzmann's constant)
q	1.6021918e-19 (electron charge)
t	Temperature in °Kelvin
Δt	t - tnom
tnom	Nominal temperature of parameter measurements in °Kelvin
vt(t)	$k \cdot t/q$: thermal voltage
vt(tnom)	$k \cdot tnom/q$: thermal voltage

Using Junction DC Equations

A basic diode device model contains three regions:

- Forward bias
- Reverse bias
- Breakdown regions

Forward Bias:

For a forward-bias diode, the anode is more positive than the cathode. The diode is turned *on* and conducts above 0.6 V. Set the R_S model parameter to limit conduction current. As the forward-bias voltage increases past 0.6 V, the limiting resistor prevents the value of the diode current from becoming too high and prevents the solution from converging.

$$v_d \geq -10 \cdot v_t$$

$$i_d = I_{S_{eff}} \cdot \left(e^{\frac{v_d}{N \cdot v_t}} - 1 \right)$$

$$v_d = v_{node1} - v_{node2}$$

Reverse Bias:

In reverse-bias, the anode (node1) is more negative than the cathode. The diode is turned *off* and conducts a small leakage current.

$$BV_{eff} < v_d < -10 \cdot v_t$$

$$i_d = -I_{S_{eff}} - J_{TUN_{eff}} \cdot \left(e^{\frac{-v_d}{N T U N \cdot v_t}} - 1 \right)$$

Breakdown:

For breakdown, set the BV (V_B) parameter which induces reverse-breakdown (or avalanche). You can see this effect in Zener diodes. It occurs when anode-cathode voltage is less than the breakdown voltage (BV). To model this action, measure the voltage (BV) and the current (I_{BV}), at the reverse-knee or at the onset of avalanche.

Note:

BV must be a positive number.

$$v_d < -BV$$

$$i_d = -I_{S_{eff}} \cdot e^{\left(\frac{v_d + BV_{eff}}{N B V \cdot v_t} \right)} - J_{TUN_{eff}} \cdot \left(e^{\frac{-v_d}{N T U N \cdot v_t}} - 1 \right)$$

The device model adjusts the BV parameter to obtain BV_{eff} :

$$i_{break} = -I_{S_{eff}} \cdot \left(e^{\frac{-BV}{N B V \cdot v_t}} - 1 \right)$$

If $I_{BV_{eff}} > i_{break}$, then:

$$BV_{eff} = BV - N B V \times v_t \times \ln \left(\frac{I_{BV_{eff}}}{i_{break}} \right)$$

Otherwise:

$$I_{BV_{eff}} = i_{break}$$

Most diodes do not behave as ideal diodes. The I_K and I_{KR} parameters are called *high-level injection* parameters. They tend to limit the exponential increase in current.

Note:

Diode models use the exponential equation in both the forward and reverse regions.

Forward Bias:

$$id = \frac{id1}{1 + \left(\frac{id1}{I_{Keff}}\right)^{\frac{1}{2}}}$$

Reverse Bias:

$$id = \frac{id1}{1 + \left(\frac{id1}{I_{KReff}}\right)^{\frac{1}{2}}}$$

For $vd \geq -BV_{eff}$:

$$id1 = I_{Seff} \cdot \left(e^{\frac{vd}{N \cdot vt}} - 1\right) - JTUN_{eff} \cdot \left(e^{\frac{-vd}{NTUN \cdot vt}} - 1\right)$$

Otherwise:

$$id1 = -I_{Seff} \cdot \left(e^{\frac{vd}{N \cdot vt}} - 1\right) - I_{Seff} \cdot e^{\left(\frac{vd + BV_{eff}}{NBV \cdot vt}\right)} - JTUN_{eff} \cdot \left(e^{\frac{-vd}{NTUN \cdot vt}} - 1\right)$$

From DC measurements of the forward-biased diode characteristics, you can estimate:

- Reverse-saturation current (I_S)
- Emission coefficient (N)
- Model parameter (R_S)

You can determine N from the slope of the diode characteristic in the ideal region. In most cases, the emission coefficient is the value of the unit, but is closer to 2 for MOS diodes.

At higher bias levels, the diode current deviates from the ideal exponential characteristic, due to ohmic resistance in the diode and the effects of high-level injection. The deviation of the actual diode voltage (from the ideal exponential characteristic) at a specific current, determines the value of R_S . In practice, the simulation of diode device models estimates R_S at several values of id and averages them, because the value of R_S depends upon the diode current.

Using Diode Capacitance Equations

In [Figure 11](#), c_d models the diode capacitance. The capacitance is a combination of diffusion ($cdiff$), depletion ($cdep$), metal ($cmetal$), and poly ($cpoly$) capacitances.

$$cd = cdiff + cdep + cmetal + cpoly$$

Using Diffusion Capacitance Equations

The transit time (TT) models the diffusion capacitance caused by injected minority carriers. In practice, the simulation of diode models estimates TT from pulsed time-delay measurements.

$$cdiff = TT \cdot \frac{\partial id}{\partial vd}$$

Using Depletion Capacitance Equations

To model depletion capacitance, diode device models use the junction bottom and junction periphery capacitances. The formulae for both bottom area and periphery capacitances are similar, but each has its own model parameters. To select either of the two equations for forward-bias junction capacitance, use `.OPTION DCAP`.

DCAP = 1

The capacitance formula for the junction bottom area is:

$$vd < FC \cdot PB$$

$$cdepa = CJeff \cdot \left(1 - \frac{vd}{PB}\right)^{-MJ}$$

$$vd \geq FC \cdot PB$$

$$cdepa = CJeff \cdot \frac{1 - FC \cdot \left(1 - \frac{vd}{PB}\right)^{-MJ} + MJ \cdot \frac{vd}{PB}}{(1 - FC)^{(1+MJ)}}$$

The capacitance formula for the junction periphery is: $vd < FCS \cdot PHP$

$$cdepp = CJPeff \cdot \left(1 - \frac{vd}{PHP}\right)^{-MJSW}$$

$$vd \geq FCS \cdot PHP$$

$$cdepp = CJPeff \cdot \frac{1 - FCS \cdot \left(1 - \frac{vd}{PHP}\right)^{-MJSW} + MJSW \cdot \frac{vd}{PHP}}{(1 - FCS)^{(1+MJSW)}}$$

$$cdep = cdepa + cdepp$$

DCAP = 2 (default)

The capacitance formula for the total depletion is:

$$vd < 0$$

$$cdep = CJ_{eff} \cdot \left(1 - \frac{vd}{PB}\right)^{-MJ} + CJP_{eff} \cdot \left(1 - \frac{vd}{PHP}\right)^{-MJ_{SW}}$$

$$vd \geq 0$$

$$cdep = CJ_{eff} \cdot \left(1 + MJ \cdot \frac{vd}{PB}\right) + CJP_{eff} \cdot \left(1 + MJ_{SW} \cdot \frac{vd}{PHP}\right)$$

DCAP = 3

Limits peak depletion capacitance to $FC \cdot CGS_{eff}$ or $FC \cdot CGS_{eff}$ with proper fall-off when the forward bias exceeds PB ($FC > 1$).

Metal and Poly Capacitance Equations (LEVEL=3 Only)

To determine the metal and poly capacitances, use the following equations:

$$c_{metal} = \left(\frac{\epsilon_{ox}}{XOI}\right) \cdot (WP_{eff} + XP_{eff}) \cdot (LP_{eff} + XP_{eff}) \cdot M$$

$$c_{poly} = \left(\frac{\epsilon_{ox}}{XOM}\right) \cdot (WM_{eff} + XM_{eff}) \cdot (LM_{eff} + XM_{eff}) \cdot M$$

Using Noise Equations

Figure 13 shows the noise model for a diode. An independent current source ($inrs$) in parallel with the resistor models the thermal noise that a resistor generates. To determine the value of $inrs$, use:

$$inrs = \left(\frac{4 \cdot k \cdot t}{RS_{eff}}\right)^{\frac{1}{2}}$$

The unit of $inrs$ is $\text{Amp}/(\text{Hz})^{1/2}$.

The ind current source models the shot and flicker noise of the diode. The following equation defines ind :

$$ind = \left(2 \cdot q \cdot id + \frac{KF \cdot id^{AF}}{f}\right)^{\frac{1}{2}}$$

Temperature Compensation Equations

This section describes the temperature-compensation equations.

Energy Gap Temperature Equations

The equations below determine the energy gap for temperature compensation:

TLEV = 0 or 1

$$egnom = 1.16 - 7.02e-4 \cdot \frac{tnom^2}{tnom + 1108.0}$$

$$eg(t) = 1.16 - 7.02e^{-4} \frac{t^2}{t + 1108.0}$$

TLEV = 2

$$egnom = EG - GAP1 \frac{tnom^2}{tnom + GAP2}$$

$$eg(t) = EG - GAP1 \frac{t^2}{t + GAP2}$$

Leakage Current Temperature Equations

$$JS(t) = JS \cdot e^{\frac{facln}{N}}$$

$$JSW(t) = JSW \cdot e^{\frac{facln}{N}}$$

TLEV = 0 or 1

$$facln = \frac{EG}{vt(tnom)} - \frac{EG}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

TLEV = 2

$$facln = \frac{egnom}{vt(tnom)} - \frac{eg(t)}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

Tunneling Current Temperature Equations

TLEV = 0 or 1

$$facInt = KEG \cdot \left(\frac{EG}{vt(tnom)} - \frac{EG}{vt(t)} \right) + XTITUN \cdot \ln\left(\frac{t}{tnom}\right)$$

TLEV = 2

$$facInt = KEG \cdot \left(\frac{egnom}{vt(tnom)} - \frac{eg(t)}{vt(t)} \right) + XTITUN \cdot \ln\left(\frac{t}{tnom}\right)$$

Breakdown-Voltage Temperature Equations

TLEV = 0

$$BV(t) = BV - TCV \Delta t$$

TLEV = 1 or 2

$$BV(t) = BV \cdot (1 - TCV \Delta t)$$

$$JTUN(t) = JTUN \cdot e^{facInt}$$

$$JTUNSW(t) = JTUNSW \cdot e^{facInt}$$

Transit-Time Temperature Equations

$$TT(t) = TT \cdot (1 + TTT1 \cdot \Delta t + TTT2 \cdot \Delta t^2)$$

Junction Built-in Potential Temperature Equations

TLEVC = 0

$$PB(t) = PB \cdot \left(\frac{t}{tnom}\right)^{-vt(t)} \cdot \left[3 \cdot \ln\left(\frac{t}{tnom}\right) + \frac{egnom}{vt(tnom)} \cdot \frac{eg(t)}{vt(t)}\right]$$

$$PHP(t) = PHP \cdot \frac{t}{tnom} \cdot \left(\frac{t}{tnom}\right)^{-vt(t)} \cdot \left[3 \cdot \ln\left(\frac{t}{tnom}\right) + \frac{egnom}{vt(tnom)} \cdot \frac{eg(t)}{vt(t)}\right]$$

TLEVC = 1 or 2

$$PB(t) = PB \cdot TPB \cdot \Delta t$$

$$PHP(t) = PHP \cdot TPHP \cdot \Delta t$$

TLEVC = 3

$$PB(t) = PB + dpbdt \cdot \Delta t$$

$$PHP(t) = PHP + dphpdt \cdot \Delta t$$

If **TLEVC = 3** and **TLEV = 0 or 1**, then:

$$dpbdt = \frac{-[egnom + 3 \cdot vt(tnom) + (1.16 - egnom) \cdot (2^{-\frac{tnom}{tnom+1108}}) \cdot PB]}{tnom}$$

$$dphpdt = \frac{-[egnom + 3 \cdot vt(tnom) + (1.16 - egnom) \cdot (2^{-\frac{tnom}{tnom+1108}}) \cdot PHB]}{tnom}$$

If **TLEV = 2**:

$$dpbdt = \frac{-[egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot (2^{-\frac{tnom}{tnom+GAP2}}) \cdot PB]}{tnom}$$

$$dphpdt = \frac{-[egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot (2^{-\frac{tnom}{tnom+GAP2}}) \cdot PHP]}{tnom}$$

Junction Capacitance Temperature Equations

TLEVC = 0

$$CJ(t) = CJ \cdot \left[1 + MJ \cdot \left(4.0e-4 \cdot \Delta t \cdot \frac{PB(t)}{PB} + 1\right)\right]$$

$$CJSW(t) = CJSW \cdot \left[1 + MJSW \cdot \left(4.0e-4 \cdot \Delta t \cdot \frac{PHP(t)}{PHP} + 1 \right) \right]$$

TLEVC = 1

$$CJ(t) = CJ \cdot (1 + CTA \cdot \Delta t)$$

$$CJSW(t) = CJSW \cdot (1 + CTP \cdot \Delta t)$$

TLEVC = 2

$$CJ(t) = CJ \cdot \left(\frac{PB}{PB(t)} \right)^{MJ}$$

$$CJSW(t) = CJSW \cdot \left(\frac{PHP}{PHP(t)} \right)^{MJSW}$$

TLEVC = 3

$$CJ(t) = CJ \cdot \left(1 - 0.5 \cdot \frac{dPB}{dt} \cdot \frac{\Delta t}{PB} \right)$$

$$CJSW(t) = CJSW \cdot \left(1 - 0.5 \cdot \frac{dPHP}{dt} \cdot \frac{\Delta t}{PHP} \right)$$

Grading Coefficient Temperature Equation

$$MJ(t) = MJ \cdot (1 + TM1 \cdot \Delta t + TM2 \cdot \Delta t^2)$$

Resistance Temperature Equations

$$RS(t) = RS \cdot (1 + TRS \cdot \Delta t)$$

Using the JUNCAP Models

This section describes the JUNCAP (junction capacitance) diode models: JUNCAP1 and JUNCAP2. These diodes models have been implemented in the PrimeSim Continuum solution as

- Level 4 = JUNCAP1
- Level 6 = JUNCAP2

For a full description of the JUNCAP models, see http://www.semiconductors.philips.com/Philips_Models/.

You can use the `.MODEL` statement to include a JUNCAP model in your PrimeSim Continuum simulator netlist. For a general description, see the `.MODEL` statement in the *PrimeSim Continuum Reference Manual: Commands and Control Options*.

Input Syntax

```
Dxxx na nb modelname [area=val] [pj=val] [pgate=val]  
+ [m=val] [dtemp=val] [off=val] [IC=val]  
.OPTION list
```

Parameter	Description
Dxxx	Diode element name. Must begin with D, followed by up to 1023 alphanumeric characters.
na	Positive terminal (anode) node name. The series resistor for the equivalent circuit is attached to this terminal.
nb	Negative terminal (cathode) node name.
mname	Diode model name reference.
area	Diode area. In the model card, AB can use this value.
pj	Length of the side-wall in the AB diffusion area, which is not under the gate. In the model card, LS uses this value.
pgate	Length of the side-wall in the AB diffusion area, which is under the gate. In the model card, LG uses this value.
m	Multiplier to simulate multiple diodes in parallel. The M setting affects all currents, capacitances, and resistances. The default is 1.
dtemp	The difference between the element temperature and the circuit temperature in degrees celsius. The default is DTA.
off	Sets initial conditions for this element to OFF in DC analysis. Default=ON.
ic	Initial voltage across a diode element. Use this value when you specify the UIC option in the .TRAN statement. The .IC statement overrides this value.
.OPTION list	Prints the updated temperature parameters for the JUNCAP diode model.

These topics are discussed in the following sections:

- [JUNCAP1 Model](#)
- [JUNCAP2 Model - Level 6](#)

JUNCAP1 Model

To use this model, specify:

```
.MODEL mname D LEVEL=4 [keyword=val]
```

Parameter	Description
mname	Model name. The diode element uses this name to refer to the model.
D	Symbol that identifies a diode model.
LEVEL	Symbol that identifies a diode model.
keyword	Model parameter keywords.

Example

```
.model MD D LEVEL=4
+ AB=2E-12 LS=2E-6 LG=1.3E-6 DTA=0 TR=30 VR=0.3 JSGBR=1.2e-3
+ JSDBR=1.3e-3 JSGBR=1.1e-3 JSDSR=1.3e-3 JSGBR=1.4e-3
+ JSDGR=1.4e-3 NB=1.6 NS=1.3 NG=1.3 VB=0.9 CJBR=1.2e-12
+ CJSR=1.2e-12 CJGR=1.3e-12 VDBR=1.6 VDSR=1.3 VGDR=1.2 PB=0.5
+ PS=0.6 PG=0.4
```

Model Parameters

The JUNCAP1 model parameters are listed in [Table 22](#).

Table 22 JUNCAP1 Model Parameters

Name	Unit	Default	Min.	Description
NOISE	-	1	0	Turns noise for JUNCAP1 models on or off. NOISE=0: turns off JUNCAP1 model noise to be consistent with other simulators NOISE=1 (default): turns on JUNCAP1 model noise for the PrimeSim Continuum simulator/ PrimeSim backward compatibility.
AB	m ²	1e-12	0.0	Diffusion area.
CJBR	Fm-2	1.0E-12	0.0	Bottom junction capacitance at V=VR.
CJGR	Fm-2	1.0E-12	0.0	Gate-edge junction capacitance at V=VR.
CJSR	Fm-2	1.0E-12	0.0	Sidewall junction capacitance at V=VR.
DTA	°C	0.0	-	Temperature offset of Juncap element with respect to TA.
JSDBR	Am-2	1.0e-3	0.0	Bottom saturation-current density due to diffusion from back-contact.
JSDGR	Am-2	1.0e-3	0.0	Gate-edge saturation-current density due to diffusion from back-contact.

Table 22 JUNCAP1 Model Parameters (Continued)

Name	Unit	Default	Min.	Description
JSDSR	Am-2	1.0e-3	0.0	Sidewall saturation-current density due to diffusion from back-contact.
JSGBR	Am-2	1.0e-3	0.0	Bottom saturation-current density due to generating electron holes at V=VR.
JSGBR	Am-2	1.0e-3	0.0	Gate-edge saturation-current density due to generating electron holes at V=VR.
JSGSR	Am-2	1.0e-3	0.0	Sidewall saturation-current density due to generating electron holes at V=VR
LG	m	0.0	0.0	Length of the side-wall for the AB diffusion area which is under the gate. (Default deviates from Philips JUNCAP =1.0e-6.)
LS	m	0.0	0.0	Length of the side-wall for the AB diffusion area which is not under the gate. (Default deviates from Philips JUNCAP =1.0e-8.)
NB	-	1.0	0.1	Emission coefficient of the bottom forward current.
NG	-	1.0	0.1	Emission coefficient of the gate-edge forward current.
NS	-	1.0	0.1	Emission coefficient of the sidewall forward current.
PB	-	0.40	0.05	Grading coefficient of the bottom junction.
PG	-	0.40	0.05	Grading coefficient of the gate-edge junction.
PS	-	0.40	0.05	Grading coefficient of the sidewall junction.
TR	°C	25	-273.15	Temperature at which simulation finds parameter values.
VB	V	0.9	-	Reverse breakdown voltage.
VDBR	V	1.00	0.05	Diffusion voltage of the bottom junction at T=TR.
VDGR	V	1.00	0.05	Diffusion voltage of the gate-edge junction.
VDSR	V	1.00	0.05	Diffusion voltage of the sidewall junction at T=TR.
VR	V	0.0	-	Voltage at which the simulation finds parameter values.

Theory

This section summarizes the elementary physics of a junction diode. Refer to semiconductor literature for additional information.

You can represent the current voltage characteristics as follows:

$$J = \left\{ J_d \left(n_i^2 + (J_g(n_i, V)) \right) \right\} \cdot \left[\exp \left(\frac{qV}{kT} \right) - 1 \right]$$

$$n_i T^{\frac{3}{2}} \cdot \exp \left(\frac{-E_g}{2kT} \right)$$

Quantity	Units	Description
J	Am-2	Total reverse-current density.
Jd	Am-2	Diffusion saturation-current density.
Jg	Am-2	Generation-current density.
ni	m-3	Intrinsic carrier concentration.
V	V	Voltage, across the diode.
Eg	J	Energy gap.
k	JK-1	Boltzmann constant.
T	K	Temperature.

For $V < V_D$, this equation describes the charge of the junction capacitance:

$$Q = Q_j \left[1 - \left(1 - \frac{V}{V_D} \right)^{1-P} \right]$$

Quantity	Units	Description
Q	C	Total diode-junction charge.
Qj	C	Junction charge at built-in voltage.
V	V	Voltage, across the diode.
Vd	V	Diffusion voltage at the junction.
P	-	Grading coefficient for the junction.

JUNCAP Model Equations

The JUNCAP model describes reverse-biasing of source, drain, or well-to-bulk junction devices. Similar to the MOS model, this model uses quasi-static approximations (in charge equations) to formulate current equations and to model AC effects.

To include the effects from differences in the sidewall, bottom, and gate-edge junction profiles, the JUNCAP model calculates these three contributions separately.

JUNCAP also models both the diffusion and the generation currents, each with individual temperature and voltage dependence.

In the JUNCAP model, the gate-edge junction (very close to the surface) provides a part of the total charge. The MOS model charge equations also includes this charge, so simulation counts it twice. However, this results in only a very minor error.

The next section shows the model equations. For the model to operate correctly in a circuit simulator environment, you must specify some numerical additions (see the [Nomenclature](#) section). You must include any fixed capacitance that is present on a node (such as metal-1-to-substrate capacitance) in either a fixed capacitor statement or INTCAP. These capacitances are not part of the JUNCAP model as they were in the old NODCAP model.

Nomenclature

[Table 23](#) lists the electrical variable parameters:

Table 23 JUNCAP Model Electrical Variable Parameters

Variable	Program Name	Units	Description
Va	VA	V	Potential applied to the anode.
Vk	VK	V	Potential applied to the cathode.
Ia	IA	A	DC current into the anode.
Ik	IK	A	DC current into the cathode.
Qa	QA	C	Charge in a device attributed to the anode.
Qk	QK	C	Charge in a device attributed to the cathode.

Note:

The model card lists the parameters. See JUNCAP model syntax earlier in this chapter.

Table 24 lists internal variables and parameters:

Table 24 JUNCAP Model Internal Variables and Parameters

Variable or Parameter	Program Name	Units	Description
Vdb	VDB	V	Diffusion voltage of the AB bottom area.
Vds	VDS	V	Diffusion voltage of the LS Locos-edge.
Vdg	VDG	V	Diffusion voltage of the LG gate-edge.
Cjb	CJB	F	Capacitance of the AB bottom area.
Cjs	CJS	F	Capacitance of the LS Locos-edge.
Cjg	CJG	F	Capacitance of the LG gate-edge.
Isdb	ISDB	A	Diffusion saturation current of AB bottom area.
Isds	ISDS	A	Diffusion saturation current of LS Locos-edge.
Isdg	ISDG	A	Diffusion saturation current of LG gate-edge.
Is gb	ISGB	A	Generation saturation current of AB bottom area.
Is gs	ISGS	A	Generation saturation current of LS Locos-edge.
Is gg	ISGG	A	Generation saturation current of LG gate-edge.
Ta	TA	C	Ambient circuit temperature.
Tkd	TKD	K	Absolute temperature of the junction/device.
V	V	V	Diode bias voltage ($V=V_A - V_K$).
I	I	A	Total DC current from anode to cathode: ($I = I_A = -I_K$)
Q	Q	C	Total junction charge: ($Q = Q_A = -Q_K$)

ON/OFF Condition

Solving a circuit involves successive calculations. The calculations start from a set of initial guesses for the electrical quantities of the nonlinear elements. The devices start in the default state.

DC Operating Point Output

The output of a DC operating point calculation contains information about the state of a device at its operation point.

Note:

G min conductance connects in parallel to the G conductance. This conductance influences the DC operating output.

Temperature, Geometry, and Voltage Dependence

The general scaling rules which apply to all three components of the JUNCAP model are:

$$T_{KR} = T_0 + T_R$$

$$T_{KD} = T_0 + T_A + DT_A$$

$$V_{TR} = k \cdot \frac{T_{KR}}{q}$$

$$V_{TD} = K \cdot \frac{T_{KD}}{q}$$

$$V_{gR} = 1.16 \cdot \frac{(7.02 \cdot 10e-4 \cdot T_{KR} \cdot T_{KR})}{(1108.0 + T_{KR})}$$

$$V_{gD} = 1.16 \cdot \frac{(7.02 \cdot 10e-4 \cdot T_{KR} \cdot T_{KD})}{(1108.0 + T_{KD})}$$

$$F_{TD} = \left(\frac{T_{KD}}{T_{KR}} \right)^{1.5} \cdot \exp \left(\frac{V_{gR}}{(2 \cdot V_{TR})} - \frac{V_{gD}}{(2 \cdot V_{TD})} \right)$$

Internal Reference

The following equations specify the internal reference parameters for the bottom component:

$$\left(V_{DB} = \frac{V_{DBR} \cdot T_{KD}}{T_{KR} - 2 \cdot V_{TD} \cdot \ln F_{TD}} \right)$$

$$C_{JB} = C_{JBR} \cdot A_B \cdot \left(\frac{(V_{DBR} - V_R)}{V_{DB}} \right)^{P_B}$$

$$I_{SGB} = J_{SGBR} \cdot F_{TD} \cdot A_B \cdot \left(\frac{V_{DB}}{(V_{DBR} - V_R)} \right)^{P_B}$$

$$I_{SDB} = J_{SDBR} \cdot F_{TD} \cdot F_{TD} \cdot A_B$$

Locos-edge and gate-edge components use similar formulations:

- Replace the _B (bottom) index with _S (locos-edge) or _G (gate-edge).
- Replace the _{AB} (bottom) area with _{LS} (locos-edge) or _{LG} (gate-edge).

For the locos-edge:

$$V_{DS} = \frac{V_{DSR} \cdot T_{KR}}{T_{KR} - 2 \cdot V_{TD} \cdot \ln F_{TD}}$$

$$C_{JS} = C_{JSR} \cdot L_S \cdot \left(\frac{(V_{DSR} - V_R)}{V_{DS}} \right)^{P_S}$$

$$I_{SGS} = J_{SGSR} \cdot F_{TD} \cdot L_S \cdot \left(\frac{V_{DS}}{(V_{DSR} - V_R)} \right)^{P_S}$$

$$I_{SDS} = J_{SDSR} \cdot F_{TD} \cdot F_{TD} \cdot L_S$$

For the gate-edge:

$$V_{DG} = \frac{V_{DGR} \cdot T_{KD}}{T_{KR} - 2 \cdot V_{TD} \cdot \ln F_{TD}}$$

$$C_{JG} = C_{JGR} \cdot L_G \cdot \left(\frac{(V_{DGR} - V_R)}{V_{DG}} \right)^{P_G}$$

$$I_{SGS} = J_{SGGR} \cdot F_{TD} \cdot L_G \cdot \left(\frac{V_{DG}}{(V_{DGR} - V_R)} \right)^{P_G}$$

$$I_{SDS} = J_{SDGR} \cdot F_{TD} \cdot F_{TD} \cdot L_G$$

Note:

The remainder of this section shows the equations only for the bottom component.

JUNCAP Capacitor and Leakage Current Model

The charge description defines the following internal parameter:

$$Q_{JDB} = C_{JB} \cdot V_{DB} (1 - P_B)$$

To prevent an unlimited increase in the voltage derivative of the charge, the charge description consists of two parts:

- Original power function
- Supplemented quadratic function

The cross-over point between these regions (indicated as V_I) defines the following parameters:

$$F_{CB} = 1 - \left(\frac{(1 + P_B)}{3} \right)^{\frac{1}{P_B}}$$

$$V_{LB} = F_{CB} \cdot V_{DB}$$

$$C_{LB} = C_{JB} (1 - F_{CB})^{-P_B}$$

$$Q_{LB} = Q_{JDB} (1 - (1 - F_{CB})^{(1-P_B)})$$

$$Q_{JBV} = Q_{JCB} \cdot \left(\frac{(1 - (1 - V))}{V_{DB}} \right)^{(1-P_B)} \quad V < V_{LB}$$

$$Q_{LB} + C_{LB} (V - V_{LB}) \cdot \left(\frac{1 + (P_B (V - V_{LB}))}{2 \cdot V_{DB} \cdot (1 - F_{CB})} \right) \quad V \geq V_{LB} (Reference)$$

Use similar expressions for the locos-edge (Q_{JSV}) and gate-edge (Q_{JGV}) charges.

The following equation describes the total charge characteristic:

$$Q = Q_{JBV} + Q_{JSV} + Q_{JGV}$$

From the Reference equation, you can use elementary mathematics to derive simple equations for the capacitance of the bottom area:

$$C_{JBV} = C_{JB} \cdot \left(1 / \left(\frac{1 - V}{V_{DB}} \right) \right)^{P_B} \quad V < V_{LB}$$

$$C_{LB} + C_{LB} \cdot P_B \cdot \left(\frac{(V - V_{LB})}{V_{DB} \cdot (1 - F_{CB})} \right) \quad V \geq V_{LB}$$

Similar expressions exist for C_{JSV} and C_{JGV} .

Total capacitance:

$$C = C_{JBV} + C_{JSV} + C_{JGV}$$

Diffusion and Generation Currents

Using the scaled parameters from the preceding section, you can express the diffusion and generation current components as:

$$I_{DB} = I_{SDB} \cdot \exp \left(\frac{V}{(N_B \cdot V_{TD})} - 1 \right)$$

$$I_{GB} = I_{SGB} \cdot \left(\frac{(V_{DB} - V)}{V_{DB}} \right)^{P_B} \cdot \left(\exp \left(\left(\frac{V}{(N_B \cdot V_{TD})} \right) - 1 \right) \right) V \leq V_{DB}$$

$$0V > V_{DB}$$

- The first relation, concerning the diffusion component, is valid over the whole operating range.
- The second relation, describing the generation current, shows an unlimited increase in the derivative of this function at $V = V_{DB}$.

Therefore, the power function merges at $V = 0.0$ with a hyperbolic function in the forward-bias range. Simulating the model then divides the exponential part by $\exp(V / (N_B \cdot V_{TD}))$. This enables a gradual decrease in the generation-current component.

This calculation uses the hyperbolic function:

$$I_{HYP} = F_{SB} (V + V_{AB})^{-B}$$

The B parameter controls the decrease in current for voltages $V > 0.0$ for all generation components. The model sets B to a fixed value of 2. The continuity constraints of the function and derivative in the merge point lead to the following relations for F_{SB} and V_{AB} :

$$V_{AB} = B \cdot \frac{V_{DB}}{P_B}$$

$$F_{SB} = I_{SGB} \cdot V_{AB} B$$

The generation current voltage characteristic in the forward region becomes:

$$I_{GB} = F_{SB} / ((V + V_{AB})^B) \cdot (1 - \exp(-v) / (N_B \cdot V_{TD}))$$

Final Model Equations

The final model equations for the currents of the bottom area are:

$$I_{DB} = I_{SDB} \cdot (\exp(V / (N_B \cdot V_{TD})) - 1)$$

$$I_{GB} = I_{SGB} \cdot \left(\frac{V_{DB} - V}{V_{DB}} \right)^{P_B} \cdot \exp \left(\frac{V}{(N_B \cdot V_{TD})} \right) - 1 V \leq 0$$

$$I_{sgb} \cdot \left(\frac{V_{ab}}{(V + V_{ab})} \right)^B \cdot \left(1 - \exp \left(\frac{-V}{(N_b \cdot V_{td})} \right) \right) V > 0.0$$

Use similar expressions for the locos-edge and gate-edge components.

The following equation expresses the total junction current:

$$I = (I_{DB} + I_{GB}) + (I_{DS} + I_{GS}) + (I_{DG} + I_{GG})$$

JUNCAP2 Model - Level 6

The JUNCAP2 model is a compact MOS model intended to describe the behavior of the diodes that are formed by the source, drain, or well-to-bulk junctions in MOSFETs. It is the successor to the JUNCAP1 model and was developed by Koninklijke Philips Electronics N.V.

Physical Effects

The following physical effects have been included in the JUNCAP2 model:

- Geometrical scaling
- Depletion capacitance
- Ideal current
- Shockley-Read-Hall current
- Trap-assisted tunneling current
- Band-to-band tunneling current
- Avalanche breakdown
- Noise

For a full description of the JUNCAP2 model, see <https://www.nxp.com/design/software/models/juncap:JUNCAP>.

JUNCAP2 Model Updates

JUNCAP2 Model 200.6.2 Update

- Addition of parameters for temperature control: TREF parameter as alias of TRJ, TRISE parameter as alias of DTEMP
- Modify of `sqrtdumax=sqrt(abs(umax))` calculation: the absolute function is removed because the variable `umax` is always positive

JUNCAP2 Model 200.6.1 Update

There were no changes made in this version.

JUNCAP2 Model 200.6 Update

Two new parameters IFACTOR and CFACTOR are added as the multiplier factor for junction current and capacitance respectively.

JUNCAP2 Model 200.5 Update

New parameter FREV is added as a coefficient parameter for reverse breakdown current limitation.

JUNCAP Model 200.3.3 Update

Fixed bug in FJUNQ-based selection-criterion in JUNCAP-express charge model.

JUNCAP Model 200.3.0 Update

The newly introduced express-option of the JUNCAP2 model, invoked by setting SWJUNEXP=1, allows the user to trade some simulation accuracy for simulation speed. In transient analysis, a simulation time reduction of up to a factor of 5 (of the simulation time associated with JUNCAP2) has been demonstrated with a very limited loss of accuracy. This is achieved by creating a strongly simplified IV-model combined with a more extensive initialization code.

JUNCAP Model 200.2.0 Update

- The band-to-band tunneling equations have been modified. At temperatures lower than the reference temperature, V_j can become lower than V_{BIR} . This caused numerical problems in the model which, in turn, at times, resulted in convergence problems in the simulator. An alternative formulation of the equations has been implemented that avoids these problems.
- Minor bug fixes.

Usage in the PrimeSim Continuum Simulator Solution

The JUNCAP2 model is `LEVEL=6` in the Synopsys diode models. Each version can be identified with model parameter `VERSION`.

To use this model, specify:

```
.MODEL mname D LEVEL=6 [keyword=val]
```

Parameter	Description
mname	Model name. The diode element uses this name to refer to the model.
D	Symbol that identifies a diode model.

Parameter	Description
LEVEL	Symbol that identifies a diode model.
keyword	Model parameter keywords.

Example

```
.MODEL NDIO D LEVEL=6 VERSION=200.33
+ TYPE=1.000E+00 TRJ=21.000E+00 DTA=0.000E+00 IMAX=1.000E+03
+ CJORBOT=1.000E-03 CJORSTI=1.000E-09 CJORGAT=1.000E-09
+ VBIRBOT=1.000E+00 VBIRSTI=1.000E+00 VBIRGAT=1.000E+00
+ PBOT=500.000E-03 PSTI=500.000E-03 PGAT=500.000E-03
+ PHIGBOT=1.160E+00 PHIGSTI=1.160E+00 PHIGGAT=1.160E+00
+ IDSATRBOT=1.000E-12 IDSATRSTI=1.000E-18 IDSATRGAT=1.000E-18
+ CSRHBOT=100.000E+00 CSRHSTI=100.000E-06 CSRHGAT=100.000E-06
+ XJUNSTI=100.000E-09 XJUNGAT=100.000E-09 CTATBOT=100.000E+00
+ CTATSTI=100.000E-06 CTATGAT=100.000E-06 MEFFTATBOT=250.000E-03
+ MEFFTATSTI=250.000E-03 MEFFTATGAT=250.000E-03
+ CBBTBOT=1.000E-12 CBBTSTI=1.000E-18 CBBTGAT=1.000E-18
+ FBBTRBOT=1.000E+09 FBBTRSTI=1.000E+09 FBBTRGAT=1.000E+09
+ STFBOT=-1.000E-03 STFBOTSTI=-1.000E-03 STFBOTGAT=-1.000E-03
+ VBRBOT=10.000E+00 VBRSTI=10.000E+00 VBRGAT=10.000E+00
+ PBRBOT=4.000E+00 PBRSTI=4.000E+00 PBRGAT=4.000E+00
```

Model Parameters

Table 25 lists the JUNCAP2 model parameters.

Table 25 JUNCAP2 Model Parameters

Name	Unit	Default	Description
LEVEL	-	1	Diode model level
TYPE	-	1	Switch (-1 or 1) to select P-N and N-P junction
VERSION	-	200.33	Version number for model update
DTA	°C	0	Temperature offset with respect to ambient temperature
IMAX	A	1000	Maximum current up to which forward current behaves exponentially
TRJ	°C	21	Reference temperature
Capacitance parameters			
CJORBOT	F/m ²	1e-3	Zero-bias capacitance per unit-of area of bottom component

Table 25 JUNCAP2 Model Parameters (Continued)

Name	Unit	Default	Description
CJORGAT	F/m	1e-9	Zero-bias capacitance per unit-of length of gate-edge component
CJORSTI	F/m	1e-9	Zero-bias capacitance per unit-of length of STI-edge component
PBOT	-	0:5	Grading coefficient of bottom component
PGAT	-	0:5	Grading coefficient of gate-edge component
PSTI	-	0:5	Grading coefficient of STI-edge component
VBIRBOT	V	1	Built-in voltage at the reference temperature of bottom component
VBIRGAT	V	1	Built-in voltage at the reference temperature of gate-edge component
VBIRSTI	V	1	Built-in voltage at the reference temperature of STI-edge component
Ideal-current parameters			
IDSATRBOT	A/m ²	1e-12	Saturation current density at the reference temperature of bottom component
IDSATRGAT	A/m	1e-18	Saturation current density at the reference temperature of gate-edge component
IDSATRSTI	A/m	1e-18	Saturation current density at the reference temperature of STI-edge component
PHIGBOT	V	1.16	Zero-temperature bandgap voltage of bottom component
PHIGGAT	V	1.16	Zero-temperature bandgap voltage of gate-edge component
PHIGSTI	V	1.16	Zero-temperature bandgap voltage of STI-edge component
Shockley-Read-Hall parameters			
CSRHBOT	A/m ³	1e+2	Shockley-Read-Hall prefactor of bottom component
CSRHGAT	A/m ²	1e-4	Shockley-Read-Hall prefactor of gate-edge component
CSRHSTI	A/m ²	1e-4	Shockley-Read-Hall prefactor of STI-edge component
XJUNGAT	m	1e-7	Junction depth of gate-edge component
XJUNSTI	m	1e-7	Junction depth of STI-edge component

Table 25 JUNCAP2 Model Parameters (Continued)

Name	Unit	Default	Description
Trap-assisted tunneling parameters			
CTATBOT	A/m^3	1e+2	Trap-assisted tunneling prefactor of bottom component
CTATGAT	A/m^2	1e-4	Trap-assisted tunneling prefactor of gate-edge component
CTATSTI	A/m^2	1e-4	Trap-assisted tunneling prefactor of STI-edge component
MEFFTATBOT	-	0:25	Effective mass (in units of m_0) for trap-assisted tunneling of bottom component
MEFFTATGAT	-	0:25	Effective mass (in units of m_0) for trap-assisted tunneling of gate-edge component
MEFFTATSTI	-	0:25	Effective mass (in units of m_0) for trap-assisted tunneling of STI-edge component
Band-to-band tunneling parameters			
CBBTBOT	AV^{-3}	1e-12	Band-to-band tunneling prefactor of bottom component
CBBTGAT	$AV^{-3}m$	1e-18	Band-to-band tunneling prefactor of gate-edge component
CBBTSTI	$AV^{-3}m$	1e-18	Band-to-band tunneling prefactor of STI-edge component
FBBTBOT	Vm^{-1}	1e+9	Normalization field at the reference temperature for band-to-band tunneling of bottom component
FBBTGAT	Vm^{-1}	1e+9	Normalization field at the reference temperature for band-to-band tunneling of gate-edge component
FBBTSTI	Vm^{-1}	1e+9	Normalization field at the reference temperature for band-to-band tunneling of STI-edge component
STFBTBOT	K^{-1}	-1e-3	Temperature scaling parameter for band-to-band tunneling of bottom component
STFBTGAT	K^{-1}	-1e-3	Temperature scaling parameter for band-to-band tunneling of gate edge component
STFBTSTI	K^{-1}	-1e-3	Temperature scaling parameter for band-to-band tunneling of STI-edge component
Avalanche and breakdown parameters			
PBRBOT	V	4	Breakdown onset tuning parameter of bottom component
PBRGAT	V	4	Breakdown onset tuning parameter of gate-edge component

Table 25 JUNCAP2 Model Parameters (Continued)

Name	Unit	Default	Description
PBRSTI	V	4	Breakdown onset tuning parameter of STI-edge component
VBRBOT	V	10	Breakdown voltage of bottom component
VBRGAT	V	10	Breakdown voltage of gate-edge component
VBRSTI	V	10	Breakdown voltage of STI-edge component

Using the Fowler-Nordheim Diode, Level 2

The `LEVEL=2` diode model parameter selects the Fowler-Nordheim model. Fowler-Nordheim diodes can be either a metal-insulator-semiconductor or a semiconductor-insulator-semiconductor layer device. The insulator is sufficiently thin (100 Angstroms) to permit tunneling of carriers. It models:

- Electrically-alterable memory cells
- Air-gap switches
- Other insulation-breakdown devices

These topics are discussed in the following sections:

- [Fowler-Nordheim Diode Model Parameters LEVEL=2](#)
- [Using Fowler-Nordheim Diode Equations](#)
- [Fowler-Nordheim Diode Capacitances](#)

Fowler-Nordheim Diode Model Parameters LEVEL=2

Table 26 shows the Fowler-Nordheim diode model parameters for `LEVEL=2`.

Table 26 Fowler-Nordheim Diode Model Parameters

Name (alias)	Units	Default	Description
EF	V/cm	1.0e8	Forward critical electric field.
ER	V/cm	EF	Reverse critical electric field.
JF	amp/V ²	1.0e-10	Forward current coefficient for Fowler-Nordheim.
JR	amp/V ²	JF	Reverse current coefficient for Fowler-Nordheim.

Table 26 Fowler-Nordheim Diode Model Parameters (Continued)

Name (alias)	Units	Default	Description
L	m	0.0	Length of the diode for calculating the current in Fowler-Nordheim. $L_{eff} = L \cdot SCALM \cdot SHRINK + XW_{eff}$
TOX	Å	100.0	Thickness of the oxide layer.
W	m	0.0	Width of the diode for calculating the current in Fowler-Nordheim. $W_{eff} = W \cdot SCALM \cdot SHRINK + XW_{eff}$
XW	m	0.0	$XW_{eff} = XW \cdot SCALM$

Using Fowler-Nordheim Diode Equations

The following forward and reverse nonlinear current source equations model the DC characteristics of the Fowler-Nordheim diode. In these equations:

$$AREA_{eff} = W_{eff} \cdot L_{eff} \cdot M$$

Forward Bias

$$v_d \geq 0$$

$$i_d = AREA_{eff} \cdot JF \cdot \left(\frac{v_d}{TOX} \right)^2 \cdot e^{-\frac{EF_{PTOX}}{v_d}}$$

Reverse Bias

$$v_d < 0$$

$$i_d = -AREA_{eff} \cdot JRP \cdot \left(\frac{v_d}{TOX} \right)^2 \cdot e^{\frac{ERTOX}{v_d}}$$

Fowler-Nordheim Diode Capacitances

The Fowler-Nordheim diode capacitance is a constant derived from:

$$cd = AREA_{eff} \cdot \frac{\epsilon_{ox}}{TOX}$$

Philips D500 Model (Advanced Diode Model), Level 5

The Diode 500 model provides a detailed description of the diode currents in forward and reverse biased Si-diodes. It is meant to be used for DC, transient, and AC analysis.

The Philips D500 model is available as Level 5 in the Synopsys diode models. For more information about the D500 model, see:

<https://www.nxp.com/design/software/models/advanced-diode-model:ADV-DIODE>

These topics are discussed in the following sections:

- [Using the Philips D-500 Model](#)
- [Equivalent Circuits and Equations](#)

Using the Philips D-500 Model

Set level=5 to identify the model as Philips D500 Model. You can use DTEMP with this model to increase the temperature of individual elements, relative to the circuit temperature. Set DTEMP on the element line.

The general syntax for the Diode element is the same as the other standard diode models.

The available parameters are described in [Table 27](#).

Table 27 Description of Philips D500 Parameters

Name	Units	Default	Clip low	Clip high	Description
Level	-	5	-	-	Model level
AF	-	1.0	0.01	-	Flicker noise exponent
CBBT	A/V	3.255	0.0	-	Band to band tunneling
CJ	F	7.0E-12	0.0	-	Zero-bias depletion capacitance
CSRH	A/cm	7.44E-7	0.0	-	Shockley-Read-Hall generation
CTAT	A/cm	3.31E-6	-0.0	-	Trap assisted tunneling
IS	A	7.13E-13	0.0	-	Saturation current
KF	-	0.0	0.0	-	Flickernoise coefficient
N	-	1.044	0.1	-	Junction emission coefficient
P	-	0.40	0.05	0.99	Grading coefficient
PTRS	-	0.0	-	-	Power for temperature dependence of RS
RS	Ohm	0.0	0.0	-	Series resistance

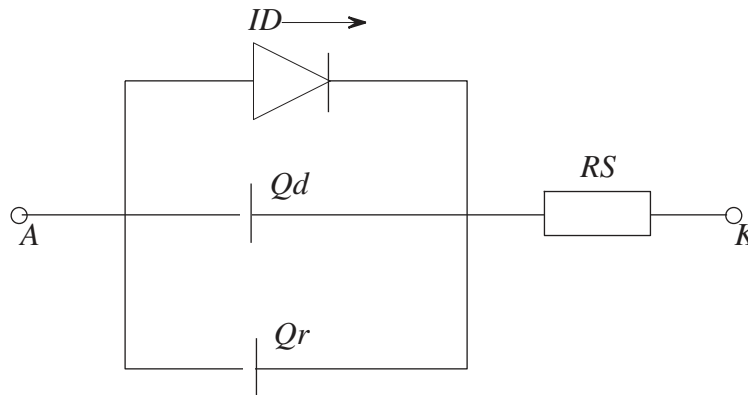
Table 27 Description of Philips D500 Parameters (Continued)

Name	Units	Default	Clip low	Clip high	Description
TAU	S	500.0E-12	0.0	-	Transit time
TREF	°C	25.0	-273.15	-	Reference temperature
EMVER	V/cm	1.36E6	1.0	-	Electric field at breakdown
VBR	V	7.459	0.1	-	Breakdown voltage
VD	V	0.90	-0.05	-	Diffusion voltage
VG	V	1.206	0.1	-	Bandgap voltage
VLC	V	0.0	-	-	Voltage dependence at low forward current

Equivalent Circuits and Equations

A full description of D level-500 for diode is provided below. The DC/transient and AC equivalent circuits are shown in [Figure 14](#) and [Figure 15](#), respectively.

Figure 14 DC/Transient Circuit for Diode



Temperature Effects

The actual simulation temperature is denoted by $TEMP$ (in °C). The temperature at which the parameters are determined is specified by $TREF$ (in °C.)

- Conversion to Kelvin

$$TK = TEMP + 273.15 + DTA$$

$$TRK = TREF + 273.15$$

- Thermal Voltages

$$k = 1.3806226 \cdot 10^{-23} JK^{-1}$$

$$q = 1.6021918 \cdot 10^{-19} C$$

$$V_T = \left(\frac{k}{q} \right) \cdot T_K$$

$$V_{TR} = \left(\frac{k}{q} \right) \cdot T_{RK}$$

- Depletion Capacitances

$$F = \left(\frac{T_K}{T_{RK}} \right)^3 \cdot \exp \left[VG \cdot \left(\frac{1}{V_{TR}} - \frac{1}{V_T} \right) \right]$$

$$VD_T = \left[\frac{VD}{V_{TR}} - \ln(F) \right] \cdot V_T$$

$$CJ_T = CJ \cdot \left(\frac{VD}{VD_T} \right)^P$$

- Transit Times

$$TAU_T = TAU \left(\frac{T_K}{T_{RK}} \right)^{1.8}$$

- Saturation Current

$$IS_T = IS \cdot \left(\frac{T_K}{T_{RK}} \right)^{1.8} \cdot \exp \left[\frac{VG}{N} \cdot \left(\frac{1}{V_{TR}} - \frac{1}{V_T} \right) \right]$$

- Shockley-Read-Hall generation and trap assisted tunneling

$$T_{up} = \left(\frac{T_K}{T_{RK}} \right)^{3/2} \cdot \exp \left[\frac{VG + VLC}{2} \cdot \left(\frac{1}{V_{TR}} - \frac{1}{V_T} \right) \right]$$

$$CSRH_T = CSRH \cdot T_{up}$$

$$CTAT_T = CTAT \cdot T_{up}$$

$$ETAT_T = 70.8 \cdot T_K^{3/2}$$

- Band-to-Band Tunneling

$$CBBT_T = CBBT \text{ (temperature independent)}$$

$$F0 = 1.9 \cdot 10^7 \cdot \left(1.04 - \frac{4.21 \cdot 10^{-4} \cdot T_K^2}{636 + T_K} \right)$$

- Avalanche Multiplication

$$dT = TEMP + DTA - 25^\circ C$$

$$Bn = 1.23 \cdot 10^6$$

$$Bn_T = Bn \cdot (1 + 7.2 \cdot 10^{-4} \cdot dT - 1.6 \cdot 10^{-6} \cdot dT^2)$$

- Breakdown

$$VBR_T = VBR \cdot \left(\frac{T_K}{T_{RK}} \right)^{0.1}$$

$$EMVBR_T = EMVBR \cdot \left(\frac{VD_T + VBR_T}{VD + VBR} \right)^{1-P}$$

- Resistance

$$RS_T = RS \cdot \left(\frac{T_K}{T_{RK}} \right)^{PTRS}$$

Model Constants and Parameter-Related Constants

$$K = 0.01$$

$$KET = 0.1$$

$$ETM = 3$$

- Maximum Electric Field and Depletion Layer width at zero bias:

$$E_0 = \frac{EMVBR_T}{\left(1 + \frac{VBR_T}{VD_T} \right)^{1-P}}$$

$$W_0 = \frac{VD_T}{E_0 \cdot (1-P)}$$

Diode Currents

First, the maximum reverse junction voltage is defined.

- Above this voltage the current is extrapolated on a logarithmic scale.

$$V_j = \begin{cases} -0.99 \cdot VBR_T, & V_{AK1} < -0.99 \cdot VBR_T \\ V_{AK1}, & V_{AK1} \geq -0.99 \cdot VBR_T \end{cases}$$

- Ideal Forward Current

$$I_{df} = I_{ST} \left\{ \exp \left(\frac{V_j}{N \cdot V_T} \right) - 1 \right\}$$

- Maximum Electric Field and Depletion Layer Width

$$VD_j = \left(\sqrt{\left(1 - \frac{V_j}{VD_T} \right)^2 + \left(\frac{V_j}{VD_T} \cdot K \right)^2} + \left(1 - \frac{V_j}{VD_T} \right) \right) / 2$$

$$E_m = E_0 \cdot VD_j^{(1-P)}$$

$$W_d = W_0 \cdot VD_j^P$$

- Shockley-Read-Hall Generation

$$I_{srh} = CSRH_T \cdot (W_d - W_0)$$

- Trap-Assisted Tunneling

$$ET_0 = \frac{\frac{E_0}{ETAT_T} + ETM - \sqrt{\left(\frac{E_0}{ETAT_T} - ETM \right)^2 + KET}}{2}$$

$$ET = \frac{\frac{E_m}{ETAT_T} + ETM - \sqrt{\left(\frac{E_m}{ETAT_T} - ETM \right)^2 + KET}}{2}$$

$$I_{tat} = CTAT_T \cdot W_d \cdot \left\{ \frac{\exp(ET^2) - \exp(ET_0^2)}{\frac{E_m}{ETAT_T}} \right\}$$

- Non-Ideal Forward Current including Tunneling

$$I_{sif} = CSRH_T \cdot \left\{ 6.28 + 38.58 \cdot \left(\frac{E_m}{ETAT_T} \cdot \exp(ET^2) \right) \right\} \cdot \frac{V_T}{E_m}$$

$$I_{lf} = I_{sif} \cdot \frac{\exp\left(\frac{V_j}{N \cdot V_T}\right) - 1}{4 \cdot \exp\left(\frac{V_j}{2N \cdot V_T}\right) + \exp\left(\frac{VLC}{2N \cdot V_T}\right)} \cdot \exp\left(\frac{VLC}{2 \cdot N \cdot V_T}\right)$$

- Band-to-Band-Tunneling

$$I_{bbt} = \frac{-CBBT_T \cdot V_j}{\left(\frac{F0}{E_m}\right)^{1.5} \cdot \exp\left(\frac{F0}{E_m}\right)}$$

- Avalanche Multiplication

$$\mu = 0.3295 \cdot \left(\frac{E_m}{EMVBR_T}\right)^2 \cdot \exp\frac{Bn_T}{EMVBR_T} - \frac{Bn_T}{E_m}$$

- Total Diode Current

$$I_d = \frac{(Id_f + I_{lf} - I_{srh}) \cdot \frac{1 + \exp(-2\mu)}{2} (I_{bbt} + I_{tat}) \cdot \exp(-\mu)}{1 - 2 \cdot \mu \cdot \{1 + \exp(-2 \cdot \mu)\}}$$

- Extrapolation of the Reverse Current

$$I_{dBR} = I_d \text{ at } V_j = -0.99VBR_T$$

$$G_{dBR} = \frac{dI_d}{dV_j} \text{ at } V_j = -0.99VBR_T$$

$$I_D = \begin{cases} I_d V_{AK1} \geq -0.99VBR_T \\ I_{dBR} \cdot \exp\left[\left(\frac{V_{AK1} + 0.99VBR_T}{I_{dBR}}\right) G_{dBR}\right] V_{AK1} < -0.99VBR_T \end{cases}$$

Transient Model

Transient behavior is modeled using the DC equations.

- Diffusion charge

$$Q_D = TAU_T \cdot Id_f$$

- Depletion charge

$$FC = 1 - \left(\frac{1+P}{3}\right)^{\left(\frac{1}{P}\right)}$$

$$Q_{AT} = CJ_T \cdot \left(\frac{VD_T}{1-P}\right)$$

$$V_L = FC \cdot VD_T$$

$$C_L = CJ_T \cdot (1-FC)^{-P}$$

$$Q_L = Q_{AT} \cdot \{1 - (1-FC)^{(1-P)}\}$$

Then, if $V_{AK1} < V_L$

$$Q_T = Q_{AT} \cdot \left[1 - \left\{ 1 - \left(\frac{V_{AK1}}{VD_T} \right) \right\}^{(1-P)} \right]$$

Or, if $V_{AK1} \geq V_L$

$$Q_T = Q_L + C_L \cdot (V_{AK1} - V_L) \cdot \left\{ 1 + \frac{P \cdot (V_{AK1} - V_L)}{2 \cdot VD_T \cdot (1-FC)} \right\}$$

AC Linearized model

Using the appropriate definitions for the various circuit elements leads to the following equation:

$$R_D = \frac{1}{dID / dV_{AK1}}$$

Where dID / dV_{AK1} is the first derivative of the total diode current with respect to the Internal voltage V_{AK1} .

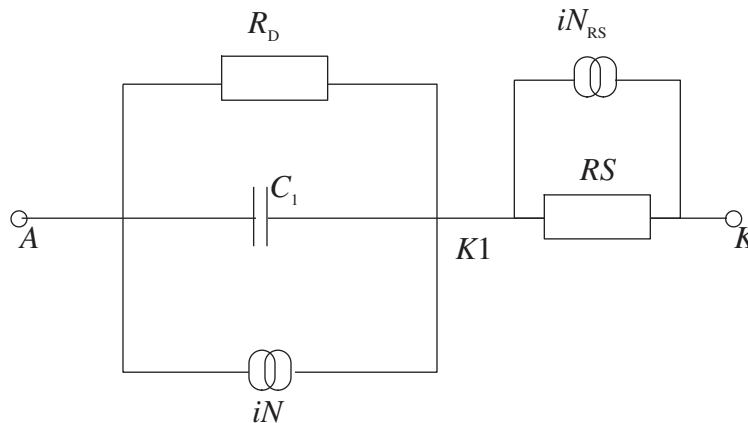
The capacitances are defined as:

$$C_T = C_{J_T} \cdot \left\{ 1 - \left(\frac{V_{AK1}}{VD_T} \right) \right\}^{-P} \text{ for } V_{AK1} < V_L$$

$$C_T = C_L \cdot \left\{ 1 + \frac{P \cdot (V_{AK1} - V_L)}{VD_T \cdot (1-FC)} \right\} \text{ for } V_{AK1} \geq V_L$$

$$C_1 = C_T + TAU_T \cdot \left(\frac{Id_f + IS_T}{N \cdot V_T} \right)$$

Figure 15 AC equivalent circuit for Diode, including Noise Sources



Noise Model

For noise analysis, noise sources are added to the small signal model as shown in [Figure 15](#). In these equations f represents the operation frequency of the transistor and Δf is the bandwidth. When Δf is taken as 1 Hz, a noise density is obtained.

- Thermal noise

$$\bar{iN}_{RS}^2 = \frac{4 \cdot K \cdot T_K \cdot \Delta f}{RS_T}$$

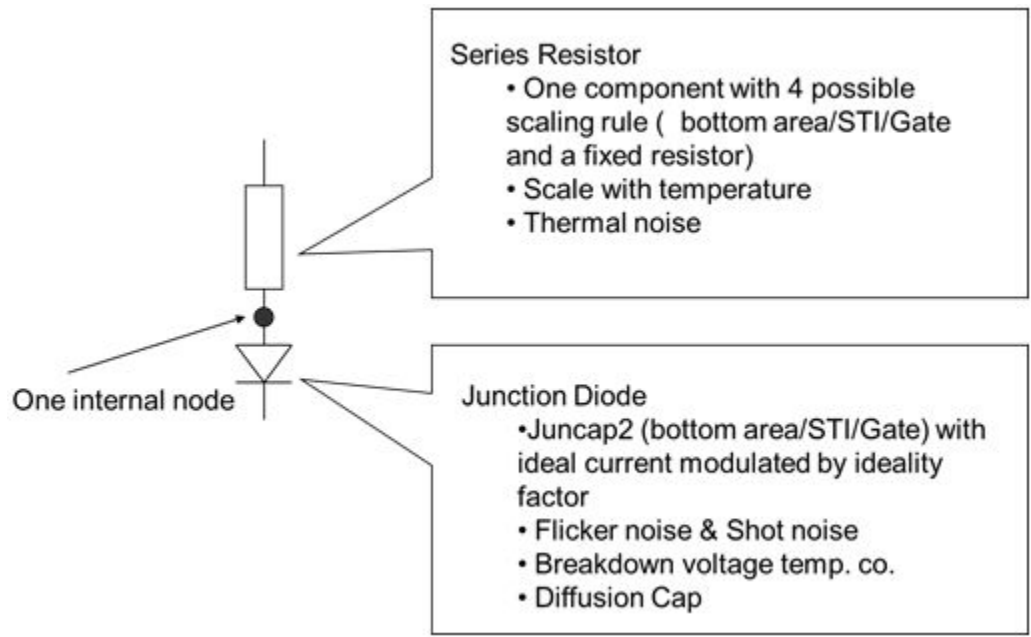
- Current noise (shot noise and 1/f noise)

$$\bar{iN}^2 = 2 \cdot q \cdot |Id_f| \cdot \Delta f + KF + MULT \cdot \left| \frac{Id_f}{MULT} \right|^{AF} \cdot \frac{\Delta f}{f}$$

DIODE_CMC Model, Level 7

The DIODE_CMC model includes following enhancement beyond JUNCAP2:

- Series resistor
- Diffusion cap
- Breakdown voltage temperature coefficient
- Ideality factor
- Noise
- Change in the default value of the instance parameter LG
- Alias parameters for AB, LS, and MULT
- Min-max parameters for warning purposes



Set level=7 to identify the diode model as DIODE_CMC.

This section covers the following topics:

- [Diode_CMC Version Updates](#)
- [Instance Parameters](#)
- [Model Parameters](#)

Diode_CMC Version Updates

Diode_CMC 2.0.0 Updates

Diode_CMC 2.0.0 implemented the following features:

- Newly added recovery model
- Improved continuity in JUNCAP current equation

Instance Parameters

Name	Unit	Default	Min	Max	Description
AB	m^2	1.00E-12	0	-	Junction area
AREA	m^2	1.00E-12	0	-	Alias of AB

Name	Unit	Default	Min	Max	Description
LS	m	1.00E-06	0	-	STI-edge part of junction perimeter
PERIM	m	1.00E-06	0	-	Alias of LS
PJ	m	1.00E-06	0	-	Alias of LS
LG	m	0	0	-	Gate-edge part of junction perimeter
MULT	-	1	0	-	Number of devices in parallel
DTEMP	K	0	-	-	Difference between the temperature of the element and circuit

Model Parameters

Name	Unit	Default	Min	Max	Description
VERSION	-	1	-	-	Model version
SUBVERSION	-	0	-	-	Model subversion
REVISION	-	0	-	-	Model revision
TYPE	-	1	-1	1	Type parameter, 1 reflects n-type, -1 reflects p-type
DTA	C	0	-	-	Temperature offset with respect to ambient temperature

Note:

If a model parameter is out of range, the PrimeSim Continuum solution prints an informational message in the *.lis file. The value of the model parameter is, however, not changed.

This section covers the following topics:

- [DIODE_CMC - Reduced Parameters](#)
- [JUNCAP2 - Express Parameters](#)
- [Diode_CMC Recovery and High-Injection Equation Parameters](#)

DIODE_CMC - Reduced Parameters

Name	Unit	Default	Min	Max	Description
IMAX	A	1000	1.00E-12	-	Maximum current up to which forward current behaves exponentially
TRJ	C	21	-250	-	Reference temperature
CJORBOT	Fm ⁻²	1.00E-03	1.00E-12	-	Zero-bias capacitance per unit-of-area of bottom component
CJORSTI	Fm ⁻¹	1.00E-09	1.00E-18	-	Zero-bias capacitance per unit-of-length of STI-edge component
CJORGAT	Fm ⁻¹	1.00E-09	1.00E-18	-	Zero-bias capacitance per unit-of-length of gate-edge component
VBIRBOT	V	1	0.05	-	Built-in voltage at the reference temperature of bottom component
VBIRSTI	V	1	0.05	-	Built-in voltage at the reference temperature of STI-edge component
VBIRGAT	V	1	0.05	-	Built-in voltage at the reference temperature of gate-edge component
PBOT	-	0.5	0.05	0.95	Grading coefficient of bottom component
PSTI	-	0.5	0.05	0.95	Grading coefficient of STI-edge component
PGAT	-	0.5	0.05	0.95	Grading coefficient of gate-edge component
PHIGBOT	V	1.16	-	-	Zero-temperature bandgap voltage of bottom component
PHIGSTI	V	1.16	-	-	Zero-temperature bandgap voltage of STI-edge component
PHIGGAT	V	1.16	-	-	Zero-temperature bandgap voltage of gate-edge component
IDSATRBOT	Am ⁻²	1.00E-12	0	-	Saturation current density at the reference temperature of bottom component

Name	Unit	Default	Min	Max	Description
IDSATRSTI	Am^{-1}	1.00E-18	0	-	Saturation current density at the reference temperature of STI-edge component
IDSATRGAT	Am^{-1}	1.00E-18	0	-	Saturation current density at the reference temperature of gate-edge component
CSRHBOT	Am^{-3}	1.00E+02	0	-	Shockley-Read-Hall prefactor of bottom component
CSRHSTI	Am^{-2}	1.00E-04	0	-	Shockley-Read-Hall prefactor of STI-edge component
CSRHGAT	Am^{-2}	1.00E-04	0	-	Shockley-Read-Hall prefactor of gate-edge component
XJUNSTI	m	1.00E-07	1.00E-09	-	Junction depth of STI-edge component
XJUNGAT	m	1.00E-07	1.00E-09	-	Junction depth of gate-edge component
CTATBOT	Am^{-3}	1.00E+02	0	-	Trap-assisted tunneling prefactor of bottom component
CTATSTI	Am^{-2}	1.00E-04	0	-	Trap-assisted tunneling prefactor of STI-edge component
CTATGAT	Am^{-2}	1.00E-04	0	-	Trap-assisted tunneling prefactor of gate-edge component
MEFFTATBOT	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of bottom component
MEFFTATSTI	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of STI-edge component
MEFFTATGAT	-	0.25	0.01	-	Effective mass (in units of m_0) for trap-assisted tunneling of gate-edge component
CBBTBOT	AV^{-3}	1.00E-12	0	-	Band-to-band tunneling prefactor of bottom component
CBBTSTI	AV^{-3}m	1.00E-18	0	-	Band-to-band tunneling prefactor of STI-edge component

Name	Unit	Default	Min	Max	Description
CBBTGAT	AV^{-3}m	1.00E-18	0	-	Band-to-band tunneling prefactor of gate-edge component
FBBTBTRBOT	Vm^{-1}	1.00E+09	-	-	Normalization field at the reference temperature for band-to-band tunneling of bottom component
FBBTBTRSTI	Vm^{-1}	1.00E+09	-	-	Normalization field at the reference temperature for band-to-band tunneling of STI-edge component
FBBTBTRGAT	Vm^{-1}	1.00E+09	-	-	Normalization field at the reference temperature for band-to-band tunneling of gate-edge component
STFBBTBTRBOT	K^{-1}	-1.00E-03	-	-	Temperature scaling parameter for band-to-band tunneling of bottom component
STFBBTBTRSTI	K^{-1}	-1.00E-03	-	-	Temperature scaling parameter for band-to-band tunneling of STI-edge component
STFBBTBTRGAT	K^{-1}	-1.00E-03	-	-	Temperature scaling parameter for band-to-band tunneling of gate-edge component
VBRBOT	V	10	0.1	-	Breakdown voltage of bottom component
VBRSTI	V	10	0.1	-	Breakdown voltage of STI-edge component
VBRGAT	V	10	0.1	-	Breakdown voltage of gate-edge component
PBRBOT	V	4	0.1	-	Breakdown onset tuning parameter of bottom component
PBRSTI	V	4	0.1	-	Breakdown onset tuning parameter of STI-edge component
PBRGAT	V	4	0.1	-	Breakdown onset tuning parameter of gate-edge component
RSBOT	VA^{-1}m^2	0	0	-	Series resistance per unit-of-area of bottom component

Name	Unit	Default	Min	Max	Description
RSSTI	VA^{-1}m	0	0	-	Series resistance per unit-of-length of STI-edge component
RSGAT	VA^{-1}m	0	0	-	Series resistance per unit-of-length of gate-edge component
RSCOM	ohm	0	0	-	Common series resistance, no scaling
STRS	-	0	0	-	Temperature scaling parameter for series resistance
KF	-	0	0	-	KF parameter for flicker noise
AF	-	1	0.1	-	AF parameter for flicker noise
TT	s	0	0	-	Transit time
STVBRBOT1	1/K	0	-	-	Temperature component of breakdown voltage bottom component
STVBRBOT2	$1/\text{K}^2$	0	-	-	Temperature component of breakdown voltage bottom component
STVBRSTI1	1/K	0	-	-	Temperature component of breakdown voltage STI-edge component
STVBRSTI2	$1/\text{K}^2$	0	-	-	Temperature component of breakdown voltage STI-edge component
STVBRGAT1	1/K	0	-	-	Temperature component of breakdown voltage gate-edge component
STVBRGAT2	$1/\text{K}^2$	0	-	-	Temperature component of breakdown voltage gate-edge component
NFABOT	-	1	0.1	-	Ideality factor bottom component
NFASTI	-	1	0.1	-	Ideality factor STI-edge component
NFAGAT	-	1	0.1	-	Ideality factor gate-edge component

Name	Unit	Default	Min	Max	Description
ABMIN	m ²	0	0	-	Minimum allowed junction area
ABMAX	m ²	1	0	-	Maximum allowed junction area
LSMIN	m	0	0	-	Minimum allowed junction STI-edge
LSMAX	m	1	0	-	Maximum allowed junction STI-edge
LGMIN	m	0	0	-	Minimum allowed junction gate-edge
LGMAX	m	1	0	-	Maximum allowed junction gate-edge
TEMPMIN	C	-55	-250	-	Minimum allowed junction temp
TEMPMAX	C	155	-250	-	Maximum allowed junction temp
VFMAX	V	0	0	-	Maximum allowed forward junction bias
VRMAX	V	0	0	-	Maximum allowed reverse junction bias
XTI	-	3	0.1	-	Temperature component of saturation current
PT	-	3	0.1	-	Alias of XTI
SCALE	-	1	0	1	Scale parameter
SHRINK	-	0	0	100	Scale parameter

JUNCAP2 - Express Parameters

Name	Unit	Default	Min	Max	Description
SWJUNEXP	-	0	0	1	Flag for JUNCAP-express; 0=full model, 1=express model
VJUNREF	-	2.5	0.5	-	Typical maximum junction voltage; usually about 2*VSUP
FJUNQ	-	0.03	0	-	Fraction below which junction capacitance components are considered negligible

Diode_CMC Recovery and High-Injection Equation Parameters

Name	Unit	Default	Min	Max	Description
CORECOVERY	-	0	0	1	Flag for recovery equations; 0=original, 1=Hiroshima model
NJH	-	1	0.5	5	High-injection emission coefficient
NJDV	1/V	0.1	0	1.00E+06	Transition slope of emission coefficient
NDIBOT	cm ⁻³	1.00E+16	1	1.00E+23	Doping concentration of drift region (bottom)
NDIGAT	cm ⁻³	1.00E+16	1	1.00E+23	Doping concentration of drift region (gate edge)
NDISTI	cm ⁻³	1.00E+16	1	1.00E+23	Doping concentration of drift region (STI edge)
INJ1	-	1	0	3	For recovery charge density
INJ2	-	10	0	20	For recovery charge density in high-injection condition
NQS	sec	5.00E-09	0	1.00E-03	Carrier delay time
TAU	sec	2.00E-07	0	1.00E-03	Carrier lifetime
WI	m	5.00E-06	0	1	Length of drift region
DEPNQS	sec	0	0	1.00E-03	Depletion delay time
TNOM	C	21	-250	inf	Alias reference temperature TRJ
TAUT	-	0	0	1.00E+02	Temperature coefficient of carrier lifetime
INJT	-	0	0	20	Temperature coefficient of carrier density in high-injection condition

Converting National Semiconductor Models

National Semiconductor's circuit simulator provides a scaled diode model, which is not the same as the diode device model. To use National Semiconductor circuit models, do the following:

1. For a subcircuit that consists of the scaled diode model, ensure that the subcircuit name is the same as the model name.

The `.PARAM` statement inside the subcircuit specifies the parameter values for the scaled diode model.
2. Add a scaled diode model inside the subcircuit and change the `.MODEL mname mtype` statement to a `.PARAM` statement.
3. Ensure that the letter X precedes the names of all the scaled diode elements.
4. Check that every parameter used in the `.MODEL` statement, inside the subcircuit, also has a value in the `.PARAM` statement.

Using the Scaled Diode Subcircuit Definition

The scaled diode subcircuit definition converts the National Semiconductor scaled diode model to a model that you can use in the PrimeSim Continuum solution. The `.PARAM` parameter, inside the `.SUBCKT`, represents the `.MODEL` parameter in the National circuit simulator.

To use this definition:

1. Replace the `.MODEL mname` statement with a `.PARAM` statement.
2. Change the model name to `SDIODE`.

The following is an example of a scaled-diode subcircuit definition:

```
.SUBCKT SDIODE NP NN SF=1 SCJA=1 SCJP=0 SIS=1 SICS=1
+ SRS=1
D NP NN SDIODE
.PARAM IS=1.10E-18 N=1.03 EG=0.8 RS=20.7E3
+ CJA=0.19E-15 PHI=0.25 CJP=0.318E-15
+ EXA=0.5 EXP=0.325 CTC=6E-4
+ TRS=2.15M M=2
*
.MODEL SDIODE D
+ IS='IS*SIS*SF' CJA='CJA*SF*SCJA' CJP='CJP*SF*SCJP'
+ RS='RS*SRS/SF' EXA=EXA EXP=EXP
+ N=N CTA=CTC CTP=CTC
+ TRS=TRS TLEV=1 TLEV=1 xti='m*n'
.ENDS SDIODE
```

You must define the values for all parameters used in this model in either a `.PARAM` statement or the `.SUBCKT` call. Circuit simulation then replaces the diode statements with the call to the `SDIODE` subcircuit; for example,

```
XDS 14 1048 SDIODE SIS=67.32 SCJA=67.32 SRS=1.2285E-2
```

DC Operating Point Output of Diodes

`id`: current across the diode.

`vd`: voltage across the diode.

`req`: equivalent resistance ($1 / \text{equivalent conductance}$).

`cap`: total diode capacitance.

4

JFET and MESFET Models

Describes how to use JFET and MESFET models in the PrimeSim Continuum solution circuit simulations.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

Three JFET/MESFET DC model levels have been provided for IC circuit simulation. These models use the same equations for gallium arsenide MESFETs and silicon-based JFETs. This is possible because these models include materials definition parameters. You can also use these models to model indium phosphide MESFETs.

- [Overview of JFETs](#)
- [Specifying a Model](#)
- [Bypassing Latent Devices \(PrimeSim HSPICE Only\)](#)
- [Overview of Capacitor Model](#)
- [JFET and MESFET Equivalent Circuits](#)
- [JFET and MESFET Model Statements](#)
- [JFET and MESFET DC Equations](#)
- [JFET and MESFET Noise Models](#)
- [JFET and MESFET Temperature Equations](#)
- [TriQuint \(TOM\) Extensions to LEVEL=3](#)
- [LEVEL=7 TOM3 \(TriQuint's Own Model III\)](#)
- [LEVEL=8 Materka Model](#)
- [Level=100 TSMC JFET Model](#)
- [References](#)

Overview of JFETs

JFETs form by diffusing a gate diode between the source and the drain. MESFETs form by applying a metal layer over the gate region and creating a Schottky diode. To control the flow of carriers, both technologies modulate the gate diode depletion region. These field effect devices are called bulk semiconductors and are in the same category as bipolar transistors. Compared to surface effect devices such as MOSFETs, bulk semiconductors have higher gain. This is because bulk semiconductor mobility is always higher than surface mobility.

Enhanced characteristics of JFETs and MESFETs, relative to surface effect devices, include lower noise generation rates and higher immunity to radiation. These advantages have created the need for newer and more advanced models.

Features for JFET and MESFET modeling include:

- Charge-conserving gate capacitors
- Backgating substrate node
- Mobility degradation due to gate field
- Computationally efficient DC model (Curtice and Statz)
- Subthreshold equation
- Physically correct width and length ($_{ACM}$)

GaAs model `LEVEL=3` [1] assumes that GaAs device velocity saturates at very low drain voltages. This model includes drain voltage induced threshold modulation and user-selectable material constants. These features let you use the model for other materials, such as silicon, indium phosphide, and gallium aluminum arsenide. The models that have been provided include a revised Curtice model [2] and a TriQuint model (TOM) that extends the earlier Statz model.

For a listing of output templates for JFET models, see Element Template Listings (PrimeSim HSPICE Only), Table 45, in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

Specifying a Model

To specify a JFET or MESFET model, use a JFET element statement and a JFET model statement. The model parameter `LEVEL` selects either the JFET or MESFET model.

`LEVEL=1` and `LEVEL=2` select the JFET, and `LEVEL=3` selects the MESFET. Different submodels for the MESFET `LEVEL=3` equations are selected using the parameter `SAT`.

Bypassing Latent Devices (PrimeSim HSPICE Only)

Use the `BYPASS` (latency) option to decrease simulation time in large designs. To speed simulation time, this option does not recalculate currents, capacitances, and conductances, if the voltages at the terminal device nodes have not changed. The `BYPASS` option applies to MOSFETs, MESFETs, JFETs, BJTs, and diodes. Use `.OPTION BYPASS` to set `BYPASS`.

`BYPASS` might reduce simulation accuracy for tightly-coupled circuits such as op-amps, high gain ring oscillators, and so on.

Parameter	Description
LEVEL=1	SPICE model
LEVEL=2	Modified SPICE model, gate modulation of LAMBDA
LEVEL=3	Hyperbolic tangent MESFET model (Curtice, Statz, Meta, TriQuint Models)
SAT=0	Curtice model (Default)
SAT=1	Curtice model with user-defined VGST exponent
SAT=2	Cubic approximation of Curtice model with gate field degradation (Statz model)
SAT=3	The PrimeSim Continuum solution variable saturation model

The `CAPOP` model parameter selects the type of capacitor model:\

Parameter	Description
CAPOP=0	SPICE depletion capacitor model
CAPOP=1	Charge conserving, symmetric capacitor model (Statz)
CAPOP=2	PrimeSim Continuum solution improvements to CAPOP=1

You can use `CAPOP=0`, `1`, or `2` for any model level. `CAPOP=1` and `2` are most often used for the MESFET `LEVEL=3` model.

The `ACM` model parameter selects the area calculation method:

JFET / MESFET Selection Parameters	
Parameter	Description
ACM=0	SPICE method (default)

JFET / MESFET Selection Parameters

Parameter	Description
ACM=1	Physically based method

Example 1

The following example selects the n channel MESFET model, LEVEL=3. It uses the SAT, ALPHA, and CAPOP=1 parameter:

```
J1 7 2 3 GAASFET
.MODEL GAASFET NJF LEVEL=3 CAPOP=1 SAT=1 VTO=-2.5
+ BETA=2.8E-3 LAMBDA=2.2M RS=70 RD=70 IS=1.7E-14
+ CGS=14P CGD=5P UCRIT=1.5 ALPHA=2
```

Example 2

The following example selects an n-channel JFET:

```
J2 7 1 4 JM1
.MODEL JM1 NJF (VTO=-1.5, BETA=5E-3, CGS=5P, CGD=1P,
+ CAPOP=1 ALPHA=2)
```

Example 3

The following example selects a p-channel JFET:

```
J3 8 3 5 JX
.MODEL JX PJF (VTO=-1.2, BETA=.179M, LAMBDA=2.2M
+ CGS=100P CGD=20P CAPOP=1 ALPHA=2)
```

Overview of Capacitor Model

The SPICE depletion capacitor model (CAPOP=0) uses a diode-like capacitance between source and gate, where the depletion region thickness (and therefore the capacitance) is determined by the gate-to-source voltage. A similar diode model is often used to describe the normally much smaller gate-to-drain capacitance.

These approximations have serious shortcomings such as:

- Zero source-to-drain voltage: The symmetry of the FET physics gives the conclusion that the gate-to-source and gate-to-drain capacitances should be equal, but they can be very different.
- Inverse-biased transistor: Where the drain acts like the source and the source acts like the drain. According to the model, the large capacitance should be between the original source and gate; but in this case, the large capacitance is between the original drain and gate.

When low source-to-drain voltage inverse-biased transistors are involved, large errors might be introduced into simulations. To overcome these limitations, use the Statz charge-conserving model by selecting model parameter `CAPOP=1`. The model selected by `CAPOP=2` contains further improvements.

Model Applications

Use MESFETs to model GaAs transistors for high speed applications. Using MESFET models, transimpedance amplifiers for fiber-optic transmitters up to 50 GHz can be designed and simulated.

Control Option	Description
DCAP	Capacitance equation selector
GMIN, GRAMP, GMINDC	Conductance options: transient or DC analysis, DC auto-convergence
SCALM	Model scaling option Note: SCALM is ignored in Level 49 and higher.
DCCAP	Invokes capacitance calculation in DC analysis

Table 28 JFET Options

Function	Control Option
capacitance	DCAP, DCCAP
conductance	GMIN, GMINDC, GRAMP
scaling	SCALM

To override a global depletion capacitance equation selection that uses the `.OPTION DCAP= val` statement in a JFET or MESFET model, include `DCAP= val` in the device's `.MODEL` statement.

Convergence

Enhance convergence for JFET and MESFET by using the GEAR method of computation (`.OPTION METHOD = GEAR`) when you include the transit time model parameter. Use the `GMIN`, `GMINDC`, and `GRAMP` options to increase the parasitic conductance value in parallel with PN junctions of the device.

Capacitor Equations

The `DCAP` option selects the equation used to calculate the gate-to-source and gate-to-drain capacitance for `CAPOP=0`. `DCAP` can be set to 1, 2 or 3. Default is 2.

JFET and MESFET Equivalent Circuits

The following sections discuss these topics:

- [Scaling](#)
- [JFET Current Conventions](#)
- [JFET Equivalent Circuits](#)

Scaling

The `AREA` and `M` element parameters, together with the `SCALE` and `SCALM` control options, control scaling. For all the three model levels, the model parameters `IS`, `CGD`, `CGS`, `RD`, `RS`, `BETA`, `LDEL`, and `WDEL`, are scaled using the same equations.

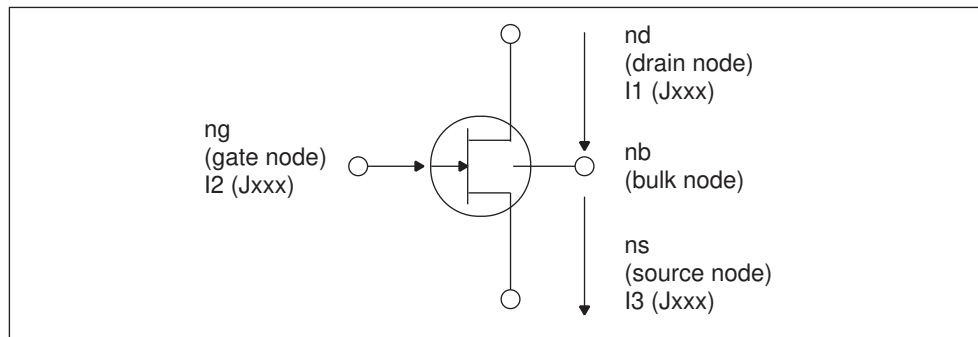
The `SCALM` option affects `A`, `L`, `W`, `LDEL`, and `WDEL` scaled parameters. `SCALM` defaults to 1.0. For example, to enter the `W` parameter with micron units, set `SCALM` to 1e-6, then enter `W=5`. The default setting is `W=5e-6` meters or 5 microns.

To override global scaling that uses the `.OPTION SCALM= val` statement in a JFET or MESFET model, include `SCALM= val` in the `.MODEL` statement.

JFET Current Conventions

[Figure 16](#) assumes the direction of current flow through the JFET. You can use either `I (Jxxx)` or `I1 (Jxxx)` syntax when printing the drain current. `I2` references the gate current and `I3` references the source current. `Jxxx` is the device name. [Figure 16](#) represents the current convention for an n-channel JFET.

Figure 16 JFET Current Convention, N-Channel



For a p-channel device, the following must be reversed:

- Polarities of the terminal voltages v_{gd} , v_{gs} , and v_{ds}
- Direction of the two gate junctions
- Direction of the nonlinear current source i_d

JFET Equivalent Circuits

Circuit simulation uses three equivalent circuits to analyze JFETs: transient, AC, and noise circuits. The components of these circuits form the basis for all element and model equation discussion.

The fundamental component in the equivalent circuit is the drain to source current (i_{ds}). For noise and AC analyses, the actual i_{ds} current is not used. Instead, the partial derivatives of i_{ds} with respect to the terminal voltages, v_{gs} , and v_{ds} are used.

The names for these partial derivatives are:

Transconductance

$$g_m = \left. \frac{\partial (i_{ds})}{\partial (v_{gs})} \right|_{v_{ds}=\text{const.}}$$

Output Conductance

$$g_{ds} = \left. \frac{\partial (i_{ds})}{\partial (v_{ds})} \right|_{v_{gs}=\text{const.}}$$

The i_{ds} equation accounts for all DC currents of the JFET. Gate capacitances are assumed to account for the transient currents of the JFET equations. The two diodes shown in Figure 17 are modeled by these ideal diode equations:

$$i_{gd} = I_{Seff} \cdot \left(e^{\frac{v_{gd}}{N \cdot v_t}} - 1 \right)$$

$$v_{gd} > -10 \cdot N \cdot v_t$$

$$i_{gd} = -I_{Seff}$$

$$v_{gd} \leq -10 \cdot N \cdot v_t$$

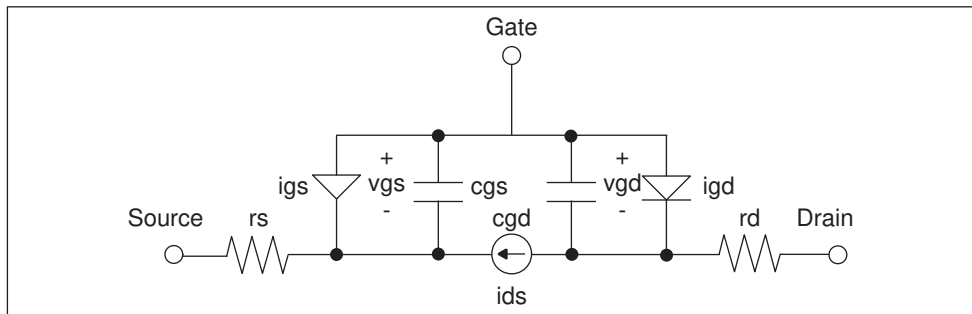
$$i_{gs} = I_{Seff} \cdot \left(e^{\frac{v_{gs}}{N \cdot v_t}} - 1 \right)$$

$$v_{gs} > -10 \cdot N \cdot v_t$$

$$i_{gs} = -I_{Seff}$$

$$v_{gs} \leq -10 \cdot N \cdot v_t$$

Figure 17 JFET/MESFET Transient Analysis



Note:

For DC analysis, the capacitances are not part of the model.

Figure 18 JFET/MESFET AC Analysis

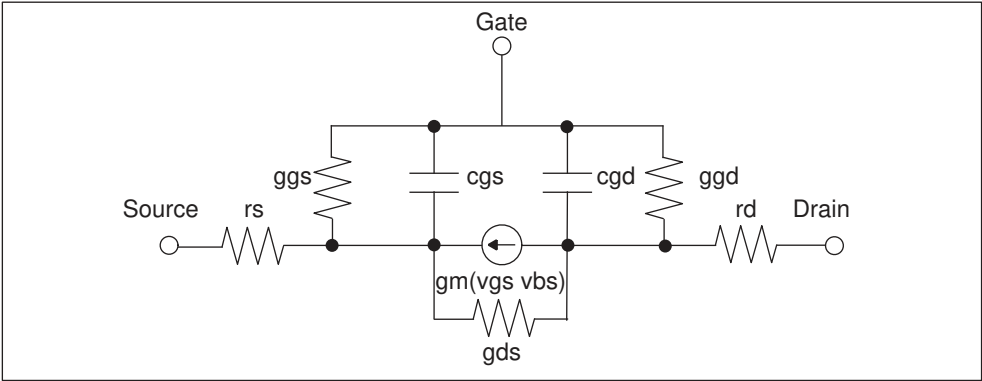


Figure 19 JFET/MESFET AC Noise Analysis

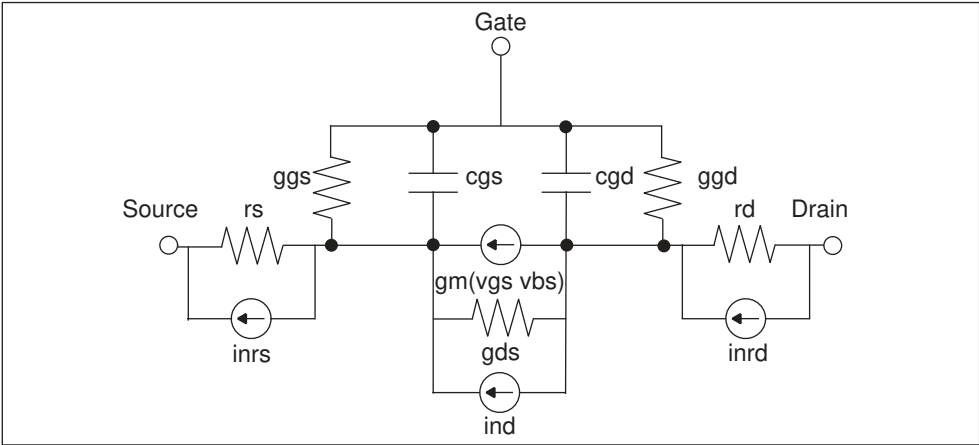


Table 29 Equation Variable Names and Constants

Variable/Quantity	Definitions
ϵ_0	Vacuum permittivity = 8.854e-12 F/m
cgd	Gate to drain capacitance
cgs	Gate to source capacitance
Dt	t - tnom
f	Frequency

Table 29 Equation Variable Names and Constants (Continued)

Variable/Quantity	Definitions
gds	Drain to source AC conductance controlled by vds
ggd	Gate to drain AC conductance
ggs	Gate to source AC conductance
gm	Drain to source AC transconductance controlled by vgs
ids	DC drain to source current
igd	Gate to drain current
igs	Gate to source current
ind	Equivalent noise current drain to source
inrd	Equivalent noise current drain resistor
inrs	Equivalent noise current source resistor
k	1.38062e-23 (Boltzmann's constant)
q	1.60212e-19 (electron charge)
rd	Drain resistance
rs	Source resistance
t	Temperature in °K
tnom	Nominal temperature of parameter measurements in °K (user-input in °C). $T_{nom} = 273.15 + T_{NOM}$
vgd	Internal gate-drain voltage
vgs	Internal gate-source voltage
vt(t)	$k \cdot t/q$
vt(tnom)	$k \cdot t_{nom}/q$

Table 30 JFET DC Operating Point Output

Quantities	Definitions
cgd	G-D capacitance

Table 30 JFET DC Operating Point Output (Continued)

Quantities	Definitions
cgs	G-S capacitance
gds	Drain-source transconductance
gm	transconductance
gmbs	drain-body (backgate) transconductance
ids	D-S current
igd	G-D current
igs	G-S current
vds	D-S voltage
vgs	G-S voltage

JFET and MESFET Model Statements

You can use the `.MODEL` statement to include a JFET or MESFET model in the PrimeSim Continuum solution netlist. For a general description of the `.MODEL` statement, see `.MODEL` in the *PrimeSim Continuum Reference Manual: Commands and Control Options*.

Syntax

```
.MODEL mname NJF [LEVEL=val] [pname1=val1] ...
.MODEL mname PJF [LEVEL=val] [pname1=val1] ...
```

Parameter	Description
mname	Model name. Elements refer to the model by this name.
NJF	Identifies an N-channel JFET or MESFET model.
LEVEL	Selects different DC model equations.
pname1=val1	Each JFET or MESFET model can include several model parameters.
PJF	Identifies a P-channel JFET or MESFET model.

JFET and MESFET Model Parameters

The DC characteristics are defined by the model parameters `VTO` and `BETA`. These parameters determine the variation of drain current with gate voltage. `LAMBDA` determines the output conductance and `IS`, the saturation current of the two gate junctions. Two ohmic resistances `RD` and `RS` are included. The charge storage is modeled by nonlinear depletion-layer capacitances for both gate junctions that vary as the $-M$ power of junction voltage, and are defined by the parameters `CGS`, `CGD`, and `PB`.

`KF` and `AF` parameters model noise, which is also a function of the series source and drain resistances (`RS` and `RD`), in addition to temperature. Use the parameters `ALPHA` and `A` to model MESFETs.

The `AREA` model parameter is common to both element and model parameters. The `AREA` element parameter always overrides the `AREA` model parameter.

Table 31 JFET and MESFET Model Parameters

Model Parameters Common to All Levels	
Capacitance	CAPOP, CGD, CGS, FC, M, PB, TT
Geometric	ACM, ALIGN, AREA, HDIF, L, LDEL, LDIF, RD, RG, RS, RSH, RSHG, RSHL, W, WDEL
Noise	AF, KF
Subthreshold	ND, NG
LEVEL=1 Model Parameters (JFET)	
DC	BETA, IS, LAMBDA, N, VTO
LEVEL=2 Model Parameters (JFET)	
DC	BETA, IS, LAMBDA, LAM1, N, VTO
LEVEL=3 Model Parameters (MESFET)	
DC	ALPHA, BETA, D, GAMDS, IS, N, K1, LAMBDA, NCHAN, SAT, SATEXP, UCRIT, VBI, VGEXP, VP, VTO

The following tables provide information about:

- Gate Diode DC Parameters, [Table 32](#)
- Gate Capacitance LEVEL=1, 2, and 3 Parameters, [Table 33](#)
- DC Model LEVEL=1 Parameters, [Table 34](#)

- DC Model LEVEL=2 Parameters, [Table 35](#)
- DC Model LEVEL=3 Parameters, [Table 36](#)

[Table 32](#) provides information about gate diode DC parameters.

Table 32 Gate Diode DC Parameters

Name (Alias)	Units	Default	Description
ACM			Area calculation method. Use this parameter to select between traditional SPICE unit less gate area calculations and the newer style of area calculations (see the ACM section). If W and L are specified, AREA becomes: $ACM=0 \text{ AREA}=W_{\text{eff}}/L_{\text{eff}}$ $ACM=1 \text{ AREA}=W_{\text{eff}} \cdot L_{\text{eff}}$
ALIGN	m	0	Misalignment of gate
AREA	-	-	Default area multiplier. This parameter affects the BETA, RD, RS, IS, CGS, and CGD model parameters. $AREA_{\text{eff}}=M \cdot AREA$ Override this parameter using the element effective area.
HDIF	m	0	Distance of the heavily diffused or low resistance region from source or drain contact to lightly doped region.
IS	amp	1.0e-14	Gate junction saturation current $IS_{\text{eff}} = IS \cdot AREA_{\text{eff}}$
L	m	0.0	Default length of FET. Override this parameter using the element L. $L_{\text{eff}} = L \cdot SCALM + LDEL_{\text{eff}}$
LDEL	m	0.0	Difference between drawn and actual or optical device length. $LDEL_{\text{eff}} = LDEL \cdot SCALM$
LDIF	m	0	Distance of the lightly doped region from heavily doped region to transistor edge.
N	-	1.0	Emission coefficient for gate-drain and gate-source diodes
RD	ohm	0.0	Drain ohmic resistance (see the ACM section) $RD_{\text{eff}} = RD / AREA_{\text{eff}}, ACM=0$
RG	ohm	0.0	Gate resistance (see the ACM section) $RG_{\text{eff}} = RG \cdot AREA_{\text{eff}}, ACM=0$
RS	ohm	0.0	Source ohmic resistance (see the ACM section) $RS_{\text{eff}} = RS / AREA_{\text{eff}}, ACM=0$
RSH	ohm/sq	0	Heavily doped region, sheet resistance
RSHG	ohm/sq	0	Gate sheet resistance

Table 32 Gate Diode DC Parameters (Continued)

Name (Alias)	Units	Default	Description
RSHL	ohm/sq	0	Lightly doped region, sheet resistance
W	m	0.0	Default FET width. The We element overrides this parameter. $W_{eff} = W \cdot SCALM + WDEL_{eff}$
WDEL	m	0.0	Difference between drawn and actual or optical device width. $WDEL_{eff} = WDEL \cdot SCALM$

Table 33 provides information about gate capacitance level 1, 2, and 3 parameters.

Table 33 Gate Capacitance LEVEL=1, 2, and 3 Parameters

Name (Alias)	Units	Default	Description
CAPOP	-	0.0	Capacitor model selector: • CAPOP=0 – default capacitance equation based on diode depletion layer • CAPOP=1 – symmetric capacitance equations (Statz) • CAPOP=2 – The PrimeSim Continuum simulator improvement to CAPOP=1
CALPHA	ALPHA	-	Saturation factor for capacitance model (CAPOP=2 only)
CAPDS	F	0	Drain to source capacitance for TriQuint model $CAPDS_{eff} = CAPDS \cdot \frac{W_{eff}}{L_{eff}} \cdot M$
CGAMDS	GAMDS	-	Threshold lowering factor for capacitance (CAPOP=2 only)
CGD	F	0.0	Zero-bias gate-drain junction capacitance $CGD_{eff} = CGD \cdot AREA_{eff}$ Override this parameter by specifying GCAP.
CGS	F	0.0	Zero-bias gate-source junction capacitance $CGS_{eff} = CGS \cdot AREA_{eff}$ Override this parameter by specifying GCAP
CRAT	-	0.666	Source fraction of gate capacitance (used with GCAP)
GCAP	F	-	Zero-bias gate capacitance. If specified, $CGS_{eff} = GCAP \cdot CRAT \cdot AREA_{eff}$ $CGD_{eff} = GCAP \cdot (1-CRAT) \cdot AREA_{eff}$

Table 33 Gate Capacitance LEVEL=1, 2, and 3 Parameters (Continued)

Name (Alias)	Units	Default	Description
FC	-	0.5	Coefficient for forward-bias depletion capacitance formulas (CAPOP=0 and 2 only)
CVTO	VTO	-	Threshold voltage for capacitance model (CAPOP=2 only)
M (MJ)	-	0.50	Grading coefficient for gate-drain and gate-source diodes (CAPOP=0 and 2 only) 0.50 - step junction 0.33 - linear graded junction
PB	V	0.8	Gate junction potential
TT	s	0	Transit time – use option METHOD=GEAR when using transit time for JFET and MESFET

Note:

Many DC parameters (such as VTO, GAMDS, ALPHA) might also affect capacitance.

Table 34 provides information about DC model LEVEL=1 parameters.

Table 34 DC Model LEVEL=1 Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	1.0	LEVEL=1 invokes the SPICE JFET model
BETA	amp/ V ²	1.0e-4	Transconductance parameter, gain $BETA_{Aeff} = BETA \cdot \frac{W_{eff} \cdot M}{L_{eff}}$
LAMBDA	1/V	0.0	Channel length modulation parameter
ND	1/V	0.0	Drain subthreshold factor (typical value=1)
NG	-	0.0	Gate subthreshold factor (typical value=1)
VTO	V	-2.0	Threshold voltage. If set, it overrides the internal calculation. A negative VTO is a depletion transistor, regardless of NJF or PJF. A positive VTO is always an enhancement transistor.

Table 35 provides information about DC model LEVEL=2 parameters.

Table 35 DC Model LEVEL=2 Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	1.0	Level of FET DC model. LEVEL=2 is a modification of the SPICE model for gate modulation of LAMBDA.
BETA	amp /V ²	1.0e-4	Transconductance parameter, gain $BETA_{eff} = BETA \cdot \frac{W_{eff} \cdot M}{L_{eff}}$
LAMBDA	1/V	0.0	Channel length modulation parameter
LAM1	1/V	0.0	Channel length modulation gate voltage parameter
ND	1/V	0.0	Drain subthreshold factor (typical value=1)
NG	-	0.0	Gate subthreshold factor (typical value=1)
VTO	V	-2.0	Threshold voltage. When set, VTO overrides the internal calculation. A negative VTO is a depletion transistor, regardless of NJF or PJF. A positive VTO is always an enhancement transistor.

Table 36 provides information about DC model level 3 parameters.

Table 36 DC Model LEVEL=3 Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	1.0	Level of FET DC model. LEVEL=3 is the Curtice MESFET model.
A	m	0.5m	Active layer thickness: $A_{eff} = A \cdot SCALM$
ALPHA	1/V	2.0	Saturation factor
BETA	amp /V ²	1.0e-4	Transconductance parameter, gain $BETA_{eff} = BETA \cdot \frac{W_{eff} \cdot M}{L_{eff}}$
D	-	11.7	Semiconductor dielectric constant: Si=11.7, GaAs=10.9
DELTA	-	0	Ids feedback parameter of TriQuint model
GAMDS (GAMMA)	-	0	Drain voltage, induced threshold voltage lowering coefficient
LAMBDA	1/V	0.0	Channel length modulation parameter
K1	V ^{1/2}	0.0	Threshold voltage sensitivity to bulk node

Table 36 DC Model LEVEL=3 Parameters (Continued)

Name (Alias)	Units	Default	Description
NCHAN	at om/cm ³	1.552e16	Effective dopant concentration in the channel
ND	1/V	0.0	Drain subthreshold factor
NG	-	0.0	Gate subthreshold factor (typical value=1)
SAT	-	0.0	Saturation factor • SAT=0 (standard Curtice model) • SAT= 1 (Curtice model with hyperbolic tangent coefficient) • SAT=2 (cubic approximation of Curtice model (Statz))
SATEXP	-	3	Drain voltage exponent
UCRIT	V/cm	0	Critical field for mobility degradation
VBI	-	1.0	Gate diode built-in voltage
VGEXP (Q)	-	2.0	Gate voltage exponent
VP	-	-	Pinch-off voltage (default is calculated)
VTO	V	-2.0	Threshold voltage. If set, it overrides internal calculation. A negative VTO is a depletion transistor regardless of NJF or PJF. A positive VTO is always an enhancement transistor.

ACM (Area Calculation Method) Parameter Equations

Use the `ACM` model parameter to select between traditional SPICE unitless gate area calculations and the newer style of area calculations.

- The `ACM=0` method (SPICE) uses the ratio of `W/L` to keep `AREA` unitless.
- The `ACM=1` (the PrimeSim Continuum solution) model requires that parameters (such as `IS`, `CGS`, `CGD`, and `BETA`) have proper physics-based units.

In the following equations, m indicates the element multiplier.

ACM=0

SPICE model, parameters determined by element areas.

$$AREA_{eff} = \frac{W_{eff}}{L_{eff}} \cdot m$$

$$R_{Defeff} = \frac{RD}{AREA_{eff}}$$

$$R_{Seff} = \frac{RS}{AREA_{eff}}$$

$$R_{Geff} = RG \cdot \frac{AREA_{eff}}{m^2}$$

ACM=1

ASPEC model, parameters are a function of element width.

$$AREA_{eff} = W_{eff} \cdot L_{eff} \cdot m$$

$$R_{Defeff} = \frac{RD}{m}$$

Or, if $RD=0$:

$$R_{Defeff} = RSH \cdot \frac{HDIF}{W_{eff} \cdot m} + RSHL \cdot \frac{LDIF + ALIGN}{W_{eff} \cdot m}$$

$$R_{Geff} = \frac{RG}{m}$$

Or, if $RG=0$:

$$R_{Geff} = RSHG \cdot \frac{W_{eff}}{L_{eff} \cdot m}$$

$$R_{Seff} = \frac{RS}{m}$$

Or, if $RS=0$:

$$R_{Seff} = RSH \cdot \frac{HDIF}{W_{eff} \cdot m} + RSHL \cdot \frac{LDIF - ALIGN}{W_{eff} \cdot m}$$

ACM=2

The PrimeSim Continuum solution model, combination of $ACM=0,1$ and provisions for lightly doped drain technology.

ACM=3

Extends $ACM=2$ model to deal with stacked devices (shared source/drains) and source/drain periphery capacitance along a gate edge.

Resulting calculations:

$$I_{Seff} = IS \cdot AREA_{eff}$$

$$CGSeff = CGS \cdot AREAeff$$

$$CGDef f = CGD \cdot AREAeff$$

$$BETAeff = BETA \cdot \frac{Weff}{Leff} \cdot m$$

Note:

The model parameter units for IS, CGS, and CGD are unitless in ACM=0 and per square meter for ACM=1.

Example

```
j1 10 20 0 40 nj_acm0 w=10u l=1u
j2a 10 20 0 41 nj_acm1 w=10u l=1u

.model nj_acm0 njf Level=3 capop=1 sat=3 acm=0
+ is=1e-14 cgs=1e-15 cgd=.3e-15
$$$note different units for is,cgs,cgd
+ rs=100 rd=100 rg=5 beta=5e-4
+ vto=.3 n=1 ng=1.4 nd=1
+ k1=.2 vgexp=2 alpha=4 ucrit=1e-4 lambda=.1
+ satexp=2
+ eg=1.5 gap1=5e-4 gap2=200 d=13

.model nj_acm1 njf Level=3 capop=1 sat=3 acm=1
+ is=1e-2 cgs=1e-3 cgd=.3e-3
$$$note different units for is,cgs,cgd
+ rs=100 rd=100 rg=5 beta=5e-4
+ vto=.3 n=1 ng=1.4 nd=1
+ k1=.2 vgexp=2 alpha=4 ucrit=1e-4 lambda=.1
+ satexp=2
+ eg=1.5 gap1=5e-4 gap2=200 d=13
```

JFET and MESFET Capacitances

Gate Capacitance CAPOP=0

The DCAP option switch selects the diode forward bias capacitance equation:

DCAP=1

Reverse Bias:

$$vgd < FC \cdot PB$$

$$cgd = CGDef f \cdot \left(1 - \frac{vgd}{PB}\right)^{-M}$$

$$vgs < FC \cdot PB$$

$$cgs = CGS_{eff} \cdot \left(1 - \frac{vgs}{PB}\right)^{-M}$$

Forward Bias:

$$vgd \geq FC \cdot PB$$

$$cgd = TT \cdot \frac{\partial igd}{\partial vgd} + CGD_{eff} \cdot \frac{1 - FC \cdot \left(1 + M\right) + M \cdot \frac{vgd}{PB}}{(1 - FC)^{M+1}}$$

$$vgs \geq FC \cdot PB$$

$$cgs = TT \cdot \frac{\partial igs}{\partial vgs} + CGS_{eff} \cdot \frac{1 - FC \cdot \left(1 + M\right) + M \cdot \frac{vgs}{PB}}{(1 - FC)^{M+1}}$$

DCAP=2 (Default)

Reverse Bias:

$$vgd < 0$$

$$cgd = CGD_{eff} \cdot \left(1 - \frac{vgd}{PB}\right)^{-M}$$

$$vgs < 0$$

$$cgs = CGS_{eff} \cdot \left(1 - \frac{vgs}{PB}\right)^{-M}$$

Forward Bias:

$$vgd \geq 0$$

$$cgd = TT \cdot \frac{\partial igd}{\partial vgd} + CGD_{eff} \cdot \left(1 + M \cdot \frac{vgd}{PB}\right)$$

$$vgs \geq 0$$

$$cgs = TT \cdot \frac{\partial igs}{\partial vgs} + CGS_{eff} \cdot \left(1 + M \cdot \frac{vgs}{PB}\right)$$

DCAP=3

Limits peak depletion capacitance to $FC \cdot CGD_{eff}$ or $FC \cdot CGS_{eff}$ with proper fall-off when forward bias exceeds PB ($FC > 1$).

Gate Capacitance CAPOP=1

Gate capacitance $CAPOP=1$ is a charge conserving symmetric capacitor model most often used for MESFET model $LEVEL=3$.

$$C_{gs} = \frac{CGS}{4\sqrt{1-\frac{v_{new}}{PB}}} \cdot \left[1 + \frac{v_{eff}-v_{te}}{\sqrt{(v_{eff}-v_{te})^2 + (0.2)^2}} \right] \cdot \left[1 + \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA}\right)^2}} \right]$$

$$\left[\frac{c_{gd}}{2} \cdot \left(1 - \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{alpha}\right)^2}} \right) \right]$$

$$C_{gd} = \left(\frac{CGS}{4\sqrt{1-\frac{v_{new}}{PB}}} \cdot \left[1 + \frac{v_{eff}-v_{te}}{\sqrt{(v_{eff}-v_{te})^2 + (0.2)^2}} \right] \cdot \left[1 - \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA}\right)^2}} \right] \right) +$$

$$\left(\frac{c_{gd}}{2} \cdot \left[1 + \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{alpha}\right)^2}} \right] \right)$$

The following equations calculate values for the preceding equations:

$$v_{te} = VTO + GAMDS \cdot v_{ds} + K1(v_{bs}) = \text{effectivethreshold}$$

$$v_{eff} = \frac{1}{2} \left[v_{gs} + v_{gd} + \sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA}\right)^2} \right]$$

$$v_{new} = \frac{1}{2} \left[v_{eff} + v_{te} + \sqrt{(v_{eff}-v_{te})^2 + (0.2)^2} \right]$$

$$CGD = \text{High } -v_{ds} \text{ Cgd at } v_{gs} = 0$$

$$CGS = \text{High } -v_{ds} \text{ Cgs at } v_{gs} = 0$$

$$CGD - CGD_{eff}$$

$$CGS - CGS_{eff}$$

Gate Capacitance CAPOP=2

Statz capacitance equations [3] ($CAPOP=1$) contain mathematical behavior that might be problematic when trying to fit data.

- For v_{gs} below the threshold voltage and $v_{ds}>0$ (normal bias condition), C_{gd} is greater than C_{gs} and rises with v_{ds} , while C_{gs} drops with v_{ds} .
- C_{gd} properly goes to a small constant representing a sidewall capacitance. However, as v_{gs} decreases, the C_{gs} curve drops along an asymptote line to zero.

(For the behavior for $V_{ds} < 0$, interchange C_{gs} and C_{gd} and replace V_{ds} with $-V_{ds}$ in the preceding descriptions.)

- It may be difficult to simultaneously fit the DC characteristics and the gate capacitances (measured by S-parameters) with the parameters that are shared between the DC model and the capacitance model.
- The capacitance model in the $CAPOP=1$ implementation also lacks a junction grading coefficient and an adjustable width for the V_{gs} transition to the threshold voltage. The width is fixed at 0.2.
- Finally, an internal parameter for limiting forward gate voltage is set to $0.8 \cdot P_B$ in the $CAPOP=1$ implementation. This is not always consistent with a good fit.

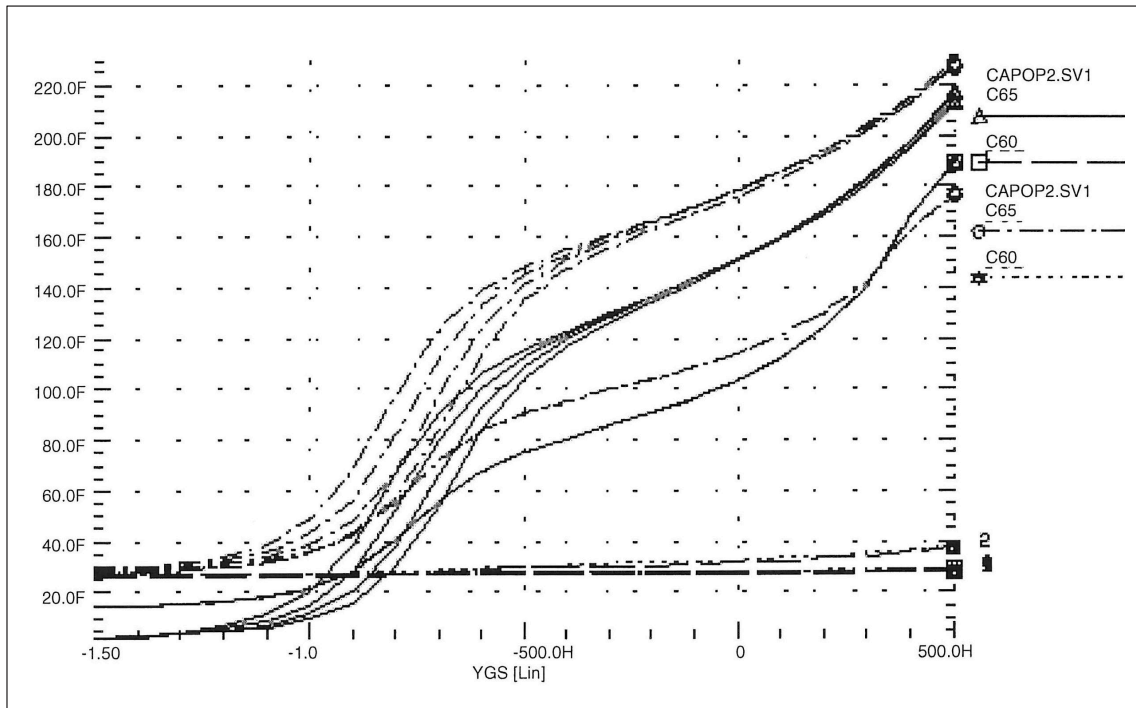
$CAPOP=2$ capacitance equations help to solve the previously-described problems.

Parameter	Default	Description
CALPHA	ALPHA	Saturation factor for capacitance model
CGAMDS	GAMDS	Threshold lowering factor for capacitance
CVTO	VTO	Threshold voltage for capacitance model
FC	0.5	P_B multiplier – typical value 0.9 gate diode limiting voltage= $FC \cdot P_B$.
M (MJ)	0.5	Junction grading coefficient
VDEL	0.2	Transition width for V_{gs}

Capacitance Comparison ($CAPOP=1$ and $CAPOP=2$)

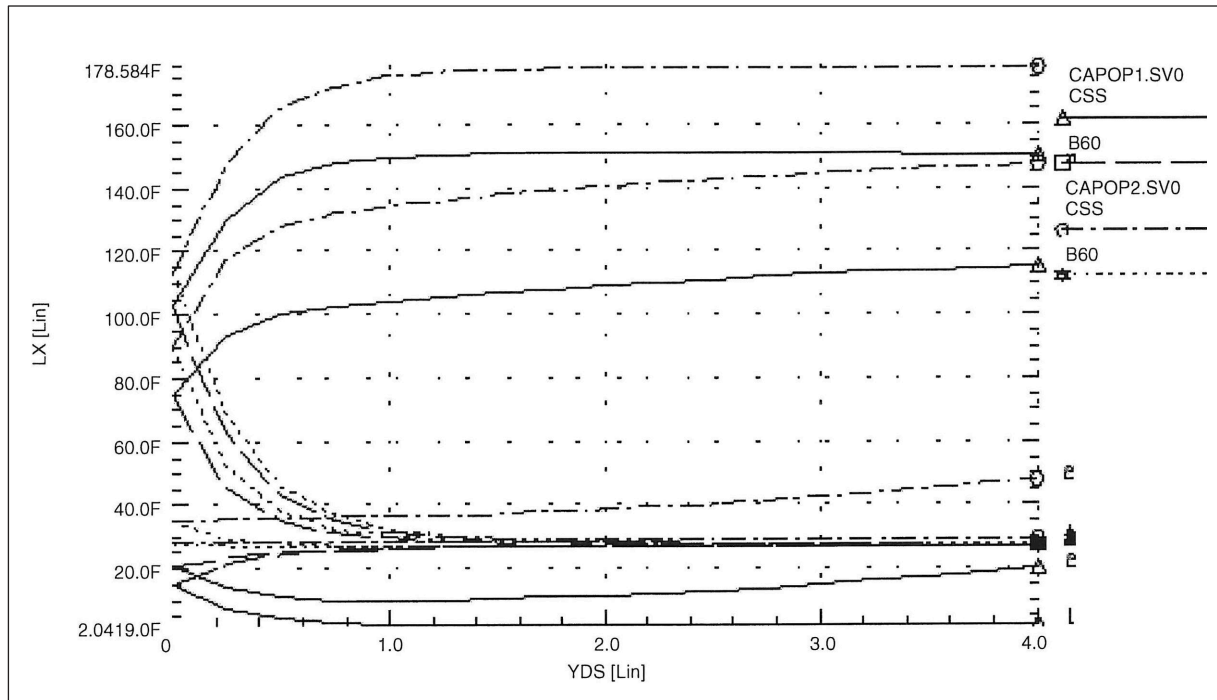
Figure 20 and Figure 21 show comparisons of $CAPOP=1$ and $CAPOP=2$. In Figure 20, below the (-0.6 v) threshold, C_{gs} for $CAPOP=2$ drops towards the same value as C_{gd} , while for $CAPOP=1$, $C_{gs} \rightarrow 0$.

Figure 20 CAPOP=1 vs. CAPOP=2. C_{gs} , C_{gd} vs. V_{gs} for $V_{ds}=0, 1, 2, 3, 4$



In Figure 21, the C_{gs} - C_{gd} characteristic curve “flips over” below the threshold for $CAPOP=1$; for $CAPOP=2$, it does not.

Figure 21 CAPOP=1 vs. CAPOP=2. C_{gs} , C_{gd} vs. V_{ds} for $V_{gs}=-1.5, -1.0, -0.5, 0$



JFET and MESFET DC Equations

The following sections discuss these topics:

- [DC Model LEVEL=1](#)
- [DC Model LEVEL=2](#)
- [DC Model LEVEL=3](#)

DC Model LEVEL=1

JFET DC characteristics are represented by the nonlinear current source (i_{ds}). The value of i_{ds} is determined by the equations:

$$v_{gst} = v_{gs} - V_{TO}$$

$v_{gst} < 0$ Channel pinched off

$$i_{ds} = 0$$

$0 < v_{gst} < v_{ds}$ Saturated region

$$ids = BETAeff \cdot vgst^2 \cdot (1 + LAMBDA \cdot vds)$$

$0 < vds < vgst$ Linear region

$$ids = BETAeff \cdot vds \cdot (2 \cdot vgst - vds) \cdot (1 + LAMBDA \cdot vds)$$

The drain current at zero vgs bias (ids) is related to VTO and $BETA$ by the equation:

$$idss = BETAeff \cdot VTO^2$$

At a given vgs , $LAMBDA$ may be determined from a pair of drain current and drain voltage points measured in the saturation region where $vgst < vds$:

$$LAMBDA = \left(\frac{ids2 - ids1}{ids1 \cdot vds2 - ids2 \cdot vds1} \right)$$

DC Model LEVEL=2

The DC characteristics of the JFET $LEVEL=2$ model are represented by the nonlinear current source (ids). The value of ids is determined by the equations:

$$vgst = vgs - VTO$$

$vgst < 0$ Channel pinched off

$$ids = 0$$

$0 < vgst < vds, vgs < 0$ Saturated region, forward bias

$$ids = BETAeff \cdot vgst^2 \cdot [1 + LAMBDA \cdot (vds - vgst) \cdot (1 + LAM1 \cdot vgs)]$$

$0 < vgst < vds, vgs > 0$ Saturated region, reverse bias

$$ids = BETAeff \cdot vgst^2 \cdot \left[1 - LAMBDA \cdot (vds - vgst) \cdot \frac{vgst}{VTO} \right]$$

$0 < vds < vgst$ Linear region

$$ids = BETAeff \cdot vds \cdot (2 \cdot vgst - vds)$$

DC Model LEVEL=3

The DC characteristics of the MESFET $LEVEL=3$ model are represented by the nonlinear hyperbolic tangent current source (ids). The value of ids is determined by the equations:

$vds > 0$ Forward region

If model parameters V_P and V_{TO} are not specified they are calculated as:

$$V_P = -\frac{q \cdot N_{CHAN} \cdot A_{eff}^2}{2 \cdot D \cdot \epsilon_o}$$

$$V_{TO} = V_P + V_{BI}$$

then:

$$v_{gst} = v_{gs} - [V_{TO} + G_{AMDS} \cdot v_{ds} + K1(v_{bs})]$$

$$bet_{eff} = \frac{BET A_{eff}}{(1 + UCRIT \cdot v_{gst})}$$

$v_{gst} < 0$ Channel pinched off

$$ids = id_{subthreshold}(NG, ND, v_{ds}, v_{gs})$$

$v_{gst} > 0$, $SAT=0$ On region

$$ids = bet_{eff} \cdot (v_{gst}^{VGEXP}) \cdot (1 + LAMBDA \cdot v_{ds}) \cdot \tanh(ALPHA \cdot v_{ds})$$

$$id_{subthreshold}(NG, ND, v_{ds}, v_{gs})$$

$v_{gst} > 0$, $SAT=1$ On region

$$ids = bet_{eff} \cdot (v_{gst}^{VGEXP}) \cdot (1 + LAMBDA \cdot v_{ds}) \cdot \tanh\left(ALPHA \cdot \frac{v_{ds}}{v_{gst}}\right)$$

$$id_{subthreshold}(NG, ND, v_{ds}, v_{gs})$$

$v_{gst} > 0$, $SAT=2$, $v_{ds} < 3/ALPHA$ On region

$$ids = bet_{eff} \cdot v_{gst}^2 \cdot (1 + LAMBDA \cdot v_{ds}) \cdot \left[1 - \left(1 - ALPHA \cdot \frac{v_{ds}}{3}\right)^3\right]$$

$$id_{subthreshold}(NG, ND, v_{ds}, v_{gs})$$

$v_{gst} > 0$, $SAT=2$, $v_{ds} > 3/ALPHA$ On region

$$(ids = bet_{eff} \cdot v_{gst}^2 \cdot (1 + LAMBDA \cdot v_{ds})) + id_{subthreshold}(N0, ND, v_{ds}, v_{gs})$$

If $v_{gst} > 0$, $SAT=3$ is the same as $SAT=2$, except exponent 3 and denominator 3 are parameterized as SAT_{EXP} , and exponent 2 of v_{gst} is parameterized as $VGEXP$.

Note:

$id_{subthreshold}$ is a special function that calculates the subthreshold currents from the $N0$ and ND model parameters.

JFET and MESFET Noise Models

Name (Alias)	Default	Description
AF	1.0	Flicker noise exponent.
KF	0.0	Flicker noise coefficient. Reasonable values for KF are in the range 1e-19 to 1e-25 V ² F.
NLEV	2.0	Noise equation selector.
GDSNOI	1.0	Channel noise coefficient. Use with NLEV=3.

Noise Equations

Figure 19 shows the JFET noise model. Thermal noise generation in the drain and source regions (R_D and R_S resistances) is modeled by the two current sources, $inrd$ and $inrs$. The $inrd$ and $inrs$ units are:

$$inrd = \left(\frac{4 \cdot k \cdot t}{rd} \right)^{1/2}$$

$$inrs = \left(\frac{4 \cdot k \cdot t}{rs} \right)^{1/2}$$

Channel thermal and flicker noise are modeled by the current source ind and defined by the equation:

$$ind = channelthermalnoise + flickernoise$$

If the model parameter $NLEV$ is less than 3, then:

$$channelthermalnoise = \left(\frac{8 \cdot k \cdot t \cdot gm}{3} \right)^{1/2}$$

The previous formula, which is used in both saturation and linear regions, might lead to wrong results in the linear region. For example, at $V_{DS}=0$, channel thermal noise is 0, because $gm=0$. This is physically impossible. If you set the $NLEV$ parameter to 3, simulation uses an equation that is valid in both linear and saturation regions. [4]

For $NLEV = 3$

Linear Region:

$$channelthermalnoise = \left(\frac{8kt}{3} \cdot 2 \cdot BETAeff \cdot (vgs-VTO) \cdot \frac{1+a+a^2}{1+a} \cdot GDSNOI \right)^{\frac{1}{2}}$$

$$\alpha = 1 - \frac{v_{ds}}{v_{gs} - V_{TO}}$$

Saturation Region:

$$channelthermalnoise = \left(\frac{8kt}{3} \cdot 2 \cdot BETAeff \cdot (v_{gs} - V_{TO}) \cdot GDSNOI \right)^{\frac{1}{2}}$$

The flicker noise is calculated as:

$$flickernoise = \left(\frac{KF \cdot ids^{AF}}{f} \right)^{1/2}$$

Parameter	Description
RD, V ² /HZ	output thermal noise due to drain resistor
RS, V ² /HZ	output thermal noise due to source resistor
RG, V ² /HZ	output thermal noise due to gate resistor
ID, V ² /HZ	output thermal noise due to channel
FN, V ² /HZ	output flicker noise
TOT, V ² /HZ	total output noise (TOT = RD + RS + RG + ID + FN)
ONoise	output noise
INoise	input noise

JFET and MESFET Temperature Equations

Table 37 lists temperature effect parameters. The temperature effect parameters apply to Levels 1, 2, and 3. They include temperature parameters for the effect of temperature on resistance, capacitance, energy gap, and a number of other model parameters. The temperature equation selectors, `TLEV` and `TLEVC`, select different temperature equations for the calculation of energy gap, saturation current, and gate capacitance. `TLEV` is either 0, 1, or 2 while `TLEVC` is either 0, 1, 2, or 3.

Table 37 Temperature Parameters (Levels 1, 2, and 3)

Function	Parameter
capacitance	CTD, CTS

Table 37 *Temperature Parameters (Levels 1, 2, and 3)*
(Continued)

Function	Parameter
DC	M, TCV, XTI
energy gap	EG, GAP1, GAP2
equation selections	TLEV, TLEVC
grading	M
mobility	BEX
resistance	TRD, TRS

Table 38 *Temperature Effect Parameters*

Name (Alias)	Units	Default	Description
BETATCE	1/°K	0.0	Beta temperature coefficient for TriQuint model
BEX	-	0.0	Mobility temperature exponent, correction for low field mobility
CTD	1/°K	0.0	Temperature coefficient for gate-drain junction capacitance. TLEVC=1 enables CTD to override the default temperature compensation.
CTS	1/°K	0.0	Temperature coefficient for gate-source junction capacitance. TLEVC=1 enables CTS to override the default temperature compensation.
EG	eV	1.16	Energy gap for the gate to drain and gate to source diodes at 0 °K • 1.17 - silicon • 0.69 - Schottky barrier diode • 0.67 - germanium • 1.52 - gallium arsenide
GAP1	eV/°K	7.02e-4	First bandgap correction factor, from Sze, alpha term • 7.02e-4 - silicon • 4.73e-4 - silicon • 4.56e-4 - germanium • 5.41e-4 - gallium arsenide

Table 38 Temperature Effect Parameters (Continued)

Name (Alias)	Units	Default	Description
GAP2	x	1108	Second bandgap correction factor, from Sze, beta term 1108 - silicon 636 - silicon 210 - germanium 204 - gallium arsenide
M (MJ)	-	0.50	Grading coefficient for gate-drain and gate-source diodes 0.50 - step junction 0.33 - linear graded junction
N	-	1.0	Emission coefficient for gate-drain and gate-source diodes
TCV (VTOTC)	1/x	0.0	Temperature compensation coefficient for VTO (threshold voltage)
TLEV	-	0.0	Temperature equation selector for junction diodes. Interacts with the TLEVC parameter.
TLEVC	-	0.0	Temperature equation selector for junction capacitances and potential. Interacts with the TLEV parameter.
TPB	V/x	0.0	Temperature coefficient for PB. TLEVC=1 or 2 overrides the default temperature compensation.
TRD (TDR1)	1/x	0.0	Temperature coefficient for drain resistance
TRG (TRG1)	1/x	0	Temperature coefficient for gate resistance
TRS (TRS1)	1/x	0.0	Temperature coefficient for source resistance
XTI	-	0.0	Saturation current temperature exponentXTI=3 for silicon diffused junctionorXTI=2 for Schottky barrier diode

Temperature Compensation Equations

The following subsections describe various types of temperature equations for JFET/ MESFET models.

Energy Gap Temperature Equations

To determine energy gap for temperature compensation, use the equations shown:

TLEV = 0 or 1

$$egnom = 1.16 - 7.02e-4 \cdot \frac{tnom^2}{tnom + 1108.0}$$

$$eg(t) = 1.16 - 7.02e-4 \cdot \frac{t^2}{t + 1108.0}$$

$$\text{TLEV} = 2$$

$$egnom = EG - GAP1 \cdot \frac{tnom^2}{tnom + GAP2}$$

$$eg(t) = EG - GAP1 \cdot \frac{t^2}{t + GAP2}$$

Saturation Current Temperature Equations

The saturation current of the gate junctions of the JFET varies with temperature according to the equation:

$$is(t) = IS \cdot e^{\frac{facln}{N}}$$

$$\text{TLEV} = 0 \text{ or } 1$$

$$facln = \frac{EG}{vt(tnom)} - \frac{EG}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

$$\text{TLEV} = 2$$

$$facln = \frac{egnom}{vt(tnom)} - \frac{eg(t)}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

Gate Capacitance Temperature Equations

Temperature equations calculate the gate capacitances. The CTS and CTD parameters are the linear coefficients. If you set $TLEV_C$ to zero, simulation uses these equations. To achieve a zero capacitance variation, set the coefficients to a very small value (such as $1e-6$) and set $TLEV_C=1$ or 2 .

$$\text{TLEV}_C = 0$$

$$CGS(t) = CGS \cdot \left[1 + M \cdot \left(4.0e-4 \cdot \Delta t - \frac{PB(t)}{PB} + 1 \right) \right]$$

$$CGD(t) = CGD \cdot \left[1 + M \cdot \left(4.0e-4 \cdot \Delta t - \frac{PB(t)}{PB} + 1 \right) \right]$$

The next equation calculates values for the preceding equations:

$$PB(t) = PB \cdot \left(\frac{t}{tnom} \right) - vt(t) \cdot \left[3 \ln\left(\frac{t}{tnom}\right) + \frac{egnom}{vt(tnom)} - \frac{eg(t)}{vt(t)} \right]$$

$$\text{TLEV}_C = 1$$

$$CGS(t) = CGS \cdot (1 + CTS \cdot \Delta t)$$

$$CGD(t) = CGD \cdot (1 + CTD \cdot \Delta t)$$

The next equation calculates values for the preceding equations:

$$PB(t) = PB - TPB \cdot \Delta t$$

TLEVC = 2

$$CGS(t) = CGS \cdot \left(\frac{PB}{PB(t)} \right)^M$$

$$CGD(t) = CGD \cdot \left(\frac{PB}{PB(t)} \right)^M$$

The next equation calculates values for the preceding equations:

$$PB(t) = PB - TPB \cdot \Delta t$$

TLEVC = 3

$$CGS(t) = CGS \cdot \left(1 - 0.5 \cdot dpbdt \cdot \frac{\Delta t}{PB} \right)$$

$$CGD(t) = CGD \cdot \left(1 - 0.5 \cdot dpbdt \cdot \frac{\Delta t}{PB} \right)$$

The next equation calculates values for the preceding equations:

$$PB(t) = PB + dpbdt \cdot \Delta t$$

TLEV = 0 or 1

$$dpbdt = \frac{-[egnom + 3 \cdot vt(tnom) + (1.16 - egnom) \cdot (2 - \frac{tnom}{tnom + 1108}) - PB]}{tnom}$$

TLEV = 2

$$dpbdt = \frac{-[egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot (2 - \frac{tnom}{tnom + GAP2}) - PB]}{tnom}$$

Threshold Voltage Temperature Equation

The threshold voltage of the JFET varies with temperature according to the equation:

$$VTO(t) = VTO - TCV \cdot \Delta t$$

$$CVTO(t) = CVTO - TCV \cdot \Delta t$$

Mobility Temperature Equation

The mobility temperature compensation equation is updated as:

$$BETA(t) = BETA \cdot \left(\frac{t}{t_{nom}} \right)^{BEX}$$

If $BETATCE=0$; otherwise (TriQuint model):

$$BETA(T) = BETA \cdot 1.01^{BETATCE(t-t_{nom})}$$

Parasitic Resistor Temperature Equations

The R_D and R_S resistances in JFET vary with temperature according to the following equations:

$$RD(t) = RD \cdot (1 + TRD \cdot \Delta t)$$

$$RS(t) = RS \cdot (1 + TRS \cdot \Delta t)$$

$$RG(t) = RG \cdot (1 + TRG \cdot \Delta t)$$

TriQuint (TOM) Extensions to LEVEL=3

TOM “TriQuint’s Own Model” [5] is implemented as part of the existing GaAs LEVEL=3 model. [6] It has a few differences from the original implementation. The PrimeSim Continuum simulators version of the TOM model takes advantage of the existing LEVEL=3 features to provide:

- Subthreshold model (NG , ND)
- Channel and source/drain resistances, geometrically derived from width and length (RD , RG , RS , RSH , $RSHG$, $RSHL$, $HDIF$, $LDIF$) ($ACM=1$)
- Photolithographic compensation ($LDEL$, $WDEL$, $ALIGN$)
- Substrate terminal
- Geometric model with width and length specified in the element ($ACM=1$)
- Automatic model selection as a function of width and length ($WMIN$, $WMAX$, $LMIN$, $LMAX$)
- User-defined band-gap coefficients (EG , $GAP1$, $GAP2$)

Several alias TOM parameters are defined for existing LEVEL=3 parameters to make the conversion easier. An alias allows the original name or the alias name to be used in the `.MODEL` statement. However, the model parameter printout is in the original name.

Please note that in two cases, a sign reversal is needed, even when using the TOM parameter name.

Alias	Printout Name	Note
Q	VGEXP	-
GAMMA	GAMDS	sign opposite of TriQuint's original
VTOTC	TCV	sign opposite of TriQuint's original
TRG1	TRG	-
TRD1	TRD	-
TRS1	TRS	-

Table 39 TOM Model Parameters

Name (Alias)	Description
BETATCE	Temperature coefficient for beta. If BETATCE is set to a nonzero value: $BETA(temp) = BETA(tnom) \cdot 1.01^{(BETATCE(temp-tnom))}$ The more common beta temperature update is: $BETA(temp) = BETA(tnom) \cdot \left(\frac{temp}{tnom}\right)^{BEX}$
DELTA	I _{ds} feedback parameter of the TOM model. This parameter is not used if its value is zero. DELTA may be negative or positive. $i_{ds} = \frac{i_{ds}}{\max[(-1 + v_{ntol}), (DELTA + v_{ds} \cdot i_{ds})]}$
CAPDS	Drain-to-source capacitance: $CAPDS_{eff} = CAPDS \cdot \frac{W_{eff}}{L_{eff}} \cdot M$

In the original TriQuint TOM implementation, LAMBDA and UCRIT parameters do not exist. Therefore, they must remain zero (their default value) in LEVEL=3 to reproduce the TOM model. Using non-zero values for these parameters with nonzero BETATCE, DELTA, or CAPDS, results in a hybrid model.

LEVEL=7 TOM3 (TriQuint's Own Model III)

TOM3 (TriQuint's Own Model III) is available as the JFET/MESFET LEVEL=7 device model. TriQuint developed it to improve the accuracy of capacitance equations by using quasi-static charge conservation in the implanted layer of a MESFET.

Using the TOM3 Model

Follow these steps to use the TOM3 model:

1. Set `LEVEL=7` to identify the model as TOM3.
2. The default room temperature is 25 in the PrimeSim Continuum simulators, but is 27 in most other simulators. When comparing to other simulators, set the simulation temperature to 27 by using either `.TEMP 27` or `.OPTION TNOM=27`.
3. The set of model parameters must include the model reference temperature, `TNOM`, which corresponds to `TREF` in other levels in the JFET/MESFET device models. The default for `TR` is 25.
4. TOM3 has its own charge-based capacitance model; so it ignores the capacitance model set in the `CAPOP` parameter.
5. The model uses analytical derivatives for conductances. This model ignores the `DERIV` parameter, which selects the finite difference method.
6. You can use `DTEMP` with this model. `DTEMP` increases the temperature of individual elements, relative to the circuit temperature. Set `DTEMP` on the element line.
7. The general syntax for this element is the same as the other standard JFET/MESFET models.
8. The model is defined by a specific sub-circuit and a set of device equations. The topology uses local feedback, decreasing the DC output conductance to model drain, model dispersion, and self-heating effects.

Note:

For more information, refer to “TOM3 Equations, Revised: 2 December 1999” by Robert B. Hallgren and David S. Smith.

Model Description

This section describes the DC and capacitance equations for the PrimeSim Continuum solution `LEVEL=7` JFET model.

DC Equations

The DC equations for the PrimeSim Continuum solution `LEVEL=7` JFET model are:

Drain to Source Current (I_{DS})

$$I_{DS} = I_O \cdot (1 + LAMBDA \cdot V_{DS})$$

$$I_O = \beta \cdot V_G^Q \cdot f_K$$

$$f_K = \frac{\alpha \cdot V_{DS}}{[1 + (\alpha \cdot V_{DS})^K]^{1,1/K}}$$

$$V_G = Q \cdot V_{ST} \cdot \log [\exp(u) + 1]$$

$$u = \frac{V_{GS} - V_{TO} + \gamma \cdot V_{DS}}{Q \cdot V_{ST}}$$

$$V_{ST} = V_{STO} \cdot (1 + M_{STO} \cdot V_{DS})$$

Transconductance

$$G_M = \left(\frac{Q \cdot \beta \cdot f_K \cdot V_G^{Q-1}}{1 + \exp(-u)} \right) \cdot (1 + LAMBDA \cdot V_{DS})$$

Output Conductance

$$G_{DS} = LAMBDA \cdot I_0 + G_M \cdot \gamma \cdot \left(\frac{(V_{GS} - V_{TO} + \gamma \cdot V_{DS}) \cdot M_{STO}}{1 + M_{STO} \cdot V_{DS}} \right) \\ \left[\left(\frac{Q \cdot I_0 \cdot M_{STO}}{1 + M_{STO} \cdot V_{DS}} \right) + \left(\frac{\alpha \cdot \beta \cdot V_G^Q}{[1 + (\alpha \cdot V_{DS})^K]^{1+1/K}} \right) \right] \cdot (1 + LAMBDA \cdot V_{DS})$$

Gate Leakage Diode Current

ILK and PLK have no temperature dependence.

$$I_{LS} = ILK \cdot \left(1 - \exp \frac{-V_{GS}}{PLK} \right)$$

$$G_{LS} = \left(\frac{ILK}{PLK} \right) \cdot \left(\exp \frac{-V_{GS}}{PLK} \right)$$

$$I_{LD} = ILK \cdot \left(1 - \exp \frac{-V_{GD}}{PLK} \right)$$

$$G_{LD} = \left(\frac{ILK}{PLK} \right) \cdot \left(\exp \frac{-V_{GD}}{PLK} \right)$$

Temperature and Geometry Dependence

$$\beta = AREA \cdot BETA \cdot 1.01^{BETATCE(T-T_{NOM})}$$

$$\alpha = ALPHA \cdot 1.01^{ALPHATCE(T-T_{NOM})}$$

$$V_{TO} = VTO + VTOC \cdot (T - T_{NOM})$$

$$\gamma = \text{GAMMA} + \text{GAMMATC} \cdot (T - T_{\text{NOM}})$$

$$V_{\text{ST0}} = V_{\text{ST}} + V_{\text{STTC}} \cdot (T - T_{\text{NOM}})$$

$$V_{\text{MT0}} = M_{\text{ST}} + M_{\text{STTC}} \cdot (T - T_{\text{NOM}})$$

Capacitance Equations

The capacitance equations for the PrimeSim Continuum solution LEVEL=7 JFET model are:

Combined Gate Charge

$$Q_{\text{GG}} = Q_{\text{GL}} \cdot f_T + Q_{\text{GH}} \cdot (1 - f_T) + Q_{\text{GG0}} \cdot (V_{\text{GSI}} + V_{\text{GDI}})$$

$$C_{\text{GS}} = C_{\text{GSL}} \cdot f_T + C_{\text{GSH}} \cdot (1 - f_T) + (Q_{\text{GL}} - Q_{\text{GH}}) \cdot \frac{\partial f_T}{\partial V_{\text{GSI}}} + Q_{\text{GG0}}$$

$$C_{\text{GD}} = C_{\text{GDL}} \cdot f_T + C_{\text{GDH}} \cdot (1 - f_T) + (Q_{\text{GL}} - Q_{\text{GH}}) \cdot \frac{\partial f_T}{\partial V_{\text{GDI}}} + Q_{\text{GG0}}$$

$$f_T = \exp(-Q_{\text{GGB}} \cdot I_{\text{DS}} \cdot V_{\text{DS}})$$

$$\frac{\partial f_T}{\partial V_{\text{GDI}}} = -Q_{\text{GGB}} \cdot [I_{\text{DS}} + (g_m + g_{\text{ds}}) \cdot V_{\text{DS}}] \cdot f_T$$

$$\frac{\partial f_T}{\partial V_{\text{GD}}} = -Q_{\text{GGB}} \cdot [I_{\text{DS}} + g_{\text{ds}} \cdot V_{\text{DS}}] \cdot f_T$$

Lower Power Gate Charge

$$Q_{\text{GL}} = qgl + Q_{\text{GCL}} \cdot (V_{\text{GS}} + V_{\text{GD}})$$

$$qgl = Q_{\text{GQL}} \cdot \exp[Q_{\text{GAG}} \cdot (V_{\text{GS}} + V_{\text{GD}})] \cdot \cosh(Q_{\text{GAD}} \cdot V_{\text{DS}})$$

$$C_{\text{GSL}} = qgl \cdot [Q_{\text{GAG}} + Q_{\text{GAD}} \cdot \tanh(Q_{\text{GAD}} \cdot V_{\text{DS}})] + Q_{\text{GCL}}$$

$$C_{\text{GDL}} = qgl \cdot [Q_{\text{GAG}} + Q_{\text{GAD}} \cdot \tanh(Q_{\text{GAD}} \cdot V_{\text{DS}})] + Q_{\text{GCL}}$$

High Power Gate Charge

$$Q_{\text{GH}} = Q_{\text{GQH}} \cdot \log\left(1 + \frac{I_{\text{DS}}}{Q_{\text{GIO}}}\right) + Q_{\text{GSH}} \cdot V_{\text{GS}} + Q_{\text{GDH}} \cdot V_{\text{GD}}$$

$$C_{\text{GSH}} = (G_{\text{M}} + G_{\text{DS}}) \cdot \left(\frac{Q_{\text{GQH}}}{I_{\text{DS}} + Q_{\text{GIO}}}\right) + Q_{\text{GSH}}$$

$$C_{GDH} = G_{DS} \cdot \left(\frac{QGQH}{I_{DS} + QGIO} \right) + QGDH$$

Table 40 TOM3 Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	1	Model Index (7 for TOM3)
TNOM	-	25	Reference temperature
AF	-	1	Flicker noise exponent
ALPHA	1/V	2	Saturation factor
ALPHATCE	K-1	0	Alpha temperature coefficient (exponential)
BETA	A/V-Q	0.1	Transconductance parameter
BETATCE	K-1	0	Linear temperature coefficient for beta
CDS	F	1E-12	Drain to source capacitance
CGDTCE	K-1	0	Linear temperature coefficient for CGD
CGSTCE	K-1	0	Linear temperature coefficient for CGS
EG	V	1.11	Barrier height at 0K(used for capacitance model)
GAMMA	-	0	Drain voltage-induced threshold voltage lowering coefficient
GAMMATC	K-1	0	Linear temperature coefficient for GAMMA
ILK	A	0	Leakage diode current parameter
IS	A	1E-14	Forward gate diode saturation current
K	-	2	Knee-function parameter
KF	-	0	Flicker noise coefficient
LAMBDA	1/V	0	Channel length modulation parameter
MST	V-1	0	Sub-threshold slope – drain parameter
MSTTC	V ⁻¹ K-1	0	Linear temperature coefficient for MST
N	-	1	Forward gate diode ideality factor
PLK	V	1	Leakage diode potential parameter

Table 40 TOM3 Parameters (Continued)

Name (Alias)	Units	Default	Description
Q	-	2	Parameter Q to model non-square-law of drain current
QGAD	V-1	1	Charge parameter
QGAG	V-1	1	Charge parameter
QGCL	F	2E-16	Charge parameter
QGDH	F	0	Sidewall capacitance
QGG0	F	0	Charge parameter
QGGB	A ⁻¹ V ⁻¹	100	Charge parameter
QGI0	A	1E-6	Charge parameter
QGQH	FV	-2E-16	Charge parameter
QGQL	FV	5E-16	Charge parameter
QGSB	F	1E-16	Sidewall capacitance
VBI	V	1	Gate diode built-in potential
VBTC	VK-1	0	Linear temperature coefficient for VBI
VST	V	1	Sub-threshold slope
VSTTC	VK-1	0	Linear temperature coefficient of VST
VTO	V	-2	Threshold voltage
VTOTC	V/K	0	Threshold voltage temperature coefficient
XTI	-	0	Diode saturation current temperature coefficient

LEVEL=8 Materka Model

This section describes the JFET/MESFET LEVEL=8 device model.

For more information about this model, see:

- Compact dc Model of GaAs FETs for Large-Signal Computer Calculation, *IEEE Journal of Solid-State Circuits*, Vol, SC-18, No.2, April 1983.
- Computer Calculation of Large-Signal GaAs FET Amplifier Characteristics, *IEEE Transactions on Microwave Theory and Techniques*, Vol. MTT-33, No. 2, February 1985.

Using the Materka Model

Follow these steps to use the Materka model:

1. Set `LEVEL=8`.
2. The default room temperature is 25 in the PrimeSim Continuum simulators, but is 27 in most other simulators. When comparing with other simulators, set the simulation temperature to 27 by using either `.TEMP 27` or `.OPTION TNOM=27`.
3. The model has its own charge-based capacitance model. This model ignores the `CAPOP` parameter, which selects difference capacitance.
4. The `ACM` parameter is not supported.

DC Model

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \tanh \left(\frac{\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right)$$

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = I_{DSS} \left[-\frac{2}{V_P} \left(1 - \frac{V_{GS}}{V_P} \right) \cdot \tanh \left(\frac{\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right) \right. \\ \left. \left(1 - \frac{V_{GS}}{V_P} \right)^2 \cdot \sec^2 \left(\frac{\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right) \cdot \frac{-\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right]$$

$$g_{DS} = \frac{\partial I_D}{\partial V_{DS}} = I_{DSS} \cdot \left(1 - \frac{V_{GS}}{V_P} \right) \cdot \left[-\frac{2\gamma V_{GS}}{V_P^2} \cdot \tanh \left(\frac{\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right) \right. \\ \left. \left(1 - \frac{V_{GS}}{V_P} \right) \cdot \sec^2 \left(\frac{\alpha 1 \cdot V_{DS}}{V_{GS} - V_P} \right) \cdot \frac{\alpha 1 \cdot (V_{GS} - V_{PO})}{(V_{GS} - V_P)^2} \right]$$

$$V_P = V_{TO} + \gamma V_{DS}$$

Table 41 DC Model Parameters

Name (Alias)	Units	Default	Description
LEVEL	-	1.0	LEVEL=8 is the Materka MESFET model.
ALPHA1	-	-	Empirical constant
GAMMA	1/V	0.0	Voltage slope parameter of pinch-off voltage
IDSS	A	0.1	Drain saturation current for Vgs=0
VP	V	-	Pinch-off voltage (default is calculated)
VTO	V	-2.0	Threshold voltage. If set, it overrides internal calculation. A negative VTO is a depletion transistor regardless of NJF or PJF. A positive VTO is always an enhancement transistor.

Gate Capacitance Model

$$C_{GS} = \frac{CGS}{\sqrt[4]{1 - \frac{v_{new}}{PB}}} \left[1 + \frac{v_{eff} - v_{te}}{\sqrt{(v_{eff} - v_{te})^2 + (0.2)^2}} \right] \cdot \left[1 + \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA1}\right)^2}} \right]$$

$$\left[\frac{CGD}{2} \left(1 - \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA1}\right)^2}} \right) \right]$$

$$C_{GD} = \frac{CGS}{\sqrt[4]{1 - \frac{v_{new}}{PB}}} \left[1 + \frac{v_{eff} - v_{te}}{\sqrt{(v_{eff} - v_{te})^2 + (0.2)^2}} \right] \cdot \left[1 - \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA1}\right)^2}} \right]$$

$$\left[\frac{CGD}{2} \left(1 + \frac{v_{ds}}{\sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA1}\right)^2}} \right) \right]$$

$$v_{te} = VTO + GAMMA \cdot v_{ds} = \text{effective threshold}$$

$$v_{eff} = \frac{1}{2} \left[v_{gs} + v_{gd} + \sqrt{v_{ds}^2 + \left(\frac{1}{ALPHA1}\right)^2} \right]$$

$$v_{new} = \frac{1}{2} \left[v_{eff} + v_{te} + \sqrt{(v_{eff} - v_{te})^2 + (0.2)^2} \right]$$

Table 42 Gate Capacitance Model Parameters

Name (Alias)	Units	Default	Description
CGS	F	0.0	Zero-bias gate-source junction capacitance
CGD	F	0.0	Zero-bias gate-drain junction capacitance
PB	V	0.8	Gate Junction Potential
N	-	1.0	Emission coefficient for gate-drain and gate-source diodes

Noise Model

Two current sources model the thermal noise generation in the drain and source regions (r_D and r_S resistances):

- $inrd$
- $inrs$

$inrd$ and $inrs$ are modeled by:

$$inrs = \left(\frac{4kt}{r_S} \right)^{1/2}$$

$$inrd = \left(\frac{4kt}{r_D} \right)^{1/2}$$

Channel thermal and flicker noise are modeled by the ind current source, and defined by the equation:

ind = channel thermal noise+ flicker noise

$$channel \text{ thermal noise} = \left(\frac{8kt \cdot g_m}{3} \right)^{1/2}$$

$$flicker \text{ noise} = \left(\frac{KF \cdot ids^{AF}}{f} \right)^{1/2}$$

Table 43 Noise Model Parameters

Name (Alias)	Units	Default	Description
AF	-	1.0	Flicker noise exponent

Table 43 Noise Model Parameters (Continued)

Name (Alias)	Units	Default	Description
KF	-	0.0	Flicker noise coefficient. Reasonable values for KF are in the range 1e-19 to 1e-25 V2 F.

Example

```
.MODEL NCH NJF LEVEL =8
+ IDSS = 69.8e-3 VTO = -2 GAMMA = 0
+ ALPHA1 = 1 RS = 0 RD = 0
+ CGS = 1e-15 CGD = 2e-16 PB = 0.8
+ IS = 5e-16 AF = 1 KF = 0
+ FC = 0.5
.END
```

Level=100 TSMC JFET Model

JFET Level=100 is a JFET compact model developed by TSMC. For detailed model formulations, contact TSMC.

JFET Level=100 Model — General Syntax

```
Jxxx nd ng ns nb jname [L=val] [W=val]
.model jname NJF [PJF] LEVEL=100 VERSION=val ...
```

The following example selects an n-channel JFET

Example 1 N-Channel JFET

```
J1 7 1 4 3 JM1
.MODEL JM1 NJF LEVEL=100 VERSION=0.3 CAPMOD=1 EG=1.16
```

The following example selects an p-channel JFET

Example 2 P-Channel JFET

```
J2 7 1 4 3 JX
.MODEL JX PJF LEVEL=100 VERSION=0.3 CAPMOD=1 EG=1.16
```

Instance Parameters

Table 44 JFET Level=100 Model Instance Parameters

Name (Alias)	Units	Default	Min	Max	Description
L	m	1.00e-06	>0.0	-	Drawn Length
W	m	1.00e-06	>0.0	-	Drawn Width
NF		1	>0.0	-	Number of fingers
AS	m ²	1.00e-12	0	-	Source diffusion area
AD	m ²	AS	0	-	Drain diffusion area
PS	m	1.00e-06	0	-	Source diffusion perimeter
PD	m	PS	0	-	Drain diffusion perimeter
NRS		1	>0.0	-	Number of squares of source diffusion for resistance calculations
NRD		NRS	>0.0	-	Number of squares of drain diffusion for resistance calculations
LGS	m	1.00e-06	>0.0	-	Space of source to gate open
LGD	m	LGS	>0.0	-	Space of drain to gate open
DTEMP		0	-	-	Temperature offset

Implementation History of JFET Level=100

Table 45 Implementation History

JFET level=100 Version	PrimeSim Continuum Release Version
0.1	H-2013.03-SP1
0.3	M-2017.03-SP2-2
0.5	P-2019.06-2

References

[1]

- GaAs FET Device and Circuit Simulation in SPICE, *IEEE Transactions on Electron Devices*, Vol. ED-34.
- [2] A MESFET Model for Use in the Design of GaAs Integrated Circuits, *IEEE Transactions on Microwave Theory*, Vol. MTT-28 No. 5.
 - [3] H. Statz, P.Newman, I.W.Smith, R.A. Pucel, and H.A. Haus, *GaAs FET Device and Circuit Simulation in Spice*.
 - [4] Tsivids, Yanis P., *Operation and Modeling of the MOS Transistor*, McGraw-Hill, 1987, p. 340.
 - [5] A.J. McCamant, G.D. Mc Cormack, and D.H.Smith, *An Improved GaAs MESFET Model for SPICE*, IEEE.
 - [6] W.Curtice, "A MESFET Model for Use in the Design of GaAs Integrated Circuits," *IEEE Tran, Microwave*, and H.Statz, P.Newman, I.W.Smith, R.A. Pucel, and H.A. Haus, "GaAs FET Device And Circuit Simulation in SPICE".

5

BJT Models

Describes how to use BJT models in the PrimeSim Continuum simulators circuit simulations.

The PrimeSim Continuum solutions ships hundreds of examples for your use; see Listing of Demonstration Input Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

The bipolar-junction transistor (BJT) model is an adaptation of the integral charge control model of Gummel and Poon. The PrimeSim Continuum solution BJT model extends the original Gummel-Poon model to include several effects at high bias levels. This model automatically simplifies to the Ebers-Moll model if you do not specify the `VA_F`, `VAR`, `IK_F`, and `IK_R` parameters.

These topics are discussed in the following sections:

- [Overview of BJT Models](#)
- [BJT Model Equations \(NPN and PNP\)](#)
- [DC Model Equations](#)
- [Substrate Current Equations](#)
- [Base Charge Equations](#)
- [Variable Base Resistance Equations](#)
- [BJT Capacitance Equations](#)
- [Defining BJT Noise Equations](#)
- [BJT Temperature Compensation Equations](#)
- [BJT Quasi-Saturation Model](#)
- [Converting National Semiconductor Models](#)
- [Defining Scaled BJT Subcircuits](#)
- [VBIC Bipolar Transistor Model](#)

- [LEVEL=6 Philips Bipolar Model \(MEXTRAM\)](#)
- [LEVEL=8 HICUM Model](#)
- [LEVEL=9 VBIC99 Model](#)
- [LEVEL=10 Philips MODELLA Bipolar Model](#)
- [LEVEL=11 UCSD HBT Model](#)
- [LEVEL=13 HICUM/L0 Model](#)
- [References](#)

Overview of BJT Models

You can use the BJT model to develop BiCMOS, TTL, and ECL circuits.

Note:

To modify high-injection effects for BiCMOS devices, use the `IKF` and `IKR` high-current Beta degradation parameters.

The `SUBS` model parameter facilitates the modeling of both vertical and lateral geometrics.

For a listing of output templates for BJT models see Element Template Listings (PrimeSim HSPICE Only), Table 44, in the *PrimeSim HSPICE User Guide: Basic Simulation and Analysis*.

The following topics are discussed in these sections:

- [Selecting Models](#)
- [BJT Model Statement](#)
- [BJT Basic Model Parameters](#)
- [BJT Model Temperature Effects](#)
- [BJT Device Equivalent Circuits](#)
- [BJT Current Conventions](#)
- [BJT Equivalent Circuits](#)

Selecting Models

To select a BJT device, use a BJT element and model statement. The element statement uses the name of the simulation device model to reference the model statement. The

following example uses the reference name MOD1. This example uses an NPN model type to describe an NPN transistor.

```
Q3 3 2 5 MOD1 parameters
.MODEL MOD1 NPN parameters
```

You can specify parameters in both element and model statements. If you specify the same parameter in both an element and a model, then the element parameter (local to the specific instance of the model) always overrides the model parameter (global default for all instances of the model, if you did not define the parameter locally). The model statement specifies the type of device—for example, for a BJT, the device type might be NPN or PNP.

BJT Control Options

The following control options affect the BJT model:

DCAP	Selects the equation that determines the BJT capacitances
DCCAP	Invokes capacitance calculations in DC analysis
GRAMP GMIN GMINDC	Place a conductance in parallel with both the base-emitter and the base-collector pn junctions

You can override global depletion capacitance equation selection that uses the `.OPTION DCAP=val` statement in a BJT model by including `DCAP=val` in the BJTs `.MODEL` statement.

Convergence

Adding a base, collector, and emitter resistance to the BJT model improves its convergence. The resistors limit the current in the device so that the forward-biased pn junctions are not overdriven.

BJT Model Statement

You can use the `.MODEL` statement to include a BJT model in the PrimeSim Continuum solution netlist. For a general description of the `.MODEL` statement, see `.MODEL` in the *PrimeSim Continuum Reference Manual: Commands and Control Options*.

Syntax

```
.MODEL mname NPN [ ( [pname1=val1] ... ) ]
.MODEL mname PNP [ ( [pname1=val1] ... ) ]
```

Parameter	Description
<i>mname</i>	Model name. Elements refer to the model by this name.

Parameter	Description
NPN	Identifies an NPN transistor model
<i>pname1</i>	Each BJT model can include several model parameters.
PNP	Identifies a PNP transistor model

Example

```
.MODEL t2n2222a NPN
+ ISS=0.   XTF= 1.   NS = 1.00000
+ CJS=0.   VJS= 0.50000   PTF= 0.
+ MJS=0.   EG = 1.10000   AF = 1.
+ ITF=0.50000   VTF= 1.00000
+ BR =40.00000   IS = 1.6339e-14   VAF=103.40529
+ VAR=17.77498   IKF= 1.00000
+ NE =1.31919   IKR= 1.00000   ISC= 3.6856e-13
+ NC =1.10024   IRB= 4.3646e-05   NF = 1.00531
+ NR =1.00688   RBM= 1.0000e-02   RB =71.82988
+ RC =0.42753   RE = 3.0503e-03   MJE= 0.32339
+ MJC=0.34700   VJE= 0.67373   VJC= 0.47372
+ TF =9.693e-10   TR =380.00e-9   CJE= 2.6734e-11
+ CJC=1.4040e-11   FC = 0.95000   XCJC=0.94518
```

BJT Basic Model Parameters

To permit the use of model parameters from earlier versions of the PrimeSim Continuum solution, many model parameters have aliases, which are included in the model parameter list in [BJT Basic Model Parameters on page 164](#). The new name is always used on printouts, even if the model statement uses an alias.

BJT model parameters are divided into several groups. The first group of DC model parameters includes the most basic Ebers-Moll parameters. This model is effective for modeling low-frequency large-signal characteristics.

Low-current Beta degradation effect parameters *ISC*, *ISE*, *NC*, and *NE* aid in modeling the drop in the observed Beta, caused by the following mechanisms:

- Recombination of carriers in the emitter-base space charge layer
- Recombination of carriers at the surface
- Formation of emitter-base channels

Low base and emitter dopant concentrations, found in some BIMOS type technologies, use the high-current Beta degradation parameters, *IKF* and *IKR*.

Use the base-width modulation parameters, that is, early effect parameters `VAF` and `VAR` to model high-gain, narrow-base devices. The model calculates the slope of the I-V curve for the model in the active region with `VAF` and `VAR`. If `VAF` and `VAR` are not specified, the slope in the active region is zero.

The `RE`, `RB`, and `RC` parasitic resistor parameters are the most frequently used second-order parameters, because they replace external resistors. This simplifies the input netlist file. All resistances are functions of the BJT multiplier `M` value. Dividing resistances by `M` simulates parallel resistances. The base resistance is a function of base current as is often the case in narrow-base technologies.

Bypassing Latent Devices (PrimeSim HSPICE Only)

Use the `BYPASS` (latency) option to decrease simulation time in large designs. To speed simulation time, this option does not recalculate currents, capacitances, and conductances, if the voltages at the terminal device nodes have not changed. The `BYPASS` option applies to MOSFETs, MESFETs, JFETs, BJTs, and diodes. Use `.OPTION BYPASS` to set `BYPASS`.

`BYPASS` might reduce simulation accuracy for tightly-coupled circuits such as op-amps, high gain ring oscillators, and so on. Use `.OPTION MBYPASS` to set `MBYPASS` to a smaller value for more-accurate results.

Parameters

Transient model parameters for BJTs are composed of two groups: junction capacitor parameters and transit time parameters.

- `CJE`, `VJE`, and `MJE` model the base-emitter junction.
- `CJC`, `VJC`, and `MJC` model base-collector junction capacitance.
- `CJS`, `VJS`, and `MJS` model the collector-substrate junction capacitance.

`TF` is the forward transit time for base charge storage. `TF` can be modified to account for bias, current, and phase, by `XTF`, `VTF`, `ITF`, and `PTF`. The base charge storage reverse transit time is set by `TR`. To select from several sets of temperature equations for the BJT model parameters, set `TLEV` and `TLEV`.

Table 46 BJT Model Parameters

Parameter	Description
DC	BF, BR, IBC, IBE, IS, ISS, NF, NR, NS, VAF, VAR
beta degradation	ISC, ISE, NC, NE, IKF, IKR
geometric	SUBS, BULK
junction capacitor	CJC, CJE, CJS, FC, MJC, MJE, MJS, VJC, VJE, VJS, XCJC

Table 46 BJT Model Parameters (Continued)

Parameter	Description
noise	KF, AF
parasitic capacitance	CBCP, CBEP, CCSP
resistor	RB, RBM, RE, RC, IRB
transit time	ITF, PTF, TF, VT, VTF, XTF

Table 47 DC Parameters for BJT Models

Name (Alias)	Unit	Default	Description
BF (BFM)		100.0	Ideal maximum forward Beta.
BR (BRM)		1.0	Ideal maximum reverse Beta.
BULK (NSUB)		0.0	Sets the bulk node to a global node name. A substrate terminal node name (ns) in the element statement overrides BULK.
EXPLI	amp	0	Current explosion model parameter. The PN junction characteristics above the explosion current area linear with the slope at the explosion point. This speeds up simulation and improves convergence. $EXPLI_{eff} = EXPLI \cdot AREA_{eff}$
IBC	amp	0.0	Reverse saturation current between base and collector. If you specify both IBE and IBC, simulation uses them in place of IS to calculate DC current and conductance; otherwise, the simulator uses IS. $IBC_{eff} = IBC \cdot AREAB \cdot MAREAC$ replaces AREAB, depending on vertical or lateral geometry.
IBE	amp	0.0	Reverse saturation current between base and emitter. If you specify both IBE and IBC, simulation uses them in place of IS to calculate DC current and conductance; otherwise, the simulator uses IS. $IBE_{eff} = IBE \cdot AREA \cdot M$
IS	amp	1.0e-16	Transport saturation current. If you specify both IBE and IBC, simulation uses them in place of IS to calculate DC current and conductance; otherwise, simulation uses IS. $IS_{eff} = IS \cdot AREA \cdot M$
ISS	amp	0.0	Reverse saturation current bulk-to-collector or bulk-to-base, depending on vertical or lateral geometry selection. $SS_{eff} = ISS \cdot AREA \cdot M$

Table 47 DC Parameters for BJT Models (Continued)

Name (Alias)	Unit	Default	Description
LEVEL		1.0	Model selector.
NF		1.0	Forward current emission coefficient.
NR		1.0	Reverse current emission coefficient.
NS		1.0	Substrate current emission coefficient.
SUBS			Substrate connection selector: • +1 for vertical geometry • -1 for lateral geometry • Default=1 for NPN • Default=-1 for PNP
UPDATE		0	UPDATE=1 uses alternate base charge equation.

Table 48 Low-Current Beta Degradation Parameters

Name (Alias)	Unit	Default	Description
ISC (C4, JLC)	amp	0.0	Base-collector leakage saturation current. If ISC is greater than 1e-4, then: $ISC = IS \cdot ISC$ otherwise: $ISC_{eff} = ISC \cdot AREAB \cdot MAREAC$ replaces AREAB, depending on vertical or lateral geometry.
ISE (C2, JLE)	amp	0.0	Base-emitter leakage saturation current. If ISE is greater than 1e-4, then: $ISE = IS \cdot ISE$ otherwise: $ISE_{eff} = ISE \cdot AREA \cdot M$
NC (NLC)		2.0	Base-collector leakage emission coefficient.
NE (NLE)		1.5	Base-emitter leakage emission coefficient.

Table 49 Base Width Modulation Parameters

Name (Alias)	Unit	Default	Description
VAF (VA, VBF)	V	0.0	Forward early voltage. Zero=infinite value.
VAR (VB, VRB, BV)	V	0.0	Reverse early voltage. Zero=infinite value.

Table 50 High-Current Beta Degradation Parameters

Name (Alias)	Unit	Default	Description
IKF (IK, JBF)	amp	0.0	Corner for forward Beta high-current roll-off. Use zero to indicate an infinite value. $IKF_{eff} = IKF \cdot AREA \cdot M$
IKR (JBR)	amp	0.0	Corner for reverse Beta high-current roll-off. Use zero to indicate an infinite value. $IKR_{eff} = IKR \cdot AREA \cdot M$
NKF		0.5	Exponent for high-current Beta roll-off.

Table 51 Parasitic Resistance Parameters

Name (Alias)	Unit	Default	Description
IRB (JRB, IOB)	amp	0.0	Base current, where base resistance falls half-way to RBM. Use zero to indicate an infinite value. $IRB_{eff} = IRB \cdot AREA \cdot M$
RB	ohm	0.0	Base resistance: $RB_{eff} = RB / (AREA \cdot M)$
RBM	ohm	RB	Minimum high-current base resistance: $RBM_{eff} = RBM / (AREA \cdot M)$
RE	ohm	0.0	Emitter resistance: $RE_{eff} = RE / (AREA \cdot M)$
RC	ohm	0.0	Collector resistance: $RC_{eff} = RC / (AREA \cdot M)$

Table 52 Junction Capacitor Parameters

Name (Alias)	Unit	Default	Description
CJC	F	0.0	Base-collector zero-bias depletion capacitance - Vertical: $CJC_{eff} = CJC \cdot AREAB \cdot M$ - Lateral: $CJC_{eff} = CJC \cdot AREAC \cdot M$ if you specify a value other than zero for ibc and ibe.
CJE	F	0.0	Base-emitter zero-bias depletion capacitance (vertical and lateral): $CJE_{eff} = CJE \cdot AREA \cdot M$
CJS (CCS, CSUB)	F	0.0	Zero-bias collector substrate capacitance - Vertical: $CJS_{eff} = CJS \cdot AREAC \cdot M$ - Lateral: $CJS_{eff} = CJS \cdot AREAB \cdot M$ If you specify a value other than zero for ibc and ibe.

Table 52 Junction Capacitor Parameters (Continued)

Name (Alias)	Unit	Default	Description
FC		0.5	Coefficient for forward bias depletion capacitance formula for $DCAP=1$ $DCAP$ Default=2 and FC are ignored.
MJC (MC)		0.33	Base-collector junction exponent (grading factor).
MJE (ME)		0.33	Base-emitter junction exponent (grading factor).
MJS(ESUB)		0.5	Substrate junction exponent (grading factor).
VJC (PC)	V	0.75	Base-collector built-in potential.
VJE (PE)	V	0.75	Base-emitter built-in potential.
VJS (PSUB)	V	0.75	Substrate junction built in potential.
XCJC (CDIS)		1.0	Internal base fraction of base-collector depletion capacitance.

Table 53 Parasitic Capacitances Parameters

Name (Alias)	Unit	Default	Description
CBCP	F	0.0	External base-collector constant capacitance: $CBCP_{eff} = CBCP \cdot AREA \cdot M$
CBEP	F	0.0	External base-emitter constant capacitance: $CBEP_{eff} = CBEP \cdot AREA \cdot M$
CCSP	F	0.0	External collector substrate constant capacitance (vertical) or base substrate (lateral): $CCSP_{eff} = CCSP \cdot AREA \cdot M$

Table 54 Transit Time Parameters

Name (Alias)	Unit	Default	Description
ITF (JTF)	amp	0.0	TF high-current parameter: $ITF_{eff} = ITF \cdot AREA \cdot M$
PTF	x	0.0	Frequency multiplier to determine excess phase.
TF	s	0.0	Base forward transit time.
TR	s	0.0	Base reverse transit time.

Table 54 Transit Time Parameters (Continued)

Name (Alias)	Unit	Default	Description
VTF	V	0.0	TF base-collector voltage dependence coefficient. Zero indicates an infinite value.
XTF		0.0	TF bias dependence coefficient.

Table 55 Noise Parameters

Name (Alias)	Unit	Default	Description
AF		1.0	Flicker-noise exponent.
KF		0.0	Flicker-noise coefficient.

Table 56 LEVEL=2 Parameters

Name (Alias)	Unit	Default	Description
BRS		1.0	Reverse beta for substrate BJT.
GAMMA		0.0	Epitaxial doping factor: $GAMMA = (2 \cdot n_i / n)^2$ In this equation, n is epitaxial impurity concentration.
NEPI		1.0	Emission coefficient.
QCO	Coul	0.0	Epitaxial charge factor: - Vertical: $QCO_{eff} = QCO \cdot AREAB \cdot M$ - Lateral: $QCO_{eff} = QCO \cdot AREAC \cdot M$ if you specify a value other than zero for ibc and ibe.
RCO (RC)	ohm	0.0	Resistance of epitaxial region under equilibrium conditions: $RC_{eff} = RC / (AREA \cdot M)$
RCC	ohm	0.0	Collector resistance
VO	V	0.0	Carrier velocity saturation voltage. Use zero to indicate an infinite value.

BJT Model Temperature Effects

Several temperature parameters control derating of the BJT model parameters. They include temperature parameters for junction capacitance, Beta degradation (DC), and base modulation (Early effect) among others.

Table 57 *BJT Temperature Parameters*

Function	Parameter
base modulation	TVAF1, TVAF2, TVAR1, TVAR2
capacitor	CTC, CTE, CTS
capacitor potentials	TVJC, TVJE, TVJS
DC	TBF1, TBF2, TBR1, TBR2, TIKF1, TIKF2, TIKR1, TIKR2, TIRB1, TIRB2, TISC1, TISC2, TIS1, TIS2, TISE1, TISE2, TISS1, TISS2, XTB, XTI
emission coefficients	TNC1, TNC2, TNE1, TNE2, TNF1, TNF2, TNR1, TNR2, TNS1, TNS2
energy gap	EG, GAP1, GAP2
equation selectors	TLEV, TLEVC
grading	MJC, MJE, MJS, TMJC1, TMJC2, TMJE1, TMJE2, TMJS1, TMJS2
resistors	TRB1, TRB2, TRC1, TRC2, TRE1, TRE2, TRM1, TRM2
transit time	TTF1, TTF2, TTR1, TTR2

Table 58 *Temperature Effect Parameters*

Name (Alias)	Unit	Default	Description
BEX		2.42	VO temperature exponent (LEVEL=2 only).
BEXV		1.90	RC temperature exponent (LEVEL=2 only).
CTC	1/°C	0.0	Temperature coefficient for zero-bias base collector capacitance. TLEVC=1 enables CTC to override the default temperature compensation.
CTE	1/°C	0.0	Temperature coefficient for zero-bias base emitter capacitance. TLEVC=1 enables CTE to override the default temperature compensation.
CTS	1/°C	0.0	Temperature coefficient for zero-bias substrate capacitance. TLEVC=1 enables CTS to override the default temperature compensation.

Table 58 Temperature Effect Parameters (Continued)

Name (Alias)	Unit	Default	Description
EG	eV		Energy gap for pn junction diode for TLEV=0 or 1, default=1.11; for TLEV=2, default=1.16: • 1.17 - silicon • 0.69 - Schottky barrier diode • 0.67 - germanium • 1.52 - gallium arsenide
GAP1	eV/x	7.02e-4	First bandgap correction factor (from Sze, alpha term): • 7.02e-4 - silicon • 4.73e-4 - silicon • 4.56e-4 - germanium • 5.41e-4 - gallium arsenide
GAP2	x	1108	Second bandgap correction factor (Sze, beta term): • 1108 - silicon • 636 - silicon • 210 - germanium • 204 - gallium arsenide
MJC(MC)		0.33	Base-collector junction exponent (grading factor).
MJE(ME)		0.33	Base-emitter junction exponent (grading factor).
MJS (ESUB)		0.5	Substrate junction exponent (grading factor).
TBF1	1/x	0.0	First-order temperature coefficient for BF.
TBF2	1/°2	0.0	Second-order temperature coefficient for BF.
TBR1	1/x	0.0	First-order temperature coefficient for BR.
TBR2	1/°2	0.0	Second-order temperature coefficient for BR.
TIKF1	1/x	0.0	First-order temperature coefficient for IKF.
TIKF2	1/°2	0.0	Second-order temperature coefficient for IKF.
TIKR1	1/x	0.0	First-order temperature coefficient for IKR.
TIKR2	1/°2		Second-order temperature coefficient for IKR.
TIRB1	1/x	0.0	First-order temperature coefficient for IRB.
TIRB2	1/°2	0.0	Second-order temperature coefficient for IRB.
TISC1	1/x	0.0	First-order temperature coefficient for ISC TLEV=3 enables TISC1.

Table 58 Temperature Effect Parameters (Continued)

Name (Alias)	Unit	Default	Description
TISC2	1/°2	0.0	Second-order temperature coefficient for ISC TLEV=3 enables TISC2.
TIS1	1/×	0.0	First-order temperature coefficient for IS or IBE and IBC TLEV=3 enables TIS1.
TIS2	1/°2	0.0	Second-order temperature coefficient for IS or IBE and IBC TLEV=3 enables TIS2.
TISE1	1/×	0.0	First-order temperature coefficient for ISE TLEV=3 enables TISE1.
TISE2	1/°2	0.0	Second-order temperature coefficient for ISE. TLEV=3 enables TISE2.
TISS1	1/×	0.0	First-order temperature coefficient for ISS TLEV=3 enables TISS1.
TISS2	1/°2	0.0	Second-order temperature coefficient for ISS TLEV=3 enables TISS2.
TITF1			First-order temperature coefficient for ITF.
TITF2			Second-order temperature coefficient for ITF.
TLEV		0.0	Temperature equation level selector for BJTs (interacts with TLEVC).
TLEVC		0.0	Temperature equation level selector: BJTs, junction capacitances, and potentials (interacts with TLEV).
TMJC1	1/×	0.0	First-order temperature coefficient for MJC.
TMJC2	1/°2	0.0	Second-order temperature coefficient for MJC.
TMJE1	1/×	0.0	First order temperature coefficient for MJE.
TMJE2	1/°2	0.0	Second-order temperature coefficient for MJE.
TMJS1	1/×	0.0	First-order temperature coefficient for MJS.
TMJS2	1/°2	0.0	Second-order temperature coefficient for MJS.
TNC1	1/×	0.0	First-order temperature coefficient for NC.
TNC2		0.0	Second-order temperature coefficient for NC.
TNE1	1/×	0.0	First-order temperature coefficient for NE.

Table 58 Temperature Effect Parameters (Continued)

Name (Alias)	Unit	Default	Description
TNE2	1/°2	0.0	Second-order temperature coefficient for NE.
TNF1	1/×	0.0	First-order temperature coefficient for NF.
TNF2	1/°2	0.0	Second-order temperature coefficient for NF.
TNR1	1/×	0.0	First-order temperature coefficient for NR.
TNR2	1/°2	0.0	Second-order temperature coefficient for NR.
TNS1	1/×	0.0	First-order temperature coefficient for NS.
TNS2	1/°2	0.0	Second-order temperature coefficient for NS.
TRB1 (TRB)	1/×	0.0	First-order temperature coefficient for RB.
TRB2	1/°2	0.0	Second-order temperature coefficient for RB.
TRC1 (TRC)	1/×	0.0	First-order temperature coefficient for RC.
TRC2	1/°2	0.0	Second-order temperature coefficient for RC.
TRE1 (TRE)	1/×	0.0	First-order temperature coefficient for RE.
TRE2	1/°2	0.0	Second-order temperature coefficient for RE.
TRM1	1/×	TRB1	First-order temperature coefficient for RBM.
TRM2	1/°2	TRB2	Second-order temperature coefficient for RBM.
TTF1	1/×	0.0	First-order temperature coefficient for TF.
TTF2	1/°2	0.0	Second-order temperature coefficient for TF.
TTR1	1/×	0.0	First-order temperature coefficient for TR.
TTR2	1/°2	0.0	Second-order temperature coefficient for TR.
TVAF1	1/×	0.0	First-order temperature coefficient for VAF.
TVAF2	1/°2	0.0	Second-order temperature coefficient for VAF.
TVAR1	1/×	0.0	First-order temperature coefficient for VAR.
TVAR2	1/°2	0.0	Second-order temperature coefficient for VAR.
TVJC	V/×	0.0	VJC temperature coefficient. TVJC uses TLEV= 1 or 2 to override default temperature compensation.

Table 58 Temperature Effect Parameters (Continued)

Name (Alias)	Unit	Default	Description
TVJE	V/×	0.0	VJE temperature coefficient. TVJE uses TLEVC= 1 or 2 to override default temperature compensation.
TVJS	V/×	0.0	VJS temperature coefficient. TVJS uses TLEVC= 1 or 2 to override default temperature compensation.
XTB(TBTCB)		0.0	Forward and reverse Beta temperature exponent (used with TLEV=0, 1, or 2).
XTI		3.0	Saturation current temperature exponent: • Use XTI=3.0 for silicon diffused junction. • Set XTI=2.0 for Schottky barrier diode.

BJT Device Equivalent Circuits

This section describes BJT scaling, current conventions, and equivalent circuits.

Scaling

Scaling is controlled by the element parameters `AREA`, `AREAB`, `AREAC`, and `M`. The `AREA` parameter, the normalized emitter area, divides all resistors and multiplies all currents and capacitors. `AREAB` and `AREAC` scale the size of the base area and collector area. Either `AREAB` or `AREAC` is used for scaling, depending on whether vertical or lateral geometry is selected (using the `SUBS` model parameter). For vertical geometry, `AREAB` is the scaling factor for `IBC`, `ISC`, and `CJC`. For lateral geometry, `AREAC` is the scaling factor. The scaling factor is `AREA` for all other parameters.

The following formula scales the DC model parameters (`IBE`, `IS`, `ISE`, `IKF`, `IKR`, and `IRB`) for both vertical and lateral BJT transistors:

$$I_{eff} = AREA \cdot M \cdot I$$

In the preceding equation, I can be `IBE`, `IS`, `ISE`, `IKF`, `IKR`, or `IRB`.

For both the vertical and lateral, the resistor model parameters, `RB`, `RBM`, `RE`, and `RC` are scaled by the following equation.

$$R_{eff} = \frac{R}{AREA \cdot M}$$

In the preceding equation, R can be `RB`, `RBM`, `RE`, or `RC`.

BJT Current Conventions

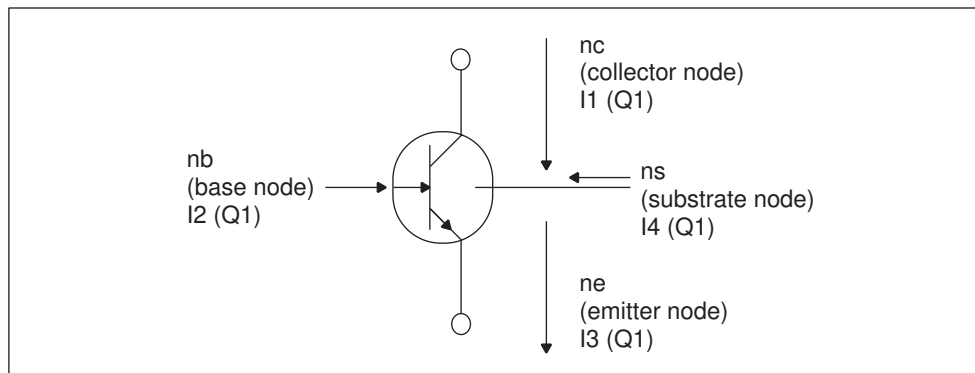
The example in [Figure 22](#) assumes the direction of current flow through the BJT. Use either `I(Q1)` or `I1(Q1)` syntax to print the collector current.

- `I2(Q1)` refers to the base current.
- `I3(Q1)` refers to the emitter current.
- `I4(Q1)` refers to the substrate current.

Note:

The above terminal currents account both DC and charge induced currents.

Figure 22 BJT Current Convention



BJT Equivalent Circuits

IC circuit simulation uses four equivalent circuits to analyze BJTs: DC, transient, AC, and AC noise circuits. The components of these circuits form the basis for all element and model equations. Because these circuits represent the entire BJT during simulation, every effort has been made to demonstrate the relationship between the equivalent circuit and the element/model parameters.

The fundamental components in the equivalent circuit are the base current (`ib`) and the collector current (`ic`). For noise and AC analyses, the actual `ib` and `ic` currents are not used. Instead, the partial derivatives of `ib` and `ic` with respect to the terminal voltages `vbe` and `vbc` are used. The names for these partial derivatives are:

Reverse Base Conductance

$$g\mu = \left. \frac{\partial i_b}{\partial v_{bc}} \right|_{v_{be}=\text{const.}}$$

Forward Base Conductance

$$g_{\pi} = \left. \frac{\partial i_b}{\partial v_{be}} \right|_{v_{bc} = \text{const.}}$$

Collector Conductance

$$g_o = \left. \frac{\partial i_c}{\partial v_{ce}} \right|_{v_{be} = \text{const.}} = - \left. \frac{\partial i_c}{\partial v_{bc}} \right|_{v_{be} = \text{const.}}$$

Transconductance

$$g_m = \left. \frac{\partial i_c}{\partial v_{be}} \right|_{v_{ce} = \text{const.}}$$

$$g_m = \frac{\partial i_c}{\partial v_{be}} + \frac{\partial i_c}{\partial v_{bc}}$$

$$g_m = \frac{\partial i_c}{\partial v_{be}} - g_o$$

The i_b and i_c equations account for all DC effects of the BJT.

Figure 23 Lateral Transistor, BJT Transient Analysis

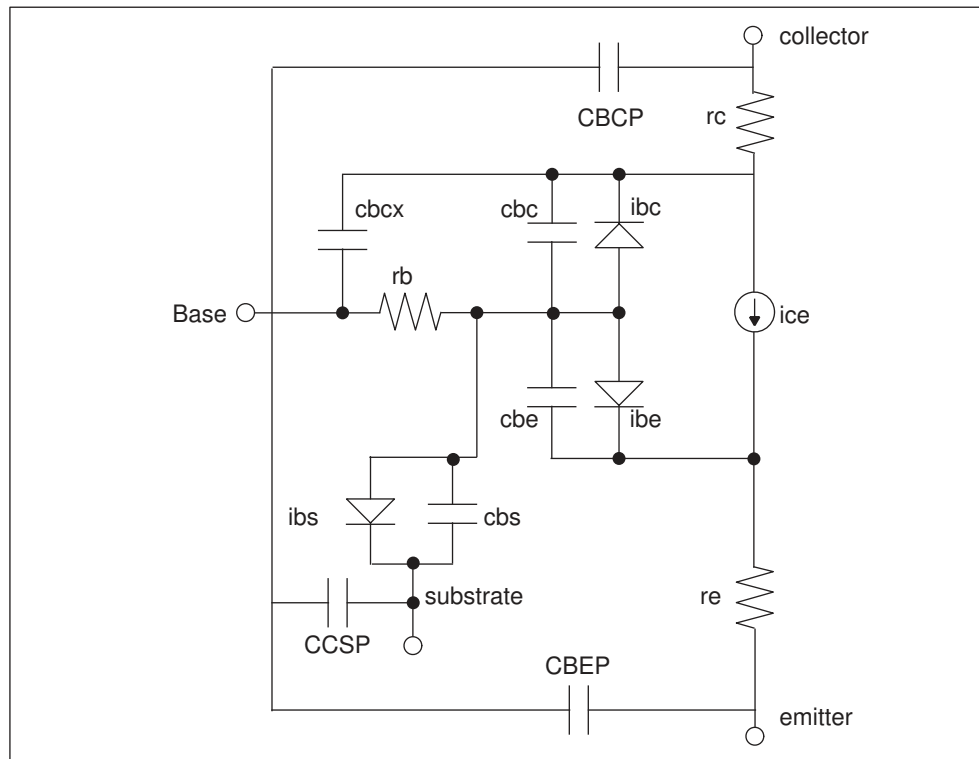


Figure 24 Vertical Transistor, BJT Transient Analysis

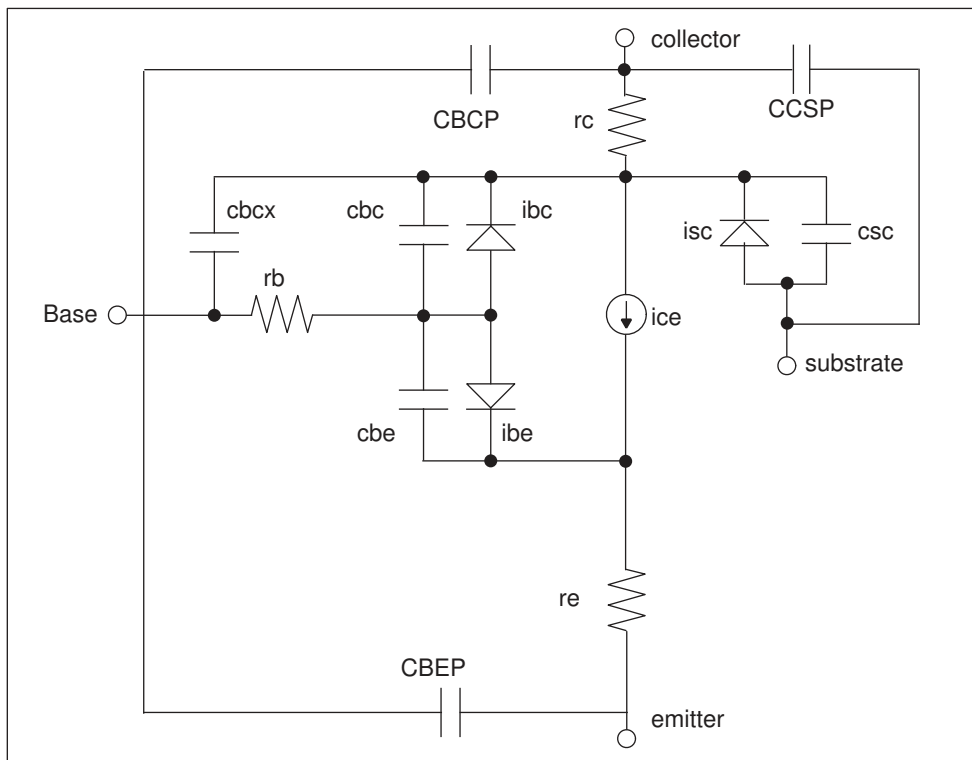


Figure 25 Lateral Transistor, BJT AC Analysis

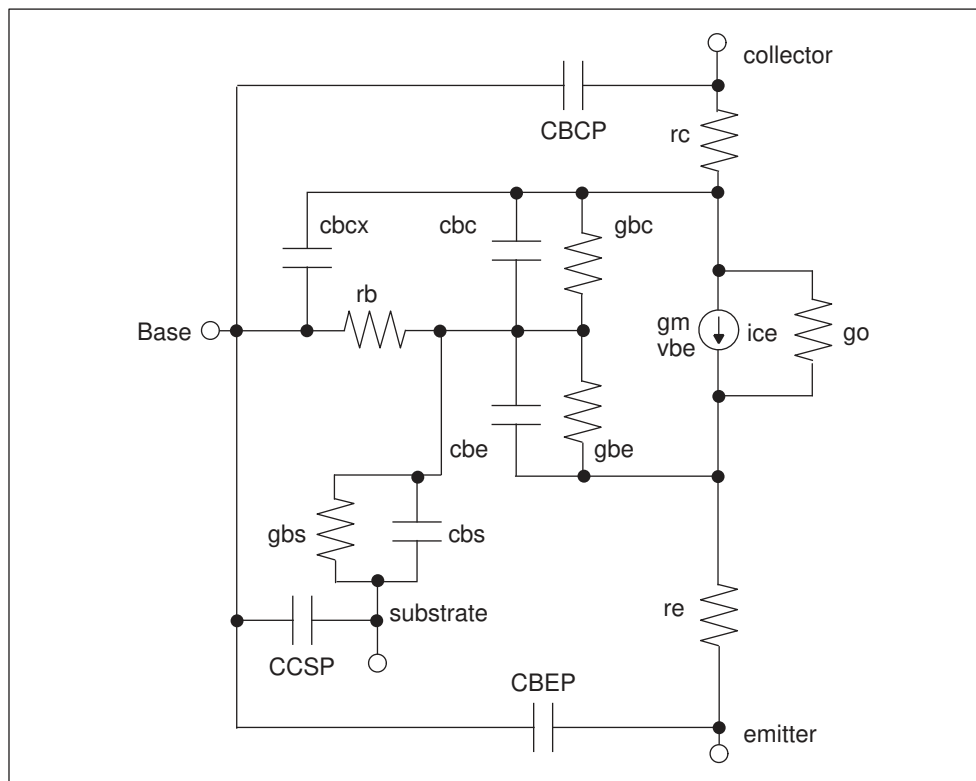


Figure 26 Vertical Transistor, BJT AC Analysis

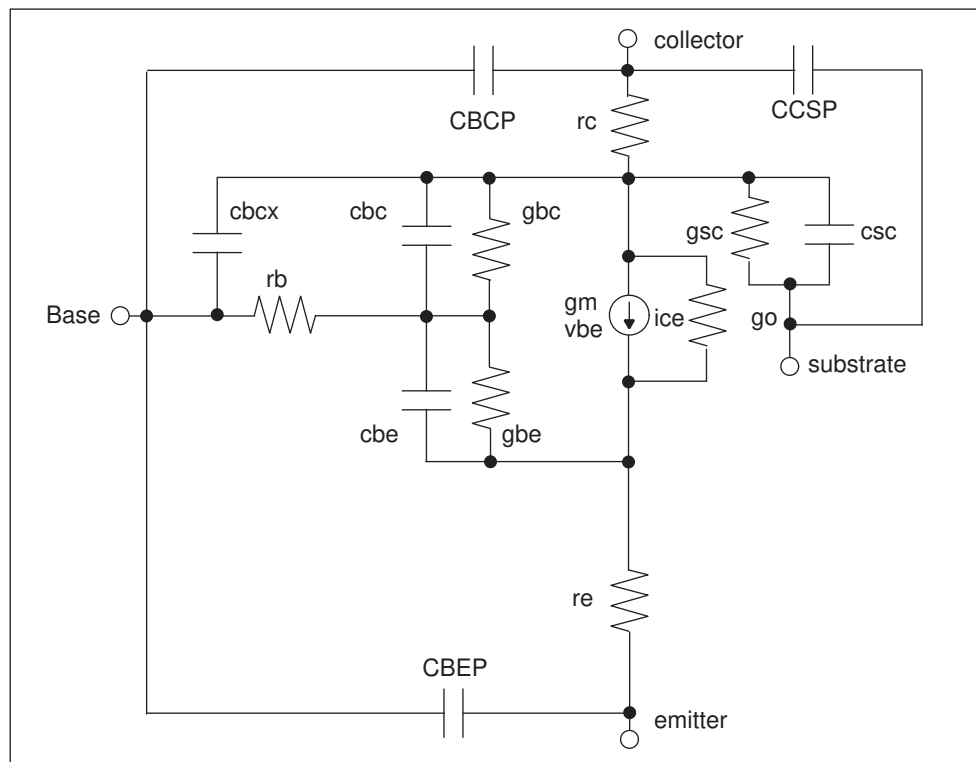


Figure 27 Lateral Transistor, BJT AC Noise Analysis

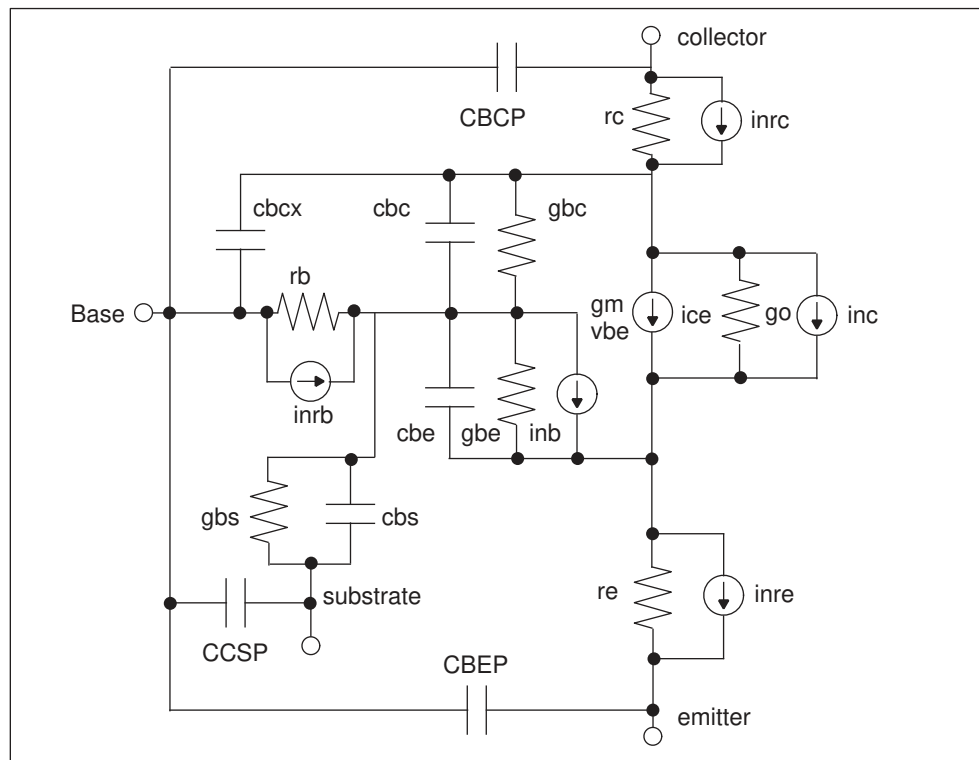


Figure 28 Vertical Transistor, BJT AC Noise Analysis

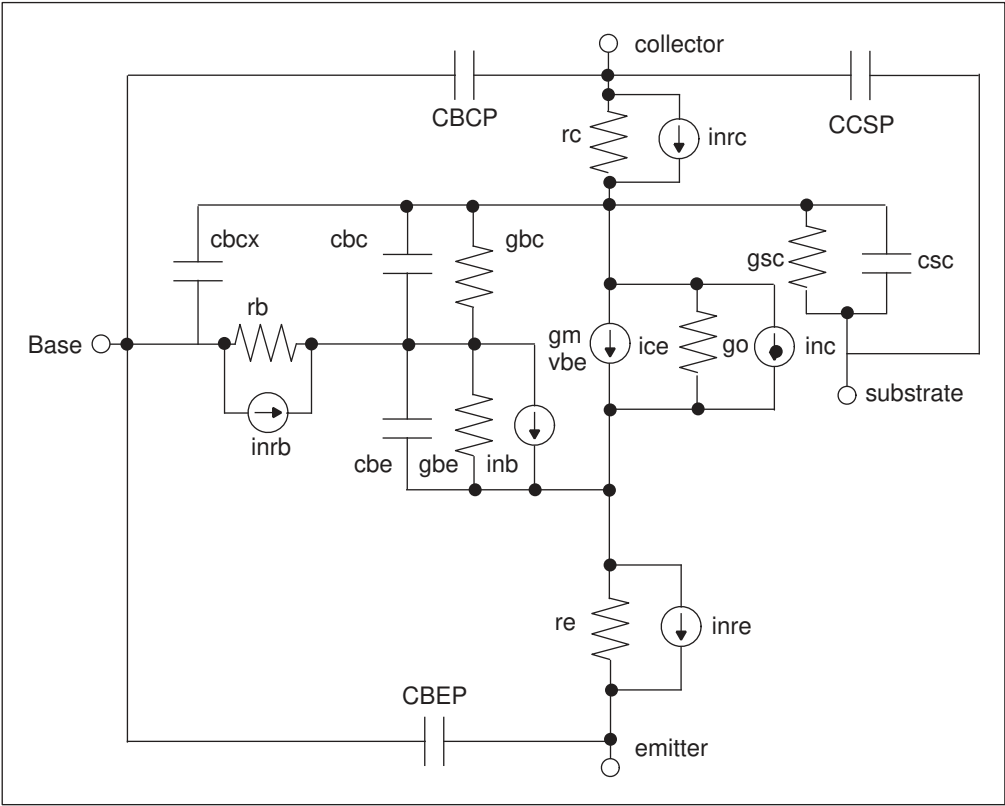


Table 59 Equation Variable Names

Variable	Definitions
cbc	Internal base to collector capacitance
cbcx	External base to collector capacitance
cbe	Internal base to emitter capacitance
csc	Substrate to collector capacitance (vertical transistor only)
cbs	Base to substrate capacitance (lateral transistor only)
f	Frequency
gbc	Reverse base conductance
gbe	Forward base conductance
gm	Transconductance

Table 59 *Equation Variable Names (Continued)*

Variable	Definitions
gsc	Substrate to collector conductance (vertical transistor only)
go	Collector conductance
gbs	Base to substrate conductance (lateral transistor only)
ib	External base terminal current
ibc	DC current base to collector
ibe	DC current base to emitter
ic	External collector terminal current
ice	DC current collector to emitter
inb	Base current equivalent noise
inc	Collector current equivalent noise
inrb	Base resistor current equivalent noise
inrc	Collector resistor equivalent noise
inre	Emitter resistor current equivalent noise
ibs	DC current base to substrate (lateral transistor only)
isc	DC current substrate to collector (vertical transistor only)
qb	Normalized base charge
rb	Base resistance
rbb	Short-circuit base resistance
vbs	Internal base substrate voltage
vsc	Internal substrate collector voltage

Table 60 *Equation Constants*

Quantities	Definitions
k	1.38062e-23 (Boltzmann's constant)
q	1.60212e-19 (Electron charge)

Table 60 Equation Constants (Continued)

Quantities	Definitions
t	Temperature in degrees Kelvin
D_t	$t - t_{nom}$
t_{nom}	$t_{nom} = 273.15 + T_{NOM}$ in degrees Kelvin
$v_t(t)$	$k \cdot t/q$
$v_t(t_{mon})$	$k \cdot t_{nom}/q$

Table 61 BJT DC Operating Point Output

Quantities	Definitions
i_b	Base current
i_c	Collector current
i_s	Substrate current
v_{be}	B-E voltage
v_{bc}	B-C voltage
v_{cs}	C-S voltage
v_s	substrate voltage
power	Power
β_{dc} (betadc)	Beta for DC analysis
g_m	Transconductance
r_{pi}	B-E input resistance
$r_{mu}(r_{muv})$	B-C input resistance
r_x	Base resistance
r_o	Collector resistance
c_{pi}	Internal B-E capacitance
c_{mu}	Internal B-C capacitance
c_{bx}	External B-C capacitance

Table 61 BJT DC Operating Point Output (Continued)

Quantities	Definitions
ccs	C-S capacitance
cbs	B-S capacitance
cxs	External substrate capacitance
betaac	Beta for AC analysis
ft	Unity gain bandwidth
*tolcc	Collector current tolerance
*tolcb	Base current tolerance

BJT Model Equations (NPN and PNP)

This section describes the NPN and PNP BJT models.

The following topics are discussed in these sections:

- [Transistor Geometry in Substrate Diodes](#)

Transistor Geometry in Substrate Diodes

The substrate diode is connected to either the collector or the base depending on whether the transistor has a lateral or vertical geometry. Lateral geometry is implied when the model parameter `SUBS=-1`, and vertical geometry when `SUBS=+1`. The lateral transistor substrate diode is connected to the internal base and the vertical transistor substrate diode is connected to the internal collector. [Figure 29](#) and [Figure 30](#) show vertical and lateral transistor geometries.

Figure 29 Vertical Transistor ($SUBS = +1$)

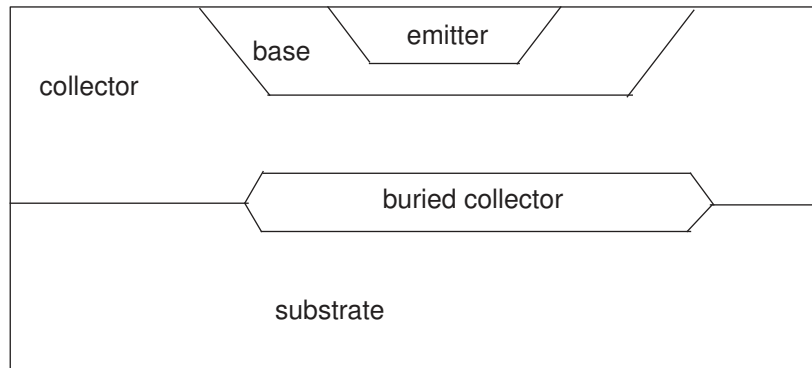
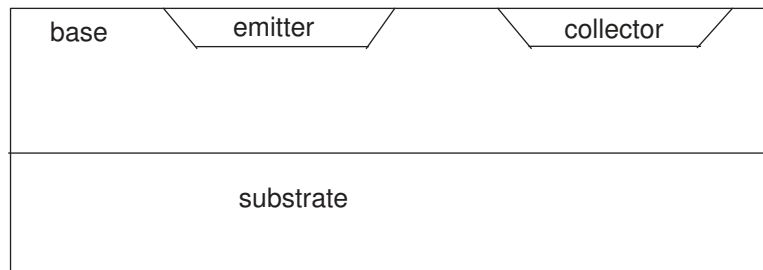
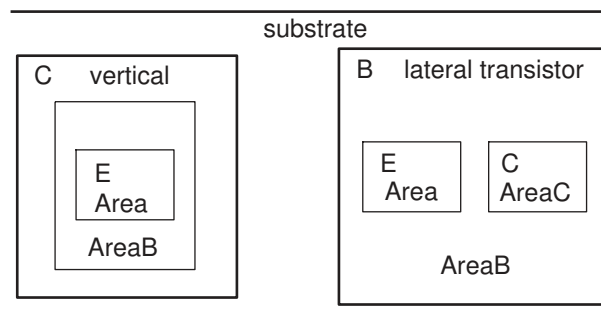


Figure 30 Lateral Transistor ($SUBS = -1$)



In [Figure 31](#), the views from the top demonstrate how I_{BE} is multiplied by either base area, $AREAB$ or collector area, $AREAC$.

Figure 31 Base, $AREAB$, Collector, $AREAC$



DC Model Equations

DC model equations are for the DC component of the collector current (i_c) and the base current (i_b).

Current Equations: IS Only

If you specify only I_S , without I_{BE} and I_{BC} :

$$i_C = \frac{I_{SEff}}{q_b} \cdot \left(e^{\frac{v_{be}}{N_F \cdot v_t}} - e^{\frac{v_{bc}}{N_R \cdot v_t}} \right) - \frac{I_{SEff}}{B_R} \cdot \left(e^{\frac{v_{bc}}{N_R \cdot v_t}} - 1 \right) - I_{SCeff} \cdot \left(e^{\frac{v_{bc}}{N_C \cdot v_t}} - 1 \right)$$

$$i_B = \frac{I_{SEff}}{B_F} \cdot \left(e^{\frac{v_{be}}{N_F \cdot v_t}} - 1 \right) + \frac{I_{SEff}}{B_R} \cdot \left(e^{\frac{v_{bc}}{N_R \cdot v_t}} - 1 \right) + I_{SEff} \cdot \left(e^{\frac{v_{be}}{N_E \cdot v_t}} - 1 \right) \cdot I_{SCeff} \cdot \left(e^{\frac{v_{bc}}{N_C \cdot v_t}} - 1 \right)$$

Current Equations: IBE and IBC

If you specify I_{BE} and I_{BC} , instead of I_S :

$$i_C = \frac{B E_{SEff}}{q_b} \cdot \left(e^{\frac{v_{be}}{N_F \cdot v_t}} - 1 \right) - \frac{I_{BCeff}}{q_b} \cdot \left(e^{\frac{v_{bc}}{N_R \cdot v_t}} - 1 \right) - \frac{I_{BCeff}}{B_R} \cdot \left(e^{\frac{v_{bc}}{N_R \cdot v_t}} - 1 \right) - I_{SCeff} \cdot \left(e^{\frac{v_{bc}}{N_C \cdot v_t}} - 1 \right)$$

$$i_B = \frac{I_{SBEff}}{B_F} \cdot \left(e^{\frac{v_{be}}{N_F \cdot v_t}} - 1 \right) + \frac{I_{BCeff}}{B_R} \cdot \left(e^{\frac{v_{bc}}{N_R \cdot v_t}} - 1 \right) + I_{SEEff} \cdot \left(e^{\frac{v_{be}}{N_E \cdot v_t}} - 1 \right) \cdot I_{SCeff} \cdot \left(e^{\frac{v_{bc}}{N_C \cdot v_t}} - 1 \right)$$

$$I_{BCeff} = I_{BC} \cdot A_{REAB} \cdot M$$

Vertical

$$I_{BCeff} = I_{BC} \cdot A_{REAC} \cdot M$$

Lateral

$$I_{BEeff} = I_{BE} \cdot A_{REA} \cdot M$$

Vertical or Lateral

$$I_{SCeff} = I_{SC} \cdot A_{REAB} \cdot M$$

Vertical

$$I_{SCeff} = I_{SC} \cdot A_{REAC} \cdot M$$

Lateral

$$I_{SEeff} = I_{SE} \cdot A_{REA} \cdot M$$

Vertical or Lateral

The last two base-current terms represent components, due to recombining the base-emitter and base-collector space charge regions, at low injection.

Substrate Current Equations

The substrate current is substrate to collector for vertical transistors and substrate to base for lateral transistors.

Vertical Transistors

$$i_{sc} = I_{S\text{eff}} \cdot \left(e^{\frac{v_{sc}}{N_S v_t}} - 1 \right)$$

$$v_{sc} > -10 \cdot N_S \cdot v_t$$

$$i_{sc} = -I_{S\text{eff}}$$

$$v_{sc} \leq -10 \cdot N_S \cdot v_t$$

Lateral Transistors

$$i_{bs} = I_{S\text{eff}} \cdot \left(e^{\frac{v_{bs}}{N_S v_t}} - 1 \right)$$

$$v_{bs} > -10 \cdot N_S \cdot v_t$$

$$i_{bs} = -I_{S\text{eff}}$$

$$v_{bs} \leq -10 \cdot N_S \cdot v_t$$

If you do not specify either I_{BE} or I_{BC} :

$$I_{S\text{eff}} = I_{SS} \cdot AREA \cdot M$$

If you specify both I_{BE} and I_{BC} :

$$I_{S\text{eff}} = I_{SS} \cdot AREAC \cdot M$$

vertical

$$I_{S\text{eff}} = I_{SS} \cdot AREAB \cdot M$$

lateral

Base Charge Equations

V_{AF} and V_{AR} are, respectively, forward and reverse early voltages. I_{KF} and I_{KR} determine the high-current Beta roll-off. I_{SE} , I_{SC} , N_E , and N_C determine the low-current Beta roll-off with i_c .

If $UPDATE=0$ or

$$\frac{vbc}{VAF} + \frac{vbe}{VAR} < 0$$

then

$$q1 = \frac{1}{\left(1 - \frac{vbc}{VAF} - \frac{vbe}{VAR}\right)}$$

Otherwise, if UPDATE=1 and

$$\frac{vbc}{VAF} + \frac{vbe}{VAR} \geq 0$$

then

$$q1 = 1 + \frac{vbc}{VAF} + \frac{vbe}{VAR}$$

$$q2 = \frac{ISeff}{IKFeff} \cdot \left(e^{\frac{vbe}{NF \cdot vt}} - 1\right) + \frac{ISeff}{IKReff} \cdot \left(e^{\frac{vbc}{NR \cdot vt}} - 1\right)$$

With IBE and IBC, the preceding equation is:

$$q2 = \frac{IBEff}{IKFeff} \cdot \left(e^{\frac{vbe}{NF \cdot vt}} - 1\right) + \frac{IBCEff}{IKReff} \cdot \left(e^{\frac{vbc}{NR \cdot vt}} - 1\right)$$

In the preceding equation:

- IBE=IS if IBE=0
- IBC=IS if IBC=0

$$qb = \frac{q1}{2} \cdot \left[1 + (1 + 4 \cdot q2)^{NKF}\right]$$

Variable Base Resistance Equations

A variable base resistance BJT model consists of a low-current maximum resistance (set using RB), and a high-current minimum resistance (set using RBM). IRB is the current when the base resistance is halfway to its minimum value. If you do not specify RBM, it is set to RB.

If you do not specify IRB:

$$rbb = RBMeff + \frac{RBeff - RBMeff}{qb}$$

If you specify IRB:

$$rbb = RBMeff + 3 \cdot (RBeff - RBMeff) \cdot \frac{\tan(z) - z}{z \cdot \tan(z) \cdot \tan(z)}$$

$$z = \frac{-1 + [1 + 144 \cdot ib / (\pi^2 \cdot IRB_{eff})]^{1/2}}{\frac{24}{\pi^2} \cdot \left(\frac{ib}{IRB_{eff}}\right)^{1/2}}$$

BJT Capacitance Equations

This section describes BJT capacitances.

The following topics are discussed in these sections:

- [Base-Emitter Capacitance Equations](#)
- [Substrate Capacitance](#)

Base-Emitter Capacitance Equations

The base-emitter capacitance contains a complex diffusion term with the standard depletion capacitance formula. The diffusion capacitance is modified by model parameters TF , XTF , ITF , and VTF .

Determine the base-emitter capacitance c_{be} by the following formula:

$$c_{be} = c_{bediff} + c_{bedep}$$

In the preceding equation, c_{bediff} is the base-emitter diffusion, and c_{bedep} is the depletion capacitance.

Note:

When you run a DC sweep on a BJT, use the `.OPTION DCCAP` command to force evaluation of the voltage-variable capacitances during the DC sweep.

The following topics are discussed in these sections:

- [Determining Base-Emitter Diffusion Capacitance](#)
- [Determining Base-Emitter Depletion Capacitance](#)
- [Determining Base Collector Capacitance](#)
- [Determining Base Collector Diffusion Capacitance](#)
- [Determining Base Collector Depletion Capacitance](#)
- [External Base — Internal Collector Junction Capacitance](#)

Determining Base-Emitter Diffusion Capacitance

Determine diffusion capacitance as follows:

ibe ≤ 0

$$cbediff = \frac{\partial}{\partial v_{be}} \left(TF \cdot \frac{ibe}{qb} \right)$$

ibe > 0

$$cbediff = \frac{\partial}{\partial v_{be}} \left[TF \cdot (1 + argtf) \cdot \frac{ibe}{qb} \right]$$

The following equation calculates the *argtf* value for the preceding equation:

$$argtf = XTF \cdot \left(\frac{ibe}{ibe + ITF} \right)^2 \cdot e^{\frac{v_{bc}}{1.44 VTF}}$$

The forward part of the collector-emitter branch current is determined as follows (*IBE=IS* if *IBE=0*):

$$ibe = IS_{eff} \cdot \left(e^{\frac{v_{be}}{NF \cdot vt}} - 1 \right)$$

Determining Base-Emitter Depletion Capacitance

There are two different equations for modeling the depletion capacitance. Select the proper equation by specifying *.OPTION DCAP*.

DCAP=1

The base-emitter depletion capacitance is determined as follows:

$$v_{be} < FC \cdot V_{JE}$$

$$cbedep = CJE_{eff} \cdot \left(1 - \frac{v_{be}}{V_{JE}} \right)^{-M_{JE}}$$

$$v_{be} \geq FC \cdot V_{JE}$$

$$cbedep = CJE_{eff} \cdot \frac{1 - FC \cdot (1 + M_{JE}) + M_{JE} \cdot \frac{v_{be}}{V_{JE}}}{(1 - FC)^{(1 + M_{JE})}}$$

DCAP=2

The base-emitter depletion capacitance is determined as follows:

$$v_{be} < 0$$

$$cbedep = CJE_{eff} \cdot \left(1 - \frac{v_{be}}{V_{JE}} \right)^{-M_{JE}}$$

$$v_{be} \geq 0$$

$$c_{bedep} = C_{JEeff} \cdot \left(1 + M_{JE} \cdot \frac{v_{be}}{V_{JE}} \right)$$

DCAP=3

Limits peak depletion capacitance to $FC \cdot C_{JCeff}$ or $FC \cdot C_{JEeff}$ with proper fall-off when forward bias exceeds PB ($FC \geq 1$).

Determining Base Collector Capacitance

Determine the base collector capacitance c_{bc} as follows:

$$c_{bc} = c_{bcdiff} + c_{bcdep}$$

In the preceding equation, c_{bcdiff} is the base-collector diffusion, and c_{bcdep} is the depletion capacitance.

Determining Base Collector Diffusion Capacitance

$$c_{bcdiff} = \frac{\partial}{\partial v_{bc}} (TR \cdot i_{bc})$$

In the preceding equation, the internal base-collector current (i_{bc}) is ($i_{bc} = I_S$ if $i_{bc} = 0$):

$$i_{bc} = I_{Seff} \cdot \left(e^{\frac{v_{bc}}{N_{Rvt}}} - 1 \right)$$

Determining Base Collector Depletion Capacitance

There are two different equations for modeling the depletion capacitance. Select the proper equation by specifying `.OPTION DCAP`.

DCAP=1

Specify `DCAP=1` to select one of the following equations:

$$v_{bc} < FC \cdot V_{JC}$$

$$c_{bcdep} = X_{CJC} \cdot C_{JCeff} \cdot \left(1 - \frac{v_{bc}}{V_{JC}} \right)^{-M_{JC}}$$

$$v_{bc} \geq FC \cdot V_{JC}$$

$$c_{bcdep} = X_{CJC} \cdot C_{JCeff} \cdot \frac{1 - FC \cdot (1 + M_{JC}) + M_{JC} \cdot \frac{v_{bc}}{V_{JC}}}{(1 - FC)^{(1 + M_{JC})}}$$

DCAP=2

Specify $DCAP=2$ to select one of the following equations:

$$v_{bc} < 0$$

$$c_{bcdep} = X_{CJC} \cdot C_{JCeff} \cdot \left(1 - \frac{v_{bc}}{V_{JC}}\right)^{-M_{JC}}$$

$$v_{bc} \geq 0$$

$$c_{bcdep} = X_{CJC} \cdot C_{JCeff} \cdot \left(1 + M_{JC} \cdot \frac{v_{bc}}{V_{JC}}\right)$$

External Base — Internal Collector Junction Capacitance

The base-collector capacitance is modeled as a distributed capacitance when the model parameter X_{CJC} is set. Since the default setting of X_{CJC} is one, the entire base-collector capacitance is on the internal base node c_{bc} .

DCAP=1

Specify $DCAP=1$ to select one of the following equations:

$$v_{bcx} < FC \cdot V_{JC}$$

$$c_{bcx} = C_{JCeff} \cdot (1 - X_{CJC}) \cdot \left(1 - \frac{v_{bcx}}{V_{JC}}\right)^{-M_{JC}}$$

$$v_{bcx} \geq FC \cdot V_{JC}$$

$$c_{bcx} = C_{JCeff} \cdot (1 - X_{CJC}) \cdot \frac{1 - FC \cdot (1 + M_{JC}) + M_{JC} \cdot \frac{v_{bcx}}{V_{JC}}}{(1 - FC)^{(1 + M_{JC})}}$$

DCAP=2

Specify $DCAP=2$ to select one of the following equations:

$$v_{bcx} < 0$$

$$c_{bcx} = C_{JCeff} \cdot (1 - X_{CJC}) \cdot \left(1 - \frac{v_{bcx}}{V_{JC}}\right)^{-M_{JC}}$$

$$v_{bcx} \geq 0$$

$$c_{bcx} = C_{JCeff} \cdot (1 - X_{CJC}) \cdot \left(1 + M_{JC} \cdot \frac{v_{bcx}}{V_{JC}}\right)$$

In the preceding equation, v_{bcx} is the voltage between the external base node and the internal collector node.

Substrate Capacitance

The function of substrate capacitance is similar to that of the substrate diode. To switch it from the collector to the base, set the `SUBS` model parameter.

The following section present these topics:

- [Substrate Capacitance Equation: Lateral](#)
- [Substrate Capacitance Equation: Vertical](#)
- [Excess Phase Equation](#)

Substrate Capacitance Equation: Lateral

Base to Substrate Diode

Reverse Bias $v_{bs} < 0$

$$cbs = CJSeff \cdot \left(1 - \frac{v_{bs}}{V_{JS}}\right)^{-MJS}$$

Forward Bias $v_{bs} \geq 0$

$$cbs = CJSeff \cdot \left(1 + MJS \cdot \frac{v_{bs}}{V_{JS}}\right)$$

Substrate Capacitance Equation: Vertical

Substrate to Collector Diode

Reverse Bias $v_{sc} < 0$

$$csc = CJSeff \cdot \left(1 - \frac{v_{sc}}{V_{JS}}\right)^{-MJS}$$

Forward Bias $v_{sc} \geq 0$

$$csc = CJSeff \cdot \left(1 + MJS \cdot \frac{v_{sc}}{V_{JS}}\right)$$

Excess Phase Equation

The model parameter, `PTF`, models excess phase. It is defined as extra degrees of phase delay (introduced by the BJT) at any frequency and is determined by the equation:

$$\text{excessphase} = \left(2 \cdot \pi \cdot PTF \cdot \frac{TF}{360}\right) \cdot (2 \cdot \pi \cdot f)$$

In the preceding equation, f is in hertz, and you can set `PTF` and `TF`. The excess phase is a delay (linear phase) in the transconductance generator for AC analysis. Use it also in transient analysis.

Defining BJT Noise Equations

Equations for modeling BJT thermal, shot, and flicker noise are as follows.

Defining Noise Equations

The mean square short-circuit base resistance noise current equation is:

$$inrb = \left(\frac{4 \cdot k \cdot t}{rbb} \right)^{\frac{1}{2}}$$

The mean square short-circuit collector resistance noise current equation is:

$$inrc = \left(\frac{4 \cdot k \cdot t}{RCeff} \right)^{\frac{1}{2}}$$

The mean square short-circuit emitter resistance noise current equation is:

$$inre = \left(\frac{4 \cdot k \cdot t}{REeff} \right)^{\frac{1}{2}}$$

The noise associated with the base current is composed of two parts: shot noise and flicker noise. Typical values for the flicker noise coefficient, K_F , are 1e-17 to 1e-12. They are calculated as:

$$2 \cdot q \cdot fknee$$

In the preceding equation, $fknee$ is the noise knee frequency (typically 100 Hz to 10 MHz), and q is electron charge.

$$inb^2 = (2 \cdot q \cdot ib) + \left(\frac{KF \cdot ib^{AF}}{f} \right)$$

$$inb^2 = \text{shotnoise}^2 + \text{flickernoise}^2$$

$$\text{shotnoise} = (2 \cdot q \cdot ib)^{\frac{1}{2}}$$

$$\text{flickernoise} = \left(\frac{KF \cdot ib^{AF}}{f} \right)^{\frac{1}{2}}$$

The noise associated with the collector current is modeled as shot noise only.

$$inc = (2 \cdot q \cdot ic)^{\frac{1}{2}}$$

Parameter	Description
RB, V2/Hz	Output thermal noise due to base resistor
RC, V2/Hz	Output thermal noise due to collector resistor
RE, V2/Hz	Output thermal noise due to emitter resistor
IB, V2/Hz	Output shot noise due to base current
FN, V2/Hz	Output flicker noise due to base current
IC, V2/Hz	Output shot noise due to collector current
TOT, V2/Hz	Total output noise: TOT=RB + RC + RE + IB + IC + FN

BJT Temperature Compensation Equations

This section describes temperature compensation equations.

Energy Gap Temperature Equations

To determine energy gap for temperature compensation, use these equations:

TLEV=0, 1 or 3

$$egnom = 1.16-7.02e-4 \cdot \frac{tnom^2}{tnom+1108.0}$$

$$eg(t) = 1.16-7.02e-4 \cdot \frac{t^2}{t+1108.0}$$

TLEV=2

$$egnom = EG-GAP1 \cdot \frac{tnom^2}{tnom+GAP2}$$

$$eg(t) = EG-GAP1 \cdot \frac{t^2}{t+GAP2}$$

Saturation/Beta Temperature Equations, TLEV=0 or 2

The basic BJT temperature compensation equations for beta and the saturation currents when TLEV=0 or 2 (default is TLEV=0):

$$BF(t) = BF \cdot \left(\frac{t}{tnom} \right)^{XTB}$$

$$BR(t) = BR \cdot \left(\frac{t}{t_{nom}} \right)^{XTB}$$

$$ISE(t) = \frac{ISE}{\left(\frac{t}{t_{nom}} \right)^{XTB}} \cdot e^{\frac{facln}{NE}}$$

$$ISC(t) = \frac{ISC}{\left(\frac{t}{t_{nom}} \right)^{XTB}} \cdot e^{\frac{facln}{NC}}$$

$$ISS(t) = \frac{ISS}{\left(\frac{t}{t_{nom}} \right)^{XTB}} \cdot e^{\frac{facln}{NS}}$$

The parameter XTB usually should be set to zero for $TLEV=2$.

$$IS(t) = IS \cdot e^{facln}$$

$$IBE(t) = IBE \cdot e^{\frac{facln}{NF}}$$

$$IBC(t) = IBC \cdot e^{\frac{facln}{NR}}$$

TLEV=0, 1 or 3

$$facln = \frac{EG}{vt(t_{nom})} - \frac{EG}{vt(t)} + XTI \cdot \ln\left(\frac{t}{t_{nom}}\right)$$

TLEV=2

$$facln = \frac{egnom}{vt(t_{nom})} - \frac{eg(t)}{vt(t)} + XTI \cdot \ln\left(\frac{t}{t_{nom}}\right)$$

Saturation and Temperature Equations, TLEV=1

The basic BJT temperature compensation equations for beta and the saturation currents when $TLEV=1$:

$$BF(t) = BF \cdot (1 + XTB \cdot \Delta t)$$

$$BR(t) = BR \cdot (1 + XTB \cdot \Delta t)$$

$$ISE(t) = \frac{ISE}{1 + XTB \cdot \Delta t} \cdot e^{\frac{facln}{NE}}$$

$$ISC(t) = \frac{ISC}{1 + XTB \cdot \Delta t} \cdot e^{\frac{facln}{NC}}$$

$$ISS(t) = \frac{ISS}{1 + XTB \cdot \Delta t} \cdot e^{\frac{facln}{NS}}$$

$$IS(t) = IS \cdot e^{facln}$$

$$IBE(t) = IBE \cdot e^{\frac{facln}{NF}}$$

$$IBC(t) = IBC \cdot e^{\frac{facln}{NR}}$$

The following equation calculates the *facln* value for the preceding equations:

$$facln = \frac{EG}{vt(tnom)} - \frac{EG}{vt(t)} + XTI \cdot \ln\left(\frac{t}{tnom}\right)$$

TLEV=0, 1, 2

The *IKF*, *IKR*, and *IRB* parameters are also modified as:

$$IKF(t) = IKF \cdot (1 + TIKF1 \cdot \Delta t + TIKF2 \cdot \Delta t^2)$$

$$IKR(t) = IKR \cdot (1 + TIKR1 \cdot \Delta t + TIKR2 \cdot \Delta t^2)$$

$$IRB(t) = IRB \cdot (1 + TIRB1 \cdot \Delta t + TIRB2 \cdot \Delta t^2)$$

Saturation Temperature Equations, TLEV=3

The basic BJT temperature compensation equations for the saturation currents when *TLEV=3* are as follows:

$$is(t) = is^{(1+tis1\Delta t+tis2\Delta t^2)}$$

$$ibe(t) = ibe^{(1+tis1\Delta t+tis2\Delta t^2)}$$

$$ibc(t) = ibc^{(1+tis1\Delta t+tis2\Delta t^2)}$$

$$ise(t) = ise^{(1+tise1\Delta t+tise2\Delta t^2)}$$

$$isc(t) = isc^{(1+tisc1\Delta t+tisc2\Delta t^2)}$$

$$iss(t) = iss^{(1+tiss1\Delta t+tiss2\Delta t^2)}$$

The *IKF*, *IKR*, and *IRB* parameters are also modified as:

$$ikf(t) = ikf^{(1+tikf1\Delta t+tikf2\Delta t^2)}$$

$$ikr(t) = ikr^{(1+tikr1\Delta t+tikr2\Delta t^2)}$$

$$irb(t) = irb^{(1+tirb1\Delta t+tirb2\Delta t^2)}$$

The following model parameters are modified only according to the following equations whenever you specify corresponding non-zero temperature coefficients, regardless of the T_{LEV} value.

$$BF(t) = BF \cdot (1 + TBF1 \cdot \Delta t + TBF2 \cdot \Delta t^2)$$

$$BR(t) = BR \cdot (1 + TBR1 \cdot \Delta t + TBR2 \cdot \Delta t^2)$$

$$VAF(t) = VAF \cdot (1 + TVAF1 \cdot \Delta t + TVAF2 \cdot \Delta t^2)$$

$$VAR(t) = VAR \cdot (1 + TVAR1 \cdot \Delta t + TVAR2 \cdot \Delta t^2)$$

$$ITF(t) = ITF \cdot (1 + TITF1 \cdot \Delta t + TITF2 \cdot \Delta t^2)$$

$$TF(t) = TF \cdot (1 + TTF1 \cdot \Delta t + TTF2 \cdot \Delta t^2)$$

$$TR(t) = TR \cdot (1 + TTR1 \cdot \Delta t + TTR2 \cdot \Delta t^2)$$

$$NF(t) = NF \cdot (1 + TNF1 \cdot \Delta t + TNF2 \cdot \Delta t^2)$$

$$NR(t) = NR \cdot (1 + TNR1 \cdot \Delta t + TNR2 \cdot \Delta t^2)$$

$$NE(t) = NE \cdot (1 + TNE1 \cdot \Delta t + TNE2 \cdot \Delta t^2)$$

$$NC(t) = NC \cdot (1 + TNC1 \cdot \Delta t + TNC2 \cdot \Delta t^2)$$

$$NS(t) = NS \cdot (1 + TNS1 \cdot \Delta t + TNS2 \cdot \Delta t^2)$$

$$MJE(t) = MJE \cdot (1 + TMJE1 \cdot \Delta t + TMJE2 \cdot \Delta t^2)$$

$$MJC(t) = MJC \cdot (1 + TMJC1 \cdot \Delta t + TMJC2 \cdot \Delta t^2)$$

$$MJS(t) = MJS \cdot (1 + TMJS1 \cdot \Delta t + TMJS2 \cdot \Delta t^2)$$

Capacitance Temperature Equations

$T_{LEV C} = 0$

$$CJE(t) = CJE \cdot \left[1 + MJE \cdot \left(4.0e-4 \cdot \Delta t - \frac{VJE(t)}{VJE} + 1 \right) \right]$$

$$CJC(t) = CJC \cdot \left[1 + MJC \cdot \left(4.0e-4 \cdot \Delta t - \frac{VJC(t)}{VJC} + 1 \right) \right]$$

$$CJS(t) = CJS \cdot \left[1 + MJS \cdot \left(4.0e-4 \cdot \Delta t - \frac{VJS(t)}{VJS} + 1 \right) \right]$$

The following equations calculate values for the preceding equations:

$$V_{JE}(t) = V_{JE} \cdot \frac{t}{t_{nom}} \cdot v_t(t) \cdot \left[3 \cdot \ln\left(\frac{t}{t_{nom}}\right) + \frac{e_{gnom}}{v_t(t_{nom})} \cdot \frac{e_g(t)}{v_t(t)} \right]$$

$$V_{JC}(t) = V_{JC} \cdot \frac{t}{t_{nom}} \cdot v_t(t) \cdot \left[3 \cdot \ln\left(\frac{t}{t_{nom}}\right) + \frac{e_{gnom}}{v_t(t_{nom})} \cdot \frac{e_g(t)}{v_t(t)} \right]$$

$$V_{JS}(t) = V_{JS} \cdot \frac{t}{t_{nom}} \cdot v_t(t) \cdot \left[3 \cdot \ln\left(\frac{t}{t_{nom}}\right) + \frac{e_{gnom}}{v_t(t_{nom})} \cdot \frac{e_g(t)}{v_t(t)} \right]$$

TLEVC=1

$$C_{JE}(t) = C_{JE} \cdot (1 + CTE \cdot \Delta t)$$

$$C_{JC}(t) = C_{JC} \cdot (1 + CTC \cdot \Delta t)$$

$$C_{JS}(t) = C_{JS} \cdot (1 + CTS \cdot \Delta t)$$

The following equations calculate the built-in potentials:

$$V_{JE}(t) = V_{JE} - TV_{JE} \cdot \Delta t$$

$$V_{JC}(t) = V_{JC} - TV_{JC} \cdot \Delta t$$

$$V_{JS}(t) = V_{JS} - TV_{JS} \cdot \Delta t$$

TLEVC=2

$$C_{JE}(t) = C_{JE} \cdot \left(\frac{V_{JE}}{V_{JE}(t)} \right)^{M_{JE}}$$

$$C_{JC}(t) = C_{JC} \cdot \left(\frac{V_{JC}}{V_{JC}(t)} \right)^{M_{JC}}$$

$$C_{JS}(t) = C_{JS} \cdot \left(\frac{V_{JS}}{V_{JS}(t)} \right)^{M_{JS}}$$

The following equations calculate values for the preceding equations:

$$V_{JE}(t) = V_{JE} - TV_{JE} \cdot \Delta t$$

$$V_{JC}(t) = V_{JC} - TV_{JC} \cdot \Delta t$$

$$V_{JS}(t) = V_{JS} - TV_{JS} \cdot \Delta t$$

TLEV=3

$$CJE(t) = CJE \cdot \left(1 - 0.5 \cdot \frac{dvjedt}{VJE} \cdot \frac{\Delta t}{VJE} \right)$$

$$CJC(t) = CJC \cdot \left(1 - 0.5 \cdot \frac{dvjcdt}{VJC} \cdot \frac{\Delta t}{VJC} \right)$$

$$CJS(t) = CJS \cdot \left(1 - 0.5 \cdot \frac{dvjsdt}{VJS} \cdot \frac{\Delta t}{VJS} \right)$$

$$VJE(t) = VJE + dvjedt \cdot \Delta t$$

$$VJC(t) = VJC + dvjcdt \cdot \Delta t$$

$$VJS(t) = VJS + dvjsdt \cdot \Delta t$$

If TLEV=0, 1, or 3, then:

$$dvjcdt = -\frac{egnom + 3 \cdot vt(tnom) + (1.16 - egnom) \cdot \left(2 - \frac{tnom}{tnom + 1108} \right) - VJC}{tnom}$$

$$dvjsdt = -\frac{egnom + 3 \cdot vt(tnom) + (1.16 - egnom) \cdot \left(2 - \frac{tnom}{tnom + 1108} \right) - VJS}{tnom}$$

If TLEV=2:

$$dvjedt = -\frac{egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot \left(2 - \frac{tnom}{tnom + GAP2} \right) - VJE}{tnom}$$

$$dvjcdt = -\frac{egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot \left(2 - \frac{tnom}{tnom + GAP2} \right) - VJC}{tnom}$$

$$dvjsdt = -\frac{egnom + 3 \cdot vt(tnom) + (EG - egnom) \cdot \left(2 - \frac{tnom}{tnom + GAP2} \right) - VJS}{tnom}$$

Parasitic Resistor Temperature Equations

The following equations determine the parasitic resistors as a function of temperature regardless of the TLEV value:

$$RE(t) = RE \cdot (1 + TRE1 \cdot \Delta t + TRE2 \cdot \Delta t^2)$$

$$RB(t) = RB \cdot (1 + TRB1 \cdot \Delta t + TRB2 \cdot \Delta t^2)$$

$$RBM(t) = RBM \cdot (1 + TRM1 \cdot \Delta t + TRM2 \cdot \Delta t^2)$$

$$RC(t) = RC \cdot (1 + TRC1 \cdot \Delta t + TRC2 \cdot \Delta t^2)$$

BJT LEVEL=2 Temperature Equations

The model parameters of BJT `LEVEL=2` model are modified for temperature compensation as:

$$GAMMA(t) = GAMMA \cdot e^{(f_{ac} \ln)}$$

$$RC(t) = RC \cdot \left(\frac{t}{t_{nom}} \right)^{BEX}$$

$$VO(t) = VO \cdot \left(\frac{t}{t_{nom}} \right)^{BEXV}$$

BJT Quasi-Saturation Model

Use the BJT quasi-saturation model (`LEVEL=2`), an extension of the Gummel-Poon model (`LEVEL=1` model) to model bipolar junction transistors that exhibit quasi-saturation or base push-out effects. When a device with lightly doped collector regions operates at high injection levels, the internal base-collector junction is forward biased, while the external base-collector junction is reverse biased; DC current gain and the unity gain frequency f_T falls sharply. Such an operation regime is referred to as quasi-saturation, and its effects have been included in this model.

[Figure 32](#) and [Figure 33](#) show the additional elements of the `LEVEL=2` model. The current source I_{epi} and charge storage elements C_i and C_x model the quasi-saturation effects. The parasitic substrate bipolar transistor is also included in the vertical transistor by the diode D and current source I_{bs} .

Figure 32 Vertical npn Bipolar Transistor (SUBS=+1)

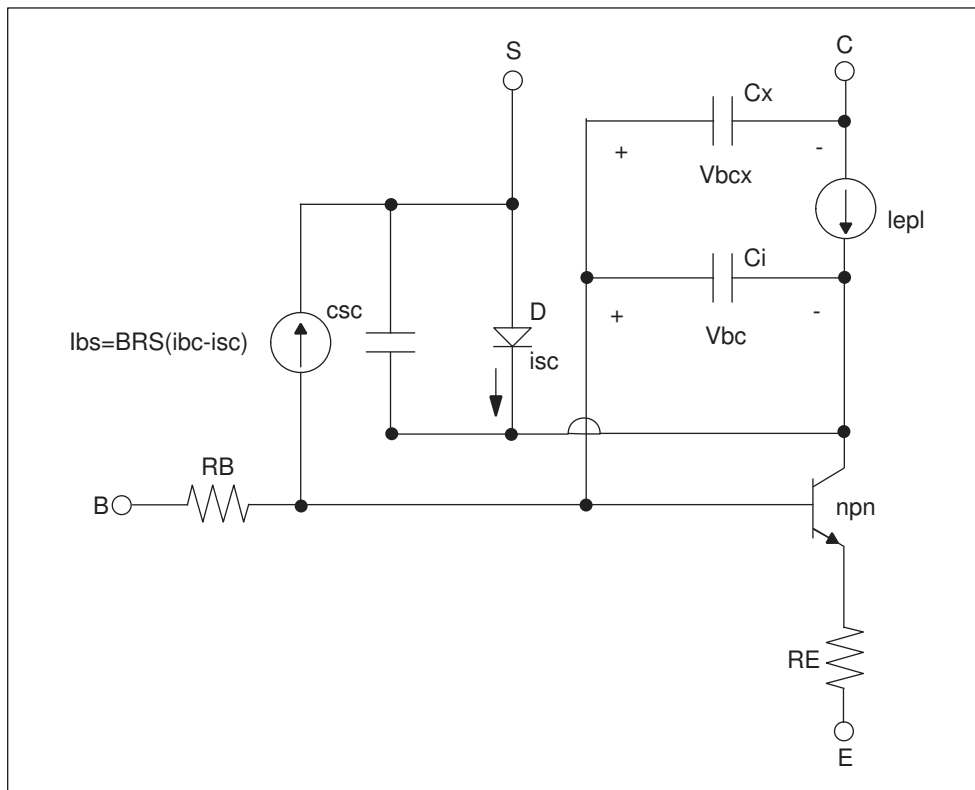
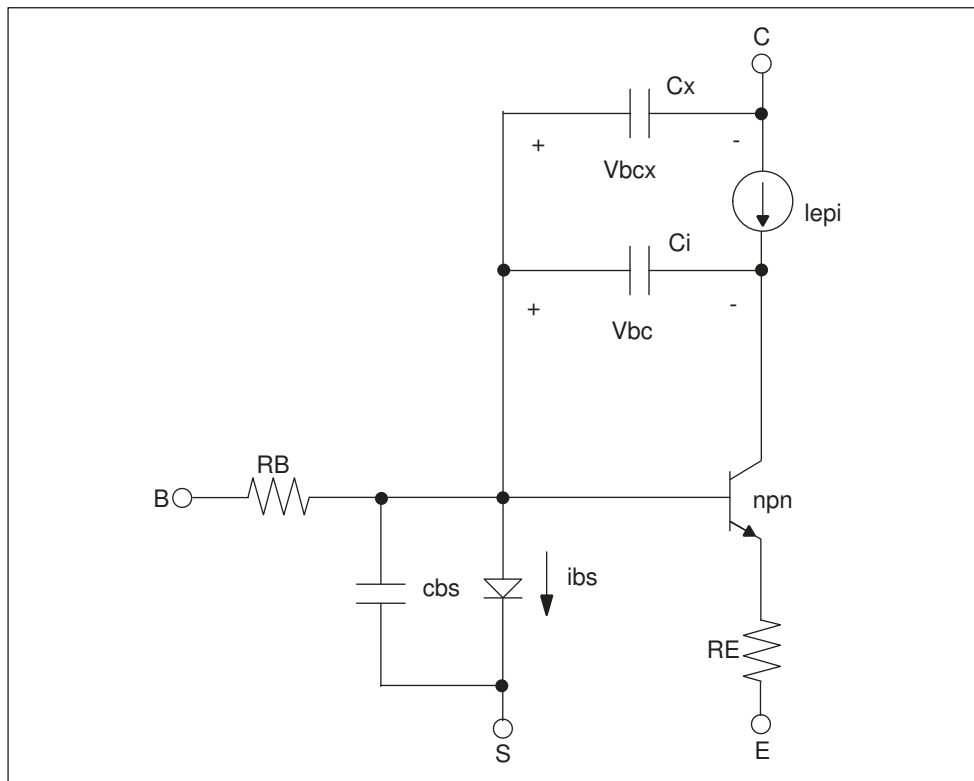


Figure 33 Lateral npn Bipolar Transistor (SUBS=-1)



Defining Noise Equations

The mean square short-circuit base resistance noise current equation is:

$$inrb = \left(\frac{4 \cdot k \cdot t}{rbb} \right)^{\frac{1}{2}}$$

The mean square short-circuit collector resistance noise current equation is:

$$inrc = \left(\frac{4 \cdot k \cdot t}{RCeff} \right)^{\frac{1}{2}}$$

The mean square short-circuit emitter resistance noise current equation is:

$$inre = \left(\frac{4 \cdot k \cdot t}{REff} \right)^{\frac{1}{2}}$$

The noise associated with the base current is composed of two parts: shot noise and flicker noise. Typical values for the flicker noise coefficient, K_F , are 1e-17 to 1e-12. They are calculated as:

$$2 \cdot q \cdot f_{knee}$$

In the preceding equation, f_{knee} is the noise knee frequency (typically 100 Hz to 10 MHz), and q is electron charge.

$$inb^2 = (2 \cdot q \cdot ib) + \left(\frac{KF \cdot ib^{AF}}{f} \right)$$

$$inb^2 = \text{shotnoise}^2 + \text{flickernoise}^2$$

$$\text{shotnoise} = (2 \cdot q \cdot ib)^{\frac{1}{2}}$$

$$\text{flickernoise} = \left(\frac{KF \cdot ib^{AF}}{f} \right)^{\frac{1}{2}}$$

The noise associated with the collector current is modeled as shot noise only.

$$inc = (2 \cdot q \cdot ic)^{\frac{1}{2}}$$

Parameter	Description
RB, V2/Hz	output thermal noise due to base resistor
RC, V2/Hz	output thermal noise due to collector resistor
RE, V2/Hz	output thermal noise due to emitter resistor
IB, V2/Hz	output shot noise due to base current
FN, V2/Hz	output flicker noise due to base current
IC, V2/Hz	output shot noise due to collector current
TOT, V2/Hz	total output noise: TOT=RB + RC + RE + IB + IC + FN

Epitaxial Current Source I_{epi}

The following equation determines the epitaxial current value, I_{epi} :

$$I_{epi} = \frac{ki - kx \cdot \ln\left(\frac{1+ki}{1+kx}\right) + \frac{vbc - vbcx}{NEPI \cdot vt}}{\left(\frac{RCeff}{NEPI \cdot vt}\right) \cdot \left(1 + \frac{|vbc - vbcx|}{VO}\right)}$$

The following equations calculate values for the preceding equations:

$$ki = \left[1 + GAMMA \cdot e^{vbc/(NEPI \cdot vt)} \right]^{\frac{1}{2}}$$

$$kx = \left[1 + GAMMA \cdot e^{vbcx/(NEPI \cdot vt)} \right]^{\frac{1}{2}}$$

If you set the `GAMMA` model parameter to zero, then the `ki` and `kx` values both become one, and:

$$Iepi = \frac{vbc - vbcx}{RCeff \cdot \left(1 + \frac{|vbc - vbcx|}{VO} \right)}$$

Epitaxial Charge Storage Elements Ci and Cx

The following equations calculate the epitaxial charges:

$$qi = QCOeff \cdot \left(ki - 1 - \frac{GAMMA}{2} \right)$$

$$qx = QCOeff \cdot \left(kx - 1 - \frac{GAMMA}{2} \right)$$

The corresponding capacitances are calculated as:

$$Ci = \frac{\partial}{\partial vbc} (qi) = \left(\frac{GAMMA \cdot QCOeff}{2 \cdot NEPI \cdot vt \cdot kx} \right) \cdot e^{vbc/(NEPI \cdot vt)}$$

$$Cx = \frac{\partial}{\partial vbcx} (qx) = \left(\frac{GAMMA \cdot QCOeff}{2 \cdot NEPI \cdot vt \cdot kx} \right) \cdot e^{vbcx/(NEPI \cdot vt)}$$

If `GAMMA=0`, then the `Ci` and `Cx` values become zero.

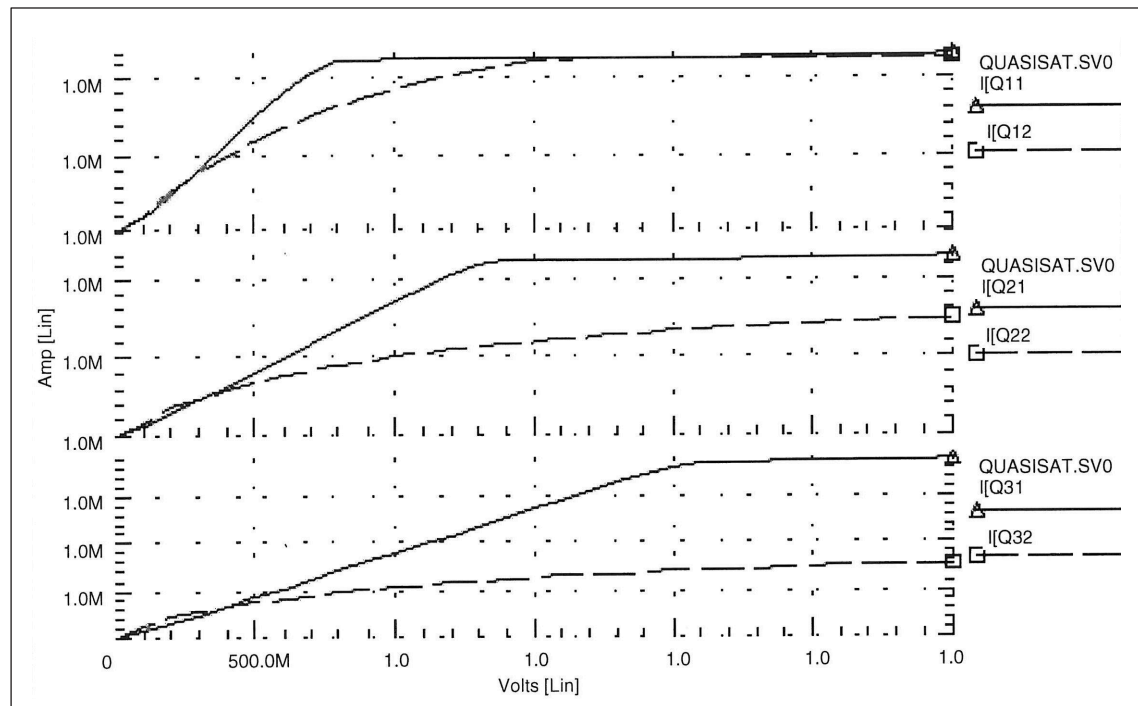
Example

This example is based on demonstration netlist `quasisat.sp`, which is available in directory `$<installdir>/demo/hspice/bjt`:

```
*quasisat.sp comparison of bjt Level1 and Level2
*model
.OPTION nomod relv=.001 reli=.001 absv=.1u absi=1p
.OPTION post
q11 10 11 0 mod1
q12 10 12 0 mod2
q21 10 21 0 mod1
q22 10 22 0 mod2
q31 10 31 0 mod1
q32 10 32 0 mod2
vcc 10 0 .7
i11 0 11 15u
i12 0 12 15u
```

```
i21 0 21 30u
i22 0 22 30u
i31 0 31 50u
i32 0 32 50u
.dc vcc 0 3 .1
.print dc vce=par('v(10)') i(q11) i(q12) i(q21)
+ i(q22) i(q31) i(q32)
.probe dc i(q11) i(q12) i(q21) i(q22)
.probe dc i(q11) i(q12)
.MODEL MOD1 NPN IS=4.0E-16 BF=75 VAF=75
+ Level=1 rc=500 SUBS=+1
.MODEL MOD2 NPN IS=4.0E-16 BF=75 VAF=75
+ Level=2 rco=500 rcc=2.599 vo=1 qco=1e-10
+ gamma=1e-9 SUBS=+1
.end
```

Figure 34 Comparing BJT LEVEL=1 and LEVEL=2 Models



Converting National Semiconductor Models

National Semiconductor's SNAP circuit simulator has a scaled BJT model that is not the same as the PrimeSim Continuum BJT models. To use this model, make the following changes.

For a subcircuit that consists of the scaled BJT model, the subcircuit name must be the same as the name of the model. Inside the subcircuit there is a `.PARAM` statement that specifies the scaled BJT model parameter values. Put a scaled BJT model inside the subcircuit, then change the `.MODEL mname mtype` statement to a `.PARAM` statement. Ensure that each parameter in the `.MODEL` statement within the subcircuit has a value in the `.PARAM` statement.

Defining Scaled BJT Subcircuits

The following subcircuit definition converts the National Semiconductor scaled BJT model to a form usable in the PrimeSim Continuum simulators. The `.PARAM` parameter inside the `.SUBCKT` represents the `.MODEL` parameter in the National circuit simulator. Therefore, replace the `.MODEL mname` statement with a `.PARAM` statement. Change the model name to SBJT.

Note:

All parameter values in the following model must come from either a `.PARAM` statement or the subcircuit call.

Example

The following is a subcircuit definition that converts the National Semiconductor scaled BJT model to a form usable in the PrimeSim Continuum solution.

```
.SUBCKT SBJT NC NB NE SF=1 SCBC=1 SCBE=1 SCCS=1 SIES=1 SICS=1
+ SRB=1 SRC=1 SRE=1 SIC=0 SVCE=0 SBET=1
Q NC NB NE SBJT IC=SIC VCE=SVCE
.PARAM IES=1.10E-18 ICS=5.77E-18 NE=1.02 NC=1.03
+ ME=3.61 MC=1.24 EG=1.12 NSUB=0
+ CJE=1E-15 CJC=1E-15 CSUB=1E-15 EXE=0.501
+ EXC=0.222 ESUB=0.709 PE=1.16 PC=0.37
+ PSUB=0.698 RE=75 RC=0.0 RB=1.0
+ TRE=2E-3 TRC=6E-3 TRB=1.9E-3 VA=25
+ FTF=2.8E9 FTR=40E6 BR=1.5 TCB=5.3E-3
+ TCB2=1.6E-6 BF1=9.93 BF2=45.7 BF3=55.1
+ BF4=56.5 BF5=53.5 BF6=33.8
+ IBF1=4.8P IBF2=1.57N IBF3=74N
+ IBF4=3.13U IBF5=64.2U IBF6=516U
*
.MODEL SBJT NPN
+ IBE='IES*SF*SIES' IBC='ICS*SF*SICS'
+ CJE='CJE*SF*SCBE' CJC='CJC*SF*SCBC'
+ CJS='CSUB*SF*SCCS' RB='RB*SRB/SF'
+ RC='RC*SRC/SF' RE='RE*SRE/SF'
+ TF='1/(6.28*FTF)' TR='1/(6.28*FTR)'
+ MJE=EXE MJC=EXC
+ MJS=ESUB VJE=PE
+ VJC=PC VJS=PSUB
+ NF=NE NR=NC
```

```
+ EG=EG BR=BR VAF=VA
+ TRE1=TRE TRC1=TRC TRB1=TRB
+ TBF1=TCB TBF2=TCB2
+ BF0=BF1 IB0=IBF1
+ BF1=BF2 IB1=IBF2
+ BF2=BF3 IB2=IBF3
+ BF3=BF4 IB3=IBF4
+ BF4=BF5 IB4=IBF5
+ BF5=BF6 IB5=IBF6
+ NSUB=0 sbet=sbet
+ TLEV=1 TLEV=1
+ XTIR='MC*NC' XTI='ME*NE'
.ENDS SBJT
```

The following replaces the BJT statement:

```
XQ1 1046 1047 8 SBJT SIES=25.5 SICS=25.5 SRC=3.92157E-2
+ SRE=3.92157E-2 SBET=3.92157E-2 SRB=4.8823E+2 SCBE=94.5234
+ SCBC=41.3745 SCCS=75.1679 SIC=1M SVCE=1
```

VBIC Bipolar Transistor Model

The VBIC (Vertical Bipolar Inter-Company) model is a bipolar transistor model. To use VBIC, specify the `LEVEL=4` parameter for the bipolar transistor model.

VBIC addresses many problems of the Gummel-Poon model:

- More accurate modeling of Early effect
- Parasitic substrate transistor
- Modulation of collector resistance
- Avalanche multiplication in collector junction, parasitic capacitances of base-emitter overlap in double poly BJTs, and self heating.

Defining Noise Equations

The mean square short-circuit base resistance noise current equation is:

$$inrb = \left(\frac{4 \cdot k \cdot t}{rbb} \right)^{\frac{1}{2}}$$

The mean square short-circuit collector resistance noise current equation is:

$$inrc = \left(\frac{4 \cdot k \cdot t}{RCeff} \right)^{\frac{1}{2}}$$

The mean square short-circuit emitter resistance noise current equation is:

$$inre = \left(\frac{4 \cdot k \cdot t}{RE_{eff}} \right)^{\frac{1}{2}}$$

The noise associated with the base current is composed of two parts: shot noise and flicker noise. Typical values for the flicker noise coefficient, K_F , are 1e-17 to 1e-12. They are calculated as:

$$2 \cdot q \cdot fknee$$

In the preceding equation, $fknee$ is the noise knee frequency (typically 100 Hz to 10 MHz), and q is electron charge.

$$inb^2 = (2 \cdot q \cdot ib) + \left(\frac{KF \cdot ib^{AF}}{f} \right)$$

$$inb^2 = \text{shotnoise}^2 + \text{flickernoise}^2$$

$$\text{shotnoise} = (2 \cdot q \cdot ib)^{\frac{1}{2}}$$

$$\text{flickernoise} = \left(\frac{KF \cdot ib^{AF}}{f} \right)^{\frac{1}{2}}$$

The noise associated with the collector current is modeled as shot noise only.

$$inc = (2 \cdot q \cdot ic)^{\frac{1}{2}}$$

Parameter	Description
RB, V2/Hz	Output thermal noise due to base resistor
RC, V2/Hz	Output thermal noise due to collector resistor
RE, V2/Hz	Output thermal noise due to emitter resistor
IB, V2/Hz	Output shot noise due to base current
FN, V2/Hz	Output flicker noise due to base current
IC, V2/Hz	Output shot noise due to collector current
TOT, V2/Hz	Total output noise: TOT=RB + RC + RE + IB + IC + FN

History of VBIC

VBIC was developed by engineers at several companies. The detailed equations for all elements are given in the referenced publication. Recent information and source code can be found on the web site:

<http://www.designers-guide.org/VBIC/>

The PrimeSim Continuum implementation complies with standard VBIC. Starting in release 2001.4 of the VBIC model, self-heating and excess phase have been implemented or enabled.

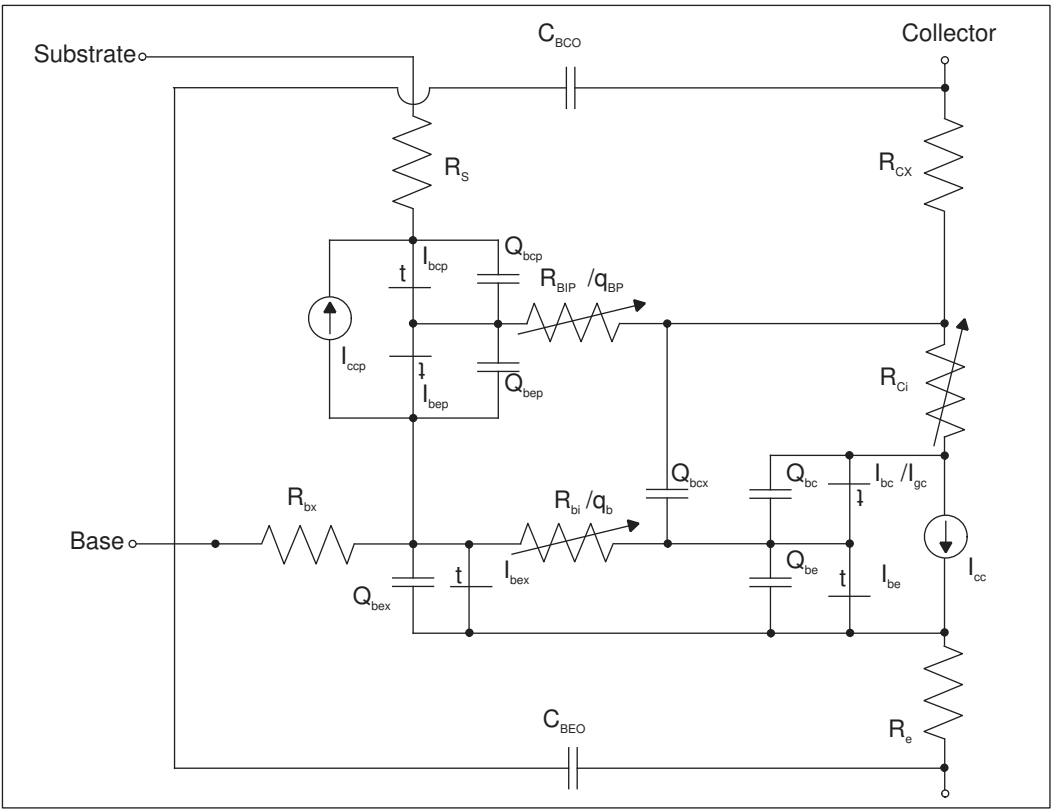
The large signal equivalent circuit for VBIC is shown in [Figure 35](#). Capacitors C_{BCO} , C_{BEO} and resistors R_{CX} , R_{BX} , R_E , and R_S are linear elements, all other elements of the equivalent circuit are nonlinear.

VBIC Parameters

[Figure 35](#) lists the parameters that you can set for the model, and shows the default values for each parameter. The same parameter names are used in the table and the previous referenced publication.

Starting in Version 2003.03, the BJT `LEVEL=4` model prints FT in the `.OP` output.

Figure 35 Transient Analysis



If you specify values for parameters beyond their ranges, those parameters are reset to new values and warnings are printed unless you set the `.OPTION NOWARN` command.

Table 62 BJT LEVEL=4 Default Model Parameters

Name (Alias)	Unit	Default	Description
AFN		1	Flicker noise exponent for current
AJC		-0.5	Base-collector capacitance switching parameter
AJE		-0.5	Base-emitter capacitance switching parameter
AJS		-0.5	Substrate-collector capacitance switching parameter
AVC1	V-1	0	Base-collector weak avalanche parameter 1
AVC2	V-1	0	Base-collector weak avalanche parameter 2
BFN		1	Flicker noise exponent for 1/f dependence

Table 62 BJT LEVEL=4 Default Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
CBCO (CBC0)	F	0	Extrinsic base-collector overlap capacitance
CBEO (CBE0)	F	0	Extrinsic base-emitter overlap capacitance
CJC	F	0	Base-collector intrinsic zero bias capacitance
CJCP	F	0	Substrate-collector zero bias capacitance
CJE	F	0	Base-emitter zero bias capacitance
CJEP	F	0	Base-collector extrinsic zero bias capacitance
CTH	J/K	0	Thermal capacitance
EA	eV	1.12	Activation energy for IS
EAIC	eV	1.12	Activation energy for IBCI/IBEIP
EAIE	eV	1.12	Activation energy for IBEI
EAIS	eV	1.12	Activation energy for IBCIP
EANC	eV	1.12	Activation energy for IBCN/IBENP
EANE	eV	1.12	Activation energy for IBEN
EANS	eV	1.12	Activation energy for IBCNP
FC		0.9	Forward bias depletion capacitance limit
GAMM		0	Epi doping parameter
HRCF		1	High-current RC factor
IBCI	A	1e-16	Ideal base-collector saturation current
IBCIP	A	0	Ideal parasitic base-collector saturation current
IBCN	A	1e-15	Non-ideal base-collector saturation current
IBCNP	A	0	Non-ideal parasitic base-collector saturation current
IBEI	A	1e-18	Ideal base-emitter saturation current
IBEIP	A	0	Ideal parasitic base-emitter saturation current
IBEN	A	1e-15	Non-ideal base-emitter saturation current
IBENP	A	0	Non-ideal parasitic base-emitter saturation current

Table 62 BJT LEVEL=4 Default Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
IKF	A	2e-3	Forward knee current
IKP	A	2e-4	Parasitic knee current
IKR	A	2e-4	Reverse knee current
IS	A	1e-16	Transport saturation current
ISMIN	A	1.0e-19	Parameter for extending the minimum value of is
ISP	A	1e-16	Parasitic transport saturation current
ISPMIN	A	1.0e-19	Parameter for extending the minimum value of isp
ITF	A	1e-3	Coefficient of TF dependence in Ic
KFN		0	Base-emitter flicker noise constant
MC		0.33	Base-collector grading coefficient
MCMIN	-	1.0d-2	Parameter for extending the minimum value of mc
ME		0.33	Base-emitter grading coefficient
MEMIN	-	1.0d-2	Parameter for extending the minimum value of me
MS		0.33	Substrate-collector grading coefficient
MSMIN	-	1.0d-2	Parameter for extending the minimum value of ms
NCI		1	Ideal base-collector emission coefficient
NCIP		1	Ideal parasitic base-collector emission coefficient
NCN		2	Non-ideal base-collector emission coefficient
NCNP		2	Non-ideal parasitic base-collector emission coefficient
NEI		1	Ideal base-emitter emission coefficient
NEN		2	Non-ideal base-emitter emission coefficient
NF		1	Forward emission coefficient
NFP		1	Parasitic forward emission coefficient
NR		1	Reverse emission coefficient

Table 62 BJT LEVEL=4 Default Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
PC	V	0.75	Base-collector built-in potential
PE	V	0.75	Base-emitter built-in potential
PS	V	0.75	Substrate-collector built-in potential
QCO (QC0)	C	0	Epi charge parameter
QTF		0	Variation of TF with base-width modulation
RBI	Ohm	1e-1	Intrinsic base resistance
RBP	Ohm	1e-1	Parasitic base resistance
RBPMIN	A	1.0e-3	Parameter for extending the minimum value of rbp.
RBX	Ohm	1e-1	Extrinsic base resistance
RCI	Ohm	1e-1	Intrinsic collector resistance
RCX	Ohm	1e-1	Extrinsic collector resistance
RE	Ohm	1e-1	Emitter resistance
RS	Ohm	1e-1	Substrate resistance
RTH	K/W	0	Thermal resistance
TAVC	1/K	0	Temperature coefficient of AVC2
TD	s	0	Forward excess-phase delay time
TF	s	1e-11	Forward transit time
TNF	1/K	0	Temperature coefficient of NF
TR	s	1e-11	Reverse transit time
TREF (TNOM)	°C	27	Nominal measurement temperature of parameters (do not use TNOM alias)
VEF	V	0	Forward Early voltage
VER	V	0	Reverse Early voltage
VO (V0)	V	0	Epi drift saturation voltage
VTF	V	0	Coefficient of TF dependence on Vbc

Table 62 BJT LEVEL=4 Default Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
WBE		1	Portion of IBEI from Vbei, 1-WBE from Vbex
WSP		1	Portion of ICCP from Vbep, 1-WSP from Vbci
XII		3	Temperature exponent of IBEI/IBCI/IBEIP/IBCIIP
XIN		3	Temperature exponent of IBEN/IBCN/IBENP/IBCNP
XIS		3	Temperature exponent of IS
XRB		1	Temperature exponent of base resistance
XRC		1	Temperature exponent of collector resistance
XRE		1	Temperature exponent of emitter resistance
XRS		1	Temperature exponent of substrate resistance
XTF		0	Coefficient of TF bias dependence
XVO (XV0)		0	Temperature exponent of VO

Noise Analysis

The following sources of noise are taken into account:

- The thermal noise of resistors RBX, RCX, RE, RS, RBP, RCI, RBI
- Shot noise of currents IBE, IBEP, ICC, ICCP
- Flicker noise due to currents IBE, IBEP

Noise due to IBEX and IGC is not included.

Defining Noise Equations

The mean square short-circuit base resistance noise current equation is:

$$inrb = \left(\frac{4 \cdot k \cdot t}{rbb} \right)^{\frac{1}{2}}$$

The mean square short-circuit collector resistance noise current equation is:

$$inrc = \left(\frac{4 \cdot k \cdot t}{RCeff} \right)^{\frac{1}{2}}$$

The mean square short-circuit emitter resistance noise current equation is:

$$inre = \left(\frac{4 \cdot k \cdot t}{REff} \right)^{\frac{1}{2}}$$

The noise associated with the base current is composed of two parts: shot noise and flicker noise. Typical values for the flicker noise coefficient, K_F , are 1e-17 to 1e-12. They are calculated as:

$$2 \cdot q \cdot fknee$$

In the preceding equation, $fknee$ is the noise knee frequency (typically 100 Hz to 10 MHz), and q is electron charge.

$$inb^2 = (2 \cdot q \cdot ib) + \left(\frac{KF \cdot ib^{AF}}{f} \right)$$

$$inb^2 = \text{shotnoise}^2 + \text{flickernoise}^2$$

$$\text{shotnoise} = (2 \cdot q \cdot ib)^{\frac{1}{2}}$$

$$\text{flickernoise} = \left(\frac{KF \cdot ib^{AF}}{f} \right)^{\frac{1}{2}}$$

The noise associated with the collector current is modeled as shot noise only.

$$inc = (2 \cdot q \cdot ic)^{\frac{1}{2}}$$

Parameter	Description
RB, V2/Hz	Output thermal noise due to base resistor
RC, V2/Hz	Output thermal noise due to collector resistor
RE, V2/Hz	Output thermal noise due to emitter resistor
IB, V2/Hz	Output shot noise due to base current
FN, V2/Hz	Output flicker noise due to base current
IC, V2/Hz	Output shot noise due to collector current
TOT, V2/Hz	Total output noise: TOT=RB + RC + RE + IB + IC + FN

Self-heating and Excess Phase

After a self-heating effect is accounted for, the device element syntax becomes:

```
Qxxx nc nb ne [ns][nt] mname [regular parameters][tnodeout]
```

In the preceding syntax, `nt` is the temperature node. If you specify this node, but not `ns`, then you must specify the `tnodeout` parameter to indicate that the fourth node is the temperature node and not the substrate node. To turn on self-heating in addition to specifying the temperature node, the `RTH` (thermal resistance) model parameter must be not zero in the model card.

Excess phase affects only AC and transient characteristics analysis. To turn on this effect, the `TD` (forward excess-phase delay time) model parameter must be non-zero. But for transient analysis, turning on excess phase is not recommended, because the model's convergence is very sensitive to the `TD` value.

Example 1

This example with a no self-heating effect is located in the following directory:

```
$installdir/demo/hspice/bjt/vbic.sp
```

Example 2

This example with self-heating effects is located in the following directory:

```
$installdir/demo/hspice/bjt/self_heat.spIn
```

the preceding example, `v(t)` uses the `T` node to print the device temperature.

Notes on Using VBIC

- If `LEVEL=4`, the model is a VBIC bipolar junction transistor.
- The `LEVEL=4` model supports Area and M factor scaling.
- Setting these parameters to zero infers a value of infinity: `HRCF`, `IKF`, `IKP`, `IKR`, `ITF`, `VEF`, `VER`, `VO`, `VTF`.
- The `CBC0`, `CBE0`, `QC0`, `TNOM`, `V0`, and `XV0` parameters are aliases for `CBCO`, `CBE0`, `QCO`, `TREF`, `VO`, and `XVO`, respectively. Do not use `TNOM` as a model parameter name, because it is the name of the default room temperature.
- The default room temperature is 25 degrees in the PrimeSim Continuum solution, but is 27 in some other simulators. If you set the VBIC bipolar junction transistor model parameters to 27 degrees, add `TREF=27` to the model so that simulation correctly interprets the model parameters. To set the nominal simulation temperature to 27, add `.OPTION TNOM=27` to the netlist. Do this when testing the PrimeSim Continuum solution versus other simulators that use 27 as the default room temperature.

- Pole-zero simulation of this model is not supported.
- For this version of implementation, all seven internal resistors should have values greater than or equal to 1.0e-3. Values smaller than this are reassigned a value of 1.0e-3.

LEVEL=6 Philips Bipolar Model (MEXTRAM)

Mextram (Most EXquisite TRAnsistor Model) is a compact model for bipolar transistor.

LEVEL=6 Element Syntax

```
Qxxx nc nb ne [ns] [nt] mname [AREA=val]  
+ [OFF] [VBE=val] [VCE=val] [M=val] [DTEMP=val] tnodeout
```

```
Qxxx nc nb ne nt mname [AREA=val]  
+ [OFF] [VBE=val] [VCE=val] [M=val] [DTEMP=val] tnodeout
```

Parameter	Description
Qxxx	BJT element name. Starts with Q, followed by up to 1023 alphanumeric characters.
nc	Collector terminal node name or number.
nb	Base node name or number.
ne	Emitter terminal node name or number.
ns	Substrate node name or number.
nt	Self-heating temperature node name or number. In the second form, nt is used as a self-heating node, but no substrate node is defined.
mname	BJT model name reference.
AREA	Normalized emitter area.
OFF	Sets initial condition to OFF for this element in DC analysis.
VBE	Initial internal base to emitter voltage.
VCE	Initial internal collector to emitter voltage.
M	Multiplier to simulate multiple BJTs in parallel (alias in an instance for the MULT model parameter). If you use MULT, set m to zero. • If (MULT > 0.0 and MULT != 1.0) and (m == 1), then the PrimeSim Continuum solution uses the MULT model parameter and displays a warning message. • Otherwise, the PrimeSim Continuum solution uses the m instance parameter and displays a warning message: MULT=1, m=1 (no warning messages) MULT=1, m=3 MULT=2, m=3 and so on...

Parameter	Description
DTEMP	Difference between element and circuit temperature.
tnodeout	Identify self heating node from substrate node.

Version Control of MEXTRAM Model

VERSION is introduced for model selection.

VERSION take precedence over VERS and INTVERS.

Use any one of the following settings to set the version for MEXTRAM 505.1.

- VERSION=505.1
- VERS=505 INTVERS=5.1

Mextram Level 503

The Philips bipolar model (MEXTRAM Level 503) is the BJT `LEVEL=6` model. See also: [MEXTRAM Level 504](#). MEXTRAM includes effects that are not included in some other BJT models (such as in the original Gummel-Poon model):

- Temperature
- Charge storage
- Substrate
- Parasitic PNP
- High-injection
- Built-in electric field in base region
- Bias-dependent Early effect
- Low-level, non-ideal base currents
- Hard- and quasi-saturation
- Weak avalanche
- Hot carrier effects in the collector epilayer
- Explicit modeling of inactive regions
- Split base-collector depletion capacitance

- Current crowding and conductivity modulation for base resistance
- First order approximation of distributed high frequency effects in the intrinsic base (high frequency current crowding and excess phase shift)

You can use either of the following two parameters to specify the difference between the circuit temperature and the ambient temperatures in the MEXTRAM model:

- `DTEMP` instance parameter as specified in the element statement.
- `DTA` (difference between the device temperature and the ambient analysis temperature) global model parameter.

`DTA` and `DTEMP` both default to zero. `DTEMP` overrides `DTA` locally, if you specify both. Simulation uses the value of `DTEMP` to de-rate the temperature in model equations and other parameters.

- If you do not specify either the `DTEMP` or the `DTA` parameter, then `DTEMP=0.0`.
- If you specify `DTA` but not `DTEMP`, then `DTEMP` uses the `DTA` value.
- If you specify `DTEMP`, then simulation uses the `DTEMP` value, and ignores the `DTA` value.

Defining Noise Equations

The mean square short-circuit base resistance noise current equation is:

$$inrb = \left(\frac{4 \cdot k \cdot t}{r_{bb}} \right)^{\frac{1}{2}}$$

The mean square short-circuit collector resistance noise current equation is:

$$inrc = \left(\frac{4 \cdot k \cdot t}{R_{Ceff}} \right)^{\frac{1}{2}}$$

The mean square short-circuit emitter resistance noise current equation is:

$$inre = \left(\frac{4 \cdot k \cdot t}{R_{Eeff}} \right)^{\frac{1}{2}}$$

The noise associated with the base current is composed of two parts: shot noise and flicker noise. Typical values for the flicker noise coefficient, `KF`, are 1e-17 to 1e-12. They are calculated as:

$$2 \cdot q \cdot fknee$$

In the preceding equation, `fknee` is the noise knee frequency (typically 100 Hz to 10 MHz), and `q` is electron charge.

$$inb^2 = (2 \cdot q \cdot ib) + \left(\frac{KF \cdot ib^{AF}}{f} \right)$$
$$inb^2 = \text{shotnoise}^2 + \text{flickernoise}^2$$
$$\text{shotnoise} = (2 \cdot q \cdot ib)^{\frac{1}{2}}$$
$$\text{flickernoise} = \left(\frac{KF \cdot ib^{AF}}{f} \right)^{\frac{1}{2}}$$

The noise associated with the collector current is modeled as shot noise only.

$$inc = (2 \cdot q \cdot ic)^{\frac{1}{2}}$$

Parameter	Description
FN, V2/Hz	Output flicker noise due to base current
IB, V2/Hz	Output shot noise due to base current
IC, V2/Hz	Output shot noise due to collector current
RB, V2/Hz	Output thermal noise due to base resistor
RC, V2/Hz	Output thermal noise due to collector resistor
RE, V2/Hz	Output thermal noise due to emitter resistor
TOT, V2/Hz	Total output noise: TOT=RB + RC + RE + IB + IC + FN

Model Parameters (503)

This section lists MEXTRAM `LEVEL=6` model parameters, including parameter names, descriptions, units, default values, and notes.

Table 63 BJT LEVEL=6 MEXTRAM 503 Flags

Name (Alias)	Unit	Default	Description
LEVEL	-	-	LEVEL=6 for MEXTRAM
EXAVL	-	0	Flag for extended modeling of avalanche currents
EXMOD	-	0	Flag for extended modeling of the reverse current gain
EXPHI	-	1	Flag for distributed high frequency effects

Table 63 BJT LEVEL=6 MEXTRAM 503 Flags (Continued)

Name (Alias)	Unit	Default	Description
SUBS	-	-	Flag for switching substrate effect
OUTFLAG			

Table 64 BJT LEVEL=6 MEXTRAM 503 Basic Parameters

Name (Alias)	Unit	Default	Description
TREF	oC	0.0	Model nominal temperature
AB	-	1.35	Temperature coefficient resistivity of the base
AC	-	0.4	Temperature coefficient resistivity of the buried layer
AEPI	-	2.15	Temperature coefficient resistivity of the epilayer
AEX	-	1.	Temperature coefficient resistivity of the extrinsic base
AF	-	1.0	Flicker noise exponent
AS	-	2.15	For a closed buried layer: AS=AC For an open buried layer: AS=AEPI
AVL	-	50.	Weak avalanche parameter
BF	A	140.0	Ideal forward current gain
BRI	-	16.0	Ideal reverse current gain
CJC	F	1.3e-13	Zero bias collector-base depletion capacitance
CJE	F	2.5e-13	Zero bias collector-base depletion capacitance
CJS	F	1.e-12	Zero bias collector-substrate depletion capacitance
EFI	-	0.7	Electric field intercept (with EXAVL=1)
ER	-	2.E-3	Temperature coefficient of VLF and VLR
ETA	-	4.0	Factor of the built-in field of the base
IBF	A	2.0E-14	Saturation current of the non-ideal forward base current
IBR	A	8.0e-15	Saturation current of the non-ideal reverse base current
IHC	A	3.e-3	Critical current for hot carriers

Table 64 BJT LEVEL=6 MEXTRAM 503 Basic Parameters (Continued)

Name (Alias)	Unit	Default	Description
IK	A	15.E-3	High-injection knee current
IKS	A	5.E-6	Knee current of the substrate
IS	A	5.E-17	Collector-emitter saturation current
ISS	A	6.E-16	Base-substrate saturation current
KF	-	2.E-16	Flicker noise coefficient ideal base current
KFN	-	2.E-16	Flicker noise coefficient non-ideal base current
MC	-	0.5	Collector current modulation coefficient
MTAU	-	1.18	Non-ideality factor of the neutral and emitter charge
NA	cm ⁻³	3.0E17	Maximum base dope concentration
PC	-	0.4	Collector-base grading coefficient variable part
PE	-	0.33	Emitter-base grading coefficient
PS	-	0.33	Collector-substrate grading coefficient
QBO	C	1.2e-12	Base charge at zero bias
RBC	ohm	50.	Constant part of the base resistance
RBV	ohm	100.	Variable part of the base resistance at zero bias
RCC	ohm	25.	Constant part of the collector resistance
RCV	ohm	750.	Resistance of the unmodulated epilayer
RE	ohm	2.0	Emitter series resistance
SCRCV	ohm	1000.0	Space charge resistance of the epilayer
SFH	-	0.6	Current spreading factor epilayer
TAUNE	s	3.e-10	Minimum delay time of neutral and emitter charge
VDC	V	0.6	Collector-base diffusion voltage
VDE	V	0.9	Emitter-base diffusion voltage
VDS	V	0.5	Collector-substrate diffusion voltage
VGB	V	1.18	Band-gap voltage of the base

Table 64 BJT LEVEL=6 MEXTRAM 503 Basic Parameters (Continued)

Name (Alias)	Unit	Default	Description
VGC	V	1.205	Band-gap voltage of the collector
VGE	V	1.01	Band-gap voltage of the emitter
VGJ	V	1.1	Band-gap voltage recombination emitter-base junction
VGS	V	1.15	Band-gap voltage of the substrate
VI	V	0.040	Ionization voltage base dope
VLF	V	0.5	Cross-over voltage of the non-ideal forward base current
VLR	V	0.5	Cross-over voltage of the non-ideal reverse base current
XCJC	-	0.1	Fraction of the collector-base depletion capacitance under the emitter area
XCJE	F	0.5	Fraction of the emitter-base depletion capacitance that belongs to the sidewall
XEXT	-	0.5	Part of I EX, Q EX, Q TEX, and I SUB that depends on the base-collector voltage VBC1
XIBI	-	0.0	Fraction of ideal base current that belongs to the sidewall
XP	F	0.2	Constant part of CJC

Example

This example is located in the following directory: `$installdir/demo/hspice/bjt/mextram.sp`

MEXTRAM Level 504

For first-order and higher-order characteristic derivatives, MEXTRAM 504 returns better results than MEXTRAM 503. This effect is noticeable in output-conductance, cut-off frequency, and low-frequency third-order distortion.

MEXTRAM Level 504 models several effects that are not included in the original Gummel-Poon model.

These effects include:

- Temperature
- Charge storage

- Substrate
- Parasitic PNP
- High-injection
- Bias-dependent early effect
- Low-level, non-ideal base currents
- Hard- and quasi-saturation (including Kirk Effect)
- Weak avalanche (optionally including snap-back behavior)
- Explicit modeling of inactive regions
- Split base-collector and base-emitter depletion capacitance
- Current crowding and conductivity modulation of the base resistance
- First order approximation of distributed high frequency effects in the intrinsic base (high frequency current crowding and excess phase shift)
- Ohmic resistance of epilayer
- Velocity saturation effects on the resistance of the epilayer
- Recombination in the base (meant for SiGe transistors)
- Early effects in the case of a graded bandgap (SiGe)
- Thermal noise, shot noise, and 1/f-noise
- Self-heating

You can use either of two parameters to specify the difference between the circuit temperature and the ambient temperatures in the MEXTRAM model:

- `DTEMP` instance parameter as specified in the element statement
- `DTA` (difference between the device temperature and the ambient analysis temperature) global model parameter.

`DTA` and `DTEMP` both default to zero. `DTEMP` overrides `DTA` locally, if you specify both. Simulation uses the value of `DTEMP` to de-rate the temperature in model equations and other parameters.

- If you do not specify either the `DTEMP` or the `DTA` parameter, then `DTEMP=0.0`.
- If you specify `DTA` but not `DTEMP`, then `DTEMP` uses the `DTA` value.
- If you specify `DTEMP`, then simulation uses the `DTEMP` value, and ignores the `DTA` value.

The following topics are covered:

- [Mextram 504 Update Releases](#)
- [Notes on Using MEXTRAM 503 or 504 Devices](#)
- [LEVEL=6 Model Parameters \(504\)](#)

Mextram 504 Update Releases

The following upgrades were made to the Mextram 504 model:

- Mextram 504, version 3 (504.3)
 - MULT has been moved in list of parameters
 - Lower clipping value of T_{ref} changed to $-273^{\circ}C$
 - Added I_C , I_B and β_{dc} to operating point information
- Mextram 504, version 4 (504.4): Noise of collector epilayer has been removed
- Mextram 504, version 5 (504.5)
 - Addition of temperature dependence of thermal resistance
 - Addition of noise due to avalanche current
- Mextram 504, version 6 (504.6)
 - Added parameter dA_{I_S} for fine tuning of temperature dependence of I_{ST}
 - "G_{EM} = 0" added to equation (4.66)
 - Upper clipping value 1.0 of K_{avl} introduced
- Mextram 504, version 7 (504.7)
 - Added resistances of buried layer R_{Cblx} and R_{Cbli} , and their temperature scaling parameter A_{Cbl} .
 - Lower clipping value of resistances R_E , R_{BC} , R_{BV} , R_{Cc} , R_{Cv} , SCR_{Cv} increased to $1m\Omega$
 - Bug fix high temperature limit B_{nT}
- Mextram 504, version 8 (504.8)
 - Zener tunneling current in emitter-base junction: Parameters: I_{zEB} and N_{zEB} .
 - Material constants, implemented as parameters: V_{gzEB} , A_{VgzEB} , and T_{VgzEB} .
- Mextram 504, version 10 (504.10) (F-2011-09-SP1)

- Collector-substrate model: Physics-based temperature scaling ideal collector-substrate current; Parameters: ICSS, ASUB
- Parasitic Base-Collector-Substrate (BCS) transistor model; Parameter: EXSUB
- Mextram 504, version 11 (504.11) (I-2013-12)
 - Extended the range of EXMOD.
 - If EXMOD=2, XIEX and XQEX do not have Ohmic asymptote.
- Mextram 504, version 12 (504.12) (M-2017-03-SP2)
 - Excess phase shift parameters, XQB and KE are added.
 - New parameter KC and FTAUN are added to improve high frequency correlated noise.
 - Avalanche noise implementation is improved.
- Mextram 504, version 13 (504.13) (P-2019.06-SP2)
 - Added a new parameter KAVLVER to provide backward compatibility of 504.12 avalanche noise with 504.11 avalanche noise.
 - Added a new parameter VEXLIM to provide upper limit of exp() function for convergence.

Notes on Using MEXTRAM 503 or 504 Devices

The following information applies to the PrimeSim Continuum device model for the MEXTRAM 503 or 504 device:

- Set LEVEL=6 to identify the model as a MEXTRAM bipolar junction transistor model.
- Set VERS parameter to 503 to use MEXTRAM 503 and to 504 to use MEXTRAM 504.
- All internal resistors are limited to greater than or equal to 1.0e-6.
- Reference temperature, TREF, is equal to 25 degrees.
- MEXTRAM does not contain extensive geometrical or process scaling rules (it has a multiplication factor to put transistors in parallel).
- MEXTRAM does not contain a substrate resistance.
- Constant overlap capacitances are not modelled within MEXTRAM.
- MEXTRAM 504 has better convergence than 503.
- MEXTRAM is more complex than Gummel-Poon (the computation time is longer and the convergence is less).

- No reverse emitter-base breakdown mechanism.
- Models the forward current of the parasitic PNP transistor.
- Output conductance dlc/dV_{ce} at the point where hard saturation starts seems to be too abrupt for high current levels, compared to measurements.
- Clarity of extrinsic current model describing X_{ie} and X_{isub} is improved by adding an extra node and an extra contact base resistance. In this case, parameter extraction would be more difficult.
- Starting in Release 2002.2:
 - Self-heating is now enabled for the MEXTRAM 504 model. You can use the `RTH` (thermal resistance) and `CTH` (thermal capacitance) model parameters, which had no effect in previous releases.
 - The `CBEO` capacitance parameter in MEXTRAM 504 models extrinsic B-E charge and capacitance effects. Also, the `CBCO` capacitance parameter models extrinsic B-C charge and capacitance effects.
 - The `SUBS` flag models the parasitic substrate effect when set to 1 (the default); `SUBS=0` does not model this effect. Both the MEXTRAM 503 and 504 models support the `SUBS` flag.
 - The MEXTRAM 504 model supports the PrimeSim Continuumsimulators specific area-scaling and multiplicity (`M` factor) features.
 - The MEXTRAM 503 model includes `KN` and `KFN` noise parameters.
- Starting in Release 2003.03, the Mextram BJT (`LEVEL=6`) model supports Philips modelkit 4.3.
- Starting in Release X-2005.09, support for avalanche current shot noise source parameters `Iavl_cc`, `Iavl_bb`, and `Iavl_bc` was added.

LEVEL=6 Model Parameters (504)

The following tables describe MEXTRAM 504 as `LEVEL=6` model parameters, including parameter names, units, default values, descriptions, and notes.

- `TAUNE` in MEXTRAM 503 acts as `TAUE` in the 504 model.
- Parameters noted with an asterisk (*) are not used in the DC model.

The following parameters used in MEXTRAM 503 are deleted in MEXTRAM 504:

• QBO	• ETA	• VI
• NA	• VLF	• EFI
• VGE	• ER	• AVL

The following parameters have been added to MEXTRAM 504:

• VEF	• TAUE	• AE	• VER
• TAUB	• DVGBF	• MLF	• TEPI
• DVGBR	• WAVL	• TAUR	• DVGTE
• VAVL	• DEG	• RTH	• AXI
• XREC	• CTH		

Table 65 BJT LEVEL=6 MEXTRAM 504 Flags

Name (Alias)	Unit	Default	Description
LEVEL	-	6	Model level
VERS	-	504	Flag to select between Mextram 503 and Mextram 504 models. Note: to set specific Mextram504 versions, such as 504.5, 504.6, 504.7, 504.8, refer to the INTVERS parameter
INTVERS	-	4.61	Flag for choosing specific Mextram 504 version Note: Supported versions include 4.3, 4.4, 4.5, 4.6, 4.61, 4.7 and 4.8.
EXMOD	-	1	Flag for extended modeling, reverse current gain
EXPHI	-	1	*Flag for distributed high frequency effects in transient
EXAVL	-	0	Flag for extended modeling of avalanche currents
TREF	°C	25.0	Reference temperature
SUBS	-	-	Flag for switching substrate effect

Table 66 BJT LEVEL=6 MEXTRAM 504 Basic Parameters

Name (Alias)	Unit	Default	Description
BF	-	215.0	Ideal forward current gain
BRI	-	7.0	Ideal reverse current gain
IBF	A	2.7e-15	Saturation current, non-ideal forward base current
IBR	A	1.0e-15	Saturation current, non-ideal reverse base current
IK	A	0.1	Collector-emitter high injection knee current
IS	A	2.2e-17	Collector-emitter saturation current

Table 66 BJT LEVEL=6 MEXTRAM 504 Basic Parameters (Continued)

Name (Alias)	Unit	Default	Description
MLF	V	2.0	Non-ideal factor of non-ideal forward base current
VEF		44.0	Forward early voltage
VER		2.5	Reverse early voltage
VEXLIM	-	400.0	Upper limit of exp() function argument for convergence.
VLR	V	0.2	Cross-over voltage, non-ideal reverse base current
XEXT	-	0.63	Part of Iex, Qex, Qtex, and Isub that depends on the base-collector voltage Vbc1
XIBI	-	0.0	Fraction of ideal base current for the sidewall

Table 67 BJT LEVEL=6 MEXTRAM 504 Avalanche Model Parameters

Name (Alias)	Unit	Default	Description
SFH	-	0.3	Current spreading factor of avalanche model (if EXAVL=1)
VAVL	V	3.0	Voltage, determines avalanche-current curvature
WAVL	m	1.1e-6	Epilayer thickness in weak-avalanche model

Table 68 BJT LEVEL=6 MEXTRAM 504 Base-Emitter Capacitances

Name (Alias)	Unit	Default	Description
CJE	F	7.3e-14	*Zero bias emitter-base depletion capacitance
PE	-	0.4	Emitter-base grading coefficient
VDE	V	0.95	Emitter-base diffusion voltage
XCJE	-	0.4	*Sidewall portion of emitter-base depletion capacitance

Table 69 BJT LEVEL=6 MEXTRAM 504 Base-Collector Capacitances

Name (Alias)	Unit	Default	Description
CJC	F	7.8e-14	*Zero bias collector-base depletion capacitance
MC	-	0.5	Coefficient for the current modulation of the collector-base depletion capacitance
PC	-	0.5	Collector-base grading coefficient
VDC	V	0.68	Collector-base diffusion voltage
XCJC	-	3.2e-2	*Fraction of the collector-base depletion capacitance under the emitter
XP	-	0.35	Constant part of CJC

Table 70 BJT LEVEL=6 MEXTRAM 504 Transit Time Parameters

Name (Alias)	Unit	Default	Description
DEG	EV	0.0	Bandgap difference over the base
MTAU	-	1.0	*Non-ideality of the emitter stored charge
TAUB	S	4.2e-12	*Transit time of stored base charge
TAUE	S	2.0e-12	*Minimum transit time of stored emitter charge
TAUR	S	5.2e-10	*Transit time, reverse extrinsic stored base charge
TEPI	S	4.1e-11	*Transit time of stored epilayer charge
XREC	-	0.0	Pre-factor of the recombination part of Ib1
XQB	-	1/3	Fraction of the total base charge supplied by the collector instead of the base (base charge partitioning)

Table 71 BJT LEVEL=6 MEXTRAM 504 Temperature Parameters

Name (Alias)	Unit	Default	Description
AB	-	1.0	Temperature coefficient of resistivity of base
AC	-	2.0	Temperature coefficient of resistivity, buried layer
AE	-	0.0	Temperature coefficient of emitter resistivity

Table 71 BJT LEVEL=6 MEXTRAM 504 Temperature Parameters (Continued)

Name (Alias)	Unit	Default	Description
AEPI	-	2.5	Temperature coefficient of resistivity of epilayer
AEX	-	0.62	Temperature coefficient of resistivity, extrinsic base
AQBO	-	0.3	Temperature coefficient, zero-bias base charge
ATH	-	0	Temperature coefficient of the thermal resistance
CVGBR	V	4.5e-2	Bandgap voltage difference, reverse current gain
DAIS	-	0	Parameter for fine tuning of temperature dependence for collector-emitter saturation current
DVGBF	V	5.0e-2	Bandgap voltage difference, forward current gain
DVGTE	V	0.05	*Bandgap voltage difference, emitter stored charge
VGB	V	1.17	Bandgap voltage of the base
VGC	V	1.18	Bandgap voltage of the collector
VGJ	V	1.15	Recombined bandgap voltage, emitter-base junction

Table 72 BJT LEVEL=6 MEXTRAM 504 Noise Parameters

Name (Alias)	Unit	Default	Description
AF	-	2.0	Exponent of the flicker-noise
KAVL	-	0	Switch for white noise contribution due to avalanche
KAVLVER	-	504.12	KAVL version, for setting KAVL behavior to 504.12 or 504.11.
KF	-	2.0e-11	Flicker-noise coefficient for ideal base current
KFN	-	2.0e-11	Flicker-noise coefficient, non-ideal base current
KC	-	0.0	Switch for RF correlation noise model selection
KE	-	0.0	Fraction of QE in excess phase shift
FTAUN	-	0.0	Fraction of noise transit time to total transit time

Table 73 *BJT LEVEL=6 MEXTRAM 504 Substrate Parameters*

Name (Alias)	Unit	Default	Description
CJS	F	3.15e-13	*Zero bias collector-substrate depletion capacitance
IKS	A	2.5e-4	Base-substrate high injection knee current
ISS	A	4.8e-17	Base-substrate saturation current
AS	-	1.58	For a closed buried layer: AS=AC For an open buried layer: AS=AEPI
PS	-	0.34	*Collector-substrate grading coefficient
VDS	V	0.62	*Collector-substrate diffusion voltage
VGS	V	1.2	Bandgap voltage of the substrate

Table 74 *BJT LEVEL=6 MEXTRAM 504 Self-Heating Parameters*

Name (Alias)	Unit	Default	Description
RTH	°C/W	0	Thermal (self-heating) resistance
CTH	J/°C	0	Thermal (self-heating) capacitance

Table 75 *BJT LEVEL=6 MEXTRAM 504 Extrinsic Capacitance Parameters*

Name (Alias)	Unit	Default	Description
CBEO	F	0	extrinsic Base-Emitter capacitance
CBCO	F	0	extrinsic Base-Collector capacitance

BJT LEVEL=6 MEXTRAM 504 DC OP Analysis Example

This example is located in the following directory: `$installdir/demo/hspice/bjt/mextram_dc.sp`

BJT LEVEL=6 MEXTRAM 504 Transient Analysis Example

This example is located in the following directory: `$installdir/demo/hspice/bjt/mextram_tran.sp`

BJT LEVEL=6 MEXTRAM 504 AC Analysis Example

This example is located in the following directory: `$installdir/demo/hspice/bjt/mextram_ac.sp`

Table 76 BJT LEVEL=6 MEXTRAM504 Noise Parameters

Name	Unit	Default	Description
ABT	-	0	Temperature coefficient of the base-emitter tunneling current parameter
AET	-	0	Temperature coefficient for the base-emitter tunneling current parameter
ALB	V	0	Exponential coefficient for the base-emitter tunneling current
BTJE	AV^{-2}	0	Pre-factor of the base-emitter tunneling current

Table 77 BJT LEVEL=6 MEXTRAM 504 Output Templates

Name	Alias	Description
VB2E1	LX26	Internal base-emitter bias
VB2C2	LX27	Internal base-collector bias
VB2C1	LX28	Internal base-collector bias including epilayer
VB1 C1	LX29	External base-collector bias without contact resistances
VE1E	LX30	Bias over emitter resistance
IN	LX31	Main current
IC1C2	LX32	Epilayer current
IB1B2	LX33	Pinched base current
IB1	LX34	Ideal forward base current
ISB1	LX35	Ideal side-wall base current
IB2	LX36	Non-ideal forward base current
IB3	LX37	Non-ideal reverse base current
IBET	LX38	Base-emitter tunneling current
IAVL	LX39	Avalanche current

Table 77 BJT LEVEL=6 MEXTRAM 504 Output Templates (Continued)

Name	Alias	Description
IEX	LX40	Extrinsic reverse base current
XIEX	LX41	Extrinsic reverse base current
XISUB	LX42	Substrate current
ISF	LX43	Substrate failure current
IRE	LX44	Current through emitter resistance
IRBC	LX45	Current through constant base resistance
IRCC	LX46	Current through constant collector resistance
QE	LX47	Emitter charge or emitter neutral charge
QTE	LX48	Base-emitter depletion charge
QSTE	LX49	Sidewall base-emitter depletion charge
QTC	LX50	Base-collector depletion charge
QEPI	LX51	Epilayer diffusion charge
QB1B2	LX52	AC current crowding charge
QTEX	LX53	Extrinsic base-collector depletion charge
XQTEX	LX54	Extrinsic base-collector depletion charge
QEX	LX55	Extrinsic base-collector diffusion charge
XQEX	LX56	Extrinsic base-collector diffusion charge
QTS	LX57	Collector-substrate depletion charge
GX	LX58	Forward transconductance
GY	LX59	Reverse transconductance
GZ	LX60	Reverse transconductance
GSPI	LX61	Conductance side-wall base-emitter junction
GPIX	LX62	Conductance floor base-emitter junction
GPIY	LX63	Early effect on recombination base current
GPIZ	LX64	Early effect on recombination base current

Table 77 BJT LEVEL=6 MEXTRAM 504 Output Templates (Continued)

Name	Alias	Description
GMUX	LX65	Early effect on avalanche current limiting
GMUY	LX66	Conductance of avalanche current
GUMZ	LX67	Conductance of avalanche current
GMUEX	LX68	Conductance of extrinsic base-collector junction
XGMUEX	LX69	Conductance of extrinsic base-collector junction
GRCVY	LX70	Conductance of epilayer current
GRCVZ	LX71	Conductance of epilayer current
RBV	LX72	Base resistance
GRBVX	LX73	Early effect on base resistance
GRBVY	LX74	Early effect on base resistance
GRBVZ	LX75	Early effect on base resistance
RE	LX76	Emitter resistance
RBC	LX77	Constant base resistance
RCC	LX78	Constant collector resistance
GS	LX79	Conductance parasitic PNP transistor
XGS	LX80	Conductance parasitic PNP transistor
GSF	LX81	Conductance substrate failure current
CSBE	LX82	Capacitance sidewall base-emitter junction
CBEX	LX83	Capacitance floor base-emitter junction
CBEY	LX84	Early effect on base-emitter diffusion charge
CBEZ	LX85	Early effect on base-emitter diffusion charge
CBCX	LX86	Early effect on base-collector diffusion charge
CBCY	LX87	Capacitance floor base-collector junction
CBCZ	LX88	Capacitance floor base-collector junction
CBCEX	LX89	Capacitance extrinsic base-collector junction

Table 77 BJT LEVEL=6 MEXTRAM 504 Output Templates (Continued)

Name	Alias	Description
XCBCEX	LX90	Capacitance extrinsic base-collector junction
CB1B2	LX91	Capacitance AC current crowding
CB1B2X	LX92	Cross-capacitance AC current crowding
CB1B2Y	LX93	Cross-capacitance AC current crowding
CB1B2Z	LX94	Cross-capacitance AC current crowding
CTS	LX95	Capacitance substrate-collector junction
GOUT	LX96	Output conductance
GMU	LX97	Feedback transconductance
CBE	LX98	Base-emitter capacitance
CBC	LX99	Base-collector capacitance
IQS	LX100	Current at onset of quasi-saturation
XIWEPI	LX101	Thickness of injection layer
VB2C2STA	LX102	Physical value of internal base-collector bias
PDISS	LX103	Dissipation
TK	LX104	Actual temperature
ISUBO	LX105	Substrate current
QBEI	LX106	Base-emitter diffusion charge
QBCI	LX107	Base- collector diffusion charge

MEXTRAM Level 505

Level 505 of the MEXTRAM model is available as BJT LEVEL=6, as is Level 503 and 504 of MEXTRAM. See [Mextram 505 Update Releases](#) for a brief description of the version 505 updates.

Mextram 505 Update Releases

Following are the upgrades to the Mextram 505 model:

- Mextram 505, Version 0 (505.0)
 - A CB junction tunneling current model is added.
 - Non-ideality factor NFF and NFR in forward and reverse transport current I_N , respectively, are added.
 - Diffusion charge and diffusion capacitance expressions are modified to maintain the same transit time.
 - Non-ideality factors NBI and NBIS in ideal base current I_{B1} and I_{B1S} , respectively, are added.
 - I_{B2S} , side-wall non-ideal base current, is added.
 - I_{BREL} , side-wall base current for reliability modeling, is added.
 - All base components have their own saturation current and non-ideality factors where needed. Current gains are no longer used.
 - Non-ideal reverse base current is now formulated the same way as forward non-ideal base current.
 - $1/f$ noise of all ideal base current is now calculated from K_f and A_f .
 - Avalanche current I_{AVL} is calculating in I_N as initiating current and I_{AVL} limits are modulated accordingly.
 - A new avalanche factor (GEM) model is added and used as default.
 - SW_{AVL} , a switch parameter for avalanche factor, is added.
 - V_{DCCTCT} , a diffusion voltage dedicated for CB depletion capacitance, is added.
 - SW_{VJUNC} , a switch for V_{junc} calculation, is added.
 - SW_{VCHC} , a switch for transition voltage width V_{ch} in CB capacitance-voltage curve smoothing, is added.
 - LEX is now corrected to described extrinsic BC junction current as hole injection into collector.
 - I_{KS} means true substrate current's knee.
 - Default value of $EXSUB$ is 1.
 - Range if $ICSS$ is changed from $(-\infty, \infty)$ to $[0.0, \infty]$.
- Mextram 505, Version 1 (505.1)

- An intrinsic substrate current ISUB_INT is introduced. ISUB and XISUB are modified accordingly. New parameter: XISUBI.
- Avalanche breakdown is added to all CB junction leakage currents. New Parameters are: SWJBRCB, VBRCB, PBRCB, and FREVCB.
- High injection effect is added to CS junction diode current. New parameter: IKCS.
- VSC induced high injection effect is added to main currents of BCS transistors. New parameter: SWVSCH.
- Upper limit of exp() function argument for convergence is added. New parameter: VEXLIM.
- Mextram 505, Version 2 (505.2)
 - New avalanche factor (Gem) model
 - A new avalanche model when swavl = 3
 - Exavl is meaningful only when swavl=2 and 3
 - New parameter : swgemlim for turning switch on and off Gem limiting
 - swgemlim = 0 turns off Gem limiting
 - swgemlim = 1 (default) turns on Gem limiting
 - New extrinsic diffusion charge model
 - swqex = 0 (default) uses original Qex model
 - swqex = 1 uses new Qex model, new parameters are vdcex and tauex
 - A new exmod = 3 option removes Fex limiting (Fex = 1.0) in both Qex and XQex
 - A new model parameter ISSR describes asymmetry of saturation current IS between forward and reverse mode
 - New model parameter DTEMP and TRISE are added as alias to DTA
 - Simulator GMIN is used and model parameter GMIN is obsoleted
 - CS junction gmin current is added when icss > 0.0
 - EB tunneling current calculation is improved
 - Izteb is set to zero for small VEB. Parameter vzmin is added to specify the minimum junction reverse voltage
 - MULT is now implemented at the branch level instead of parameter level

- Mextram 505, Version 3 (505.3)
 - A new NBR base current model
 - `swib1 = 0` (default) uses `Ib1` and `Ib1s` model in 505.2 and earlier versions.
 - `swib1 = 1` uses new `Ib1` and `Ib1s` model, new parameters are `ibinbr`, `ibinbrs`, `vkibr` and `ibinbrqs`, new temperature parameters are `vgnbrqs`, `vgnbr`, `vgnbrs` and `vgknbr`.
 - A limitation is added on `Ibinbrqs` to avoid large quasi-saturation component on reverse operation and breakdown.
 - CB breakdown current is implemented between B2 and C1 when `swib1 = 1`, as well as power dissipation and noise.
 - A minimum resistance `minr` is added.
 - Unphysical noise from GMIN currents in `Ib2` and `Ib3` are removed. The corresponding `opinfo` has been changed.
 - Limits for `p0star` and `pw` are added, new parameters are `p0starlim` and `pwlim`.
 - EB tunneling current calculation is improved with a new approach (from `nxp`) to improve stability and convergence at very low reverse bias. Parameter `vzmin` is kept for backward compatible, but it is no longer used. A small number `1.0e-30` is added to limit the value of V_{b2e1}/v_{de_t} when calculating `dE0EB`. Using Taylor approximation for small `dzeb`.
 - CB tunneling current calculation is improved in a similar manner.
- Mextram 505, Version 4 (505.4)
 - Parameter `nex` for reverse extrinsic stored epilayer charge when `swqex = 1`
 - Parameter `dtmax` for maximum temperature rise by self-heating
 - Variable base resistance `Rb2` is applied MINR limiting
 - New non-linear self-heating model
 - `swnlsh=0` (default) : normal self-heating
 - `swnlsh=1` : nonlinear self-heating turned on

LEVEL=6 Model Parameters (505)

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters

Name (Alias)	Unit	Default	Description
LEVEL	-	6	Model level.
VERS	-	504	Flag to select between Mextram 503, Mextram 504, or Mextram 505 models.
INTVERS	-	4.7	Flag for choosing specific Mextram 505 version.
TREF	°C	25	Reference temperature.
DTA	°C	0	Difference between local ambient and global ambient temperature.
MULT	-	1.0	Multiplication factor.
EXMOD	-	1	Flag for extended modeling of the reverse current gain.
EXPHI	-	1	Flag for distributed high-frequency effect in transient.
EXAVL	-	0	Flag for extended modeling of avalanche currents.
EXDUB	-	1	Flag for extended modeling of substrate currents.
IS	A	22.0e-18	CE saturation current.
NFF	-	1.0	Non-ideality factor of forward main current.
NFR	-	1.0	Non-ideality factor of reverse main current.
IK	A	0.1	CE high injection knee current.
VER	V	2.5	Reverse early voltage.
VEF	V	44.0	Forward early voltage.
IBI	A	0.1e-18	Saturation current of ideal base current.
NBI	-	1.0	Non-ideality factor of ideal base current.
IBIS	A	0.0	Saturation current of ideal side wall base current.
NBIS	-	1.0	Non-ideality factor of ideal side wall base current.
IBF	A	2.70e-15	Saturation current of non-ideal forward base current.
MLF	-	2.0	Non-ideality factor of non-ideal forward base current.

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
IBFS	A	0.0	Saturation current of non-ideal side wall forward base current.
MLFS	-	2.0	Non-ideality factor of non-ideal side wall forward base current.
IBX	A	3.14e-18	Saturation current of extrinsic reverse bias current.
IKBX	A	14.29e-03	Extrinsic CE high injection knee current.
IBR	A	1.00e-15	Saturation current of non-ideal reverse base current.
MLR	V	2.0	Non-ideality factor of non-ideal reverse base current.
XEXT	-	0.63	Part of I_{ex} , Q_{tex} , Q_{ex} , and I_{sub} that depends on VBC3 instead of VB1C4.
IZEB	A	0.0	Pre-factor of EB Zener tunneling current.
NZEB	-	22.0	Coefficient of EB Zener tunneling current.
IZCB	A	1	Pre-factor of CB Zener tunneling current.
NZCB	-	50.0	Coefficient of CB Zener tunneling current.
SWAVL	-	1	Switch of avalanche factor GEM model.
AAVL	-	50.0	Impact ionization A of GEM when SWAVL=1.
CAVL	-	-0.5	Exponent in GEM model when SWAVL=1.
ITOAVL	-	500e-03	Current temperature parameter of avalanche when SWAVL=1.
BAVL	-	10.0	Impact ionization B of GEM when SWAVL=1.
VDCAVL	-	0.68	CB diffusion voltage dedicated for GEM model when SWAVL=1.
WAVL	M	1.10e-06	Epilayer thickness used in weak-avalanche model.
VAVL	V	3.0	Voltage determining curvature of avalanche current.
SFH	-	0.3	Current spreading factor of avalanche model (when EXAVL=1).
RE	Ω	5.0	Emitter resistance.
RBC	Ω	23.0	Constant part of base resistance.

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
RBV	Ω	18.0	Zero-bias value of variable part of the base resistance.
RCC	Ω	12.0	Constant part of collector resistance.
RCBLX	Ω	0.0	Resistance of Collector Buried Layer: Extrinsic part.
RCBLI	Ω	0.0	Resistance of Collector Buried Layer: Intrinsic part.
RCV	Ω	150.0	Resistance of un-modulated epilayer.
SCRCV	Ω	1250.0	Space charge resistance of epilayer.
IHC	A	4.00e-03	Critical
AXI	-	0.3	Smoothness parameter for onset of quasi-saturation.
VDC	V	0.68	CB diffusion voltage.
CJE	F	73.0e-15	Zero-bias EB depletion capacitance.
VDE	V	0.95	EB diffusion voltage of depletion capacitance.
PE	-	0.4	EB grading coefficient.
XCJE	-	0.4	Sidewall fraction of the EB depletion capacitance.
CBEO	F	0.0	EB overlap capacitance.
CJC	F	78.0e-15	Zero-bias CB depletion capacitance.
VDCCTC	V	0.68	CB diffusion voltage of depletion capacitance.
PC	-	0.5	CB grading coefficient.
SWVCHC	-	0	Switch of Vch for CB depletion capacitance.
SWVJUNC	-	0	Switch of Vjunc for CB depletion capacitance.
XP	-	0.35	Constant part of CJC.
MC	-	0.5	Coefficient for current modulation of CB depletion capacitance.
XCJC	-	32.0e-03	Fraction of CB depletion capacitance under the emitter.
CBCO	F	0.0	CB overlap capacitance.
MTAU	-	1.0	Non-ideality factor of emitter stored charge.

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
TAUE	s	2.00e-12	Minimum transit time of stored emitter charge.
TAUB	s	4.20e-12	Transit time of stored base charge.
TEPI	s	41.0e-12	Transit time of stored epilayer charge.
TAUR	s	520.0e-12	Transit time of reverse extrinsic stored base charge.
DEG	eV	0.0	Bandgap difference over the base.
XREC	-	0.0	Pre-factor of recombination part of IB1.
XQB	-	1/3	Emitter-fraction of base diffusion charge.
KE	-	0.0	Fraction of QE in excess phase shift.
AQBO	-	0.3	Temperature coefficient of zero-bias base charge.
AE	-	0.0	Temperature coefficient of resistivity of emitter.
AB	-	1.0	Temperature coefficient of resistivity of base.
AEPI	-	2.5	Temperature coefficient of resistivity of epilayer.
AEX	-	0.62	Temperature coefficient of resistivity of extrinsic base.
AC	-	2.0	Temperature coefficient of resistivity of collector contact.
ACX	-	1.3	Temperature coefficient of extrinsic reverse base current.
ACBL	-	2.0	Temperature coefficient of resistivity of collector buried layer.
VGB	V	1.17	Band-gap voltage of base.
VGC	V	1.18	Band-gap voltage of collector.
VGE	V	1.12	Band-gap voltage of emitter.
VG CX	V	1.125	Band-gap voltage of extrinsic collector.
VGJ	V	1.15	Band-gap voltage of recombination EB junction.
VGZEB	V	1.15	Band-gap voltage at Tref for EB tunneling.
AVGEB	V/K	4.73e-04	Temperature coefficient of band-gap voltage for EB tunneling.

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
TVGEB	K	636.0	Temperature coefficient of band-gap voltage for EB tunneling.
VGZCB	V	1.15	Band-gap voltage at Tref for CB tunneling.
AVGCB	V/K	4.73e-04	Temperature coefficient of band-gap voltage for CB tunneling.
TVGCB	K	636.0	Temperature coefficient of band-gap voltage for CB tunneling.
DVGTE	V	0.05	Band-gap voltage difference of emitter stored charge.
DAIS	-	0.0	Fine tuning of temperature dependence of C-E saturation current.
TNFF	/K	0.0	Temperature coefficient of NFF.
TNFR	/K	0.0	Temperature coefficient of NFR.
TBAVL	-	400e-06	Temperature scaling parameter of BAVL when SWAVL=1.
AF	-	2.0	Exponent of Flicker-noise of ideal base current.
AFN	-	2.0	Exponent of Flicker-noise of non-ideal base current.
KF	-	20.0e-12	Flicker-noise coefficient of ideal base current.
KFN	-	20.0e12	Flicker-noise coefficient of non-ideal base current.
KAVL	-	0.0	Switch for white noise contribution due to avalanche.
KC	-	0.0	Switch for RF correlation noise model selection.
FTAUN	-	0.0	Fraction of noise transit time to total transit time.
ISS	A	48.0e-18	Saturation current of parasitic BCS transistor main current.
ICSS	A	0.0	CS junction ideal saturation current.
IKS	A	545.5e-06	Knee current for BCS transistor main current.
CJS	F	315.0e-15	Zero-bias CS depletion capacitance.
VDS	V	0.62	CS diffusion voltage.
PS	-	0.34	CS grading coefficient.

Table 78 BJT LEVEL=6 MEXTRAM 505 Model Parameters (Continued)

Name (Alias)	Unit	Default	Description
VGS	V	1.2	Band-gap voltage of the substrate.
AS	-	1.58	For a closed buried layer: AS=AC, and for an open buried layer: AS=AEPI.
ASUB	-	2.0	Temperature coefficient for mobility of minorities in the substrate.
RTH	°C/W	0.0	Thermal resistance.
CTH	J/°C	0.0	Thermal capacitance.
ATH	-	0.0	Temperature coefficient of thermal resistance.
ISIBREL	A	0.0	Saturation current of base current for reliability simulation.
NFIBREL	-	2.0	Non-ideality factor of base current for reliability simulation.
GMIN	-	1.00e-13	Minimum conductance.
XISUBI	-	0	Part of substrate current that belongs to the intrinsic region.
SWJBRCB	-	0	Switch for breakdown in CB junction leakage.
VBRCB	V	100	Breakdown voltage for CB junction leakage.
PBRCB	V	4	Breakdown onset tuning parameter for CB junction leakage.
FREVCB	-	1000	Coefficient for limiting CB junction breakdown leakage current.
IKCS	A	5.00E-05	Knee current for CS junction diode current.
SWVSCH	-	0	Switch for VSC induced high injection in main currents of BCS transistors and CS diode current.
VEXLIM	-	400	Upper limit of exp() function argument for convergence.

LEVEL=8 HICUM Model

The High Current Model (HICUM) is an advanced transistor model for bipolar transistors with a primary emphasis on circuit design for high-speed/high-frequency applications. HICUM development was spurred by the SPICE Gummel-Poon model's (SGPM)

inadequate level of accuracy for high-speed, large-signal transient applications and the required high-collector current densities. Other major disadvantages of the SGPM are:

- A lack of sufficient physical background
- Poor descriptions of base resistance and junction capacitances in the regions of interest
- Inadequate description of both Si- and III-V material-based HBTs.

The HICUM model is implemented as `LEVEL=8` in the BJT models and currently the PrimeSim Continuum solution supports HICUM version 2.4. The new parameters for version 2.32, 2.31 and 2.3 are listed in [Table 104](#). The parameters for version 2.2 include those of version 2.1 are in [Table 80](#).

These topics are covered in the following sections:

- [HICUM Model Advantages](#)
- [PrimeSim Continuum HICUM Model vs. Public HICUM Model](#)
- [HICUM/L2 Version 3.1 Updates](#)
- [HICUM/L2 Version 3.0 Updates](#)
- [HICUM/L2 Version 2.4 Updates](#)
- [HICUM/L2 Version 2.34 Updates](#)
- [HICUM/L2 Version 2.33 Updates](#)
- [LEVEL=8 Element Syntax](#)
- [LEVEL=8 Model Parameters](#)

HICUM Model Advantages

Major features of HICUM are:

- Accurate description of the high-current operating region (including quasi-saturation and saturation).
- Distributed modeling of external base-collector region.
- Proper handling of emitter periphery injection and charge storage.
- Internal base resistance as a function of operating point (conductivity modulation and emitter current crowding), and emitter geometry.
- Sufficiently physical model equations allowing predictions of temperature and process variations, as well as scalability, even at high current densities.

- Parasitic capacitances, independent on operating point, are available in the equivalent circuit, representing base-emitter and base-collector oxide overlaps, that become significant for small-size transistors.
- Weak avalanche breakdown is available.
- Self-heating effects are included. Non-quasi-static effects, resulting in a delay of collector current AND stored minority charge, are modelled as function of bias.
- Collector current spreading is included in minority charge and collector current formulation.
- Extensions for graded-base SiGe HBTs have been derived using the Generalized Integral Charge-Control Relation (GICCR); the GICCR also permits modelling of HBTs with (graded) bandgap differences within the junctions.
- Base-emitter tunneling model is available (for example, for simulation of varactor leakage).
- Simple parasitic substrate transistor is included in the equivalent circuit.
- Simple parallel RC network taking into account the frequency dependent coupling between buried layer and substrate terminal.
- Parameter extraction is closely related to the process enabling parametric yield simulation; parameter extraction procedure and list of test structures are available; HICUM parameters can be determined using standard measurement equipment and mostly simple, decoupled extraction procedures.
- Simple equivalent circuit and numerical formulation of model equations result in easy implementation and relatively fast execution time.
- Vertical NQS effects have been implemented in HICUM through “additional delay times” for both minority charge and forward transfer current.
- The correlation between base and collector noise is included in the HICUM compact model.

If you use these features with easily-measurable basic variables (such as junction capacitances and transit time), the results are more accurate than if you use SGPM. This improved accuracy applies to digital circuit, small-signal high-frequency, and especially high-speed large-signal transient simulation. Also, you can laterally scale HICUM over a wide range of emitter widths and lengths, up to high collector current densities. The scaling algorithm is generic, and has been applied to the SGPM (within its validity limits).

In summary, HICUM's major advantages over other bipolar compact models are:

- Scalability
- Process-based and relatively simple parameter extraction

- Predictive capability in terms of process and layout variations
- Fairly simple numerical formulation facilitating easy implementation and resulting in still reasonable simulation time compared to the (too) simple SGPM at high current densities

PrimeSim Continuum HICUM Model vs. Public HICUM Model

The PrimeSim Continuum solution LEVEL=8 model is based on version 2.2 of the public HICUM model. There are two versions: version 2.1 and 2.2.

- Version 2.1, based on the Technology University of Dresden's HICUM standalone model kit source code
- Version 2.2, based on the Technology University of Dresden's HICUM Verilog-A code (current version 2.24)

To maintain flexibility, the LEVEL=8 HICUM model uses `IS`, `MCF`, and `ZETACX` as additional model parameters. For more information on these parameters, see [Table 103](#).

For a complete description of HICUM model, see http://www.iee.et.tu-dresden.de/iee/eb/hic_new/hic_doc.html

HICUM/L2 Version 3.1 Updates

The following changes are made with the release of v3.1:

- Changed the ICK smoothing function for hard saturation modeling.
- Defined `ajci`, `ajcx`, `ajs` as model parameters.
- Base current model extension: added `TAT` current, extended `BtBT` current evaluation to `VBE=0.7V`.
- Implemented backward compatibility (use `flcomp=300`).
- Fixed the bug for `hf0` inconsistency.
- Defined the physical parameters `q`, `kB` in model code.
- Added the GMIN element between `c'` and `s'`.
- Added the model/simulator parameter `minr` to separate `ddt()` operation for `rbx`, `rbi`, `rcx`, `re`, `rsu`, and `rth`.
- Removed the conditional statement for the thermal update without self-heating.
- Made some code efficiency improvements.

HICUM/L2 Version 3.0 Updates

For detailed update, refer to HiCUM L2 manual released by Compact Model Coalition (CMC).

HICUM/L2 Version 2.4 Updates

The following changes are made with the release of v2.4:

- Addition of model parameters KAVL and ALKAV for avalanche computing.
 - Modification to the value range for model parameters AHC and IBETS.
-

HICUM/L2 Version 2.34 Updates

The following changes are made with the release of v2.34:

- Change range for grading factors of capacitances (excluding 1).
 - Change conditional statement for noise correlation evaluation.
 - Add parameter AICK for smoothing the ICK curve.
 - Implement the peripheral substrate capacitance.
 - Add a term to avoid numerical issues in the QJCMOD.
 - Correct output resistance calculation (OP only).
-

HICUM/L2 Version 2.33 Updates

The following changes are made with the release of version 2.33:

- Addition of TK and DTSH to the operating point values.
 - Changes to ranges of ALIT and ALQF back to including 0.
 - Bug fixes for versions 2.32 and 2.31.
-

LEVEL=8 Element Syntax

This section provides the syntax for BJT `LEVEL=8`, and an example of an input netlist and output format.

Syntax

```
Qxxx nc nb ne ns> <nt>mnamearea [M=val] [DTEMP=val]  
+ [tnodeout]
```

Parameter	Description
Qxxx	BJT element name
nc	Collector terminal node
nb	Base terminal node, connected to 1 => 2
ne	Emitter terminal node, connected to 1 => 0
ns	Substrate terminal node
nt	Temperature node
mname	BJT model name reference
area	Emitter area multiplying factor. Affects current, resistance, capacitance. The default is 1.
M	Multiplier to simulate multiple BJTs in parallel. The default is 1.
DTEMP	Difference between the element temperature and the circuit temperature in degrees Celsius. The default is 0.0.
tnodeout	Identify self heating node from substrate node.

Example

The following is an example of a BJT Q1 model:

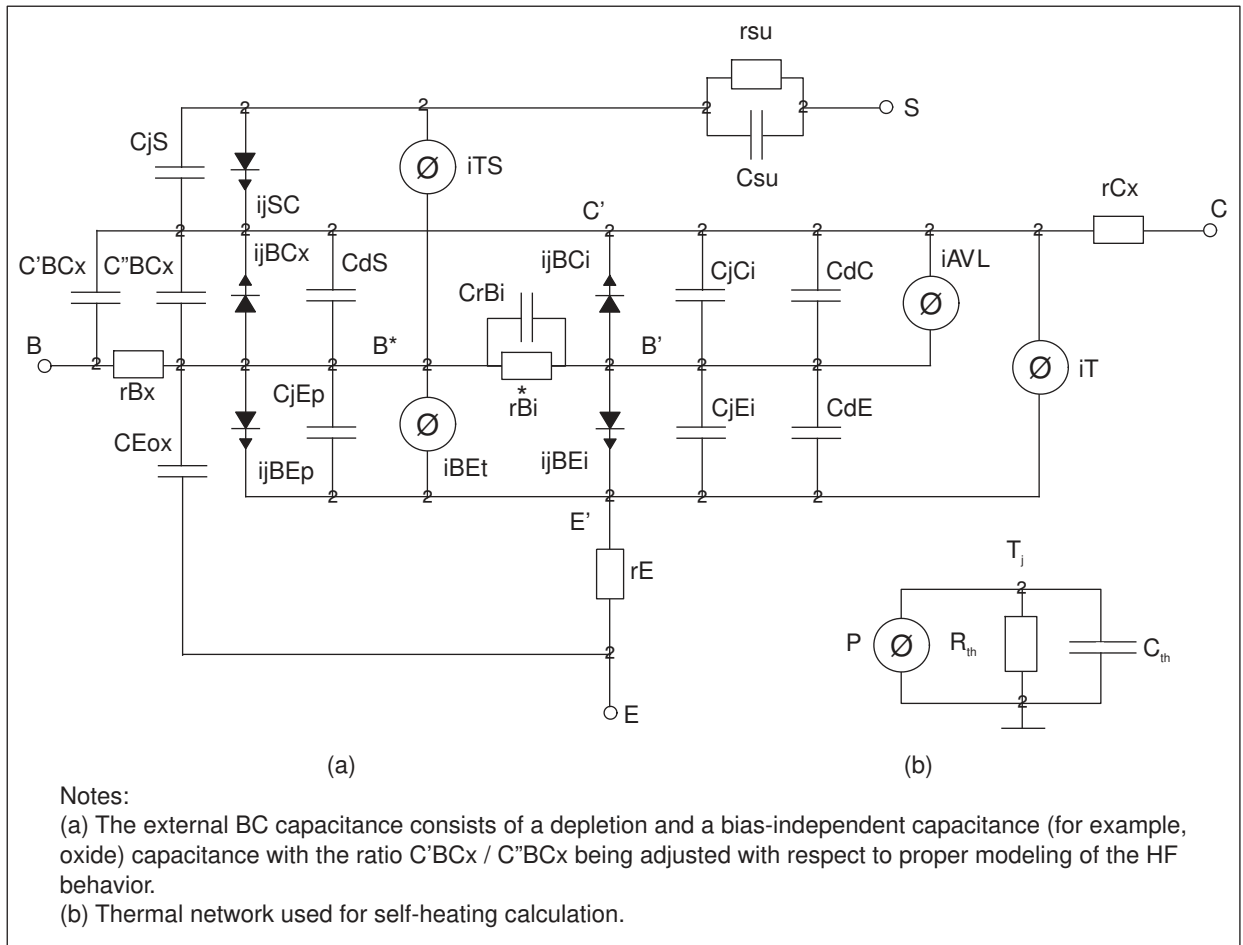
```
Q1 1 2 0 4 QM area=1*0.5*5 dtemp=0.002
```

The preceding example includes the following connections:

- Collector is connected to node 1.
- Base is connected to node 2.
- Emitter is connected to node 0.
- Substrate is connected to node 4.
- QM references the name of the BJT model.

HICUM LEVEL=2 Circuit Diagram

Figure 36 Large-signal HICUM LEVEL=2 Equivalent Circuit



Input Netlist

```
.DATA test_data vbe vce vsub
0.0 0.0 0.0
0.1 0.0 0.0
0.2 0.0 0.0
0.3 0.0 0.0
0.4 0.0 0.0
0.5 0.0 0.0
0.6 0.0 0.0
0.7 0.0 0.0
0.8 0.0 0.0
0.9 0.0 0.0
1.0 0.0 0.0
```

```
.ENDDATA

.OPTION
.TEMP 26.85
VIN 2 0 vbe
VC 1 0 vce
VS 4 0 vsub
VE 3 0 0
Q1 1 2 3 4 hicum
.DC data=test_data
.PRINT DC I(VIN) i2(q1) I(VC) i1(q1) I(VCS) i4(q1)
.MODEL hicum NPN Level=8
+ tref =26.85
+ c10=.3760000E-31 qp0=.2780000E-13 ich=.2090000E-01
+ hfc=.1000000E+01
+ hfe=1.0000000E+00 hjei=.0000000E+00
+ hjci=.1000000E+01 tr=1.000000000E-9
+ cjei0=.81100E-14 vdei=.9500000E+00 zei=.5000000E+00
+ aljei=.18000E+01
+ cjci0=.11600E-14 vdc1=.8000000E+00 zci=.3330000E+00
+ vptci=.41600E+03
+ rci0=.127800E+03 vlim=.700000E+00 vpt=.5000000E+01
+ vces=.100000E+00
+ t0=.47500000E-11 dt0h=.210000E-11 tbvl=.400000E-11
+ tef0=.180000E-11 gtfe=.140000E+01 thcs=.300000E-10
+ alhc=.750000E+00
+ fthc=.600000E+00
+ latb=.376500E+01 latl=.342000E+00 fqi=.9055000E+00
+ alit=.450000E+00 alqf=.225000E+00
+ favl=.118600E+01 qavl=.111000E-13 alfav=.82500E-04
+ alqav=.19600E-03
+ ibeis=.11600E-19 mbei=.101500E+01 ibeps=.10000E-29
+ mbep=.200000E+01
+ ireis=.11600E-15 mre1=.200000E+01 ireps=.10000E-29
+ mrep=.200000E+01
+ rbi0=.000000E+00 fdqr0=.000000E+00 fgeo=.730000E+00
+ fcrbi=.000000E+00
+ cjep0=.00000E+00 vdep=.105000E+01 zep=.4000000E+00
+ aljep=.24000E+01
+ ceox=.000000E+00
+ cjc0=.00000E+00 vdcx=.700000E+00 zcx=.3330000E+00
+ vptcx=.10000E+03
+ ccox=.000000E+00 fbc=.1526000E+00
+ ibcx=.10000E-29 mbcx=.200000E+01 ibcis=.11600E-19
+ mbci=.101500E+01
+ cjs0=.000000E+00 vds=.6000000E+00 zs=.44700000E+00
+ vpts=.100000E+04
+ rcx=.00000000E+00 rbx=.00000000E+00 re=.00000000E+00
+ kf=.00000000E+00 af=.00000000E+00
+ vgb=.1170000E+01 alb=.6300000E-02 alt0=.000000E+00
+ kt0=.0000000E+00
+ zetaci=.1600E+01 alvs=.100000E-02 alces=.40000E-03
+ zetarbi=0.5880E+00 zetarcx=0.2230E+00
```

```
+ zetarbx=0.2060E+00          zetare=0.0000E+00
+ rth=0.0 cth=0.0
+ ibets=.000000E+00 abet=.000000E+00
+ itss=.000000E+00 msf=.000000E+00 tsf=0.000000E+00
+ iscs=.000000E+00
+ msc=.000000E+00
+ rsu=.000000E+00 csu=.000000E+00
.END
```

LEVEL=8 Model Parameters

This section lists the HICUM `LEVEL=8` model parameters, internal transistor parameters, peripheral element parameters, and external element parameters for version 2.1. The parameters for version 2.2 include those of version 2.1 and new parameters listed in [Table 80](#). This includes parameter names, descriptions, units, default values, factors, and notes.

Table 79 BJT LEVEL=8 Model Parameters

Name (Alias)	Unit	Default	Description
LEVEL	-	8	HICUM BJT level
TREF	C	26.85	Temperature in simulation
SYNOPSYS_UPDATE	-	1	Flag for bug fixing. • 1: Bug fixed by Synopsys • 0: No bug fix
CDCI_UPDATE		1	Flag for CDCI equation; This flag can be used for SYNOPSYS_UPDATE = 1 • 1=Latest equation • 0=Old equation
VERSION	-	2.24	HICUM version

Table 80 BJT LEVEL=8 Parameters Added in Version 2.2

Name	Unit	Default	Factor	Description
CBEPAR	-	1.0		Total parasitic BE capacitance (spacer and metal)
F1VG	V/K	1.02377e-4		Coefficient K1 in T dependent bandgap equation
F2VG	V/K	4.3215e-4		Coefficient K2 in T dependent bandgap equation
FBEPAR	-	0		Partitioning factor of parasitic BE capacitance

Table 80 BJT LEVEL=8 Parameters Added in Version 2.2 (Continued)

Name	Unit	Default	Factor	Description
FLSH	-	0	M	Self-heating flag, also has instance parameter. If FLSH is not 0 and RTH>0, ON; otherwise OFF.
TBHREC	ps	0		BC recombination time constant at the BC barrier for high forward injection
TUNODE	-	1		Tunneling flag (1=on peripheral node; 0=on internal node)
VGC	V	1.17		Effective collector bandgap voltage V_gC_{eff}
VGE	V	1.17		Effective emitter bandgap voltage V_gE_{eff}
VGS	V	1.17		Effective substrate bandgap voltage V_gS_{eff}
ZETABET	-	5		Exponent coefficient in BE junction current temperature dependence
ZETACT	-	4.5		Exponent coefficient in transfer current temperature dependence

The default parameter values for HICUM version 2.2 are as follows:

```

****
.DATA test_data vbe vce vsub
0.0 0.0 0.0
0.1 0.0 0.0
0.2 0.0 0.0
0.3 0.0 0.0
0.4 0.0 0.0
0.5 0.0 0.0
0.6 0.0 0.0
0.7 0.0 0.0
0.8 0.0 0.0
0.9 0.0 0.0
1.0 0.0 0.0
.ENDDATA
.OPTION post
.TEMP 26.85
VIN 2 0 vbe
VC 1 0 vce
VS 4 0 vsub
VE 3 0 0
Q1 1 2 3 4 hicum
.DC data= test_data
.PRINT DC I(VIN) i2(q1) I(VC) i1(q1) I(VCS) i4(q1)
.MODEL hicum NPN Level=8
+ tref = 26.85

```

```
+ c10=.3760000E-31 qp0=.2780000E-13 ich=.2090000E-01
+ hfc=.1000000E+01
+ hfe=1.0000000E+00 hjei=.0000000E+00
+ hjci=.1000000E+01 tr=1.00000000E-9
+ cjei0=.81100E-14 vdei=.950000E+00 zei=.5000000E+00
+ aljei=.18000E+01
+ cjci0=.11600E-14 vdc1=.800000E+00 zci=.3330000E+00
+ vptci=.41600E+03
+ rci0=.127800E+03 vlim=.700000E+00 vpt=.5000000E+01
+ vces=.100000E+00
+ t0=.47500000E-11 dt0h=.210000E-11 tbvl=.400000E-11
+ tef0=.180000E-11 gtfe=.140000E+01 thcs=.300000E-10
+ alhc=.750000E+00
+ fthc=.600000E+00
+ latb=.376500E+01 latl=.342000E+00 fqi=.9055000E+00
+ alit=.450000E+00 alqf=.225000E+00
+ favl=.118600E+01 qavl=.111000E-13 alfav=.82500E-04
+ alqav=.19600E-03
+ ibeis=.11600E-19 mbei=.101500E+01 ibeps=.10000E-29
+ mbep=.200000E+01
+ ireis=.11600E-15 mre1=.200000E+01 ireps=.10000E-29
+ mrep=.200000E+01
+ rbi0=.000000E+00 fdqr0=.00000E+00 fgeo=.730000E+00
+ fcrbi=.00000E+00
+ cjep0=.00000E+00 vdep=.105000E+01 zep=.4000000E+00
+ aljep=.24000E+01
+ ceox=.000000E+00
+ cjcx0=.00000E+00 vdcx=.700000E+00 zcx=.3330000E+00
+ vptcx=.10000E+03
+ ccox=.000000E+00 fbc=.1526000E+00
+ ibcx0=.10000E-29 mbcx=.200000E+01 ibcis=.11600E-19
+ mbci=.101500E+01
+ cjs0=.000000E+00 vds=.6000000E+00 zs=.44700000E+00
+ vpts=.100000E+04
+ rcx=.0000000E+00 rbx=.0000000E+00 re=.00000000E+00
+ kf=.00000000E+00 af=.00000000E+00
+ vgb=.1170000E+01 alb=.6300000E-02 alt0=.000000E+00
+ kt0=.0000000E+00
+ zetaci=.1600E+01 alvs=.100000E-02 alces=.40000E-03
+ zetarbi=0.5880E+00 zetarcx=0.2230E+00
+ zetarbx=0.2060E+00 zetare=0.0000E+00
+ rth=0.0 cth=0.0
+ ibets=.00000E+00 abet=.000000E+00
+ itss=.000000E+00 msf=.0000000E+00 tsf=0.000000E+00
+ iscs=.000000E+00
+ msc=.0000000E+00
+ rsu=.0000000E+00 csu=.0000000E+00
.END
```

Internal Transistors

This section lists the HICUM LEVEL=8 internal transistor parameters. This includes parameter names, descriptions, units, default values, factors, and notes.

Table 81 *BJT LEVEL=8 Transfer Current Parameters*

Name (Alias)	Unit	Default	Factor	Description
C10	A ^{2s}	2e-30	M ²	Constant. The IS setting determines the C10 parameter value. If IS > 0, then C10=IS * QP0; otherwise, C10=C10. IS=1.0e-16; C10=2.0e-30.
QP0	As	2e-14		Zero-bias hole charge
ICH	A	1e+20		High-current correction for 2D/3D
HFC	-	1		Weighting factor for Qfc (mainly for HBTs)
HFE	-	1		Weighting factor for Qef in HBTs
HJCI	-	1		Weighting factor for Qjci in HBTs
HJEI	-	1		Weighting factor for Qjei in HBTs

Table 82 *BJT LEVEL=8 BE Depletion Capacitance Parameters*

Name (Alias)	Unit	Default	Factor	Description
ALJEI (AJEI)	-	2.5		Ratio of max. to zero-bias value
CJEI0	F	0		Zero-bias value
VDEI	V	0.9		Built-in voltage
ZEI	-	0.5		Exponent coefficient

Table 83 *BJT LEVEL=8 BC Depletion Capacitance Parameters*

Name (Alias)	Unit	Default	Factor	Description
ALJEP (AJEP)	-	2.5		Ratio of max. to zero-bias value
CJCI0	F	0	M	Zero-bias value
VDCI	V	0.7		Built-in voltage
ZCI	-	0.4		Exponent coefficient

Table 83 BJT LEVEL=8 BC Depletion Capacitance Parameters (Continued)

Name (Alias)	Unit	Default	Factor	Description
VPTCI	V	1e+20		Punch-through voltage ($=q N_{ci} w_{ci}^2 / (2\epsilon_{silicon})$)

Table 84 BJT LEVEL=8 Forward Transit Time Parameters

Name (Alias)	Unit	Default	Factor	Description
ALHC	-	0.1		Smoothing factor for current dep. C and B transit time
ALQF	-	0		Factor for additional delay time of Q_f
DT0H	s	0		Time constant for base and BC SCR width modulation
FLNQS	-	0		NQS flag (NQS supported in V2.23); also has instance parameter
FTHC	-	0		Partitioning factor for base and collection portion
GTFE	-	1		Exponent factor for current dep. emitter transit time
T0	s	0		Low current transit time at $V_{BC}=0$
TBVL	s	0		Voltage for modeling carrier jam at low $V_{C'E'}$
TEF0	s	0		Storage time in neutral emitter
THCS	s	0		Saturation time constant at high current densities

Table 85 BJT LEVEL=8 Critical Current Parameters

Name (Alias)	Unit	Default	Factor	Description
RCI0	Ohm	150	1/M	Low-field resistance of internal collector region
VLIM	V	0.5		Voltage separating ohmic and SCR regime
VPT	V	1e+20		Epi punch-through vtg. of BC SCR
VCES	V	0.1		Internal CE sat. vtg.

Table 86 BJT LEVEL=8 Inverse Transit Time Parameter

Name (Alias)	Unit	Default	Factor	Description
TR	s	0		Time constant for inverse operation

Table 87 BJT LEVEL=8 Base Current Component Parameters

Name (Alias)	Unit	Default	Factor	Description
IBEIS	A	1e-18	M	BE saturation current
MBEI	-	1		BE saturation current
IREIS	A	0	M	BE recombination saturation current
MREI	-	2		BE recombination non-ideality factor
TBHREC	ps	0		BC recombination time constant at the BC barrier for high forward injection
IBCIS	A	1e-16	M	BC saturation current
MBCI	-	1		BC non-ideality factor

Table 88 BJT LEVEL=8 Weak BC Avalanche Breakdown Parameters

Name (Alias)	Unit	Default	Factor	Description
FAVL	1/V	0		Prefactor for CB avalanche effect
QAVL	C	0	M	Exponent factor for CB avalanche effect

Table 89 BJT LEVEL=8 Internal Base Resistance Parameters

Name (Alias)	Unit	Default	Factor	Description
FCRBI	-	0		Ratio of HF shunt to total internal capacitance.
FDQR0	-	0		Correction factor for BE and BC SCR modulation
FGEO	-	0.6557		Geometry factor (corresponds to long emitter stripe)
FQI	-	1.0		Ratio of internal to total minority charge
RBIO	Ohm	0	1/M	Value at zero-bias

Table 89 BJT LEVEL=8 Internal Base Resistance Parameters (Continued)

Name (Alias)	Unit	Default	Factor	Description
RBX	Ohm	0	1/M	External base series resistance
RCX	Ohm	0	1/M	External collector series resistance
RE	Ohm	0	1/M	Emitter series resistance

Table 90 BJT LEVEL=8 Lateral Scaling

Name (Alias)	Unit	Default	Factor	Description
LATB	-	0		Scaling factor for Q _{fc} in I _E ("I" is the letter L: not the number 1)
LATL	-	0		Scaling factor for Q _{fc} in I _E direction ("I" is the letter L: not the number 1)

Peripheral Elements

This section lists the HICUM LEVEL=8 model peripheral element parameters. This includes parameter names, descriptions, units, default values, factors, and notes.

Table 91 BJT LEVEL=8 BE Depletion Capacitance

Name (Alias)	Unit	Default	Factor	Description
ALJEP	-	2.5		Ratio of max. to zero-bias value
CJEP0	F	0	M	Zero-bias value
VDEP	V	0.9		Built-in voltage
ZEP	-	0.5		Depletion coeff

Table 92 BJT LEVEL=8 Base Current

Name (Alias)	Unit	Default	Factor	Description
IBEPS	A	0	M	Saturation current
IREPS	A	0	M	Recombination saturation factor
MBEP	-	1		Non-ideality factor

Table 92 BJT LEVEL=8 Base Current (Continued)

Name (Alias)	Unit	Default	Factor	Description
MREP	-	2		Recombination non-ideality factor

Table 93 BJT LEVEL=8 BE Tunneling

Name (Alias)	Unit	Default	Factor	Description
ABET	-	40		Exponent coefficient
IBETS	A	0	M	Saturation current
TUNODE	-	1		Tunneling flag • 1=on peripheral node • 0=on internal node

External Elements

This section lists the HICUM LEVEL=8 model external element parameters. This includes parameter names, descriptions, units, default values, factors, and notes.

Table 94 BJT LEVEL=8 BC Capacitance

Name (Alias)	Unit	Default	Factor	Description
CBEPAR	-	1.0		Total parasitic BE capacitance (spacer and metal)
CCOX (CBCPAR)	F	0	M	Collector oxide capacitance
CJCX0	F	0	M	Zero-bias depletion value
FBC (FDCPAR)	-	0		Partitioning factor for $C_{BCX} = C'_{BCX} + C''_{BCX}$
FBEPAR	-	0		Partitioning factor of parasitic BE capacitance
VDCX	V	0.7		Built-in voltage
VPTCX	V	1e+20		Punch-through voltage
ZCX	-	0.4		Exponent coefficient

Table 95 *BJT LEVEL=8 BC Base Current Component*

Name (Alias)	Unit	Default	Factor	Description
IBCXS	A	0	M	Saturation current
MBCX	-	1		Non-ideality factor

Table 96 *BJT LEVEL=8 Other External Elements*

Name (Alias)	Unit	Default	Factor	Description
RBX	Ohm	0	1/M	External base series resistance
RE	Ohm	0	1/M	Emitter series resistance
RCX	Ohm	0	1/M	External collector series resistance

Table 97 *BJT LEVEL=8 Substrate Transistor Parameters*

Name (Alias)	Unit	Default	Factor	Description
ISCS	A	0	M	Saturation current of CS diode
ITSS	A	0	M	Transfer saturation current
MSC	-	1		Non-ideality factor of CS diode
MSF	-	1		Non-ideality factor (forward transfer current)
TSF	s	0		Minority charge storage transit time

Table 98 *BJT LEVEL=8 Collector-Substrate Depletion Capacitance*

Name (Alias)	Unit	Default	Factor	Description
CJS0	F	0	M	Zero-bias value of CS depletion cap
VDS	V	0.6		Built-in voltage
VPTS	V	1e+20		Punch-through voltage
ZS	-	0.5		Exponent coefficient

Table 99 BJT LEVEL=8 Substrate Coupling Network

Name (Alias)	Unit	Default	Factor	Description
RSU	Ohm	0	1/M	Substrate series resistance
CSU	F	0		Substrate capacitance from permittivity of bulk material

Table 100 BJT LEVEL=8 Noise Parameters

Name (Alias)	Unit	Default	Factor	Description
KF	-	0	M^{1-AF}	Flicker noise factor (no unit only for AF=2!)
AF	-	2		Flicker noise exponent factor

Table 101 BJT LEVEL=8 Temperature Dependence Parameters

Name (Alias)	Unit	Default	Factor	Description
ALCES	1/K	0		Relative temperature coefficient of VCES
ALFAV	1/K	0		Relative temperature coefficient for avalanche breakdown
ALQAV	1/K	0		Relative temperature coefficient for avalanche breakdown
ALT0	1/K	0		First-order relative temperature coefficient, TEFO
ALVS	1/K	0		Relative temperature coefficient of saturation drift velocity
F1VG	V/K	1.02377e-4		Coefficient K1 in T dependent bandgap equation
F2VG	V/K	4.3215e-4		Coefficient K2 in T dependent bandgap equation
KT0	1/K ²	0		Second-order relative temperature coefficient, TEFO
VGB	V	1.17		Bandgap voltage
VGC	V	1.17		Effective collector bandgap voltage $V_g C_{eff}$
VGE	V	1.17		Effective emitter bandgap voltage $V_g E_{eff}$
VGS	V	1.17		Effective substrate bandgap voltage $V_g S_{eff}$

Table 101 BJT LEVEL=8 Temperature Dependence Parameters (Continued)

Name (Alias)	Unit	Default	Factor	Description
ZETABET	-	5		Exponent coefficient in BE junction current temperature dependence
ZETACT	-	4.5		Exponent coefficient in transfer current temperature dependence
ZETARBI	-	0		Temperature exponent factor of RBI0
ZETARBX	-	0		Temperature exponent factor of RBX
ZETARCX	-	0		Temperature exponent factor of RCX
ZETARE	-	0		Temperature exponent factor of RE

To use the self-heating HICUM feature (in BJT LEVEL=8), set VERS=2.1 and set an RTH parameter value other than 0. If you use vers=2.0 or RTH=0, then self-heating is OFF.

The self-heating effect also applies to the circuit temperature as an increased self-heating temperature. $T = T_{ckt}(\text{circuit temperature}) + T_{sh}(\text{self heating temperature}) + dtemp$ (difference between circuit temperature and ambient temperature).

Table 102 BJT LEVEL=8 Self-Heating Parameters

Name (Alias)	Unit	Default	Factor	Description
RTH	K/W	0	1/M	Thermal resistance (not supported in v2000.4)
CTH	Ws/K	0	M	Thermal resistance (not supported in v2000.4)
FLSH	-	0	M	Self-heating flag, also has instance parameter. If FLSH is not 0 and RTH>0, ON; otherwise OFF.

Table 103 BJT LEVEL=8 Other Parameters

Name (Alias)	Unit	Default	Factor	Description
IS	A	-1.0		Ideal saturation current
MCF	-	1.0		Non-ideal factor of reverse current between base and collector. $V_T = V_T * MCF$
ZETACX	-	1.0		Temperature exponent factor (epi-layer)

HICUM v2.3 Added Parameters

Table 104 BJT LEVEL=8 Version 2.32, 2.31 and 2.3 Added Parameters

Name (Alias)	Unit	Default	Description
DELCK	-	1	Fitting factor for ICK
AHJEI	-	0	Parameter describing the slope of $h_{JEI}(V_{BE})$
RHJEI	-	1	Smoothing parameter for $h_{JEI}(V_{BE})$ at high voltage
DVGBE	V	0	Bandgap difference between base and BE-junction, used for h_{JEI0} and $hf0$.
ZETAHJEI	-	1	Temperature coefficient for ah_{JEI}
ZETA VGBE	-	1	Temperature coefficient for h_{JEI0}
HF0	-	1	Weight factor for the low current minority charge
VCBAR	V	0	Barrier voltage
ACBAR	-	1	Smoothing parameter for barrier voltage
ICBAR	A	1	Normalization parameter
ZETARTH	K/W	1	Temperature coefficient for R_{th}
KFRE	-	0	RE flicker noise coefficient
AFRE	-	2	RE flicker noise exponent factor

HICUM v2.4 Added Parameters

Table 105 BJT LEVEL=8 Version 2.34 Added Parameters

Name (Alias)	Unit	Default	Description
cscp0	F	0.0	Perimeter S-C zero-bias depletion capacitance
vdsp	V	0.6	Perimeter S-C built-in voltage
zsp	-	0.5	Perimeter S-C grading coefficient
vptsp	V	100	Perimeter S-C punch-through voltage
aick	-	1e-3	Smoothing term for ICK

LEVEL=9 VBIC99 Model

The VBIC 95 (Vertical Bipolar Inter-Company Model) for Motorola bipolar transistor device is installed in the device models as BJT `LEVEL=4`. VBIC99 is a newer version of the VBIC model, and is implemented in the device models as BJT `LEVEL=9`.

To use the VBIC99 model, set the `LEVEL` parameter to 9 for the bipolar transistor model.

The VBIC99 model includes several effects that are improved compared to the VBIC95 model.

- In VBIC99, the temperature coefficients of the base and collector resistances are split.
- The temperature dependence of the built-in potential is also improved.

The following topics are discussed in these sections:

- [Level 9 Usage Notes](#)
- [LEVEL=9 Element Syntax](#)
- [Effects of VBIC99](#)
- [Model Implementation](#)
- [LEVEL=9 Model Parameters](#)

Level 9 Usage Notes

The following information applies to the PrimeSim Continuum device model for the VBIC99 device:

- Set `LEVEL` to 9 to identify the model as a VBIC99 bipolar junction transistor model.
- The reference temperature, `TREF`, equals 27 degrees.
- The VBIC99 model supports `AREA` and `M` factor scaling.
- This model supports self-heating. Model parameters are `RTH` and `CTH`.

LEVEL=9 Element Syntax

```
Qxxx nc nb ne [ns] mname[AREA=val] [OFF] [VBE=val] [VCE=val]  
+ [M=val] [DTEMP=val]
```

Parameter	Description
Qxxx	BJT element name. Must begin with Q, followed by up to 1023 alphanumeric characters.
Nc	Collector terminal node name or number.
Nb	Base terminal node name and number.
Ne	Emitter terminal node name or number.
Ns	Substrate node name or number.
t	Self heating node name or number.
Mname	BJT model name reference.
AREA	The normalized emitter area. VBIC99 LEVEL=9 model has no area effect. Default value=1. Area is used only as an alias of the multiplication factor (M).
OFF	Sets the initial condition to OFF for this element in DC analysis. You cannot use OFF with VBE or VCE.
VBE	Initial internal base-emitter voltage.
VCE	Initial internal collector-emitter voltage.
M	Multiplier to simulate multiple BJTs in parallel.
DTEMP	The temperature difference between the element and circuit.

Effects of VBIC99

The VBIC99 model includes several effects that are improved compared to the VBIC95 model:

- Addition of temperature dependency for several parameters
- Base-emitter breakdown model
- Reach-through model for base-collector depletion capacitance
- High-current beta rolloff effect
- Fixed collector-substrate capacitance
- Reverse transport saturation current

Model Implementation

The following parameters were added to the VBIC99 model and are not in the VBIC95 model.

ISRR	IKF	VRT	ART	QBM
DEAR	EAP	VBBE	NBBE	IBBE
TVBBE1	TVBBE2	TNBBE	EBBE	CCSO
XRCX	XR BX	XRBP	XIXF	XISR

LEVEL=9 Model Parameters

These tables describe VBIC99 as the PrimeSim Continuum simulator BJT $\text{LEVEL}=9$ model parameters, including parameter names, descriptions, units, default values, and notes. Parameters with an asterisk (*) are not used in the DC model.

Table 106 LEVEL=9 VBIC99 Basic Parameters

Parameter	Unit	Default	Description
LEVEL	-	9	Model level
TREF	W	27.0	Nominal measurement temperature of parameters
ART	-	0.1	*Smoothing parameter for reach-through
AVC1	1/V	0.0	Base-collector avalanche parameter 1
AVC2	1/V	0.0	Base-collector avalanche parameter 2
DEAR	-	0.0	*Delta activation energy for ISRR
EA	EV	1.12	Activation energy for IS
EAIC	EV	1.12	Activation energy for IBCI/IBEIP
EAIE	EV	1.12	Activation energy for IBEI
EAIS	EV	1.12	Activation energy for IBCIP
EANC	EV	1.12	Activation energy for IBCN/IBENP
EANE	EV	1.12	Activation energy for IBEN
EANS	EV	1.12	Activation energy for IBCNP

Table 106 LEVEL=9 VBIC99 Basic Parameters (Continued)

Parameter	Unit	Default	Description
EAP	-	1.12	*Activation energy for ISP
EBBE	-	0.0	$\exp(-V_{BBE}/(N_{BBE} \cdot V_{tv}))$
GAMM	-	0.0	Epi doping parameter
HRCF	-	0.0	High current RC factor
IBBE	-	1.0e-6	*Base-emitter breakdown current
IBCI	A	1.0e-16	Ideal base-collector saturation current
IBCIIP	A	0.0	Ideal parasitic base-collector saturation current
IBCN	A	0.0	Non-Ideal base-collector saturation current
IBCNIP	A	0.0	Non-Ideal base-collector saturation current
IBEI	A	1.0e-18	Ideal base-emitter saturation current
IBEIIP	A	0.0	Ideal parasitic base-emitter saturation current
IBEN	A	0.0	Non-Ideal base-emitter saturation current
IBENIP	A	0.0	Non-Ideal parasitic base-emitter saturation current
IKF	A	0.0	Forward knee current, zero means infinity
IKP	A	0.0	Parasitic knee current, zero means infinity
IKR	A	0.0	Reverse knee current, zero means infinity
IS	A	1.0e-16	Transport saturation current
ISP	A	0.0	Parasitic transport saturation current
ISRR	A	1.0	*Reverse transport saturation current
ITF	A	0.0	Coefficient of TF dependence on I_c
MC	-	0.33	Base-collector Grading coefficient
ME	-	0.33	Base-emitter Grading coefficient
MS	-	0.33	Substrate-collector Grading coefficient
NBBE	-	1.0	*Base-emitter breakdown emission coefficient
NCI	-	1.0	Ideal base-collector emission coefficient

Table 106 LEVEL=9 VBIC99 Basic Parameters (Continued)

Parameter	Unit	Default	Description
NCIP	-	1.0	Ideal parasitic base-collector emission coefficient
NCN	-	2.0	Non-ideal base-collector emission coefficient
NCNP	-	2.0	Ideal parasitic base-collector emission coefficient
NEI	-	1.0	Ideal base-emitter emission coefficient
NEN	-	2.0	Non-ideal base-emitter emission coefficient
NF	-	1.0	Forward emission coefficient
NFP	-	1.0	Parasitic forward emission coefficient
NKF	-	0.5	*High current beta roll off parameter
NR	-	1.0	Reverse emission coefficient
PC	V	0.75	Base-collector built-in potential
PE	V	0.75	Base-emitter built-in potential
PS	V	0.75	Substrate-collector built-in potential
QBM	-	0.0	*Base charge model selection
QTF	-	0.0	Variation of TF with base-width modulation
RBI	W	0.0	Intrinsic collector Resistance
RBP	W	0.0	Parasitic base Resistance
RBX	W	0.0	Extrinsic collector Resistance
RCI	W	0.0	Intrinsic collector Resistance
RCX	W	0.0	Extrinsic collector Resistance
RE	W	0.0	Emitter Resistance
RS	W	0.0	Substrate Resistance
TF	S	0.0	Forward transit time
TNBBE	-	0.0	*Temperature coefficient of NBBE
TR	S	0.0	Reverse transit time
TVBBE1	-	0.0	*Linear temperature coefficient of VBBE

Table 106 *LEVEL=9 VBIC99 Basic Parameters (Continued)*

Parameter	Unit	Default	Description
TVBBE2	-	0.0	*Quadratic temperature coefficient of VBBE
VBBE	-	0.0	*Base-emitter breakdown voltage
VEF	V	0.0	Forward early voltage, zero means infinity
VER	V	0.0	Reverse early voltage, zero means infinity
VO	V	0.0	Epi drift saturation voltage
VRT	V	0.0	*Reach-through voltage for Cbc limiting
VTF	V	0.0	Coefficient of TF dependence on Vbc
WBE	-	1.0	Portion of IBEI from Vbei, 1-WBE from Vbex
WSP	-	1.0	Portion of ICCP from Vbep, 1-WBE from Vbci
XTF	-	0.0	Coefficient of TF bias dependence

Table 107 *LEVEL=9 VBIC99 Capacitance/Charge Parameters*

Parameter	Unit	Default	Description
AJC	-	-0.5	Base-collector cap smoothing factor
AJE	-	-0.5	Base-emitter cap. Smoothing factor
AJS	-	-0.5	Substrate-collector cap. Smoothing factor
CBCO	F	0.0	Extrinsic base-collector overlap cap
CBEO	F	0.0	Extrinsic base-emitter overlap cap
CCSO	F	0.0	*Fixed collector-substrate capacitance
CJC	F	0.0	Base-collector zero bias cap
CJCP	F	0.0	Substrate-collector zero bias cap
CJE	F	0.0	Base-emitter zero bias cap
CJEP	F	0.0	Base-collector extrinsic zero bias cap
FC	-	0.9	Forward bias depletion cap limit
QCO	Coul	0.0	Epi charge parameter

Table 108 *LEVEL=9 VBIC99 Temperature Coefficients*

Parameter	Unit	Default	Description
TAVC	1/K	0.0	Temperature coefficient of AVC2
TNF	1/K	0.0	Temperature exponent of NF
XII	-	3.0	Temperature exponent of IBEI/IBCI/IBEIP/IBCIP
XIKF	-	0.0	*Temperature exponent of IKF
XIN	-	3.0	Temperature exponent, IBEN/IBCN/IBENP/IBCNP
XIS	-	3.0	Temperature exponent of IS
XISR	-	0.0	*Temperature exponent of ISRR
XRBI	-	0.0	Temperature exponent of intrinsic base resistance
XRBP	-	0.0	*Temperature exponent of parasitic base resistance
XRBX	-	0.0	*Temperature exponent, extrinsic collector resistance
XRCI	-	0.0	Temperature exponent, intrinsic collector resistance
XRCX	-	0.0	*Temperature exponent of extrinsic base resistance
XRE	-	0.0	Temperature exponent of emitter resistance
XRS	-	0.0	Temperature exponent of substrate resistance
XVO	-	0.0	Temperature exponent of VO

Table 109 *LEVEL=9 VBIC99 Noise Parameters*

Parameter	Unit	Default	Description
AFN	-	1.0	Base-emitter Flicker noise exponent
KFN	-	0.0	Base-emitter Flicker noise constant
BFN	-	1.0	Base-emitter Flicker noise 1/f dependence

Table 110 LEVEL=9 VBIC99 Self-heating Parameters

Parameter	Unit	Default	Description
RTH	K/W	0.0	Thermal resistance
CTH	J/K	0.0	Thermal capacitance

Table 111 LEVEL=9 VBIC99 Excess Phase Parameter

Parameter	Unit	Default	Description
TD	S	0.0	Forward excess-phase delay time

VBIC99 LEVEL=9 AC Analysis Example

The VBIC99 level9 AC test example is located in the following directory:

`$installdir/demo/hspice/bjt/vbic99_ac.sp`

VBIC99 LEVEL=9 DC Analysis Example

The VBIC99 level9 DC test example is located in the following directory:

`$installdir/demo/hspice/bjt/vbic99_dc.sp`

VBIC99 LEVEL=9 TRAN Analysis Example

The VBIC99 level9 transient test example is located in the following directory:

`$installdir/demo/hspice/bjt/vbic99_tran.sp`

LEVEL=10 Philips MODELLA Bipolar Model

The Philips MODELLA LEVEL=10 provides a highly-accurate compact model for lateral pnp integrated circuit transistors. This model is based directly on device physics. It uses a physical modelling approach where the main currents and charges are independently related to bias-dependent minority carrier concentrations. It also models current crowding effects, high injection effect, and a bias-dependent output impedance. [Table 112](#) describes the transistor parameters for this model.

Table 112 BJT LEVEL=10 Transistor Parameters

Name (Alias)	Unit	Default	Description
LEVEL		10	Model level
AE		4.48	Temperature coefficient of BF

Table 112 BJT LEVEL=10 Transistor Parameters (Continued)

Name (Alias)	Unit	Default	Description
AF		1.00	Flicker noise exponent
BF		131.00	Ideal forward common-emitter current gain
BR		25.00	Ideal reverse common-emitter current gain
CJC	F	3.90e-13	Zero-bias collector-base depletion capacitance
CJE	F	6.10e-14	Zero-bias emitter-base depletion capacitance
CJS	F	1.30e-12	Zero-bias substrate-base depletion capacitance
DTA	oC	0.00	Difference between the device temperature and the ambient analysis temperature
EAFL	V	20.50	Early voltage of the lateral forward current component at zero collector-base bias
EAfv	V	75.00	Early voltage of the vertical forward current component at zero collector-base bias
EARL	V	13.10	Early voltage of the lateral reverse current component at zero emitter-base bias
EARV	V	104.00	Early voltage of the vertical reverse current component at zero emitter-base bias
EXPHI		0.00	rad Excess phase shift
IBF	A	2.60e-14	Saturation current of non-ideal forward base current
IBR	A	1.20e-13	Saturation current of non-ideal reverse base current
IK	A	1.10e-4	High injection knee current
IS	A	1.80e-16	Collector-emitter saturation current
ISS	A	4.00e-13	Saturation current of substrate-base diode
KF		0.00	Flicker noise coefficient
PC		0.36	Collector-base grading coefficient
PE		0.30	Emitter-base grading coefficient
PS		0.35	Substrate-base grading coefficient
RBCC	W	10.00	Constant part of the base resistance RBC

Table 112 BJT LEVEL=10 Transistor Parameters (Continued)

Name (Alias)	Unit	Default	Description
RBCV	W	10.00	Variable part of the base resistance RBC
RBEC	W	10.00	Constant part of the base resistance RBE
RBEV	W	50.00	Variable part of the base resistance RBE
RCEX	W	5.00	External part of the collector resistance
RCIN	W	47.00	Internal part of the collector resistance
REEX	W	27.00	External part of the emitter resistance
REIN	W	66.00	Internal part of the emitter resistance
RSB	W	1.00e15	Substrate-base leakage resistance
SNB		2.60	Temperature coefficient of epitaxial base electron mobility
SNBN		0.30	Temperature coefficient, buried layer electron mobility
SPB		2.853	Temperature coefficient, epitaxial base hole mobility
SPC		0.73	Temperature coefficient of collector hole mobility
SPE		0.73	Temperature coefficient of emitter hole mobility
SX		1.00	Temperature coefficient of combined minority carrier mobilities in emitter and buried layer
TFN	S	2.00e-10	Low injection forward transit time due to charge stored in emitter, and buried layer under the emitter
TFVR	S	3.00e-8	Low injection forward transit time due to charge stored in the epilayer under the emitter
TLAT	S	2.40e-9	Low injection (forward/reverse) transit time of charge stored in epilayer between emitter and collector
TREF	oC	25.00	Reference temperature of the parameter set
TRN	S	3.00e-9	Low injection reverse transit time due to charge stored in collector, and buried layer under collector
TRVR	S	1.00e-9	Low injection reverse transit time due to charge stored in the epilayer under the collector
VDC	V	0.57	Collector-base diffusion voltage

Table 112 BJT LEVEL=10 Transistor Parameters (Continued)

Name (Alias)	Unit	Default	Description
VDE	V	0.52	Emitter-base diffusion voltage
VDS	V	0.52	Substrate-base diffusion voltage
VGB	V	1.206	Bandgap voltage, base between emitter and collector
VGCB	V	1.206	Bandgap voltage of collector-base depletion region
VGE	V	1.206	Bandgap voltage of the emitter
VGEB	V	1.206	Bandgap voltage of the emitter-base depletion region
VGJE	V	1.123	Bandgap voltage recombination emitter-base junction
VGSB	V	1.206	Bandgap voltage of substrate-base depletion region
VLF	V	0.54	V Cross-over voltage, non-ideal forward base current
VLR	V	0.48	Cross-over voltage of non-ideal reverse base current
XCS		3.00	Ratio between the saturation current of c-b-s transistor and c-b-e transistor
XES		2.70e-3	Ratio between saturation current of e-b-s transistor and e-b-c transistor
XHCS		1.00	Fraction of substrate current of c-b-s transistor subject to high injection
XHES		0.70	Fraction of substrate current of e-b-s transistor subject to high injection
XIFV		0.43	Vertical fraction of forward current
XIRV		0.43	Vertical fraction of reverse current

The following topics are discussed in these sections:

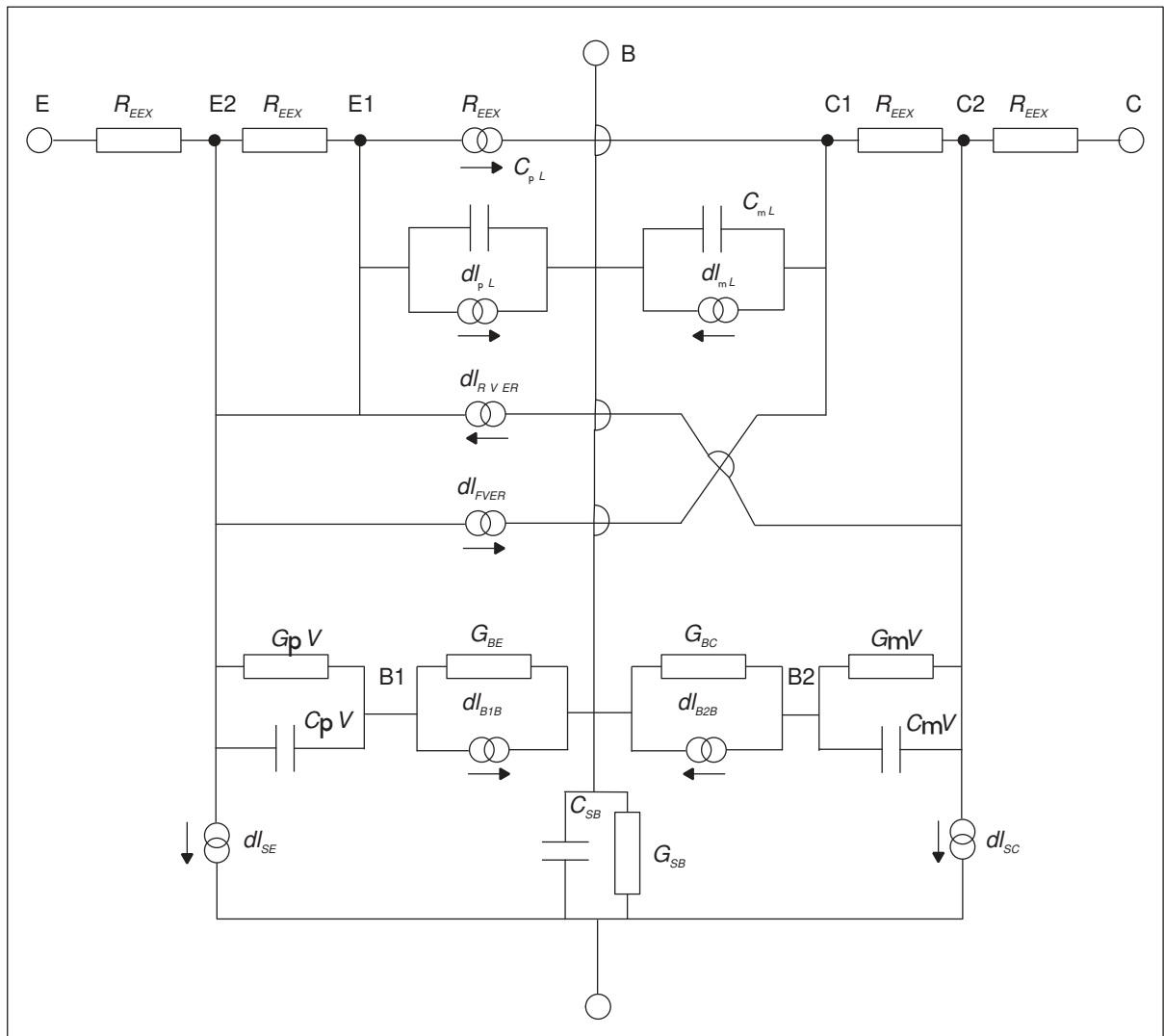
- [Equivalent Circuits](#)
- [DC Operating Point Output](#)

Chapter 5: BJT Models

LEVEL=10 Philips MODELLA Bipolar Model

- [Model Equations](#)
- [Temperature Dependence of Parameters](#)

Figure 38 Small-signal Equivalent Circuit



DC Operating Point Output

The DC operating point output facility gives information on the state of a device at its operation point. Figure 1 shows the DC large signal equivalent circuit. Figure 38 shows the small signal equivalent circuit. R_{EEX} , R_{EIN} , R_{CIN} and R_{CEX} are constant resistors.

$$dI_{LAT} = g_{fL} \times dV_{E1B} - g_{rL} \times dV_{C1B}$$

$$dI_{FVER} = g_{11} \times dV_{E2B1} + g_{12} \times dV_{C1B}$$

$$dI_{RVER} = g_{21} \times dV_{E1B} + g_{22} \times dV_{C2B2}$$

$dI_{B1B} = G_{IBE} \times dV_{E2B1}$
 $dI_{B2B} = G_{IBC} \times dV_{C2B2}$
 $dI_{PL} = j_w \times C_{IPL} \times dV_{C1B}$
 $dI_{mL} = j_w \times C_{ImL} \times dV_{E1B}$
 $dI_{SE} = G_{ISE} \times dV_{E2B1}$
 $dI_{SC} = G_{ISC} \times dV_{C2B2}$

Table 113 BJT LEVEL=10, DC Operating Point Parameters

Name (Alias)	Description
CMUL	Reverse diffusion capacitance, lateral path: $\partial Q_{RLAT} / \partial V_{C1B}$
CMUV	Reverse total capacitance, vertical path: $\partial (Q_{tc} + Q_{rver} + Q_{rn}) / \partial V_{C2B2}$
CPIL	Forward diffusion cap., lateral path: $\partial Q_{FLAT} / \partial V_{E1B}$
CPIV	Forward total capacitance, vertical path: $\partial (Q_{TE} + Q_{FVER} + Q_{FN}) / \partial V_{E2B1}$
CSB	Total capacitance s-b junction: $\partial Q_{TS} / \partial V_{SB} + \partial Q_{SD} / \partial V_{SB}$
G11	Forward conductance, vertical path.: $\partial I_{FVER} / \partial V_{E2B1}$
G12	Collector Early-effect on I_{FVER} : $\partial I_{FVER} / \partial V_{C1B}$
G21	Emitter Early-effect on I_{RVER} : $\partial I_{RVER} / \partial V_{E1B}$
G22	Reverse conductance, vertical path.: $\partial I_{RVER} / \partial V_{C2B2}$
GFL	Forward conductance, lateral path.: $\partial I_{FLAT} / \partial V_{E1B1}$
GMU	Conductance c-b junction: $\partial (I_{RC} + I_{LC}) / \partial V_{C2B2}$
GPI	Conductance e-b junction: $\partial (I_{RE} + I_{LE}) / \partial V_{E2B1}$
GRL	Reverse conductance, lateral path.: $\partial I_{RLAT} / \partial V_{C1B}$
GSB	Conductance s-b junction: $\partial I_{SF} / \partial V_{SB} + 1/R_{SB}$

Model Equations

Early Factors

The Early factors for the components of the main current I_p are derived from the variation of the depletion widths in the base relative to the base width itself.

Early factor of the lateral current components

$$FLAT = hyp_1 \left\{ 1 - \frac{\sqrt{\left(1 - \frac{V}{VD}\right) + \delta}}{1 + \frac{EARL}{2VD}} - \frac{\sqrt{\left(1 - \frac{V_1}{VD}\right)^2 + \delta}}{1 + \frac{EAFL}{2VD}} \right\} \cdot \delta_E$$

Early factor of the forward vertical current component

$$FFVR = hyp_1 \left\{ 1 - \frac{\sqrt[4]{\left(1 - \frac{V_{E2B1}}{VD_T}\right)^2 + \delta}}{1 + \frac{EARV}{2VD_T}} - \frac{\sqrt[4]{\left(1 - \frac{V_{CLB}}{VD_T}\right)^2 + \delta}}{1 + \frac{EAFV}{2VD_T}} \right\} \cdot \delta_E$$

Early factor of the reverse vertical current component

$$FRVER = hyp_1 \left\{ 1 - \frac{\sqrt[4]{\left(1 - \frac{V_{E1B}}{VD_T}\right)^2 + \delta}}{1 + \frac{EARV}{2VD_T}} - \frac{\sqrt[4]{\left(1 - \frac{V_{C2B2}}{VD_T}\right)^2 + \delta}}{1 + \frac{EAFV}{2VD_T}} \right\} \cdot \delta_E$$

Model Parameters

- EAFL
- EAFV
- EARL
- EARV

Currents

The ideal diode equations are as follows.

$$I_1 = I_S (e^{V_{E1B}/V_{t1}} - 1)$$

$$I_2 = I_S (e^{V_{E2B1}/V_{t1}} - 1)$$

$$I_r = I_S (e^{V_{C1B}/V_{t1}} - 1)$$

$$I_{r2} = I_S (e^{V_{c2b2}/V_t} - 1)$$

model parameter: I_S

The I_p main current is as follows.

$$I_p = I_{flat} + I_{fver} - I_{rlat} - I_{rver}$$

Forward currents— I_{flat} and I_{fver}

The main forward current is separated into lateral and vertical components, originating from the emitter-base junction sidewall and bottom, respectively. These formulations include Early and high injection effects. Because the two currents depend on different internal emitter-base junction voltages, emitter current crowding is also modelled.

The lateral forward current component (I_{flat}) is:

$$I_{flat} = \left(\frac{4 \times (1 - X_{ifv}) \times I_{f1}}{3 + \sqrt{1 + 16 \times \frac{I_{f1}}{I_k}}} \right) \div F_{flat}$$

The vertical forward current component (I_{fver}) is:

$$I_{fver} = \left(\frac{4 \times X_{ifv} \times I_{f2}}{3 + \sqrt{1 + 16 \times \frac{I_{f2}}{I_k}}} \right) \div F_{fver}$$

Model parameters:

- X_{ifv}
- I_k

Reverse currents— I_{rlat} and I_{rver}

The main reverse current contains lateral and vertical components, originating from the collector-base junction sidewall and bottom, respectively. These formulations include Early and high injection effects. The two currents depend on different internal collector-base junction voltages, collector current crowding is also modelled.

The lateral reverse current component (I_{rlat}) is:

$$I_{rlatt} = \left(\frac{4 \times (1 - X_{irv}) \times I_{r1}}{3 + \sqrt{1 + 16 \times \frac{I_{r1}}{I_k}}} \right) \div F_{lat}$$

The vertical reverse current component (I_{rver}) is:

$$I_{rver} = \left(\frac{4 \times X_{irv} \times I_{r2}}{3 + \sqrt{1 + 16 \times \frac{I_{r2}}{I_k}}} \right) \div F_{rver}$$

Model parameter: X_{irv}

Base Current

Forward components

The total forward base current is composed of an ideal and a non-ideal component. Both components depend on the bottom part of the emitter-base junction.

Ideal component:

$$I_{re} = \frac{I_f}{B_f}$$

Non-ideal component:

$$I_{le} = \frac{I_{bf} \times (e^{V_{e2b1}/V_t} - 1)}{e^{V_{e2b1}/V_t} + e^{V_{lf}/2V_t}}$$

Model parameters:

- B_f
- I_{bf}
- V_{lf}

Reverse components

The total reverse base current is composed of an ideal and a non-ideal component. Both components depend on the bottom part of the collector-base junction.

Ideal component:

$$I_{rc} = \frac{I_r}{B_r}$$

Non-ideal component:

$$I_{lc} = \frac{I_{br} \times (e^{V_{c2b2}/V_t} - 1)}{e^{V_{c2b2}/2V_t} + e^{V_{lr}/2V_t}}$$

Model parameters:

- B_r
- I_{br}
- V_{lr}

Substrate current

Forward components

The forward substrate component depends on the bottom part of the emitter-base junction. It consists of an ideal component, and a component subject to high injection effects. The X_{HES} parameter determines the fraction that is subject to high injection.

$$I_{se} = (1 - X_{hes}) \times X_{es} \times I_f + \frac{4 \times X_{hes} \times X_{es} \times I_f^2}{3 + \sqrt{1 + 16 \times \frac{I_f^2}{I_k}}}$$

Model parameters:

- X_{es}
- X_{hes}

Reverse components

The reverse substrate component depends on the bottom part of the collector-base junction. It consists of an ideal component, and a component subject to high injection effects. The X_{HCS} parameter determines the fraction that is subject to high injection.

$$I_{sc} = (1 - X_{hcs}) \times X_{cs} \times I_r + \frac{4 \times X_{hcs} \times X_{cs} \times I_r^2}{3 + \sqrt{1 + 16 \times \frac{I_r^2}{I_k}}}$$

Model parameters:

- X_{cs}
- X_{hcs}

Additional Substrate and Base current

An ideal diode models the substrate-base junction. You can use the reverse leakage current of this junction to model the zero-crossover phenomena, sometimes observed in the base current at low bias conditions and high temperatures.

$$I_{sf} = I_{ss} \times (e^{V_{sb}/V_t} - 1)$$

Model parameter: I_{ss}

Charges

Depletion Charges

The Poon-Gummel formulation models the depletion charges.

Emitter-base depletion charge

$$Q_{te} = \frac{-C_{je}}{1-P_e} \times \left\{ \frac{V_{de}-V_{e2b1}}{\left[\left(1-\frac{V_{e2b1}}{V_{de}} \right)^2 + \delta \right]^{\frac{P_e}{2}}} \right\}$$

Model parameters:

- C_{je}
- V_{de}
- P_e

Collector-base depletion charge

$$Q_{tc} = \frac{-C_{jc}}{1-P_c} \times \left\{ \frac{V_{dc}-V_{c2b2}}{\left[\left(1-\frac{V_{c2b2}}{V_{dc}} \right)^2 + \delta \right]^{\frac{P_c}{2}}} \right\}$$

Model parameters:

- C_{jc}
- V_{dc}
- P_c

Substrate-base depletion charge

$$Q_{ts} = \frac{-C_{js}}{1-P_s} \times \left\{ \frac{V_{ds}-V_{sb}}{\left[\left(1-\frac{V_{sb}}{V_{ds}} \right)^2 + \delta \right]^{\frac{P_s}{2}}} \right\}$$

Model parameters:

- C_{js}
- V_{ds}
- P_s

Forward Stored Charges

Storing forward-active charges consists of three main components.

- Charge stored in epitaxial base region between emitter and collector:

$$Q_{flat} = T_{lat} \times I_k \times \left(\sqrt{1 + 16 \times \frac{I_{f1}}{I_k}} - 1 \right) \times \frac{F_{lat}}{8}$$

- Charge stored in epitaxial base region under emitter:

$$Q_{fver} = T_{fvr} \times I_k \times \left(\sqrt{1 + 16 \times \frac{I_{f2}}{I_k}} - 1 \right) \times \frac{1}{8}$$

- Charge stored in emitter and buried layer under emitter:

$$Q_{fn} = T_{fn} \times I_{f2}$$

Reverse Stored Charges

Storing reverse-active charges consists of three main components.

- Charge stored in epitaxial base region between emitter and collector:

$$Q_{rlat} = T_{lat} \times I_k \times \left(\sqrt{1 + 16 \times \frac{I_{r1}}{I_k}} - 1 \right) \times \frac{F_{lat}}{8}$$

- Charge stored in epitaxial base region under collector:

$$Q_{rver} = T_{rvr} \times I_k \times \left(\sqrt{1 + 16 \times \frac{I_{r2}}{I_k}} - 1 \right) \times \frac{1}{8}$$

- Charge stored in collector and buried layer under collector:

$$Q_{rn} = T_{rn} \times I_{r2}$$

Substrate-base Stored Charge

Charge stored in the substrate and base, due to the substrate-base junction. This charge storage occurs only when the substrate-base junction is forward biased:

$$Q_{sd} = T_{sd} \times I_{sf}$$

Note:

T_{sd} is a constant.

Series Resistances

The emitter includes the following series resistance:

- R_{eex} —constant
- R_{ein} —constant

The collector includes the following series resistance:

- R_{cex} —constant
- R_{cin} —constant

The conductivity modulation of the base resistances is derived from the fact that the voltage drop across the epitaxial layer, is inversely proportional to the electron concentration under the emitter and collector.

Base resistance under the emitter:

$$R_{be} = R_{bec} + \frac{2 \times R_{bev}}{\sqrt{1 + 16 \times \frac{I_f^2}{I_k}}}$$

Base resistance under the collector:

$$R_{bc} = R_{bcc} + \frac{2 \times R_{bcv}}{\sqrt{1 + 16 \times \frac{I_r^2}{I_k}}}$$

The R_b resistance models the ohmic leakage, across the substrate-base junction.

Noise Equations

For noise analysis current sources are added to the small signal equivalent circuit. In these equations:

- f represents the operation frequency of the transistor.
- Δf is the bandwidth.

When measured at 1 Hz, a noise density is obtained.

Thermal Noise

$$iN_{REEX}^2 = \frac{4 \cdot k \cdot T_k}{R_{EEX}} \cdot \Delta f$$

$$iN_{REIN}^2 = \frac{4 \cdot k \cdot T_k}{R_{EIN}} \cdot \Delta f$$

$$iN_{RCIN}^2 = \frac{4 \cdot k \cdot T_k}{R_{CIN}} \cdot \Delta f$$

$$iN_{RCEX}^2 = \frac{4 \cdot k \cdot T_k}{R_{CEX}} \cdot \Delta f$$

$$iN_{RBE}^2 = \frac{4 \cdot k \cdot T_k}{R_{BE}} \cdot \Delta f$$

$$iN_{RBC}^2 = \frac{4 \cdot k \cdot T_k}{R_{BC}} \cdot \Delta f$$

$$iN_{RSB}^2 = \frac{4 \cdot k \cdot Tk}{RSB} \cdot \Delta f$$

Lateral Collector Current Shot Noise

$$iN_{CLAT}^2 = 2 \cdot q \cdot |I_{FLAT} - I_{RLAT}| \cdot \Delta f$$

Vertical Collector Current Shot Noise

$$iN_{CVER}^2 = 2 \cdot q \cdot |I_{FVER} - I_{RVER}| \cdot \Delta f$$

Forward-base Current Shot Noise and 1/f Noise

$$iN_B^2 = 2 \cdot q \cdot |I_{RE} - I_{LE}| \cdot \Delta f + \frac{KF \cdot MULTI^{1-AF} \cdot |I_{RE} \cdot I_{LE}|^{AF}}{f} \cdot \Delta f$$

Temperature Dependence of Parameters

$$Tk = T_{ref} + 273.16$$

$$Tn = \frac{Temp}{T_{ref} + 273.16}$$

$$Ti = \frac{1}{T_{ref} + 273.16} - \frac{1}{Temp}$$

Series Resistance

$$RCIN_T = RCIN \times T_N^{SPC}$$

$$RBCC_T = RBCC \times T_N^{SNBN}$$

$$RBCV_T = RBCV \times T_N^{SNB}$$

$$RBEC_T = RBEC \times T_N^{SNBN}$$

$$RBEV_T = RBEV \times T_N^{SNB}$$

The BJT LEVEL=10 model assumes that REEX and RCEX are temperature independent.

Depletion Capacitances

$$VDxt = -3k \frac{TEMP}{q} \cdot \ln(T_N) + VDx \cdot T_N + (1 - T_N) \cdot Vgap$$

$$CJ_{XT} = CJx \cdot \left(\frac{VDx}{VD_{XT}} \right)^{PX}$$

Emitter-base Junction

Vgap=VGEB, x=E

Collector-base Junction

Vgap=VGCB, x=C

Substrate-base Junction

Vgap=VGSB, x=S

Temperature Dependence of Other Parameters

$$VD_T = -3k \frac{TEMP}{q} \cdot \ln(T_N) + VD \cdot T_N + (1-T_N) \cdot VGB$$

$$EAF_{LT} = EAF_L \cdot \sqrt{\frac{VD_T}{VD}}$$

$$EAR_{LT} = EAR_L \cdot \sqrt{\frac{VD_T}{VD}}$$

$$EAF_{VT} = EAF_V \cdot \sqrt{\frac{VD_T}{VD}}$$

$$EAR_{VT} = EAR_V \cdot \sqrt{\frac{VD_T}{VD}}$$

$$IS_T = IS \cdot T_{N(4.0-SPB)} \cdot \exp(q \cdot VGB \cdot T_I / k)$$

$$BF_T = BF \cdot T_{N(AE-SPB)} \cdot \exp\{q \cdot (VGB-VGE) \cdot T_I / k\}$$

$$IBF_T = IBF \cdot T_N^2 \cdot \exp\{q \cdot (VGJE / 2) \cdot T_I / k\}$$

$$IK_T = IK \cdot T_{N(1-SPB)}$$

$$BR_T = BR \cdot \frac{BF_T}{BF}$$

$$IBR_T = IBR \cdot \frac{IBF_T}{IBF}$$

$$ISS_T = ISS \cdot T_N^2 \cdot \exp\{q \cdot VGSB \cdot T_I / k\}$$

$$TLAT_T = TLAT \cdot T_{N(SP-1.0)}$$

$$TFVR_T = TFVR \cdot \frac{TLAT_T}{TLAT}$$

$$TFN_T = TFN \cdot T_{N(SX-1.0)}$$

$$TRVR_T = TRVR \cdot \frac{TLAT_T}{TLAT}$$

$$TRN_T = TRN \cdot \frac{TFN_T}{TFN}$$

All other model parameters are temperature-independent.

LEVEL=11 UCSD HBT Model

The UCSD High Speed Devices Group in collaboration with the HBT Model Working Group, has been developing better SPICE models for heterojunction bipolar transistors (HBTs). The PrimeSim Continuum simulator implementation of the UCSD HBT MODEL is based on the website: <http://hbt.ucsd.edu>

- [Usage Notes](#)
- [LEVEL=11 Element Syntax](#)
- [Model Equations](#)
- [Equivalent Circuit](#)
- [Example Model Statement for BJT LEVEL=11](#)

Usage Notes

The following information applies to the PrimeSim Continuum device model for the UCSD HBT device:

- Set BJT `LEVEL=11`.
- The default room temperature is 25o C in the PrimeSim Continuum simulator, but is 27o C in most other simulators. When comparing to other simulators, do one of the following:
 - set the simulation temperature to 27, or
 - `set TEMP 27`, or
 - `set .OPTION TNOM=27`
- The set model parameter should always include the model reference temperature, `TREF`. The default value for `TREF` is 27.

- You can use `DTEMP` with this model to increase the temperature of individual elements, relative to the circuit temperature. Set its value on the element line.
- The HBT (BJT `LEVEL=11`) model includes self-heating effects. If you turn on self-heating, then set `RTH` to more than zero and `SELFT` to 1 in the model card.

LEVEL=11 Element Syntax

```
Qxxx nc nb ne [ns] mname [AREA=val] [OFF] [VBE=val]  
+ [VCE=val] [M=val] [DTEMP=val]
```

Parameter	Description
Qxxx	BJT element name. Must begin with Q, which can be followed by up to 1023 alphanumeric characters.
nc	Collector terminal node name or number.
nb	Base terminal node name and number.
ne	Emitter terminal node name or number.
ns	Substrate node name or number.
t	Self-heating node name or number.
mname	BJT model name reference.
AREA	Normalized emitter area.
OFF	Sets initial condition to OFF for this element in DC analysis. You cannot use OFF with VBE or VCE.
VBE	Initial internal base-emitter voltage.
VCE	Initial internal collector-emitter voltage.
M	Multiplier to simulate multiple BJTs in parallel.
DTEMP	Difference between the temperature of the element and circuit.

Table 114 BJT LEVEL=11 Parameters

Parameter	Unit	Default	Description
AFN	-	1	BE flicker noise exponent for current
BF	-	10000	Forward ideal current gain
BFN	-	1	BE flicker noise exponent for frequency

Table 114 BJT LEVEL=11 Parameters (Continued)

Parameter	Unit	Default	Description
BKDN	logic	false	Flag indicating to include BC breakdown
BR	-	10000	Reverse ideal current gain
BVC	V	1000	Collector-base breakdown voltage BVcbo
CCMIN	F	0	Minimum value of intrinsic BC Cj
CEMIN	F	0	Minimum BE capacitance
CJC	F	0	Intrinsic BC depletion capacitance at zero bias
CJCX	F	0	Extrinsic BC depletion capacitance at zero bias
CJE	F	0	BE depletion capacitance at zero bias
CJS	F	0	Collector-substrate depletion capacitance (0 bias)
CTH	C/Joule	0	Thermal capacitance of device.
CXMIN	F	0	Minimum extrinsic Cbc
EAA	V	0	Added activation energy, ISE temp dependence
EAB	V	0	Added activation energy, ISC temp dependence
EAC	V	0	Activation energy, ISB temperature dependence
EAE	V	0	Activation energy, ISA temperature dependence
EAX	V	0	Added activation energy, ISEX temp dependence
EG	V	1.5	Activation energy for IS temperature dependence
FA	-	0.9	Factor for specification of avalanche voltage
FC	-	0.8	Factor for start of high bias BC Cj approximation
FCE	-	0.8	Factor for start of high bias BE Cj approximation
FEX	-	0	Factor to determine excess phase
ICRIT0	A	1e3	Critical current for intrinsic Cj variation
ICS	A	1e-30	Saturation value for collector-substrate current
IK	A	1e10	Knee current for dc high injection effect
IKRK	A	1e3	Characteristic current for Kirk effect

Table 114 BJT LEVEL=11 Parameters (Continued)

Parameter	Unit	Default	Description
IS	A	1e-25	Saturation value for forward collector current
ISA	A	1e10	Collector current EB barrier limiting current
ISB	A	1e10	Collector current BC barrier limiting current
ISC	A	1e-30	Saturation value for intrinsic bc junction current
ISCX	A	1e-30	Saturation value for extrinsic bc junction current
ISE	A	1e-30	Saturation value for non-ideal base current
ISEX	A	1e-30	Saturation value for emitter leakage diode
ITC	A	0	Characteristic current for TFC
ITC2	A	0	Characteristic current for TFC
KFN	-	0	BE flicker noise constant
MJC	-	0.33	Exponent for voltage variation of Intrinsic BC Cj
MJCX	-	0.33	Exponent for voltage variation, Extrinsic BC Cj
MJE	-	0.5	Exponent for voltage variation of BE Cj
MJS	-	0.5	Exponent for voltage variation of CS Cj
NA	-	2	Collector current EB barrier ideality factor
NB	-	2	Collector current BC barrier ideality factor
NBC	-	8	Exponent for BC multiplication factor vs voltage
NC	-	2	Ideality factor for intrinsic bc junction current
NCS	-	2	Ideality factor for collector-substrate current
NCX	-	2	Ideality factor for extrinsic bc junction current
NE	-	2	Ideality factor for non-ideal forward base current
NEX	-	2	Ideality factor for emitter leakage diode
NF	-	1	Forward collector current ideality factor
NR	-	1	Reverse current ideality factor
RBI	ohm	0	Intrinsic base resistance

Table 114 BJT LEVEL=11 Parameters (Continued)

Parameter	Unit	Default	Description
RBX	ohm	0	Extrinsic base resistance
RCI	ohm	0	Intrinsic collector resistance
RCX	ohm	0	Extrinsic collector resistance
RE	ohm	0	Emitter resistance
REX	ohm	0	Extrinsic emitter leakage diode series resistance
RTH	C/W	0	Thermal resistance, device to thermal ground
SELFT	-	0	Flag. Indicates whether to use self-heating. • 0 (default) does not use self-heating • 1 turns on the self-heating feature
TBCXS	S	0	Excess BC heterojunction transit time
TBEXS	S	0	Excess BE heterojunction transit time
TFB	S	0	Base transit time
TFC0	S	0	Collector forward transit time
TKRK	S	0	Forward transit time for Kirk effect
TNC	-	0	Coefficient for NC temperature dependence
TNE	-	0	Coefficient for NE temperature dependence
TNEX	-	0	Coefficient for NEX temperature dependence
TR	S	0	Reverse charge storage time, intrinsic BC diode
TREF	C	27	Temperature at which model parameters are given
TRX	S	0	Reverse charge storage time, extrinsic BC diode
TVJC	V/C	0	Coefficient for VJC temperature dependence
TVJCX	V/C	0	Coefficient for VJCX temperature dependence
TVJE	V/C	0	Coefficient for VJE temperature dependence
TVJS	V/C	0	Coefficient for VJS temperature dependence
VAF	V	1000	Forward Early voltage
VAR	V	1000	Reverse Early voltage

Table 114 BJT LEVEL=11 Parameters (Continued)

Parameter	Unit	Default	Description
VJC	V	1.4	Intrinsic BC diode built-in potential for Cj estimation
VJCX	V	1.4	Extrinsic BC diode built-in potential for Cj estimation
VJE	V	1.6	BE diode built-in potential for Cj estimation
VJS	V	1.4	CS diode built-in potential for Cj estimation
VKRK	V	1e3	Characteristic Voltage for Kirk effect
VTC	V	1e3	Characteristic voltage for TFC
XCJC	-	1	Factor for partitioning extrinsic BC Cj
XRB	-	0	Exponent for RB temperature dependence
XRC	-	0	Exponent for RC temperature dependence
XRE	-	0	Exponent for RE temperature dependence
XREX	-	0	Exponent for REX temperature dependence
XTB	-	2	Exponent for beta temperature dependence
XTI	-	2	Exponent for IS temperature dependence
XTIKRK	-	0	Exponent for IKRK temperature dependence
XTITC	-	0	Exponent for ITC temperature dependence
XTITC2	-	0	Exponent for ITC2 temperature dependence
XTTF	-	0	Exponent for TF temperature dependence
XTTKRK	-	0	Exponent for TKRK temperature dependence
XTVKRK	-	0	Exponent for VKRK temperature dependence

Model Equations

This section describes the model equations for the PrimeSim Continuum simulator BJT LEVEL=11 model.

Current Flow

There are seven different current flow calculations for the BJT LEVEL=11 device model.

- Intrinsic collector current contributions. This model computes the electron flow between Ei and Ci nodes by using equations similar to the Gummel-Poon model with modifications to take into account the potential spike that can appear at the base-emitter or base-collector junctions of HBTs. This model separates the electron current into forward and reverse components, I_{cf} and I_{cr} .

$$I_{cf} = I_S * [\exp(qV_{bei}/NF/KT) - 1] / D \quad I_{cr} = I_S * [\exp(qV_{bci}/NR/KT) - 1] / D$$

In these equations:

$$D = q_b + I_S * \exp(qV_{bei}/NA/KT) / I_{SA} + I_S * \exp(qV_{bci}/NB/KT) / I_{SB}$$

I_{SA} , I_{SB} , NA and NB are new parameters. I_{SA} and I_{SB} approximate the transition currents, from base-transport controlled to potential-barrier controlled, current flow.

q_b partially retains the standard BJT model form (a fractional increase in the base charge associated with the bias changes).

$$q_b = q_{1/2} * [1 + (1 + 4 * q_2^2)^{0.5}] \quad q_1 = 1 / [1 - V_{bci}/V_{AF} - V_{bei}/V_{AR}] \quad q_2 = I_S / I_K * [\exp(qV_{bei}/NF/KT) - 1]$$

In the preceding equations, q_b omits the reverse knee current contribution. As noted in the following, q_b is not used to define the ac model in the fashion of the Gummel-Poon model.

The total collector current I_{cc} is:

$$I_{cc} = I_{cf} - I_{cr}$$

This formulation uses the I_S , NF , V_{AF} , V_{AR} , and I_K parameters, established in the SPICE BJT model in addition to the I_{SA} , I_{SB} , NA , and NB parameters described above.

- Intrinsic Base-Emitter Diode. Ideal and non-ideal components are included:

$$I_{bei} = I_{cf} / BF + I_{SE} * [\exp(q V_{bei} / NE / KT) - 1]$$

- Extrinsic Base-Emitter Diode. The LEVEL=11 model includes a diode connected between the Ex and E nodes, and an associated series resistance (R_{ex}). You can use the diode and its resistance to model contributions from emitter edges.

$$I_{bex} = I_{SEX} * [\exp(q V_{bex} / NEX / KT) - 1]$$

- Intrinsic Base-Collector Diode. Ideal and non-ideal components are included:

$$I_{bci} = I_{cr} / BR + I_{SC} * [\exp(q V_{bci} / NC / KT) - 1]$$

- Intrinsic Base-Collector Breakdown Current. I_{bk} is current between the collector and base nodes, generated due to avalanche breakdown of the base-collector junction. If you set the $BKDN$ parameter to true, then I_{bk} is determined according to:

$$I_{bk} = (M_f - 1) * I_{cf} \text{ Otherwise, } I_{bk} = 0$$

The preceding equations use the following definitions:

- M_f is the multiplication factor associated with the BC junction at the specified voltage.
- I_{cf} is the forward electron current (as computed above in the absence of multiplication).

M_f is calculated with a physically based expression, modified to avoid the singularity at $V_{bci} = -V_{BC}$.

M_f depends exclusively on the intrinsic base-intrinsic collector voltage, V_{bci} . If $-V_{bci}$ closely approaches or exceeds V_{BC} ($-V_{bci} > F_A * V_{BC}$ with F_A typically chosen to be 0.95), then the multiplication factor is computed according to a constant slope expression.

$M_f = 1 / [1 - (-V_{bci}/V_{BC})^{NBC}]$ for $K_{Top}/q < -V_{bci} < F_A * V_{BC}$ $M_f = 1$ for $-V_{bci} > K_{Top}/q$
 $M_f = M_{fl} + g_l * (-V_{bci} - F_A * V_{BC})$ for $-V_{bci} > F_A * V_{BC}$

In the preceding equations, M_{fl} and g_l are the values of M_f and its derivative with respect to voltage, evaluated at the voltage $-V_{bci} = F_A * V_{BC}$:

$M_{fl} = 1 / (1 - F_A^{NBC})$ $g_l = M_{fl} * (M_{fl} - 1) * NBC / (F_A * V_{BC})$

- Extrinsic Base-Collector Diode. This diode has customary I-V characteristics with its own saturation current and ideality factor.

$I_{bcx} = I_{SCX} * [\exp(q V_{bcx} / N_{CX} / K_{Top}) - 1]$

- Substrate-Extrinsic Collector Diode. This diode allows for conducting substrates. Use it primarily for SiGe HBTs.

$I_{cs} = I_{CS} * [\exp(-q V_{cs} / N_{CS} / K_{Top}) - 1]$

In accordance with the model topology, the external currents through the E, B, and C nodes are:

$I_b = I_{bei} + I_{bex} - I_{bk} + I_{bci} + I_{bcx}$ $I_c = I_{cc} + I_{bk} - I_{bci} - I_{bcx} - I_{cs}$

Charge Storage

This section describes the following different charge storage calculations for the PrimeSim Continuum simulator BJT LEVEL=11 device model.

- **Base-Emitter Charge.** The overall charge stored at the base-emitter junction has components associated with the base-emitter depletion layer:
 - Q_{bej} , which is current-independent.
 - Q_{bediff} , a collector current-dependent charge. Q_{bediff} corresponds to a portion of the base charge, and the (collector current-dependent) base-collector charge.

$$Q_{be} = Q_{bej} + Q_{bediff}$$

- **Base-Emitter Depletion Charge, Q_{bej} .** The depletion charge, Q_{bej} , follows equations standard for SPICE, modified to allow specification of a minimum capacitance C_{EMIN} (corresponding to reach-through to an n+ layer).

There is often an extra component of charge storage at the base-emitter heterojunction of HBTs, associated with a minimum in the conduction band energy profile.

Q_{bej} is computed using DepletionCapMod.

Define:

$$V_{min} = V_{JE} * [1 - (C_{JE}/C_{EMIN}) (1/M_{JE})]$$

(the critical voltage for attaining the minimum capacitance value)

If $V_{bei} < F_{CE} * V_{JE}$ and $V_{bei} < V_{min}$:

$$Q_{bej} = C_{EMIN} * (V_{bei} - V_{JE}) + C_{EMIN} * V_{JE} * M_{JE} / (M_{JE} - 1) * (C_{JE}/C_{EMIN}) (1/M_{JE})$$

$$C_{bej} = dQ_{bej} / dV_{bei} = C_{EMIN}$$

If $V_{bei} < F_{CE} * V_{JE}$ and $V_{bei} > V_{min}$:

$$Q_{bej} = -C_{JE} * V_{JE} * (1 - V_{bei}/V_{JE}) (1 - M_{JE}) / (1 - M_{JE})$$

$$C_{bej} = C_{JE} * (1 - V_{bei}/V_{JE}) (-M_{JE})$$

If $V_{bei} > F_{CE} * V_{JE}$, and $C_{JE} > C_{EMIN} * (1 - F_{CE}) M_{JE}$:

$$Q_{bej} = -C_{JE} * V_{JE} / (1 - F_{CE}) M_{JE} * [(1 - F_{CE}) / (1 - M_{JE}) + F_{CE} - V_{bei}/V_{JE}$$

$$- M_{JE} * (F_{CE} - V_{bei}/V_{JE})^2 / 2 / (1 - F_{CE})]$$

$$C_{bej} = C_{JE} / (1 - F_{CE}) M_{JE} * [1 + M_{JE} * (V_{bei}/V_{JE} - F_{CE}) / (1 - F_{CE})]$$

If $V_{bei} > F_{CE} * V_{JE}$, and $C_{JE} < C_{EMIN} * (1 - F_{CE}) M_{JE}$,

$$Q_{bej} = C_{EMIN} * (V_{bei} - V_{JE}) + C_{EMIN} * V_{JE} * M_{JE} / (M_{JE} - 1)$$

$$+ (C_{JE}/C_{EMIN}) (1/M_{JE}) + C_{JE} * V_{JE} * (V_{bei}/V_{JE} - F_{CE})^2 * M_{JE} / 2 / (1 - F_{CE}) (M_{JE} + 1)$$

$$C_{bej} = C_{EMIN} + C_{JE} * V_{JE} * M_{JE} * (V_{bei}/V_{JE} - F_{CE}) / (1 - F_{CE}) (M_{JE} + 1)$$

- **Base-Emitter Diffusion Charge, Qbediff.** The diffusion charge in HBTs is associated with contributions from minority carriers in the base, and from mobile charge in the collector depletion region. In homojunction transistors, diffusion charge storage in the emitter is also present. The LEVEL=11 model evaluates the base and collector-depletion region contributions separately (if necessary, the emitter charge storage can be associated with the base contribution).
 - Specify the base charge through the base transit time, T_{FB} . This transit time varies with bias through several mechanisms:
 - The Early effect causes a change in transit time with junction voltage.
 - In heterojunction transistors, there is frequently a minimum in the conduction band, on the base side of the base-emitter (and potentially base-collector) heterojunction. Minority carriers tend to accumulate in these potential wells.

The stored charge adds to the base charge (to a good approximation). In the lowest order, the charge stored is directly proportional to the collector current, and thus contributes to T_{FB} . For a greater degree of accuracy, the depth of the potential well on the emitter side varies with V_{be} . Similarly, the amount of charge stored at the base-collector side varies with V_{bc} .

The equations used to describe the effects are:

$$T_{FBt} = T_{FB} * (1 + V_{bei} / V_{AR} + V_{bci} / V_{AF}) + T_{BEXS} * \exp(-q(V_{bei} - V_{JE}) / N_A / K_{Top}) + T_{BCXS} * \exp(q(V_{bci} - V_{JC}) / N_B / K_{Top})$$

Note:

Different signs are associated with the BE and BC junction effects. The value of the T temperature to describe these effects is assumed to be T_{op} .

You can use any of these methods to specify collector charge:

- A part is specified by the T_{FC0} transit time parameter, modified by the q_{cc} velocity modulation factor to account for voltage and current dependencies.
- A part of the mobile charge is specified in the calculation of base-collector depletion region charge. To calculate this part, Q_{bcm} , an expression for the collector current-dependent base-collector depletion charge is developed. Then the current-independent part is subtracted off (as discussed in the next section).
- A separate charge term, Q_{krk} , is associated with the Kirk effect.

$$Q_{fdiff} = I_{cf} * f_{tt} * (T_{FBt} + T_{FC0} / q_{cc}) + Q_{bcm} + Q_{krk}$$

$$f_{tt} = r_{TXTTF}$$

q_{cc} is a factor describing bias dependence of electron velocity in the BC depletion region:

$$q_{cc} = [1 + (I_{cf}/I_{TC})^2] / [1 + (I_{cf}/I_{TC2})^3 + (V_{JCI} - V_{bci})/V_{TC}]$$

- I_{TC} is the threshold current for the velocity profile modulation effect.
- I_{TC2} is a higher current at which the velocity profile modulation peaks (and the cutoff frequency begins to roll-off).
- V_{TC} provides a voltage (or electric field) dependence of the carrier velocity.

$$I_{TC} = I_{TC@Tnom} * r_{TXTI_{TC}} \quad I_{TC2} = I_{TC2@Tnom} * r_{TXTI_{TC2}}$$

The following expression calculates the charge storage associated with the Kirk effect:

$$Q_{krk} = T_{KRK} * I_{cf} * \exp[V_{bci}/V_{KRK} + I_{cf}/I_{KRK}]$$

To account for excess phase, a fraction $(1-FEX)$ of the current-dependent forward charge (Q_{fdiff}) is associated with the BE junction, while the remainder is associated with the intrinsic BC junction.

$$Q_{bediff} = (1-FEX) * Q_{fdiff}$$

Note:

Q_{fdiff} (and thus Q_{bediff}) depends on V_{bci} , through the terms involving I_{cf} , q_{cc} , Q_{krk} , and Q_{bcm} . As a result, a trans-capacitance is implied in the ac model. Similarly, Q_{bcdiff} depends on V_{bei} , implying another trans-capacitance.

- Intrinsic Base-Collector Charge, Q_{bci} . Charge stored at the intrinsic base-collector junction includes:
 - Depletion charge from the junction region.
 - Diffusion charge associated with normal operation of the transistor.
 - Diffusion charge associated with reverse operation of the device.

$$Q_{bci} = Q_{bcj} + TRI * I_{cr} + FEX * Q_{fdiff}$$

Although the charge in the depletion region depends on I_c , this section describes the portion corresponding to the $I_c=0$ condition. Subsequently, the proper I_c dependent contribution is considered, and included in Q_{bcm} (a charge that is part of Q_{fdiff}).

- Intrinsic base-collector depletion charge, Q_{bcj} . When $I_c=0$, the depletion charge is calculated using the same algorithm as applied to Q_{bej} (which accounts for a minimum of capacitance when the n- collector is depleted).

- Intrinsic base-collector diffusion charge. For reverse operation, a diffusion capacitance is implied by the TRI term in the Q_{bci} equation. Here TRI is the effective reverse transit time, which is assumed to be bias-independent. The associated reverse diffusion capacitance is:

$$C_{bcrdiff} = TRI * dI_{bci} / dV_{bci}$$

For operation also includes diffusion capacitance in a manner similar to base-emitter capacitance with a partitioning specified by the excess phase factor, FEX.

The terms associated with $I_{cf} * f_{tt} * (TFC0 / q_{cc}) + Q_{krk}$ have already been discussed above for calculating Q_{bediff} . The next section describes the Q_{bcm} portion.

- Q_{bcm} . This charge is the difference between the “proper” I_{cf} -dependent charge in the BCi depletion region (called Q_{bcf}), and the BCi depletion charge computed above (Q_{bcj}), assuming that $I_{cf}=0$.

$$Q_{bcm} = Q_{bcf} - Q_{bcj}$$

To properly compute Q_{bcf} , a formulation of the depletion region charge (similar to that used above) is used with the modification that the C_J parameter (zero bias capacitance) can depend on the I_{cf} collector current. This corresponds to the physical phenomenon of varying charge density in the depletion region as a result of the mobile electron charge in that region.

The current-dependent C_J parameter is termed C_{JCH} ; its form is:

$$C_{JCH} = C_{JC} * \text{sign}(1 - I_{cf} / I_{CRIT}) * \text{ABS}(1 - I_{cf} / I_{CRIT})^{MJC}$$

In this equation, I_{CRIT} is a critical current, at which the effective charge density in the BC depletion region vanishes (and the capacitance C_{bci} drops dramatically). I_{CRIT} is dependent on temperature and bias conditions, according to:

$$I_{CRIT} = I_{CRIT0} * q_{cc} / f_{tt}$$

In the preceding equation, f_{tt} and q_{cc} are the temperature-dependence, and I_{cf} and V_{cb} are the dependence parameters described above.

Using this formulation, the current dependence of the BC capacitance is included (although it is partially assigned to the BE junction charge, and partially to the BC junction, through the FEX excess-phase parameter).

You can extract I_{CRIT} and associated parameters from measurements of C_{bc} versus I_c .

Note:

These parameters also control some of the components of the forward transit time.

A delay time is associated with specifying `ICRIT`:

$$TFC1 = CJC * VJC * MJC / (MJC - 1) / ICRIT$$

Use the `ICRIT` parameter carefully, generally in conjunction with selecting `TFC0` and `CJCI` in such a way that the sum `TFB + TFC0 + TFC1` provide a reasonable estimate of charge storage, similar to `TF` in Gummel-Poon SPICE.

- **Extrinsic Base-Collector Charge, `Qbcx`.** The `Qbcx` stored charge consists of a depletion charge and a diffusion charge.

Standard SPICE does not use the diffusion charge component. However, this component can be an important contribution to saturation stored-charge in many HBTs (in addition to the contribution associated with the intrinsic base-collector junction).

The corresponding charge storage time, `TRX`, might be different from the intrinsic time, `TRI`. This difference occurs because of implant-induced recombination, surfaces, or other structural changes.

The depletion charge corresponds to a standard depletion region expression (without considering charge density modulation due to current), modified to allow for a minimum value of capacitance under a reach-through condition.

Furthermore, as indicated in the following, if you assign a value other than unity to the `XCJC` variable, then the depletion charge is partitioned between the Bx-Cx capacitance and the B-Cx capacitance.

$$Qbcx = TRX * Ibcx + XCJC * Qbcxo$$

In the preceding equation, `Qbcxo` is the depletion charge.

As a result the dependences of `Ibcx` on `Vbcx`, a diffusion capacitance results from the formulation:

$$Cbcxdiff = TRX * dIbcx / dVbcx$$

Base-Extrinsic Collector Charge (`Qbcxx`), and Treatment of `XCJC`

In standard SPICE, `XCJC` indicates the fraction of overall `Cbc` depletion capacitance that should be associated with the intrinsic base node. The remaining fraction (`1 - XCJC`) is attached to the base terminal. HBT Spice uses a similar assignment: the depletion charge associated with the extrinsic base-collector junction is partitioned between the Bx node and the B node:

$$Qbcx = TRX * Ibcx + XCJC * Qbcxo$$

has been defined above, between the Bx and Cx nodes, and charge

$$Qbcxx = (1 - XCJC) * Qbcxo$$

is assigned between the B and Cx nodes. The Qbcxxo charge is computed with the same algorithm as for Qbcxo by using Vbcxx (rather than Vbcx) as the voltage.

- **Collector-Substrate Charge, Q_{cs} .** This corresponds to a depletion charge, formulated in the standard SPICE fashion:

For $V_{cs} > -FC * V_{JS}$,

$$Q_{cs} = -CJS * V_{JS} * (1 + V_{cs}/V_{JS}) (1 - MJS) / (1 - MJS) \quad Ccs = CJS * (1 + V_{cs}/V_{JS}) (-MJS)$$

For $V_{cs} < -FC * V_{JS}$,

$$Q_{cs} = -CJS * V_{JS} / (1 - FC) MJS * [(1 - FC) / (1 - MJS) + FC + V_{cs}/V_{JS} - MJS/2 / (1 - FC) * (FC + V_{cs}/V_{JS})^2]$$

$$Ccs = CJS * (1 - FC) (-MJS) * [1 - MJS / (1 - FC) * (FC + V_{cs}/V_{JS})]$$

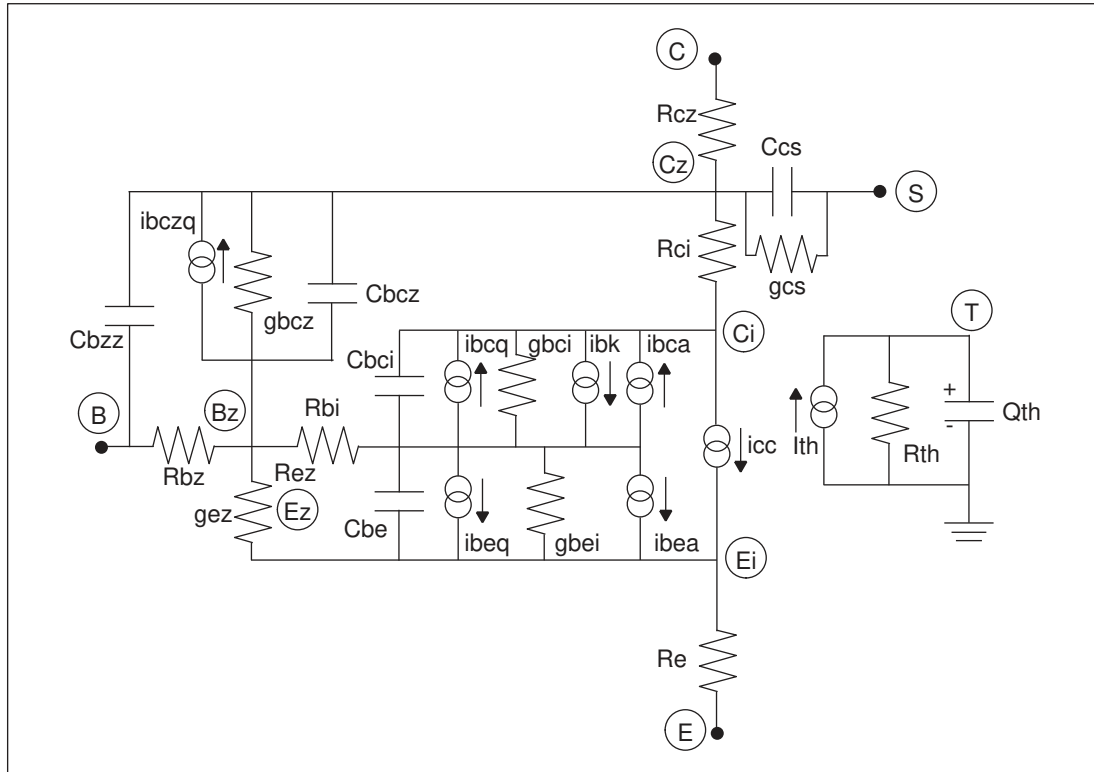
Noise

The LEVEL=11 model includes noise current generators, similar to those in standard Spice. The noise current generators have magnitudes in units of A²/Hz, and are computed based on 1Hz bandwidth. The noise sources are placed in parallel with corresponding linearized elements in the small signal model. Sources of 1/f noise have magnitudes that vary with the frequency (f); you can use a BFN exponent, if you do not see the exact f⁻¹ behavior.

```
inc2=2*q*Icc
inb2= 2 * q * Ibe + KFN * IbeAFN / f BFN
inre2= 4 * K * Td / RE
inrbx2= 4 * K * Td / RBX
inrbi2= 4 * K * Td / RBI
inrcx2= 4 * K * Td / RCX
inrci2= 4 * K * Td / RCI
inrex2= 4 * K * Td / REX
```

Equivalent Circuit

Figure 39 Circuit Diagram for Small-signal HBT Model



Example Model Statement for BJT LEVEL=11

```
.model hbt npn level=11
+ IS=1.2E-18 NF=1 NR=1 BF=200
+ BR=5 VAF=60 VAR=20 ISE=1E-17
+ NE=1.4 ISEX=4E-24 NEX=1.3 ISCX=1E-14
+ NCX=2 ISC=1E-16 NC=2 NA=10
+ ISA=2.18E-10 NB=10 ISB=1E10 RE=16
+ REX=20 RBI=20 BVC=28 NBC=6
+ FA=0.995 RCX=30 RCI=20 CJE=1.8E-14
+ VJE=1.45 CXMIN=1E-16 MJE=0.5 FC=0.8
+ ICRIT0=0.23 CCMIN=3E-15 TR=3.5E-10 VJCX=1.4
+ CJCX=8E-15 MJCX=0.35 XCJC=1 VJS=1.4
+ CJS=5E-16 MJS=0.01 CTH=1E-6 RTH=0
+ EG=1.645 XTI=0 XTB=-1.8 EAA=-0.495
+ EAB=-0.1 EAE=0.105 TNE=0 EAC=0.34
+ XTTF=1.5 ICS=1E-30 NCS=2 CEMIN=1E-15
+ FCE=0.8 TFB=2E-12 TFC0=2.5E-11 TBEXS=1E-14
+ ITC=7E-3 ITC2=0.014 VTC=40 TKRK=5E-13
```

```
+ VKRK=10   IKRK=0.012   TRX=3.5E-10   FEX=0
+ XTITC=1.5   XTITC2=1   TREF=25   CJC=7E-15
+ VJC=1.4   MJC=0.35
```

LEVEL=13 HICUM/L0 Model

HICUM/L0 is a simplified bipolar transistor model that combines the simplicity of the SPICE Gummel-Poon Model (SGPM) with various improvements from HICUM. The HICUM/L0 model is implemented as `LEVEL=13` in the PrimeSim Continuum BJT models.

- [HICUM/L0 Model Advantages](#)
- [HICUM/L0 Model vs. HICUM LEVEL=2 Model](#)
- [HICUM/L0 Version 2.10 Updates](#)
- [HICUM/L0 Version 2.00 Updates](#)
- [HICUM/L0 Version 1.32 Updates](#)
- [HICUM/L0 Version 1.31 Updates](#)
- [HICUM/L0 Version 1.3 Updates](#)
- [LEVEL=13 Element Syntax](#)
- [LEVEL=13 Model Parameters](#)

HICUM/L0 Model Advantages

Major features of HICUM/L0 are:

- Strongly circuit-design oriented and easy to understand for circuit designers
- Sufficiently accurate for many applications
- Computationally efficient and fast
- Allows a fast parameter extraction for single transistors
- Makes use of the advanced capabilities of HICUM `LEVEL=2` and the related knowledge base for generating geometry scalable and statistical models
- Offers an easy migration path from a conventional, single-transistor-based, to a process-based geometry scalable parameter extraction and model usage to meet today's requirements for advanced integrated circuit design.

The model parameters have a clear (physical) meaning and many of them are similar to HICUM `LEVEL=2` parameters.

HICUM/L0 Model vs. HICUM LEVEL=2 Model

The following are the differences between HICUM/L0 and HICUM `LEVEL=2`:

- The perimeter base node has been eliminated by properly merging the respective internal and external counterparts of the BE depletion capacitance (C_{JE}), the base resistance (R_B), the BC depletion capacitance (C_{JC}), and the base current components across the BE and BC junction
- BE tunneling current, substrate coupling network, parasitic substrate transistor, and capacitance for modeling AC emitter current crowding in HICUM `LEVEL=2` have been omitted.

HICUM/L0 Version 2.10 Updates

The following changes are made with the release of v2.10:

- Changed the IBCIS default value from 0 to 1e-16, to be compatible with HICUM/L2 and SPICE GP model.
- Turned off the temperature dependence of emitter storage time `TEF0` by default, for compatibility with HICUM/L2.
- Changed the ZEDC upper limit to 1.0, to avoid possible arithmetic overflow.
- Phased out “DC” base-emitter charge.
- `MinR`, as defined by CMC guidelines.
- New temperature dependence for IQFH (new parameter `ZETA IQFH`).

HICUM/L0 Version 2.00 Updates

Refer the official document for detailed update.

HICUM/L0 Version 1.32 Updates

The following changes are made with the release of v1.32:

- Change in the type selection (NPN or PNP) : supply a single type parameter (`type = 1` for NPN and `type=-1` for PNP)
- Parasitic PNP transistor also takes the device type into account
- Usage of `rth` instead of `rth_t` for power calculation

HICUM/L0 Version 1.31 Updates

The following change is made with the release of v1.31:

- Set model parameter FLNQS=1 to support vertical non-quasi-static (NQS) effect

HICUM/L0 Version 1.3 Updates

The following changes were made with the release of v1.3.:

- Improvement to transfer current, including bias and temperature dependencies.
- Bug fixes, as follows:
 - Fixing of model parameter typos:
 - AHQ was given proper SPICE name "AHQ"
 - ZETAIQF was given proper SPICE name "ZETAIQF"
 - Addition of "gmin" (=1e-12) between internal collector emitter node of transfer current iT for convergence enhancement in case of reverse biased junctions
 - The CC function is defined as $cc = c_{jci0_t_ii}/c_{jcii}$, which may cause numerical instability; Solution: a conditional loop as in L2 model.

```
if(cjcii > 0.0 && cjc_i0_t_ii > 0.0) begin
    cc = cjc_i0_t_ii/cjcii;
    qjci = qjci/cjc_i0_t_ii;
end else begin
    cc = 1.0;
    qjci = 0;
end
```

- Calculation for base resistance conductivity modulation
 - In the previous L0v1.2 model the calculation was done using DC charge (qje), which is physically incorrect => AC charge is being used.
 - "QJMODF" macro has been re-evaluated with AC parameters only.
 - Could lead to version incompatibility problem
- Limitation of junction related charge qj
 - junction related charge qj could go negative at $v_{be} < 0$ with the typically low VER values
 - Solution: Smoothing function as in L2 model has been applied to limit the charge

```
qj      = (1+qjci/vef+qjei/ver);  
a_bpt  = 0.05;  
b_q    = qj/a_bpt-1;  
qj_2   = 0.025*(1+(b_q +sqrt(b_q*b_q+1.921812))/2);
```

For complete details, see *Improving HICUM L0 v1.2 transfer current, including bias and temperature dependencies* at http://www.iee.et.tu-dresden.de/iee/eb/comp_mod.html

LEVEL=13 Element Syntax

Syntax

```
Qxxx nc nb ne [ns][nt]mname [area] [M=val] [DTEMP=val]
```

Parameter	Description
Qxxx	BJT element name
nc	Collector terminal node name or number
nb	Base terminal node name or number
ne	Emitter terminal node name or number
ns	Substrate terminal node name or number
nt	Self-heating node name or number
mname	BJT model name reference
area	Emitter area multiplying factor. Affects current, resistance, capacitance. Default is 1.
M	Multiplier to simulate multiple BJTs in parallel. Default is 1.
DTEMP / DT	Difference between the element temperature and the circuit temperature in degrees Celsius. Default is DTA (difference between the device temperature and the ambient analysis temperature). If you do not specify DTEMP, then DTEMP uses the DTA value.

LEVEL=13 Model Parameters

Table 115 lists the HICUM/L0 LEVEL=13 model parameters.

Table 115 BJT LEVEL=13 Model Parameters

Parameter	Unit	Default	Description
IS	A	1e-16	Transform saturation current

Table 115 BJT LEVEL=13 Model Parameters (Continued)

Parameter	Unit	Default	Description
MCF	-	1	Non-ideality coefficient of forward collector current
MCR	-	1	Non-ideality coefficient of inverse collector current
VEF	V	∞	Forward Early voltage
IQF	A	∞	Forward DC high-injection roll-off current
IQR	A	∞	Inverse DC high-injection roll-off current
IQFH	A	∞	High-injection correction current
TFH	-	∞	High-injection correction factor
CJE0	F	0	BE zero-bias depletion capacitance
VDE	V	0.9	BE built-in voltage
ZE	-	0.5	BE exponent factor
AJE	-	2.5	BE ratio of maximum to zero-bias value
CJC10	F	0	BC total zero-bias depletion capacitance
VDCI	V	0.7	BC built-in voltage
ZCI	-	0.4	BC exponent factor
VPTCI	V	1e+20	BC punch-through voltage
T0	sec	0	Low current transit time at $V_{BC}=0$
DT0H	sec	0	Base width modulation contribution
TBVL	sec	0	SCR width modulation contribution
TEF0	sec	0	Storage time in neutral emitter
GTE	-	1	Exponent factor for emitter transit time
THCS	sec	0	Saturation time at high current densities
AHC	-	0.1	Smoothing factor for current dependence
TR	sec	0	Storage time at inverse operation
RCI0	Ohm	150	Low-field collector resistance under emitter
VLIM	V	0.5	Voltage dividing ohmic and saturation region

Table 115 BJT LEVEL=13 Model Parameters (Continued)

Parameter	Unit	Default	Description
VPT	V	∞	Punch-through voltage
VCES	V	0.1	Saturation voltage
IBES	A	1e-18	BE saturation current
MBE	-	1	BE non-ideality factor
IRES	A	0	BE recombination saturation current
MRE	-	2	BE recombination non-ideality factor
IBCS	A	0	BC saturation current
MBC	-	1	BC non-ideality factor
KAVL	-	0	Avalanche prefactor
EAVL	-	1	Avalanche exponent factor
RBIO	Ohm	0	Internal base resistance value at zero-bias
VR0E	V	2.5	Forward Early voltage (normalization voltage)
VR0C	V	∞	Reverse Early voltage (normalization voltage)
FGEO	-	0.656	Geometry factor
RBX	Ohm	0	External base series resistance
CJCX0	F	0	Zero-base external BC depletion capacitance
VDCX	V	0.7	External BC built-in voltage
ZCX	-	0.4	External BC exponent factor
VPTX	V	1e+20	Punch-through voltage
FBC	-	1	Split factor= C_{JC10}/C_{JC0}
RE	Ohm	0	Emitter series resistance
RCX	Ohm	0	External collector series resistance
CBEPAR(CEOX)	F	0	Emitter-base isolation (overlap) capacitance
CBCPAR(CCOX)	F	0	Collector-base oxide capacitance
ISCS	A	0	SC saturation current

Table 115 BJT LEVEL=13 Model Parameters (Continued)

Parameter	Unit	Default	Description
AF	-	2	Flicker noise exponent factor
AHQ	-	0	Smoothing factor for the DC injection width
ALB	1/K	0	Relative temperature coefficient of forward current gain
ALCES	1/K	0	Relative temperature coefficient of VCES
ALEAV	1/K	0	Temperature coefficient of avalanche exponent factor
ALKAV	1/K	0	Temperature coefficient of avalanche prefactor
ALT0	1/K	0	First-order relative temperature coefficient of T0
ALVS	1/K	0	Relative temperature coefficient of saturation drift velocity
CJS0	F	0	Zero-bias SC depletion capacitance
CTH	Ws/K	0	Thermal capacitance
DT	°C	0	Temperature change for particular transistor
F1VG	V/K	-8.46e-5	Coefficient K1 in temperature-dependent bandgap equation
F2VG	V/K	3.042e-4	Coefficient K2 in temperature-dependent bandgap equation
FIQF	-	0	Flag to turn on voltage dependence of base-related critical current
KF	M^{1-AF}	0	Flicker noise coefficient (no unit only if AF=2)
KT0	1/K	0	Second-order relative temperature coefficient of T0
MSC	-	1	SC non-ideality factor
RTH	K/W	0	Thermal resistance
TNOM	°C	27	Temperature for which parameters are valid
VDS	V	0.3	SC built-in voltage
VER	V	∞	Reverse Early voltage
VGB	V	1.2	Bandgap voltage
VGC	V	1.2	Effective collector bandgap voltage
VGE	V	1.2	Effective emitter bandgap voltage

Table 115 BJT LEVEL=13 Model Parameters (Continued)

Parameter	Unit	Default	Description
VGS	V	1.2	Effective substrate bandgap voltage
VPTS	V	1e+20	SC punch-through voltage
ZETABET	-	5	Exponent coefficient in BE junction current temperature dependence
ZETACI	-	0	Temperature coefficient of epi-collector diffusivity
ZETACT	-	4.5	Exponent coefficient in transfer current temperature dependence
ZETARBI	-	0	Temperature coefficient of internal base resistance
ZETARBX	-	0	Temperature coefficient of external base resistance
ZETARCX	-	0	Temperature coefficient of external collector resistance
ZETARE	-	0	Temperature coefficient of emitter resistance
ZS	-	0.3	External SC exponent factor
VDEDC	V	0.9	BE charge built-in voltage for DC. transfer current
ZEDC	-	0.5	BE charge exponent factor for DC. transfer current
AJEDC	-	2.5	BE capacitance ratio (maximum to zero-bias value) for DC transfer current
ZETAIQF	-	0.0	Temperature coefficient of IQF
ZETARTH	-	0.0	Exponent factor for temperature dependent thermal resistance
ALQF	-	0.167	Factor for additional delay time of minority charge
ALIT	-	0.333	Factor for additional delay time of transfer current
FLNQS	-	0	Flag for turning on and off of vertical NQS effect (set FLNQS=1 to turn on NQS)
TYPE	-	1	Transistor type NPN (TYPE=+1) or PNP (TYPE=-1)

For a complete description of the HICUM/L0 model, see:

http://www.iee.et.tu-dresden.de/iee/eb/comp_mod.html

References

- [1] C. McAndrew, J. Seitchik, D. Bowers, M. Dunn, M. Foisy, I. Getreu, M. McSwain, S. Moinian, J. Parker, D. Roulston, M. Schroter, P. van Wijnen, and L. Wagner, "VBIC95: The vertical bipolar intercompany model," *IEEE Journal of Solid State Circuits*, vol.31, p.1476-1483, 1996.

6

Magnetic Tunnel Junction Model

Describes how to use Magnetic Tunnel Junction Model (MTJ) models in the PrimeSim Continuum circuit simulations.

These topics are discussed in the following sections:

- [Overview of MTJ Model](#)
- [Level1 BSIM BDMC STT-MRAM Model](#)

Overview of MTJ Model

Magnetic tunnel junction (MTJ) is a two-terminal device with an oxide tunnel barrier sandwiched by two ferromagnetic (FM) layers. One of the FM layers is with fixed magnetization (fixed or reference or pinned layer), while the other can be flipped by the input (free layer). The bi-state current characteristics due to the tunneling magnetoresistance (TMR) effect make MTJ device ideal for random-access memory component.

Level1 BSIM BDMC STT-MRAM Model

BDMC STT-MRAM model focuses on Perpendicular MTJ (PMT) whose magnetization of ferromagnetic layers are normal to surface.

- [Using the STT-MRAM Model](#)
- [Model Revisions](#)
- [STT-MRAM Model — General Syntax](#)
- [STT-MRAM Model Parameters and Equations](#)

Using the STT-MRAM Model

To simulate using BDMC STT-MRAM model in the PrimeSim Continuum simulator. Set LEVEL=1 for device type MTJ.

Model Revisions

The first supported version is Version 1.0.

STT-MRAM Model — General Syntax

The syntax for SST-MRAM model is:

```
Xxxx fi fr wer mname [STATE=_val_] [R0=_val_] [AREA=_val_] [TOX=_val_]
+ [TFL=_val_]
+ [WERMOD=_val_]
+ .model mname MTJ level=1
```

- STT-MRAM instance name must start with *x*
- *fi* indicates fixed layer
- *fr* indicates free layer
- *wer* indicates write error. Use *i(wer)* to get the write error rate

Table 116 STT-MRAM Instance Parameters

Parameter	Unit	Default	Min	Max	Description
STATE		0	0	1	Initial state,
RO		2e-6			Resistivity factor
AREA	m^2	3e-15			Area of MTJ
TOX	m	1e-9			Thickness of barrier oxide
TFL	m	2.5e-9			Thickness of free layer
WERMOD		0	0	1	Switch for write error rate
resistor	RB, RBM, RE, RC, IRB				
transit time	ITF, PTF, TF, VT, VTF, XTF				

STT-MRAM Model Parameters and Equations

For the STT-MRAM model parameters and equations, see the STT-MRAM manual released by BDMC.

A

Finding Device Libraries

Lists the device libraries you can use in the PrimeSim Continuum solution.

The PrimeSim Continuum solution ships hundreds of examples for your use; see Listing of Input Demonstration Files in *PrimeSim HSPICE User Guide: Demonstration Netlists* for paths to demo files.

For libraries with multiple models of a specific active or passive device element, you can use the automatic model selector in the PrimeSim Continuum simulator to automatically find the proper model for each transistor size.

This appendix lists device libraries that you can use. It includes the following topics:

- [Overview of Library Listings](#)
- [Analog Device Models](#)
- [Behavioral Device Models](#)
- [Bipolar Transistor Models](#)
- [Diode Models](#)
- [JFET and MESFET Models](#)

Overview of Library Listings

The following sections list the names of the models provided with the PrimeSim Continuum solution. Models are stored by type in directories named after the model type, such as *dio* for diodes and *bjt* for bipolar junction transistors. The directory path is shown for each model type. You specify the directory path in the `.OPTION SEARCH` statement, as in the following example:

```
.OPTION SEARCH '$installdir/96/parts/dio'
```

In the preceding syntax, `$installdir` is the environment variable set to the path to the software installation directory and `96` is the PrimeSim Continuum simulator release number. All model directories are under the `parts` directory.

Analog Device Models

The search path for analog device models is:

```
$installdir/parts/ad
```

Table 117 *Analog Model Names*

AD581	ad581j	ad581k	ad581l	ad581s
ad581t	ad581u	ad584	ad584j	ad584k
ad584l	ad584s	ad584t	ad587	ad587j
ad587k	ad587l	ad587s	ad587t	ad587u
ad600	ad600j	ad602	ad602j	ad620
ad620a	ad620b	ad620s	ad624	ad624a
ad624b	ad624c	ad624s	ad630	ad630a
ad630b	ad630j	ad630k	ad630s	ad633
ad633j	ad645	ad645a	ad645b	ad645j
ad645k	ad645s	ad704	ad704a	ad704b
ad704j	ad704k	ad704t	ad705	ad705a
ad705b	ad705j	ad705k	ad705t	ad706
ad706a	ad706b	ad706j	ad706k	ad706t
ad711	ad711a	ad711b	ad711c	ad711j
ad711k	ad711s	ad711t	ad712	ad712a
ad712b	ad712c	ad712j	ad712k	ad712s
ad712t	ad713	ad713a	ad713b	ad713j
ad713k	ad713s	ad713t	ad734	ad734a
ad734b	ad734s	ad743	ad743a	ad743b
ad743j	ad743k	ad743s	ad744	ad744a
ad744b	ad744c	ad744j	ad744k	ad744s
ad744t	ad745	ad745a	ad745b	ad745j

Table 117 *Analog Model Names (Continued)*

ad745k	ad745s	ad746	ad746a	ad746b
ad746j	ad746s	ad780	ad780a	ad780b
ad780s	ad797	ad797a	ad797b	ad797s
ad810	ad810a	ad810s	ad811	ad812
ad812a	ad813	ad813a	ad817	ad817a
ad818	ad818a	ad820	ad826	ad826a
ad828	ad828a	ad829	ad829a	ad829j
ad829s	ad830	ad830a	ad830j	ad830s
ad840	ad840j	ad840k	ad840s	ad843
ad843a	ad843b	ad843j	ad843k	ad843s
ad844	ad844a	ad844b	ad844s	ad845
ad845a	ad845b	ad845j	ad845k	ad845s
ad846	ad846a	ad846b	ad846s	ad847
ad847a	ad847j	ad847s	ad848	ad848a
ad848j	ad848s	ad9617	ad9618	ad9621
ad9622	ad9623	ad9624	ad9630	adg411
adg411b	adg411t	adg412	adg412b	adg412t
adg413	adg413b	adg413t	amp01	amp02
buf04	mat02	mat03	mat04	mlt04
mlt04g	op160	op160a	op160f	op160g
op176	op176g	op177	op177a	op177b
op177e	op177f	op177g	op20	op200
op200a	op200e	op200f	op200g	op20b
op20c	op20f	op20g	op20h	op21
op213	op215	op215a	op215b	op215c
op215e	op215f	op215g	op21a	op21e

Table 117 *Analog Model Names (Continued)*

op21f	op21g	op21h	op220	op220a
op220c	op220e	op220f	op220g	op221
op221a	op221b	op221c	op221e	op221g
op249	op249a	op249e	op249f	op249g
op260	op27	op275	op275g	op27a
op27b	op27c	op27e	op27f	op27g
op282	op282g	op283	op285	op285g
op290	op290a	op290e	op290f	op290g
op292	op295	op297	op297a	op297e
op297f	op297g	op37	op37a	op37b
op37c	op37e	op37f	op37g	op400
op400a	op400e	op400f	op400g	op400h
op41	op41a	op41b	op41e	op41f
op41g	op42	op420	op420b	op420c
op420f	op420g	op420h	op421	op421b
op421c	op421f	op421g	op421h	op42a
op42e	op42f	op42g	op43	op43a
op43b	op43e	op43f	op43g	op44
op467	op467g	op470	op482	op482g
op490	op490a	op490e	op490f	op490g
op492	op497	op497a	op497b	op497c
op497f	op497g	op61	op64	op77
op77a	op77b	op77e	op77f	op77g
op80	op80b	op80e	op80f	op80g
op90	op90a	op90e	op90f	op90g
op97	op97a	op97e	op97f	pm1012

Table 117 *Analog Model Names (Continued)*

ref01	ref01a	ref01c	ref01e	ref01h
ref02	ref02a	ref02c	ref02d	ref02e
ref02h	ref05	ref05a	ref05b	ref10
ref10a	ref10b	ssm2017	ssm2017p	ssm2131
ssm2210	ssm2220			

Behavioral Device Models

The search path for behavioral device models is:

```
$installdir/parts/behave
```

The required element syntax is:

```
Xyyyyy in- in+ out vcc vee modelname
```

Optional parameters are:

```
vos=value, ibos=value, av=value
```

Table 118 *Behavioral Model Names*

ad4bit	ad8bit	alf155	alf156	alf157
alf255	alf347	alf351	alf353	alf355
alf356	alf357	alf3741	alm101a	alm107
alm108	alm108a	alm111	alm118	alm124
alm124a	alm139a	alm1458	alm1558	alm158
alm158a	alm201a	alm207	alm208	alm208a
alm224	alm258	alm258a	alm2901	alm2902
alm2904	alm301a	alm307	alm308	alm308a
alm318	alm324	alm3302	alm339	alm358
alm358a	alm725	alm741	alm747	alm747c

Table 118 Behavioral Model Names (Continued)

amc1458	amc1536	amc1741	amc1747	ane5534p
anjm4558	anjm4559	anjm4560	aop04	aop07
aop14	aop15b	aop16b	at094cns	atl071c
atl072c	atl074c	atl081c	atl082c	atl084c
atl092cp	atl094cn	aupc1251	aupc358	ga201
rcfilt	tline			

Bipolar Transistor Models

The search path for bipolar transistor models is:

```
$installdir/parts/bjt
```

The required element syntax is:

```
Xyyyy coll base emit modelname
```

Optional parameters are:

```
betaf=value, tauwf=value
```

Table 119 Bipolar Transistor Model Names

t2n1132a	t2n2102	t2n2219a	t2n2222	t2n2222a
t2n2369	t2n2369a	t2n2501	t2n2605	t2n2642
t2n2857	t2n2894	t2n2904	t2n2904a	t2n2905
t2n2905a	t2n2906	t2n2907	t2n2907a	t2n2945a
t2n3013	t2n3227	t2n3250	t2n3250a	t2n3251
t2n3251a	t2n3467	t2n3501	t2n3546	t2n3637
t2n3742	t2n3743	t2n3866	t2n3904	t2n3906
t2n3946	t2n3947	t2n3962	t2n4261	t2n4449
t2n5058	t2n5059	t2n5179	t2n6341	t2n6438

Table 119 Bipolar Transistor Model Names (Continued)

t2n706	t2n708	t2n869	t2n869a	t2n918
t2n930	t2sa1015	t2sa950	t2sa965	t2sa970
t2sc1815	t2sc1923	t2sc2120	t2sc2235	t2sc2669
tmpr6595	tne741	tne901		

Diode Models

The search path for diode models is:

```
$installdir/parts/dio
```

The required element syntax is:

```
Xyyyyy anode cathode modelname
```

Optional parameters are:

```
isat=value, tt=value
```

Table 120 Diode Model Names

d12bg11	d12bh11	d12dg11	d12dh11	d12fg11
d12fh11	d12gg11	d12gh11	d12jg11	d12jh11
d1n3016	d1n3017	d1n3018	d1n3019	d1n3020
d1n3021	d1n3022	d1n3023	d1n3024	d1n3025
d1n3026	d1n3027	d1n3028	d1n3029	d1n3030
d1n3031	d1n3032	d1n3033	d1n3034	d1n3035
d1n3036	d1n3037	d1n3038	d1n3039	d1n3040
d1n3041	d1n3042	d1n3043	d1n3044	d1n3045
d1n3046	d1n3047	d1n3048	d1n3049	d1n3050
d1n3051	d1n3821	d1n3822	d1n3823	d1n3824
d1n3825	d1n3826	d1n3827	d1n3828	d1n3829

Table 120 Diode Model Names (Continued)

d1n3830	d1n4001	d1n4002	d1n4003	d1n4004
d1n4005	d1n4006	d1n4007	d1n4148	d1n4149
d1n4150	d1n4370	d1n4371	d1n4372	d1n4446
d1n4447	d1n4448	d1n4449	d1n4728	d1n4729
d1n4730	d1n4731	d1n4732	d1n4733	d1n4734
d1n4735	d1n4736	d1n4737	d1n4738	d1n4739
d1n4740	d1n4741	d1n4742	d1n4743	d1n4744
d1n4745	d1n4746	d1n4747	d1n4748	d1n4749
d1n4750	d1n4751	d1n4752	d1n4753	d1n4754
d1n4755	d1n4756	d1n4757	d1n4758	d1n4759
d1n4760	d1n4761	d1n4762	d1n4763	d1n4764
d1n5221	d1n5222	d1n5223	d1n5224	d1n5225
d1n5226	d1n5227	d1n5228	d1n5229	d1n5230
d1n5231	d1n5232	d1n5233	d1n5234	d1n5235
d1n5236	d1n5237	d1n5238	d1n5239	d1n5240
d1n5241	d1n5242	d1n5243	d1n5244	d1n5245
d1n5246	d1n5247	d1n5248	d1n5249	d1n5250
d1n5251	d1n5252	d1n5253	d1n5254	d1n5255
d1n5256	d1n5257	d1n5258	d1n5259	d1n5260
d1n5261	d1n5262	d1n5263	d1n5264	d1n5265
d1n5266	d1n5267	d1n5268	d1n5269	d1n5270
d1n5271	d1n5272	d1n5333	d1n5334	d1n5335
d1n5336	d1n5337	d1n5338	d1n5339	d1n5340
d1n5341	d1n5342	d1n5343	d1n5344	d1n5345
d1n5346	d1n5347	d1n5348	d1n5349	d1n5350
d1n5351	d1n5352	d1n5353	d1n5354	d1n5355

Table 120 Diode Model Names (Continued)

d1n5356	d1n5357	d1n5358	d1n5359	d1n5360
d1n5361	d1n5362	d1n5363	d1n5364	d1n5365
d1n5366	d1n5367	d1n5368	d1n5369	d1n5370
d1n5371	d1n5372	d1n5373	d1n5374	d1n5375
d1n5376	d1n5377	d1n5378	d1n5379	d1n5380
d1n5381	d1n5382	d1n5383	d1n5384	d1n5385
d1n5386	d1n5387	d1n5388	d1n5817	d1n5818
d1n5819	d1n5913	d1n5914	d1n5915	d1n5916
d1n5917	d1n5918	d1n5919	d1n5920	d1n5921
d1n5922	d1n5923	d1n5924	d1n5925	d1n5926
d1n5927	d1n5928	d1n5929	d1n5930	d1n5931
d1n5932	d1n5933	d1n5934	d1n5935	d1n5936
d1n5937	d1n5938	d1n5939	d1n5940	d1n5941
d1n5942	d1n5943	d1n5944	d1n5945	d1n5946
d1n5947	d1n5948	d1n5949	d1n5950	d1n5951
d1n5952	d1n5953	d1n5954	d1n5955	d1n5956
d1n746	d1n747	d1n748	d1n749	d1n750
d1n751	d1n752	d1n753	d1n754	d1n755
d1n756	d1n757	d1n758	d1n759	d1n914
d1n957	d1n958	d1n959	d1n960	d1n961
d1n962	d1n963	d1n964	d1n965	d1n966
d1n967	d1n968	d1n969	d1n970	d1n971
d1n972	d1n973	d1n974	d1n975	d1n976
d1n977	d1n978	d1n979	d1n980	d1n981
d1n982	d1n983	d1n984	d1n985	d1n986
d1s1585	d1s1586	d1s1587	d1s1588	d1sv147

Table 120 Diode Model Names (Continued)

d1sv149	dmbr115p	dmbr120p	dmbr130p	dmbr140p
dsk4a3				

JFET and MESFET Models

The search path for JFET and MESFET models is:

```
$installdir/parts/fet
```

The required element syntax is:

```
Xyyyyy drain gate source modelname
```

Optional parameters are:

```
vt=value, betaf=value
```

Table 121 FET Model Names

J108	J109	J110	J111	J112
j113	j2n3330	j2n3460	j2n3824	j2n4391
j2n4392	j2n4393	j2n4856	j2n4857	j2n5457
j2n5458	j2n5459	j2n5460	j2n5461	j2n5462
j2n5463	j2n5465	j309	j511	j557
jsj74	jsk170	m2n6755	m2n6756	m2n6757
m2n6758	m2n6759	m2n6760	m2n6761	m2n6762
m2n6763	m2n6764	m2n6765	m2n6766	m2n6767
m2n6768	m2n6769	m2n6770	m2n6787	m2n6788
m2n6789	m2n6790	m2n6791	m2n6792	m2n6793
m2n6794	m2n6795	m2n6796	m2n6797	m2n6798
m2n6799	m2n6800	m2n6801	m2n6802	mbuz10
mbuz20	mbuz23	mbuz24	mbuz32	mbuz35

Table 121 FET Model Names (Continued)

mbuz36	mbuz42	mbuz45	mbuz46	mbuz60
mbuz63	mbuz64	mbuz71	mbuz72a	mbuz74
mbuz76	mirf120	mirf121	mirf122	mirf123
mirf130	mirf131	mirf132	mirf133	mirf140
mirf141	mirf142	mirf143	mirf150	mirf151
mirf152	mirf153	mirf220	mirf221	mirf222
mirf223	mirf230	mirf231	mirf232	mirf233
mirf240	mirf241	mirf242	mirf243	mirf250
mirf251	mirf252	mirf253	mirf320	mirf321
mirf322	mirf323	mirf330	mirf331	mirf332
mirf333	mirf340	mirf341	mirf342	mirf343
mirf350	mirf351	mirf352	mirf353	mirf420
mirf421	mirf422	mirf423	mirf430	mirf431
mirf432	mirf433	mirf440	mirf441	mirf442
mirf443	mirf450	mirf451	mirf452	mirf453
mirf510	mirf511	mirf512	mirf513	mirf520
mirf521	mirf522	mirf523	mirf530	mirf531
mirf532	mirf533	mirf540	mirf541	mirf542
mirf543	mirf610	mirf611	mirf612	mirf613
mirf620	mirf621	mirf622	mirf623	mirf630
mirf631	mirf632	mirf633	mirf640	mirf641
mirf642	mirf643	mirf710	mirf711	mirf712
mirf713	mirf720	mirf721	mirf722	mirf723
mirf730	mirf731	mirf732	mirf733	mirf740
mirf741	mirf742	mirf743	mirf810	mirf811
mirf812	mirf813	mirf820	mirf821	mirf822

Table 121 *FET Model Names (Continued)*

mirf823	mirf830	mirf831	mirf832	mirf833
mirf840	mirf841	mirf842	mirf843	mirf9020
mirff110	mirff111	mirff112	mirff113	mirff120
mirff121	mirff122	mirff123	mirff130	mirff131
mirff132	mirff133	mirff210	mirff211	mirff212
mirff213	mirff220	mirff221	mirff222	mirff223
mirff230	mirff231	mirff232	mirff233	mirff310
mirff311	mirff312	mirff313	mirff320	mirff321
mirff322	mirff323	mirff330	mirff331	mirff332
mirff333	mirff430	mirff431	mirff432	mirff433

Transmission Line Models

[Table 122](#) lists the Transmission Line models that you can use in the PrimeSim Continuum simulator.

Search path: `$installdir/parts/tline`

Table 122 *Transmission Line Model Names*

rcfilt	rg11_u	rg11a_u	rg15_u	rg180b_u
rg188a_u	rg53_u	rg54a_u	rg58a_u	rg58c_u
rg59b_u	rg62_u	rg62b_u	rg71_u	rg71b_u
rg9_u	rg9b_u	tw_sh_u	tw_un_u	